



Transportation Management (TM)

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Transportation Management (TM)

Use

Transportation Management (TM) supports you in all activities connected with the physical transportation of goods from one location to another. You can use TM to perform the following activities, for example:

- Work with TM-relevant orders and deliveries
- Create forwarding orders for your ordering parties
- Create freight units
- Create delivery proposals
- Create freight orders and freight bookings (manually or automatically using transportation planning)
- Perform carrier selection
- Create agreements
- Tender transportation services
- Dispatch and monitor the transportation
- Calculate the transportation charges for both the ordering party and the supplier side
- Consider foreign trade and dangerous goods regulations
- Use integrated analytics functions and KPIs for real-time performance visibility

You can use TM to create and monitor an efficient transportation plan that fulfills the relevant constraints (for example, service level, costs, and resource availability). You can determine options to save costs and to optimize the use of available resources. You can react to transportation events and find solutions to possible deviations from the original transportation plan.

Creating an efficient transportation plan and optimizing the use of available resources also enables you to reduce greenhouse gas (GHG) emissions. TM also allows you to specify GHG emissions for your trucks and take them into account during planning. For more information, see [Sustainability](#).

Integration

TM is integrated with the following applications and services:

Application/Service	Integration Type
Extended Warehouse Management (EWM)	In addition to the integration with Extended Warehouse Management (EWM), you can integrate TM with SAP Extended Warehouse Management (SAP EWM). For more information, see Integration with Extended Warehouse Management .
Advanced Shipping and Receiving	You can use this integrated process to send, receive, and transport products in an end-to-end process. Information moves freely between the components EWM, Stock Room Management, Inventory Management (MM-IM), and Logistics Execution (LE). For more information, see Advanced Shipping and Receiving .

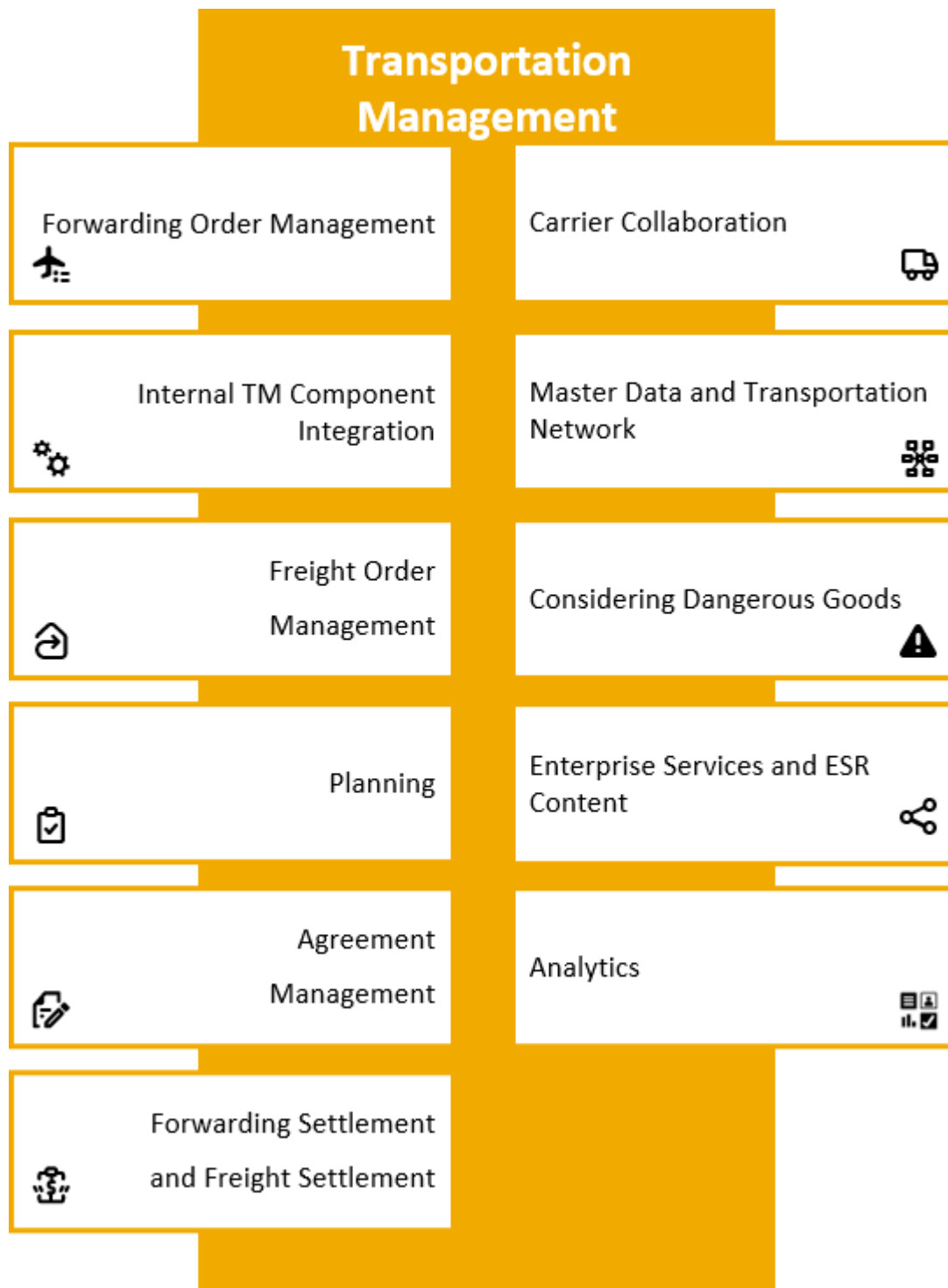
Application/Service	Integration Type
SAP Event Management	You can use event tracking to monitor the execution status of your transportation.
SAP Global Trade Services	You can perform customs processing for your TM business documents in a connected global trade services (GTS) system. For example, you can request export declarations for freight orders or freight bookings. The data is transferred using enterprise services.
Document Information Extraction	You can use this service to extract data from delivery documents attached to a consignment order. For more information, see AI-Assisted Goods Receipt Analysis .

Note also the following:

- You use PS&S Services in TM to ensure the safe transportation of dangerous goods in accordance with legal regulations (see [Considering Dangerous Goods](#)).
- You can integrate TM with external systems by using enterprise services. For example, you can automatically create and confirm forwarding orders based on data received from your ordering parties.

Features

The following interactive graphic shows the subcomponents or functions of TM. Hover over the graphic and click highlighted areas for more information.



Please note that image maps are not interactive in PDF outputs.

Introductory Information

Before you start working with Transportation Management (TM), we recommend that you familiarize yourself with the following important information.

Working with Web Applications

The SAP Fiori launchpad is the access point for all web-based applications (apps) for TM. Apps are displayed as tiles, and you can launch each app by clicking the corresponding tile on the launchpad. The tiles displayed on your launchpad home page are arranged into groups, with the group title displayed at the top of each group.

The set of tiles initially visible to you on the launchpad depends on the business catalogs assigned to the business roles that your business user is assigned to. The business catalogs and the corresponding tile groups are delivered as part of SAP S/4HANA.

In the Fiori launchpad frontend system, you can use the roles SAP_BR_TRANSPORTATION_MANAGER, SAP_BR_TRANSPORTATION_PLANNER, SAP_BR_FRGHT_CTRC_SPECIALIST, and SAP_BR_FRGHT_SETTLEMENT_CLERK as templates. These contain references to the Fiori tile catalogs that are listed in the following table. This enables you to access all applications that are relevant for TM.

Business Catalogs for Transportation Manager Role

Business Catalog	Technical Name
Transportation – Order Management	SAP_TM_BC_OM
Transportation – Planning	SAP_TM_BC_PLN
Transportation – Settlement	SAP_TM_BC_STLM
Transportation – Contract Management	SAP_TM_BC_CTRM
Transportation – Charge Management	SAP_TM_BC_CHGM
Transportation – Master Data	SAP_TM_BC_MD
Transportation – Profiles and Settings	SAP_TM_BC_PRFS
Settlement Management - Manage Freight Cost Allocation Documents	SAP_SM_BC_FCADOC_MNG
Settlement Management - Monitor Freight Cost Allocation Document Items	SAP_SM_BC_FCADOC_ITM_MON
Settlement Management - Freight Cost Allocation Document Release Scheduling	SAP_SM_BC_FCADOC_REL_JOB
Sales – Billing Document Display	SAP_SD_BC_BIL_DOC_DISPL
Sales – Billing Document	SAP_SD_BC_BIL_DOC
Sales – Sales Order Display	SAP_SD_BC_SO_DISPL
Sales – Sales Order Processing	SAP_SD_BC_SO_PROC_OP
Purchasing – Purchase Order Processing	SAP_PRC_BC_PURCHASER_PO
Purchasing – Contract Management	SAP_PRC_BC_PURCHASER_PUC
Logistics Execution – Inbound Deliveries Display	SAP_LE_BC_IBDLV_DISPLAY
Logistic Execution – Inbound Delivery Processing	SAP_LE_BC_ID_PROC
Logistic Execution – Outbound Deliveries Display	SAP_LE_BC_OBDLV_DISPLAY
Logistic Execution – Outbound Delivery Processing	SAP_LE_BC_OD_PROC
Accounts Payable – Supplier Invoice Processing	SAP_PRC_BC_INVOICER
Project Management – Commercial Project Management	SAP_CPA_BC_COMMPRJ_MANAGE
Asset Accounting – Master Data	SAP_SFIN_BC_AA_MASTER_DATA

You can rearrange the tiles within a group by dragging and dropping them. By using [Edit Home Page](#), you can add tiles, remove tiles, or relocate tiles between groups. We recommend to create own roles, referencing only the necessary catalogues required by the various groups of business users.

If you want to use apps on the launchpad, you need to set up a role in role maintenance. To do so, on the [SAP Easy Access](#) screen, choose [Tools](#) > [Administration](#) > [User Maintenance](#) > [Role Administration](#) > [Roles](#).

i Note

- FPM BOPF Integration (FBI) offers configurable and codeless integration of Floor Plan Manager (FPM) technology and BOPF-based business objects (BOPF = Business Objects Processing Framework).
- If you press the [Refresh](#) button on the user interface of a business document, Customizing data and master data is loaded from the buffer and not read from the database. That means, changes made to the database are not taken into account. However, if you use the browser refresh or reload, the system reopens the business document and reads Customizing data and master data from the database.

For more information about SAP Fiori launchpad, see [SAP Fiori Launchpad](#).

Delivery Customizing

If you want to use delivery Customizing for TM, see SAP Note [2516155](#) .

Displaying Application Help

For displaying context-sensitive help in WebDynpro apps and in Fiori apps, select the upper question mark in the upper right corner.

For displaying the application help on the internet, we recommend to use the quick link https://help.sap.com/s4_op_tm.

Displaying Field Help

To view field help in the SAP ABAP system, choose the field and press F1.

For displaying in-app help in WebDynpro apps and in Fiori apps, select the upper question mark in the upper right corner.

To view a screen area explanation in an app, right-click anywhere on the screen and choose [Display Quick Help](#).

Described Feature Scope of the Product Assistance

For some of the features that are described in this product assistance for TM, you need a separate licence. For more information, see the Feature Scope Description for SAP S/4HANA and the following SAP Note: [3065464](#) .

Furthermore, some features that are described in this product assistance are currently not released. For more information, see the notes in the corresponding chapters of this product assistance and the following SAP Note: <https://me.sap.com/notes/> .

Master Data

You use this component to manage master data that is used in a variety of business transactions in Transportation Management (TM). The following master data objects are part of this component:

- Business partner (see [Business Partner](#))
- Organizational management (see [Organizational Management](#))

- Transportation network (see [Transportation Network](#))
- Transportation mode (see [Transportation Mode](#))
- Product (see [Product](#))
- Carrier profile (see [Carrier Profile](#))
- Equipment (see [Equipment Groups and Equipment Types](#))
- Resources (see [Definition of Resources](#))
- Transportation charges (see [Setup of MD for Charge Management and SP Catalogs](#))

i Note

Before you define any master data objects, you **must** create an active model in Customizing. For more information, see Customizing for [Transportation Management](#) under ► [Master Data](#) ► [Create Active Version and Model](#) ►.

The following functions that are described below are currently not released:

- Enterprise Asset Management Integration

Equipment and Functional Location data are not integrated with the TM vehicle resource.

Business Partner

Definition

A person, organization, group of persons, or group of organizations in which a company has a business interest.

Use

You use this business object for a variety of business transactions.

You create and manage your business partners centrally along with the roles they assume for your company. To do so, you define the general data of the business partner once and you assign business partner roles to the business partner. Specific data is stored for each business partner role. In this way, you don't store redundant data since the general data is independent of a business partner's function or of application-specific extensions.

When you first create a business partner in the system, the business partner role **General Business Partner** is automatically assigned to the business partner.

Structure

You identify each business partner with a unique business partner number.

You create and manage the specific data for a business partner role once you've assigned a specific role to the business partner.

You create and manage the following general elements of a business partner:

General Data

Here, you enter data (or the system displays data) that is valid for all business partner roles. This data includes, for example, the name, address, communication data, identification numbers, and business partner status.

TM-Specific Data

The following business partner roles are most commonly used in TM:

- 000000 Business Partner (General)
- FLCU01 Customer
- FLCU00 FI Customer
- FLVN01 Vendor
- FLVN00 FI Vendor
- CRM010 Carrier
- SLLCOF Global Trade Services: Customs Office
- TM0001 Driver
- BUP004 Organizational Unit
- BUP001 Contact Person
- BUP002 Prospect
- BUP003 Employee
- BUP005 Internet User

The role CRM010 (Carrier) contains the data of role FLVN01 (Vendor) plus additional TM specific data such as:

- Regulated Agent Code
- IATA Agent Code
- CASS Account
- SCAC Codes
- Airline Codes
- Service Level Codes
- Carrier Service Codes

The role FLVN01 (Vendor) and the role CRM010 (Carrier) allow maintenance of data on the level of a purchasing organization.

The roles FLCU001 (FI Customer) and FLVN00 (FI Vendor) allow maintenance of data on the level of a company code.

i Note

FLCU01 contains data that can be maintained on sales area level (a sales area is a combination of sales organization, distribution channel, and division.)

As TM documents such as forwarding orders and forwarding quotations don't contain distribution channel and division, you need to maintain a default value for distribution channel and division in Customizing for **Transportation Management** under **Master Data** > **Basic Functions** > **General Settings** > **Define General Settings for TM** . These default values are used to

access business partner master data that are stored on sales area level by adding these as defaults for the sales organization defined for TM.

i Note

In TM you need to create organizational units to represent sales organizations, purchasing organizations and company codes in TM.

For more information, see [Organizational Management](#).

Integration

In an integrated system landscape (side-by-side scenario), you can replicate master data from a source system, such as an ECC system or an SAP S/4HANA system, to the Transportation Management (TM) component of an SAP S/4HANA system (target system). To replicate business partner master data, you set up a replication model in your source system using the DRF customizing transaction DRFIMG and define the target system as part of the replication model.

In business transaction documents, such as a forwarding order, the business partner master data object appears as a party that takes part in transportation processes. The party assumes partner functions, such as ship-to party, shipper, or sold-to party.

i Note

You don't need to define special business partner roles in Customizing for the partner functions (see [Definition of Business Partners](#)). You can assign any business partner to the different partner functions.

You can configure the system to assign business partner roles to allowed business document partner functions.

For all business documents that you work with in TM:

- The system checks whether a business partner is compatible with a business document partner function. If you enter a business partner with a particular partner function in a business document, the system verifies whether that role is allowed for that business partner.
- You can activate the checking of the organizational data of a business partner's company or the sales or purchasing organizational data. These checks ensure that all business partners with the carrier role or customer role involved in certain business documents are assigned to one or both of the following:
 - To a sales organization or purchasing organization in the business partner's company
 - To the sales organization or purchasing organization in the header of the business document

These checks are relevant for the following business documents:

- Forwarding orders, forwarding agreements, and forwarding settlement documents
- Freight orders, freight bookings, freight agreements, and freight settlement documents

You activate these checks in Customizing for [Transportation Management](#) under [Basic Functions](#) > [General Settings](#) > [Define General Settings for TM](#) .

- The search help for business partners reflects this configuration of business partners for the allowed business document partner functions by only returning business partners in the result list that can be used for a specific business document partner function.

More Information

Definition of Business Partners

Use

You use this process to define business partners (BPs) in Transportation Management (TM).

Process

1. You define a BP in the the following apps, which you open from the launchpad:

- [Create Customer \(Sales\)](#)
- [Create Vendor \(Purchasing\)](#)
- [Create Forwarding Agent](#)

2. You assign additional BP roles to the BP.

TM delivers the following BP role:

- TM0001 - Driver

The assigned business partner role category [TM0001 - Driver](#) allows business partner categories Organization and Person

3. You can create relationships between BPs.

i Note

The relationship category [Has Contact Person](#) is used in the tendering process.

The relationship category [Has Subsidiary / Agent](#) is used to describe subsidiaries and agents of a carrier organization.

Additional Texts

Use

You can define additional texts for the business partner master and use them as printing addresses or signatures on different documents.

Prerequisites

You have defined additional text types in Customizing for [Transportation Management](#) under [►Master Data►Business Partners►Define Text Types for Business Partner Master►](#).

Features

You can enter additional texts for the business partner role [Business Partner \(Gen.\)](#) on the [Maintain Business Partner](#) user interface (transaction BP) on the [Additional Texts](#) tab page. Texts from this tab page (for example, addresses or signatures) are used by Transportation Management (TM) as printing addresses or signatures on several printing documents such as air waybills.

To ensure that the texts have the correct dimensions for the designated areas, you specify text types with maximum numbers of characters and lines. To support printing in several languages, you can define the language of the texts and enter additional texts for one text type in different languages for one business partner instance.

i **Note**

You can also use additional texts to define formatted printing addresses or paragraphs for the related business partner and edit them so that they are letter of credit-compliant. To use this function, you must define the default text type for printing addresses in Customizing for [Transportation Management](#) under [Master Data > Business Partners > Assign BP Text Types to Partner Functions](#) .

Business Partner Blocking

Use

With this function, you can block the assignment of a business partner to a forwarding quotation or to a forwarding order. Additionally, you can block the business partner, so that within a forwarding order the system prevents forwarding settlement documents from being created.

The blocking options are structured hierarchically in the business partner master record. You can set the blocks at the following levels:

- Complete business partner
- Company code
- Purchasing organization
- Sales organization

Features

You can block the business partner manually.

You can set the blocks in TM by opening one of the following apps from the launchpad:

- [Create Customer \(Sales\)](#)
- [Create Vendor \(Purchasing\)](#)
- [Create Forwarding Agent](#)

Block	Role	Description	Document	Tab Page
Posting Block	FLCU00	• Blocks the assignment of business partners to a forwarding order document or a forwarding quotation document	Forwarding Order System checks the following: • Sold-to Party • Prepaid Agreement Party	Choose the Company Code button then see the Customer: Status tab.

Block	Role	Description	Document	Tab Page
		- Blocks the creation of forwarding settlement documents - Blocks the creation of freight orders	Collect Agreement Party	
Posting Block	FLVN00	- Blocks the assignment of business partners to a forwarding order document or a forwarding quotation document - Blocks the creation of forwarding settlement documents	Forwarding Order System checks all partner functions.	Choose the Company Code button then see the Vendor: Status tab.
Sales Order Block	FLCU01	- Blocks the assignment of business partners to forwarding order documents or forwarding quotation documents - Blocks the creation of forwarding settlement documents of the specific sales organization	Forwarding Order System checks the following: - Sold-to Party - Prepaid Agreement Party - Collect Agreement Party	Choose the Sales and Distribution button then see the Customer: Status tab.
Billing Block	FLCU01	Blocks the creation of forwarding settlement documents of the specific sales organization	Forwarding Settlement Document	Choose the Sales and Distribution button then see the Customer: Status tab.
Purchasing Block	FLVN01	Blocks the creation of freight orders	Freight order, Freight Booking System checks all partner functions.	Choose the Purchasing button then see the Purchasing Data tab.

Activities

On the corresponding tab page, select the check box with the block you want to set or, in case of specific sales organizations, enter block reason codes.

Example

You would like to block the business partner **Kulinarisch Weltweit**, located in Cologne. To do so, on the **Customer Company Org Data** tab page, you select the **Central Posting Block** check box. From now on, the system blocks all sales organizations of the respective business partner and you cannot assign the business partner to a forwarding order.

If a forwarding order document already exists in the system when you set the block, you cannot create forwarding settlement documents. This applies to the following blocks:

- Central billing block
- Central order block
- Central posting block

Business Partner Determination

Use

You can enable the system to determine the business partners that you use in different business document types. You can also enable the system to automatically determine business partners for forwarding orders with specific Incoterms. This allows you easily and efficiently enter business partners in a business document. It cuts down the chance of manual errors and reduces the need to enter a business partner for each partner function.

In the **Define Partner Determination Profiles** Customizing activity, you can create a partner determination profile that the system uses to automatically determine the following features:

- The partner functions available in a business document
You specify the list of partner functions that the system makes available in a business document.
- The level of control the user has in entering the partner function
Depending on the settings you define in Customizing, you change or delete a partner function, or add a partner function from a list of available partner functions. You can also specify settings so that you cannot change or delete specific partner functions.
- How the system determines business partners for the partner functions

The system uses one of the following relationships that you specify in Customizing, to determine a business partner:

- Partner function and source partner function
In Customizing, you can specify a source partner function for a particular partner function. In the application, when you enter a business partner in the field that you have specified as the source partner function, the system automatically copies the business partner to the associated partner function.

For example, in Customizing, you specify the partner function as **Invoicing Party**, the source type as **Partner Function**, and the source function as **Sold-to Party**. In the application, you specify the sold-to party as **Customer01**. The system assigns Customer01 as the business partner for the invoicing party.

- Partner function and the source partner function in business partner master data

The system uses the settings that you specify on the **Partner Function** tab page in the **BP – Maintain Business Partner** transaction to determine a business partner in a business document.

For example, you are a logistics service provider (LSP), and you have a business partner APP_1. In the **BP – Maintain Business Partner** transaction, you specify APP_1 with the BP role of FLCU01 (Customer). On the **Partner Function** tab page of the BP transaction for APP_1, you specify an organizational unit and a partner function for the business partner APP_INV. APP_INV fulfills the role of invoicing party for APP_1. In Customizing, you specify the partner function as Invoicing Party, the source type as BP Relationship, and the source role as Sold-to Party.

In a forwarding order, you specify the organization as 561234 on the **General Data** tab page and the sold-to party as APP_1 on the **Business Partner** tab page. The system checks the business partner master data for APP_1. On the **Partner Function** tab page, it finds that APP_1 uses APP_INV as the invoicing party for organization 561234. It assigns APP_INV as the business partner for the invoicing party.

- Partner function and business partner

In Customizing, you directly assign a business partner to a partner function. In the application, the system automatically assigns this business partner to the partner function.

For example, in Customizing, you specify the partner function as Invoicing Party, the source type as Business Partner, and the business partner as CSI_CARR1. The system automatically assigns CSI_CARR1 as the business partner for the invoicing party in the application.

Note that you can enter a business partner that does not exist in the current system. This enables you to assign a business partner to a partner function in one system and transport it to another system where the business partner exists. For example, in a Customizing system, you have not defined any business partner master data. You have assigned business partner LUF to a partner function of carrier. You transport this data to a production system where LUF is defined. In a business document of the production system, the system determines that the business partner for the carrier is LUF.

- Relationship category

The system can determine the business partner for a partner function based on the partner relationship you specify in the **Maintain Business Partner** transaction (transaction BP).

To do so, you must specify the **Source Type** as **Relationship Category** in the Customizing for **Transportation Management** under **Master Data** **Business Partners** **Define Partner Determination Profiles**. The system determines the corresponding business partner based on the partner relationship you specify in the **Maintain Business Partner** transaction (transaction BP) (such as **Has Invoicing Party**, **Has Contact Person**, and so on).

You can also choose to have a reverse relationship between the partner functions by selecting the **Rev. Rel.** checkbox in the Customizing.

For example, in the **Maintain Business Partner** transaction (transaction BP), you have performed the following:

- Created two business partners INV and SHP.
- Defined a relationship that SHP has the invoicing party INV

You have performed the following in Customizing:

- Specified the partner function as RS (**Invoicing Party**)
- Specified the source type as **Relationship Category**
- Specified the source function as U6 (**Shipper**)
- Specified the relationship category as BBP002 (**Has Invoicing Party**)

- Selected the [Rev. Rel.](#) checkbox

The system considers the following during business partner determination in a business document such as a forwarding order or a freight order:

- Direct relationship - Shipper **SHP** has invoicing party **INV**.
- Reverse relationship - Invoicing party **INV** is invoicing party of shipper **SHP**.

In Customizing, you can assign different profiles to different types of business documents. For example, you can specify a profile that controls how the system determines business partners in forwarding orders, and another profile for business partner determination in forwarding settlement documents.

In Customizing, you can assign a profile to the following business document types:

- Forwarding orders
- Forwarding quotations
- Forwarding settlement documents
- Freight orders
- Freight bookings
- Freight settlement documents
- Freight agreement RFQs
- Forwarding agreement quotations
- Freight agreements
- Forwarding agreements
- Internal agreements
- Mutual agreements

When you use a particular forwarding order type to create a forwarding order and enter an Incoterm, the system uses the information you specify in the [Assign Partner Determination Profiles Based on Incoterms](#) Customizing activity to determine the business partner. It overwrites the partner determination profile you have assigned to the forwarding order type in the [Define Forwarding Order Types](#) Customizing activity.

Prerequisites

You have made the settings in Customizing to create a partner determination profile that the system uses to automatically determine business partners. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) [Business Partners](#) [Define Partner Determination Profiles](#).

If you want to automatically determine business partners in a forwarding order with a specific Incoterm, you have assigned the partner determination profile to forwarding order type and Incoterm combination in Customizing. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) [Business Partners](#) [Assign Partner Determination Profiles Based on Incoterms](#).

You have created relationships between organizations and business partners in the transaction [BP – Maintain Business Partner](#) under [SAP Menu](#) [Transportation Management](#) [Master Data](#).

If you want to assign a particular business partner to a particular partner function in a business document, you have performed the following actions:

- Entered Business Add-In in the **Src Type** (Source Type) field in the **Define Partner Determination Profiles** Customizing activity
- Implemented the Business Add-In **BAdI: Specific Partner Assignment** in Customizing for **Transportation Management** under **Business Add-Ins for Transportation Management** > **Master Data** > **Business Partners** >

More Information

[Settings for the Forwarding Order](#)

[Creation and Editing of a Forwarding Quotation](#)

[Forwarding Settlement](#)

[Creation and Editing of Freight Orders](#)

[Creation and Editing of Freight Bookings](#)

[Freight Settlement](#)

Definition of Carrier Service Codes

Use

A carrier can offer services to shippers or logistics service providers (LSP). The shippers or LSPs contact the carrier and obtain the list of services. Subsequently, the shippers or LSPs manually enter these services in the master data of the system to use it in a freight document with the carrier.

You can enter the services that a carrier offers to a shipper or LSP in the **Maintain Business Partner** transaction (transaction BP). When you enter the carrier in a freight agreement, the system makes these services available in the document.

Activities

In the **Maintain Business Partner** transaction, perform the following steps to enter the services:

1. Enter the business partner.
2. Change **BP role** to **Carrier**.
3. In the **Carrier Service Codes** screen area of the **Transportation Data** tab page, enter the carrier service code and the service type.

You can define service types in Customizing for **Transportation Management** under **Basic Functions** > **General Settings** > **Define Service Types** >

Example

In the **Maintain Business Partner** transaction, you enter carrier service codes and service types for services **Fumigation**, **Loading**, and **Unloading** for a business partner **LUF_1** with a **BP role** of **Carrier**. In a freight agreement, you enter the carrier

LUF_1. The system automatically makes the carrier service codes for Fumigation, Loading, and Unloading available in the input help of the field **Carrier Service Code** in the agreement.

Carrier Profile

Definition

A profile with which you characterize transportation capabilities of a carrier.

Use

You use this object to store information that is used in a similar constellation for multiple carriers. Carriers are business partners (BPs) to which you have assigned the business partner role **Carrier**. You can only create one profile for each carrier.

Structure

You can store the following objects in a carrier profile:

- Freight code sets
- Transportation lane and carrier-specific parameters
- Product freight groups
- Transportation groups
- Equipment groups and equipment types
- Fixed transportation costs
- Dimension costs

Integration

You make use of the carrier profile when creating transportation proposals (see [Generation of Transportation Proposals](#)) and when selecting a carrier.

Definition of Carrier Profiles

Use

You use this process to define a profile for a carrier.

Prerequisites

- You have assigned the business partner role **Carrier** to the business partner for which you want to create a carrier profile (see [Definition of Business Partners](#)).
- You have defined transportation lanes in Customizing (see [Definition of Transportation Lanes](#)).
- You have defined freight code sets in Customizing. For more information, see Customizing for **Transportation Management** under ► **Master Data** ► **Transportation Lane** ► **Carrier Profile** ► **Define Freight Code Sets, Freight Codes, and**

Determination .

- You have defined product freight groups in Customizing. For more information, see Customizing for **Transportation Management** under **Master Data > Transportation Lane > Carrier Profile > Define Product Freight Groups** .
- You have defined transportation groups in Customizing. For more information, see Customizing for **Transportation Management** under **Master Data > Product > Maintain Transportation Group** .
- You have defined equipment groups and equipment types in Customizing. For more information, see Customizing for **Transportation Management** under **Master Data > General Settings > Define Equipment Groups and Equipment Types** .
- You have defined means of transport in Customizing. For more information, see Customizing for **Transportation Management** under **Master Data > Resources > Define Means of Transport** .

Process

1. On the SAP Easy Access screen, choose **Master Data > General > Define Carrier Profile**  or, from the launchpad, open the **Define Carrier Profile** app.
2. You choose the carrier for which you want to define a profile and save your entry.

You can set the following attributes:

- Performance
- **Cont. Move** (Continuous Move) checkbox

3. You make settings on the following tab pages:

- **Transportation Lane**

You can enter any attributes that are relevant for the carrier, such as the business share or the priority.

- **Freight Code Set**

You must specify the country or region and the means of transport to which the freight code set relates.

- **Product Freight Group**

- **Transportation Group**

- **Equipment**

You can enter information about the location of the equipment or the means of transport to which this equipment is restricted.

- **Fixed Transportation Costs**

You can define fixed transportation costs for the means of transport of the carrier.

- **Dimension Costs**

You can define dimension costs for the means of transport of the carrier.

4. Save your entry.

Result

You have defined a profile for a carrier. You can search for settings that you have made in the carrier profile and identify all carriers that have those settings.

Organizational Management

Definition

A means for creating and managing the organizational and staffing structures in your company that uses an organizational model as its basis.

In Transportation Management (TM), the organizational structure can be based on the structure of the ERP components or a connected system (SAP ERP). To enable this, you can create the following organizational units:

- Corporate organization

This organization unit is optional and serves as a single entry point into the organizations structure in TM.

- Company organization

This organization unit corresponds to the ERP company code containing the local currency.

i Note

The company organization unit is used by transportation charge management (TCM) for invoicing and charging.

- Sales organization, office, and group

A sales organization in TM is represented as an organizational unit maintained within the organizational management application.

A sales organization is an organization unit that has function "4" (Sales) and role "1" (Organization).

For each sales organization that you maintain in organizational management, you need to assign a BSG Org. Unit. This is the ID of a sales organization that is used to retrieve sales organization-dependent business partner data and is maintained in Customizing.

- Purchase organization and group

A purchasing organization in TM is represented as an organizational unit maintained within the organizational management application.

A purchasing organization is an organizational unit that has function "2" (Purchasing) and role "1" (Organization).

For each purchasing organization that you maintain in organizational management, you need to assign a BSG Org. Unit. This is the ID of a purchasing organization that is used to retrieve purchasing organization-dependent business partner data and is maintained in Customizing.

- Forwarding house

A forwarding house in TM is represented as an organizational unit maintained within the organizational management application.

A forwarding house is an organizational unit that has function "89" (Forwarding House) and role "1" (Organization).

In TM, you can use a forwarding house as a sales organization and as a purchasing organization.

For each forwarding house that you maintain in organizational management, you need to assign a BSG Org. Unit.

This organizational unit is the ID of both a purchasing organization and a sales organization. Both are maintained in Customizing. This means that the ID of the sales organization and the purchasing organization are identical.

If the forwarding house is used as a sales organization, the BSG Org. Unit ID that is assigned to it is used to retrieve sales organization-dependent business partner data.

If the forwarding house is used as a purchasing organization, the BSG Org. Unit ID that is assigned to it is used to retrieve purchasing organization-dependent business partner data.

- Planning and execution organization and group

i Note

You can activate the checking of the organizational data of a business partner's company or the sales or purchasing organizational data. These checks ensure that all business partners with the carrier role or customer role involved in certain business documents are assigned to one or both of the following:

- To a sales organization or purchasing organization in the business partner's company
- To the sales organization or purchasing organization in the header of the business document

These checks are relevant for the following business documents:

- Forwarding orders, forwarding agreements, and forwarding settlement documents
- Freight orders, freight bookings, freight agreements, and freight settlement documents

You activate these checks in Customizing for **Transportation Management** under ►► **Basic Functions** ► **General Settings** ► **Define General Settings for TM** ►.

More Information

For more information, go to <http://help.sap.com/s4hana>, choose the relevant release, and navigate to ►► **Product Assistance** ► **Enterprise Business Applications** ► **Human Resources** ► **Personnel Management (PA)** ►.

Creation of Organizational Models

Use

In Transportation Management (TM), you use this process to create organizational models for the organizations that take part in transportation processes. You can create the organizational models according to a predefined hierarchical structure. You can assign employees who take part in workflow processes to the organizational units.

Process

1. You define an organizational model. To do so, on the SAP Easy Access screen, choose ►► **Customer Service** ► **Service Processing** ► **Environment** ► **Organization** ► **Organizational Plan** ► **Organization and Staffing** ► **PPOCE - Create** ►.
2. You define organizational units (see [Organizational Unit](#)) and assign an appropriate organizational unit function and organizational unit role to them.

→ Recommendation

You can use transaction /SCMTMS/ORG_INT to transfer organizational units from core components of SAP S/4HANA to TM.

3. You relate organizational units to other organizational units according to their organizational unit role and depending on the organizational unit function.

i Note

The hierarchical relationship of organizational units is fixed.

For organizational units with a **Sales** unit function, you can define the unit roles **Organization**, **Office**, and **Group**. Organizational units with unit role **Group** can be assigned to organizational units with a unit role **Office** or **Organization**, and organizational units with a unit role **Office** to organizational units with a unit role **Organization**.

For organizational units with a **Purchasing** or **Planning and Execution** unit function, you can only assign organizational units with a unit role **Group** to organizational units with a unit role **Organization**.

4. You assign employees to organizational units, see [Assigning Employees to Organizational Units](#).

You can also create an TM organization structure. For more information, see Customizing for **Transportation Management** under **► Master Data ► Organizational Management ► Organizational Model ► Create and Merge Organizational Hierarchy ►**.

Example

You want to re-create the structure of a purchasing organization or a planning and execution organization. You proceed as follows:

1. You create an organizational unit with a **Purchasing** or **Planning and Execution** unit function and an **Organization** unit role.
2. You create an organizational unit with a **Purchasing** or **Planning and Execution** unit function and a **Group** unit role.
3. You assign the organizational unit with the **Group** unit role to the organizational unit with the **Organization** unit role.

Organizational Unit

Definition

An object that is used for mapping the organizational structure of a company in an organizational model. Each organizational unit is defined in more detail by its unit function and unit role.

Use

You can use this object as the basis for an organizational model by arranging organizational units hierarchically and therefore mirroring the structure of your company.

i Note

You may only use the **Organization** or **Group** organizational unit roles for organizational units with the **Purchasing** or **Planning and Execution** organizational unit functions.

Integration

You use organizational units with a **Sales** organizational unit function to create the following:

- Forwarding orders
- Forwarding quotations
- Forwarding settlements

You use organizational units with a **Purchasing** organizational unit function to create the following:

- Freight orders
- Freight bookings
- Freight settlements

You use organizational units with an **Execution** organizational unit function to create the following:

- Resources (see [Definition of Resources](#))
- Freight orders

For settlement integration, the system uses the business system group (BSG) assigned to an organization to determine whether the settlement document should be posted to an external system, or to the internal Sales and Distribution (SD) or Materials Management (MM) component of the SAP S/4 HANA system. For example, for posting the forwarding settlement document, the system uses the sales organization specified in the forwarding settlement document to determine the BSG. The BSG indicates the logical system from which the sales organization is replicated in the organization model. If this logical system is different to the current client logical system, the settlement document is posted to the external SD using enterprise services. If the logical system is the same as the current client logical system, the settlement document is posted to the internal SD component.

Assigning Employees to Organizational Units

Prerequisites

- You have defined a business partner (BP) of the type **Person** (see [Definition of Business Partners](#)).
- You have assigned the business partner role (BP role) **Employee** to the BP and maintained any data relevant for this BP role.

i Note

As a minimum, you need to enter the country or region in which this business partner resides.

Context

You use this procedure to assign employees to organizational units in Transportation Management (TM).

Procedure

1. Access the organizational model on the SAP Easy Access screen by entering PPOME (**Change Organization and Staffing**) in the command field.
2. Select an organizational unit and choose the **Create** button.
3. Choose **Incorporates Position** and enter a name and a description in the **Position** input field.
4. In the context menu of the employee symbol, choose **Assign**.

The **Choose Relationship** screen appears.

5. Choose **Holder Central Person** and enter the name of the business partner that you want to assign to the position in the search dialog.

Results

You have assigned an employee to an organizational unit by assigning a business partner with the role **Employee** to the organizational unit.

The system establishes a correlation between the master data objects **Organizational Unit** and **Business Partner**.

Changing of Organizational Models

Use

You can use this process to change an organizational model in Transportation Management (TM).

Process

1. You change an organizational model. To do so, on the SAP Easy Access menu, choose ► **Customer Service** ► **Service Processing** ► **Environment** ► **Organization** ► **Organizational Plan** ► **Organization and Staffing** ► **PPOME - Change** ►.
2. You search for an existing organizational unit.
3. You change attributes of the organizational unit, such as:
 - Organizational unit ID
 - Reference organizational unit ID
 - Organizational unit function
 - Organizational unit role
 - Local currency (company organizations only)
 - Relationship with other organizational units
4. You save your entry.

The system automatically adds a technical business partner and sets the business system group.

Workflow in Transportation Management

Definition

Sequence of steps processed either by people or by an SAP system.

Use

Transportation Management (TM) uses **SAP Business Workflow**, which is a cross-application tool that enables the integration of business tasks across whole application areas.

For more information, go to <http://help.sap.com/s4hana>, choose the relevant release, and navigate to ► **Product Assistance** ► **Enterprise Technology** ► **SAP Business Workflow** ►.

An example of the use of the workflow capability in TM is the approval check that you can define. For more information, see [Approval Check](#).

i Note

You have to assign employees to organizational units to be able to use the workflow capabilities. For more information, see [Assigning Employees to Organizational Units](#).

Approval Check

Use

You use this function to do the following:

- Check whether a forwarding order or a forwarding quotation meets the value ranges defined in a condition for specific attributes of the business document.
- Trigger the workflow to get approval for the processing of a business document, freight agreement RFQ, agreement, or validity period in a rate table.

When you save a business document, freight agreement RFQ, agreement, or validity period in a rate table, the system triggers the approval workflow.

Integration

You can use this workflow with forwarding orders, forwarding quotations, freight agreement RFQs, agreements, and rate tables.

Prerequisites

- You have customized the workflow engine. For more information, on the [SAP Easy Access](#) screen, choose [Tools](#) > [Business Workflow](#) > [Development](#) > [Utilities](#) > [Automatic Workflow Customizing](#).
- You have enabled the approval workflow functionality as relevant in the following activities in Customizing for [Transportation Management](#):
 - [Master Data](#) > [Agreement RFQs and Quotations](#) > [Define Freight Agreement RFQ Types](#)
 - [Forwarding Order Management](#) > [Forwarding Order](#) > [Define Forwarding Order Types](#)
 - [Forwarding Order Management](#) > [Forwarding Quotation](#) > [Define Forwarding Quotation Types](#)
 - [Master Data](#) > [Agreements and Service Products](#) > [Define Freight Agreement Types](#)
 - [Master Data](#) > [Agreements and Service Products](#) > [Define FWA and Service Product Catalog Types](#)
 - [Master Data](#) > [Agreements and Service Products](#) > [Define Internal Agreement Types](#)
 - [Master Data](#) > [Rate Tables](#) > [Define Rate Table Types](#)
- If you want to be able to override the approval check for forwarding orders and forwarding quotations, you must implement the [Specific Checks for Approval Workflow in FWO Management](#) (TRQ_APPR_CHECK_B0) BAdI. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Forwarding Order Management](#) > [BAdI: Specific Checks for Approval Workflow in FWO Management](#).
- If you want to check mandatory fields in freight agreement RFQs before proceeding with the approval workflow, you must implement the [Specific Checks for Approval Workflow in Freight Agreement RFQs](#) (/SCMTMS/RFQ_APPR_CHECK_B0) BAdI. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for](#)

[Transportation Management](#) > [Master Data](#) > [Agreement RFQs](#) > [BAdI: Specific Checks for Approval Workflow in Freight Agreement RFQs](#).

- If you want to check mandatory fields in agreements before proceeding with the approval workflow, you must implement the [Specific Checks for Approval Workflow in Agreements](#) (/SCMTMS/FAG_APPR_CHECK_B0) BAdI. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Master Data](#) > [Agreements](#) > [BAdI: Specific Checks for Approval Workflow in Agreements](#).
- If you want to check mandatory fields in rate tables before proceeding with the approval workflow, you must implement the [Specific Checks for Approval Workflow in Rate Tables](#) (/SCMTMS/RATES_APPR_CHECK) BAdI. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Master Data](#) > [Rate Tables](#) > [BAdI: Specific Checks for Approval Workflow in Rate Tables](#).
- You have created an organizational unit, assigned a position to it, for example, [Customer Agent](#), and assigned an employee to this position. For more information, see [Creation of Organizational Models](#). When creating the organizational unit, you must have specified the following settings:
 - Marked the appropriate position as [Head of own organizational unit](#).
 - Assigned the [Sales](#) organizational unit function to the organizational unit for forwarding orders and forwarding quotations.
- You have defined agents for the workflow in the Workflow Builder and you have defined the standard task as a general task in the workflow. For more information, on the [SAP Easy Access](#) screen choose [Tools](#) > [Business Workflow](#) > [Development](#) > [Definition tools](#) > [Workflow Builder](#) > [Workflow Builder](#).
- For business documents (not freight agreement RFQs, agreements, or validity periods in rate tables), you have assigned conditions to the /SCMTMS/APPROV_TRQ condition type:

A condition assigned to this type specifies certain fields and the exact limits the system must consider for starting the approval workflow. For example, a condition says that the gross weight must be bigger than 5.000 kg. At 5.001 kg the approval workflow starts.

For more information about defining conditions, see [Definition of Conditions](#).

Features

When you create a freight agreement RFQ, agreement, validity period in a rate table, or a business document, the system sets the status as follows:

- RFQ status in a freight agreement RFQ to [In Process](#)
- Approval status in an agreement, or validity period in the rate table, to [In Process](#)
- Approval status in a business document to [Not Checked](#)

Freight Agreement RFQs

If you have enabled the approval workflow for RFQ publication, when you create individual freight agreement RFQs, the system changes the status to [Freight Agreement RFQs Created](#). In the freight agreement RFQ master under [Follow Up](#), you can request publishing approval. The system changes the status to [Awaiting Approval](#).

The customer agent or approver can approve or reject the publishing request. The system updates the status as follows:

- If the agent or approver approves, the status changes to [Published](#). The system publishes an individual freight agreement RFQ to each of the relevant carriers.

- If the agent or approver rejects, the status changes to **Freight Agreement RFQs Created**. You can change the details of the RFQ master and publish the RFQ master again.

If you have enabled the approval workflow for agreement creation, in the freight agreement RFQ master under **Follow Up**, you can request agreement creation approval. The system changes the status from **Published** to **Awaiting Approval**.

The customer agent or approver can approve or reject the agreement creation request. The system updates the status as follows:

- If the agent or approver approves, the status changes to **Ready for Agreement Creation**.
- If the agent or approver rejects, the status changes to **Published**.

You must create a new negotiation round or change the business share allocation.

Agreements, Validity Periods in Rate Tables, and Business Documents

If you have enabled the approval workflow for agreements, validity periods in rate tables, and business documents, the system changes the status as follows:

- For an agreement, or a validity period in a rate table, when you request approval, the system changes the status of the agreement or validity period in the rate table to **Awaiting Approval**. When you then save the agreement, or validity period in the rate table, the system triggers the workflow process.
- For business documents, when you first save the business document, the system checks the document against the value ranges defined in the conditions and takes one of the following actions:
 - If the business document meets the value ranges defined in the conditions, the system changes the approval status from **Not Checked** to **No Approval Needed**. You can then process the business documents further.
 - If the business document does not meet the value ranges defined in the conditions, the system changes the status of the business document from **Not Checked** to **Approval Needed**. The system then triggers the workflow and sends a message to the customer agent or approver.

i Note

An action in the Post Processing Framework (PPF) triggers the workflow. For more information about the PPF, see [Post Processing Framework \(BC-SRV-GBT\)](#).

- The customer agent or approver can approve or reject the agreement, validity period, or business document as follows:
 - If the agent or approver approves the business document, the approval status changes to **Approved**.
You can then process the business document further.
 - If the agent or approver approves the agreement or validity period, the approval status changes to **Released**.
You can then process the agreement or validity period further.
 - If the agent or approver rejects the business document, the approval status changes to **Not Approved**.
The system blocks the business document for further processing and triggers the workflow.
 - If the agent or approver rejects the agreement or validity period, the approval status changes to **In Process**.

i Note

The customer agent or approver can cancel the processing of the workflow item. The workflow item then stays in the personal inbox of the customer agent or approver and the system cannot process the agreement RFQ, business document, agreement, or validity period further.

More Information

[Definition of Conditions](#)

[Post Processing Framework \(BC-SRV-GBT\)](#)

[Workflow in Transportation Management](#)

Transportation Network

You use this component in Transportation Management (TM) to define the following master data objects:

- [Location](#)
- [Transportation Zone](#)
- [Transportation Lane](#)
- [Schedule](#)

In conjunction with the means of transport, the transportation network (TN) defines the direct reachability of transportation zones and locations for a particular vehicle resource.

Transportation Network Cockpit

Use

You can use this function to display the following master data of the transportation network on a map:

- Locations
- Transportation zones
- Transportation lanes
- Trade lanes
- Schedules
- Default routes
- Connections
- Business partner networks

Features

General Map Functions

For more information, see [Map Functions](#).

Object Relationships

For the individual master data objects, you can determine other master data objects that have a relationship to them. Select a master data object and then one of the search functions available in the context menu. The system adds the master data objects

determined to the map view.

Display Options

You can use the display profile to select all the objects that are to be displayed on the map. The display profile is available in the toolbar

You can set the display status for the individual master data objects. Select a master data object and then one of the functions available for hiding objects in the context menu. The system then removes the master data objects determined from the map view. You can hide the master data object itself, all the other master data objects, and the master data objects that do not have a relationship.

Activities

To call the transportation network cockpit, open the [Transportation Network Cockpit](#) app on the launchpad.

Path Finder

Use

You can use this function to search for legs in your transportation network and take the following into account in the process:

- Locations
- Transshipment locations
- Transportation lanes
- Schedules
- Transportation zones

This enables you to easily determine whether a connection exists between two locations and, therefore, whether you have correctly configured your master data.

i Note

The path finder does not take incompatibilities into account. For more information about incompatibilities, see [Incompatibilities](#).

Integration

This function is available both in the transportation network cockpit and on the user interface for transportation proposals. In both cases, the result of the path finder is displayed on the geographical map.

Prerequisites

You have defined the following objects:

- Locations [mandatory] (see [Definition of Locations](#))
- Transshipment locations that are assigned to locations [optional] (see [Transshipment Location](#))

- Connecting transportation lanes [optional] (see [Definition of Transportation Lanes](#))
- Schedules [optional] (see [Schedule Creation](#))

Features

You can execute this function and check the result in either the transportation network cockpit or on the transportation proposals user interface:

- Transportation Network Cockpit

When you open the context menu for a location in the transportation network cockpit, you can choose: **Find > Connection To Location...** and enter a destination in the dialog box displayed. The system then calculates all of the available connections in the transportation network and displays them on the map. The color of the connections indicates whether a valid transportation lane or schedule exists.

- Transportation Proposals

When you call the user interface for transportation proposals, you can choose the appropriate pushbutton to show or hide the available connections on the map. As in the transportation network cockpit, the color of the connections indicates whether a valid transportation lane or schedule exists.

- Explanation Tool

When you select one or more freight units, you can choose the **Show Network Path** pushbutton to analyze the underlying transportation network that is taken into account when planning the selected freight units.

More Information

[Transportation Network Cockpit](#)

[Generation of Transportation Proposals](#)

Location

Definition

A logical or physical place in which products or resources are managed on a quantity basis.

Use

You use this business object as the basis for transportation processes. In transportation processes, you name a source location, a destination location, and any transshipment locations necessary to complete the transportation process. For this purpose, you define locations.

Structure

A location is identified by a name. To define a location, you have to specify its name and type. Transportation Management (TM) provides standard location types such as the following:

- Production plant (1001)
- Distribution center (1002)

- Shipping point (1003)
- Transportation zone (1005)
- Business Partner (1021)
- Terminal (1030)
- Store (1040)
- Port (1100)
- Airport (1110)
- Railway station (1120)
- Container freight station (1130)
- Hub (1140)
- Gateway (1150)
- Container yard (1160)
- Warehouse (1170)
- Carrier warehouse (1180)
- Loading point (1200)

In TM, you use the location type as a means for categorizing locations. Location types are not used for controlling business processes, they are only used as a means of reference.

Integration

This business object is used throughout TM in almost every business process, including the following:

- Definition of more complex master data objects that are part of the transportation network, such as transportation zones, transportation lanes, or schedules.
- Definition of other master data objects, such as resources.
- Definition of loading requirements. You can specify for vehicle resources and locations whether they can be loaded or unloaded from the left, right, back, or top. Further, you can specify the equipment that is attached to a vehicle resource and the equipment that is required for loading and unloading at a location, for example, a forklift. A vehicle resource can only be loaded or unloaded at a location if it fulfills at least one of the loading direction requirements and all of the attached equipment requirements of this location. For more information see [Loading Requirements for Locations](#).
- Definition of incompatibilities. For example, a location may be marked as having a restriction in regards to the delivery of goods by means of a specific vehicle resource.
- Creation of forwarding orders, freight units, planned transportation activities, executed transportation activities, and so on.

Locations of Business Partners

A business partner can have multiple standard addresses and multiple delivery addresses. One standard address is valid at any one time. In TM, one business partner can have multiple locations. For every relevant address of a business partner, you can create a respective location. The address number of the location is the same as the address number of the business partner. When TM receives a sales document or outbound delivery, it receives address data which it uses to determine the correct location, taking into account the validity dates of the address.

More Information

[Definition of Locations](#)

[Transshipment Location](#)

One-Time Location

Definition

A location that is stored as a transient master data object in the system.

Use

You use one-time locations in business processes that require the input of locations that have not been defined as master data objects, such as the creation of forwarding orders. The system creates one-time locations based on one-time address details that you can specify on the user interface.

Structure

A one-time location is specified by the name of the organization, its address data, its communication data, or a combination of these.

You can create a note for each one-time location. When you create a note, you can only define one note per location and language.

Integration

- To find addresses for one-time locations using the input help (F4), complete the Customizing activities for SAP NetWeaver under [Application Server](#) > [Basis Services](#) > [Address Management](#) > [Regional Structure/City File](#).
- In Customizing, you can activate an address search that determines whether one-time locations match locations that already exist. For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [General Settings](#) > [Define General Settings for TM](#).
- In the standard system, all user roles have been configured so that users can read the details of one-time locations. If you reconfigure the roles to suite your business needs, we recommend that you do the following:
 - Configure all roles available in Transportation Management (TM) so that users assigned to them can read the details of one-time locations.
 - Configure the roles that are used for creating or changing forwarding orders (see [Forwarding Order](#)), such as the transportation booking agent, so that they can create one-time locations.

More Information

You can activate the following BAdI implementations for one-time locations:

- A customer-specific address search for one-time locations. For a possible implementation, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Master Data](#) > [General Settings](#) > [One-Time Locations](#) > [BAdI: Customer-Specific Search Logic for Locations Based on Address Data](#).
- The adjustment of location data created for one-time locations. For a possible implementation, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAdIs\) for Transportation Management](#) > [Master Data](#) > [General](#)

Transshipment Location

Definition

A location that is used for unloading goods from one vehicle resource and loading it onto another vehicle resource during the transportation process.

Use

You use transshipment locations when different means of transport or different carriers have to be used in the transportation process. You can also use transshipment locations when consolidating or de-consolidating goods to be transported.

Structure

A transshipment location is defined by assigning a location to **either** another location **or** to a transportation zone (see [Transportation Zone](#)).

When you assign a transshipment location to a transportation zone, it can be used as a transshipment location by all locations that are part of that transportation zone.

i Note

You cannot define a transshipment location by assigning a location to itself.

→ Recommendation

For a better system performance, we recommend that you define only transshipment locations that you use.

Integration

If you define a transshipment location and want to use it in your transportation processes, you must define transportation lanes (see [Transportation Lane](#)) that take the transshipment location into account.

❖ Example

You want to transport goods from location A to location C. You have to unload some goods in location B and change your means of transport. Location B is therefore a transshipment location.

For this transportation process to take place, you have to define the following:

1. A transshipment location B
2. A transportation lane from A to B and a transportation lane from B to C

For more information, see [Examples for the Transshipment Location](#).

Define Transshipment Location Assignments

Use

You, as a system administrator, can use this report to display, create, and delete transshipment location assignments as follows:

- You can display transshipment location assignments for selected locations, transportation zones, and transshipment locations. You can display all transshipment location assignments in the system, by selecting the **Display All Assignments** checkbox.
- You can assign locations and/or transportation zones to one or more transshipment locations.
- You can delete transshipment location assignments for selected locations, transportation zones, and transshipment locations. To simulate the deletion of transshipment location assignments before deleting the assignments from the system, you select the **Simulate Deletion** checkbox.

Activities

To define transshipment location assignments, on the **SAP Easy Access** screen choose ► **Logistics** ► **Master Data** ► **Transportation Network** ► **Location** ► **Define Transshipment Location Assignments** ►.

Examples for the Transshipment Location

Example 1

You want to transport goods from location A to location C. You have defined the transshipment location B for the location A. This situation is depicted in the following figure (the transshipment location is displayed with a blue background):



Examples for the Transshipment Location

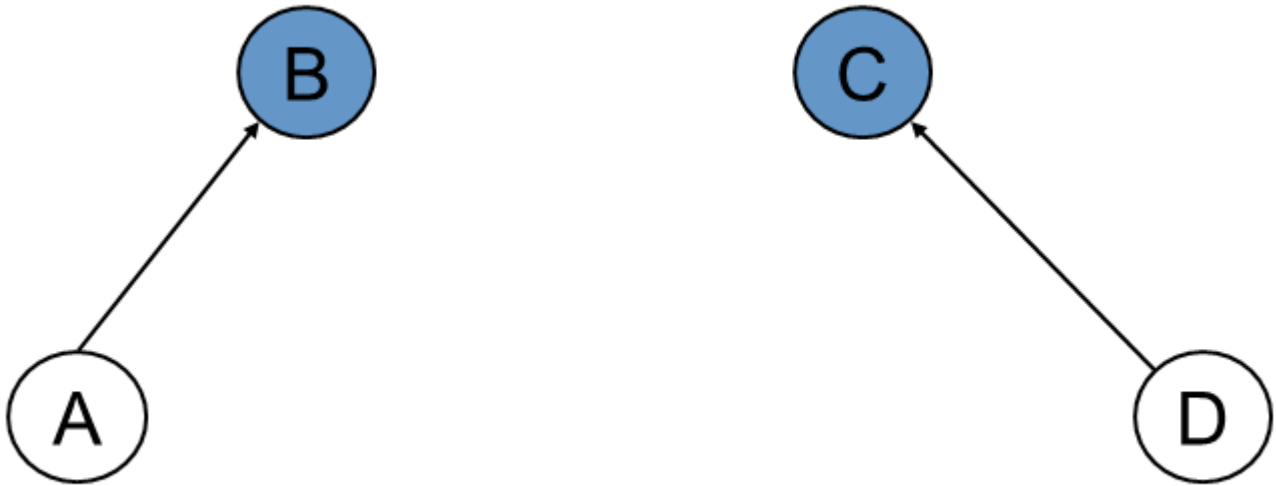
You have the following options for transporting goods from location A to location C (provided you have defined the respective transportation lanes):

- A to C
You can transport the goods directly from location A to location C.
- A to B to C

You can transport the goods to location B first, and from location B to location C.

Example 2

You want to transport goods from location A to location D. You have defined the transshipment location B for the location A, and the transshipment location C for the location D. This situation is depicted in the following figure (the transshipment locations are displayed with a blue background):



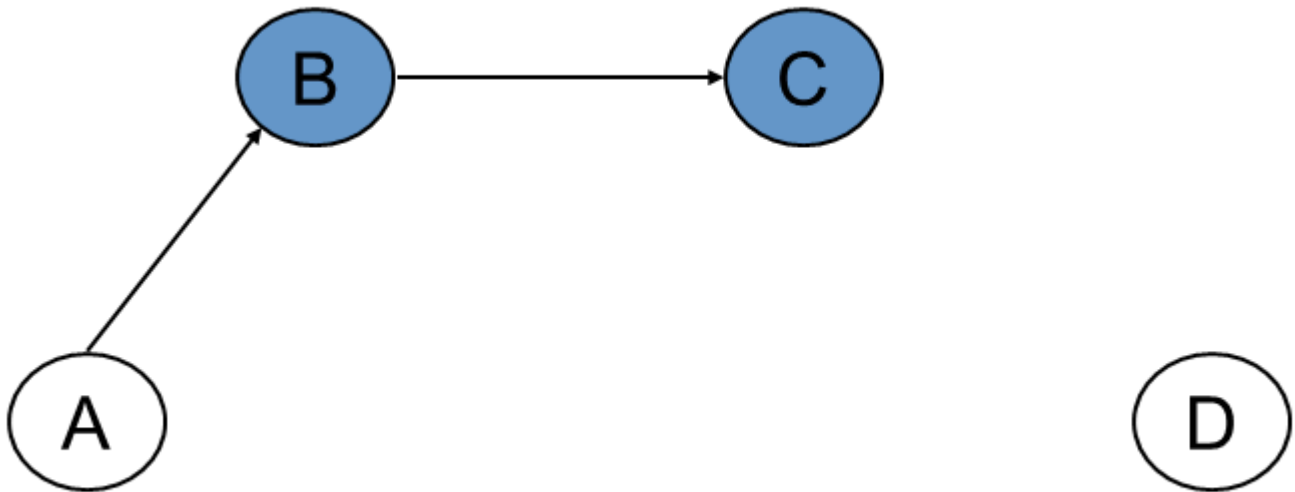
Examples for the Transshipment Location

You have the following options for transporting goods from location A to location D (provided you have defined the respective transportation lanes):

- A to D
- A to B to D
- A to C to D
- A to B to C to D

Example 3

You want to transport goods from location A to location D. You have defined the transshipment location B for the location A, and the transshipment location C for the location B. This situation is depicted in the following figure (the transshipment locations are displayed with a blue background):



Examples for the Transshipment Location

You have the following options for transporting goods from location A to location D (provided you have defined the respective transportation lanes):

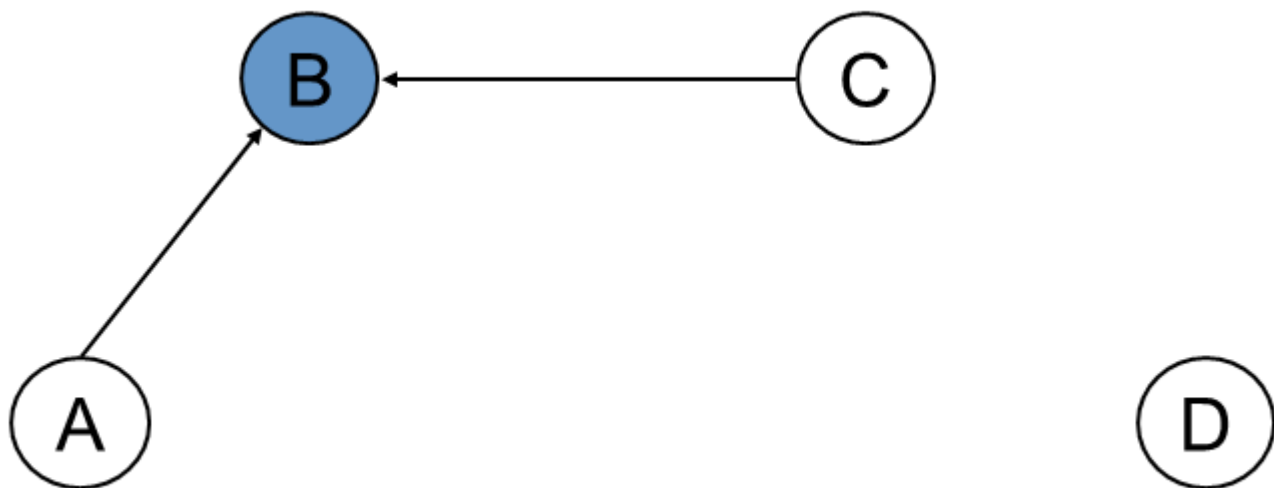
- A to D
- A to B to D
- A to B to C to D

i Note

In this example, you cannot transport goods from A to C to D, since C is not a transshipment location for the location A.

Example 4

You want to transport goods from location A to location D. You have defined the transshipment location B for the location A and the location C. This situation is depicted in the following figure (the transshipment location is displayed with a blue background):



Examples for the Transshipment Location

You have the following options for transporting goods from location A to location D (provided you have defined the respective transportation lanes):

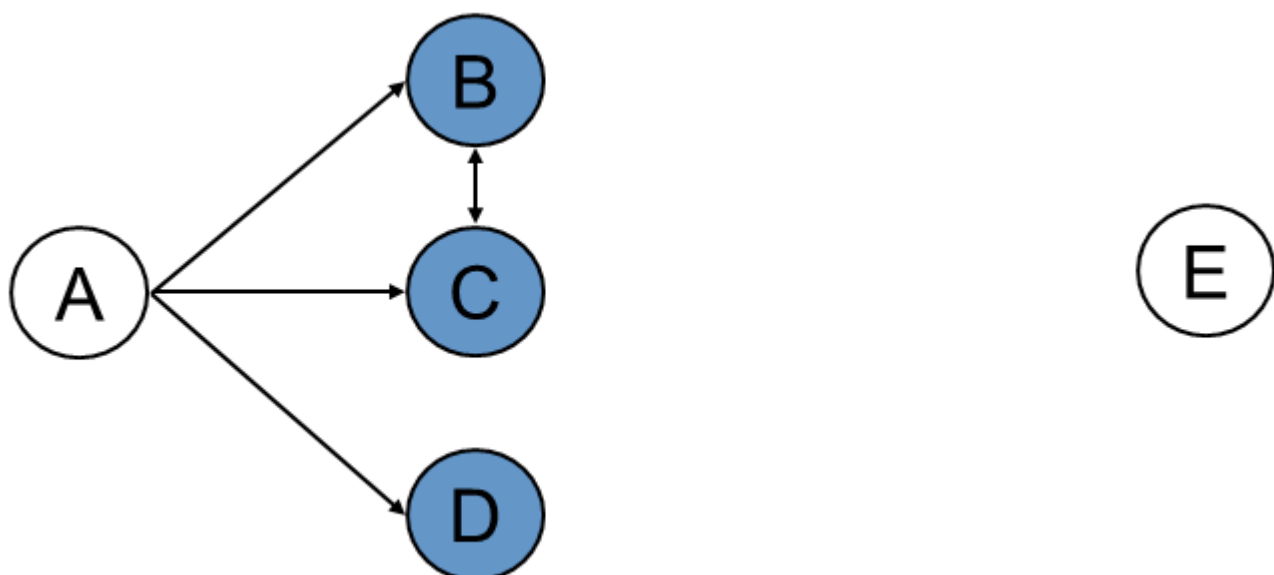
- A to D
- A to B to D

i Note

In this example, you cannot transport goods from A to B to C to D, since C is not a transshipment location for the location B.

Example 5

You want to transport goods from location A to location E. You have defined the transshipment locations B, C, and D for the location A, and specified that B and C are mutual transshipment locations. This situation is depicted in the following figure (the transshipment locations are displayed with a blue background):



You have the following options for transporting goods from location A to location E (provided you have defined the respective transportation lanes):

- A to E
- A to B to E
- A to C to E
- A to D to E
- A to B to C to E
- A to C to B to E

Definition of Locations

Use

You use this process to define locations in Transportation Management (TM).

i Note

If you create locations for a referring entity like business partners, shipping points, and plants, you should use the report `/SAPAPO/CREATE_LOCATION`, to ensure that the addresses of locations and the referring entity are consistent. For more information, see [Creating Locations](#).

Process

1. From the launchpad, you open the [Define Location](#) app (`/SCMTMS/LOC3`).
2. You specify your location, choose the location type, and choose [Create](#).
3. You describe your location and enter location-specific data on the following tab pages:

- [General](#)

This tab page contains information about standard identification numbers for the location and the reference object of the location, for example, the business partner. It also contains geographical data of the location, the business partner that owns the location, and the priority of the location for business processes.

- [Address](#)

This tab page contains information about the contact details for the location, such as street address, PO box address, and communication data.

i Note

If you enter data on the [Address](#) tab page, the system calculates the geographical data for the location and enters the values in the corresponding fields on the [General](#) tab page. For this process to take place, as a minimum you must specify the country or region code of the location.

The system can use a third-party geocoding program to calculate the geographical data. For more information, see [Geocoding](#).

Note that with TM, we do not deliver a third-party geocoding program.

- **Alt. Identifiers**

This tab page contains information about the alternative location identifiers defined for the location. For more information, see [Definition of Alternative Location Identifiers](#)

- **TM**

This tab page contains information about the goods wait time of products for the location. You can also enter information related to air cargo security (for example, the shipper security status for the location). For more information, see [Configuration of Air Cargo Security](#).

- **Resources**

This tab page contains information about restrictions that apply for this location in regards to resources and operating times of resources for the inbound and outbound processes.

- **Addit.** (Additional)

This tab page can contain distinctive information that you want to store with locations.

4. You save the location.

Result

You have created locations. You can change or display locations that you have defined using the **Define Location** app, which you can open from the launchpad.

i Note

If you search for an existing location by entering a business partner, and the business partner has multiple addresses, a popup appears where you can choose the address for which you want to display the location.

You can use the locations in transportation processes, such as the creation of forwarding orders or the definition of resources.

Operating Times

Definition

Time stream that describes the availability of a location.

Use

You use this business object to specify at which times loading and unloading activities can take place at a specific location, and at which times transportation activities can start or end at this location.

You can consider a multiple-shift operation when specifying operating times for a location.

Integration

You specify operating times for a location by using a calendar resource (see [Calendar Resource](#)). On the **Resources** tab page of the **Change Location** screen, you can assign calendar resources for the inbound and outbound delivery to the location.

i Note

- Several locations can refer to a single calendar resource.
- VSR optimization (VSR = vehicle scheduling and routing) considers operating times only if you have specified values greater than zero for loading and unloading durations (see [Loading and Unloading Durations](#)).
- The business add-in (BAI) /SCMTMS/EX_PLN_OT lets you adjust operating times so that vehicle scheduling and routing (VSR) optimization, transportation requirement routing and scheduling (TRS), and embedded scheduling can consider them.

For more information, see Customizing for Transportation Management under Business Add-Ins (BAIs) for Transportation Management -> Planning -> BAI: Adjustment of Operating Times.

Creating Locations

Creating Locations Using a Report

You can use report /SAPAPO/CREATE_LOCATION to create new locations for plants, shipping points and business partners. This is useful if you have a lot of entities for which you require locations.

To run the report you select from plants, shipping points and business partners. You can choose whether the report creates locations only for the standard addresses of the entity, or for both the standard addresses and delivery addresses. The report identifies all selected entities which have no location, and the system:

- Creates a location using the existing address of the entity
- Calculates geocodes
- Creates a transportation zone location if needed
- Updates the location to zone assignment

Creating Locations out of Business Documents

If you create or update a document that is relevant for Transportation Management (TM), the system checks that all necessary locations needed to represent plants, shipping points and business partners in TM are present in the system. If a location has not already been created, the system creates it. You can create locations, on demand, from the following TM-relevant business document categories:

- SD Documents
 - Sales Order (SO)
- MM Documents:
 - Purchase Order (PO)
 - Stock Transfer Order (STO)
- Deliveries
 - Outbound Deliveries
 - Inbound Deliveries

Creating Locations out of Business Partner Master

For business partners, the system can be configured to immediately create locations when you create and save a business partner.

To do so you need to implement business add-in (BAI) /SAPAPO/LOC_CREATE which allows you to specify the conditions for creating or updating a location when a business partner is saved. Once you create a location, the system triggers any updates as you make a change to the business partner address.

Related Information

[Creation of Locations](#)

Loading Requirements for Locations

You can define loading requirements for locations.

These requirements can be taken into account in VSR optimization, load consolidation, and for the generation of transportation proposals. The system considers the requirements specified for locations and resources and can identify incompatibilities that would, for example, make a resource unsuitable for use at a specific location. For manual planning, you can specify how the system should handle, for example, a plan that includes a resource that is not compatible with the specified location.

Features

Loading Direction Profiles

You can use loading direction profiles to specify:

- the direction from which loading or unloading is possible
- a preference for loading direction, if required (not yet supported)

For more information see Customizing for **Transportation Management** under **Master Data > Transportation Network > Location > Define Loading Direction Profile**.

You can also specify loading direction profiles for resources. For more information see [Loading Requirements for Vehicle Resources](#).

Attached Equipment Profiles

You can define attached equipment profiles that describe the equipment provided by the location for loading/unloading. For more information see Customizing for **Transportation Management** under **Master Data > Transportation Network > Location > Define Attached Equipment Profile**.

You can also define attached equipment profiles to describe the equipment provided by a resource for loading/unloading. For more information see [Loading Requirements for Vehicle Resources](#).

In assessing compatibility, the system compares the location's attached equipment profile with the resource's attached equipment profile. Other equipment that is assigned to the resource, for example, loading equipment in a vehicle loading profile, is not relevant in this context.

Location Loading Profiles

You can use location loading profiles to define specific loading constraints for a location as follows:

- Loading Direction Profile
- Loading Equipment Required (not yet supported)

Defines for which loads a resource must provide a loading equipment at this location:

- For no loads
- For all loads
- For loads in double-deck resources

For more information see Customizing for [Transportation Management](#) under [Master Data](#) [Transportation Network](#) [Location](#) [Define Location Loading Profile](#).

Planning Profile

In the incompatibility settings of your planning profile, you can define constraints for loading direction and attached equipment. Note that once you define these constraints, we recommend that you check and adapt your master data. That means, add the loading direction profile and the attached equipment profile to the locations and vehicle resources that should consider these constraints.

Example: If a location has no loading direction profile defined, no restrictions with regards to this constraint are applied. That would mean, vehicle resources can be loaded or unloaded from all sides at this location.

For more information, see [Planning Profile](#).

Related Information

[Loading Requirements for Vehicle Resources](#)
[Planning Profile](#)

Updating Locations Using a Report

The following reports are available for you to use to update your locations.

- You can use the report `/SAPAPO/UPD_LOC_SP_PL` to update locations for plants and shipping points. You can trigger this report in the following situations:
 - After making changes in the address data that are relevant for the geocodes of plants and shipping points
 - On a scheduled basis, to capture any changes made during a specific period
- It may happen that users create multiple instances of the same location in the system, for example, for business partners, plants or shipping points, and that the system assigns different address numbers to these instances. You can use the report `/SAPAPO/UPDATE_LOC_ADDRNR` to overwrite the address numbers of such locations. For locations of business partners, this is only possible if the business partner has just one standard address.

The report assigns the location the same address number as the original entity. Furthermore, location details like geocodes or transportation zone assignments are updated according to the new address. Details are written into the application log.

i Note

You can also use the report `/SAPAPO/UPDATE_LOC_ADDRNR` after executing the report `/SAPAPO/FILL_LOCASGMT`, to correct any errors in cases where the address number of the original entity is not equal to the address number of the corresponding location.

Related Information

[Updating Locations](#)

Deleting Locations

Use

You can use this procedure to delete locations that have become obsolete in Transportation Management (TM).

Prerequisites

1. You have checked if dependencies exist by generating a [Where-Used List](#). For example, forwarding orders or resources may depend on locations.

To generate a [Where-Used List](#), do the following:

- a. On the [SAP Easy Access](#) screen, choose ► [Logistics](#) ► [Transportation Management](#) ► [Master Data](#) ► [WUF – Where-Used List](#) ►.
- b. On the [Where-Used List](#) screen, enter the object type you want to check. In the [Selection Criteria](#) screen area, enter the object value you want to check. For example, enter a location ID in a location check..
- c. Under [Object Name](#) on the left hand side of the screen, select the object you want to check. On the right hand side of the screen, choose the appropriate hosting object. For example, choose [Transportation Document](#) or [Model](#) for a location.

The system displays the where-used information for the object you select.

2. You have eliminated any dependencies involving the locations.

i Note

Do not remove the technical dependency on the default model.

Procedure

1. In the launchpad, open the [Define Locations](#) app and enter or choose the location you want to delete.
2. Select the [Deletion Flag](#) checkbox for the location by choosing ► [More](#) ► [Location](#) ► [Deletion Flag](#) ►.

i Note

If you have selected the [Deletion Flag](#) checkbox for a location and you no longer want to delete the location, you can deselect it at any time before deletion by choosing ► [More](#) ► [Location](#) ► [Deletion Flag](#) ►.

3. If necessary, repeat steps number 1 and 2 to select the [Deletion Flag](#) checkbox for other locations.
4. Delete all locations that have been flagged for deletion by choosing ► [More](#) ► [Extras](#) ► [Delete Locations](#) ►.

i Note

The system administrator may have scheduled a background job that deletes locations that have been flagged for deletion periodically. In this case, you do not have to delete the locations manually as described in step number 4.

More Information

Deleting One-Time Locations

The procedure described above is not valid for deleting one-time locations (see [One-Time Location](#)).

The system can delete one-time locations as soon as all dependencies of the individual one-time location to business processes have been eliminated.

The system starts the deletion process for one-time locations when you carry out step number 4 as described above or when the system starts a scheduled background job.

❖ Example

The system deletes one-time locations of a forwarding order if you do not save the forwarding order or if you delete the forwarding order.

Application Logs for Locations

Use

You use this function to collect messages, exceptions, and errors that have been stored in the system for a specific object, such as a location. Logs can be displayed and deleted from the database.

Features

The application log shows you messages issued by specific programs, such as the following:

- The deletion program for locations

The application log for this program provides you with an overview of where references to the deleted location still exist. You can then remove these.

- All programs capable of running in the background

Activities

- You can display logs for locations using the [Application Logs](#) app, which you open from the launchpad.

Use of Change Documents for Locations

Use

You use this process to trace changes that are made to locations in Transportation Management (TM).

The changes are logged in change documents, which allow you to find out at any time, what was changed, when the changes occurred, and how the changes were made. Change documents facilitate the analysis of errors. To use change documents for locations, you have to define the business object as a change object in the system.

Process

1. You activate change documents in Customizing. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Location](#) > [Activate Change Documents](#).
2. You display the change documents for locations by opening the [Display Change Documents](#) app from the launchpad.

Definition of Alternative Location Identifiers

Use

You use this function to define alternative location identifiers (ALI) in Transportation Management (TM).

Once defined, you can use an ALI in transportation processes such as the creation of forwarding orders or the definition of resources.

Prerequisites

- You have defined at least one ALI type in Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Location](#) > [Configuration for Alternative Location Identifiers](#).
- If you want to use an ALI type with a strict code list, you must have defined a code list and type in Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Location](#) > [Configuration of Alternative Location Identifiers](#).
- If you want to add an ALI to an existing location, this location must have been previously defined.

For more information, see [Definition of Locations](#).

Activities

1. You open the [Define Locations](#) app from the launchpad, then you either navigate to an existing location where you want to add an ALI, or you select the [Alt. Identifiers](#) tab while you are defining a new location.
2. You specify the [ALI Type](#) and [Alternate ID](#) in the ALI table.

i Note

If you are using an ALI type with a defined code list, the value of the [Alternate ID](#) table entry must be a member of the list.

3. You now repeat step 2 to add as many additional alternative location identifiers as you require.
4. You save the location.

Geocoding

You can determine geographical coordinates (geocoordinates) based on address data.

Use

The term geocoding comes from the area of geographical data processing. It describes the process for determining the geocoordinates (longitude, latitude, and altitude) from the address data given. There are various commercial geocoding programs that determine the required data using the address attributes (country, region, postal code, city, street, and so on).

You can use geocoordinates to do the following:

- Display of business objects on a map, for example a location (localization of a business object on a map using address data)
- Determination of the distance (distances can be calculated on the earth's surface from the geocoordinates determined; for example, as a first approximation as the crow flies or with the relevant map material on a street basis)
- Routing (for example, determination of a route on a street basis during transportation planning)

Prerequisites

- If you want to calculate simple geocoordinates based on country and region, you need to make the respective settings in Customizing for ABAP Platform under [General Settings](#) > [Set Geocoding](#).
- If you want to calculate more precise geocoordinates, you need to set up the integration with SAP HANA spatial services. For more information, see [Standard Integration with SAP HANA spatial services](#).
- If you have set up the integration with SAP HANA spatial services, you need to use the process controller framework to ensure a more flexible approach to the determination of geocoordinates. The process controller framework is a flexible framework that enables you to create different geocoding strategies so you can connect to external geographical information systems (GIS) different from the standard geocoding program. This flexibility allows you to use third-party GIS providers if required. You configure the process controller for geocoding in Customizing. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Location](#) > [Geocoding](#) > [Configure Process Controller for Geocoding](#).

Features

Geocoding Levels

You use geocoding levels to store geocoordinates for the address of a location on different levels of detail, for example by house number, postal code, or region. Having different levels of detail for the geocoordinates of a location allows you to improve the performance of the distance and duration determination between two locations. This is important if, for example, you have a large number of locations geocoded on a high level of detail thereby making the distance and duration determination a performance critical process. The lower the accuracy level the more locations share the same geocoordinates.

You can define the geocoding strategy that a geocoding level uses as well as which geocoding program to use and which address fields to pass. Moreover, you can specify a GIS zone. You define geocoding levels in Customizing (see Customizing activity [Define Geocoding Levels](#)).

❖ Example

You have two transportation lanes. Transportation lane A contains the start location WALLDORF and the destination location MUNICH. Transportation lane B contains the start location WALLDORF and the destination location REGENSBURG. In the location master, MUNICH and REGENSBURG are part of the same region and geocoded to be part of the same region.

When determining the distances and durations, the system first determines the distance and duration for transportation lane A and saves the data in the buffering table. As transportation lane B ends in the same region, the system can reuse the data in the buffering table instead of running a second determination of distance and duration thereby increasing the performance of the system. The system only needs to call the distance and duration determination program once. You can define the required accuracy level in Customizing (see Customizing activity [Define Strategy for Distance and Duration Determination and Georouting](#)).

i Note

If you delete geocoding levels in the system, you can delete geocoordinates that have become obsolete as a consequence of this. To delete obsolete geocoordinates, you can use the [Delete Obsolete Geocoordinates](#) report. For more information, see [Deletion of Obsolete Geocoordinates](#).

Mass Geocoding

If you want to maintain geocoding for multiple locations, you can use the [Mass Geocoding](#) report. For more information, see [Mass Geocoding](#).

Validity of Geocoordinates

In the Customizing activity [Define GIS Providers](#), you can indicate the validity of geocoordinates based on the GIS strategies for a given GIS provider. The validity specifies the number of days or hours that geocoordinates from a GIS provider are stored in the system before they are automatically deleted and redetermined. You can also indicate whether and how GIS data is to be stored. It can be stored based on the validity of the GIS provider (in days) or the validity of the GIS strategy (in hours).

For more information, see Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Geographical Services](#) > [Define GIS Providers](#).

Related Information

[Distance and Duration Determination \(DDD\)](#)

Calculation of Transportation Zone Coordinates

You can calculate geographical coordinates (geocoordinates) for transportation zones, allowing you to determine distances between business partners, for example.

Use

You can use this program to calculate geocoordinates that represent the geolocation of a transportation zone. A geolocation is defined by the following data:

- Latitude
- Longitude
- Time zone
- Precision of the geolocation (for example, continent, country, region, postal code)

The system calculates the transportation zone coordinates by means of a geocoding program. The program determines the geographical latitude and longitude of the transportation zone. You can use this data to calculate the distance between two business partners, for example.

The geocoding is carried out according to the required precision, which may be defined at the level of country, region, postal code, place, street, or house number.

Prerequisites

You have entered the necessary address data. The system calculates the transportation zone coordinates on the basis of this address data.

Activities

On the SAP Easy Access screen, choose [Logistics](#) > [Transportation Management](#) > [Master Data](#) > [Transportation Network](#) > [Transportation Zone](#) > [Calculate Transportation Zone Coordinates](#).

Choose the transportation zones for which you wish to calculate the coordinates from the list of results and then choose [Calculate Coordinates](#). The system calculates the coordinates of the selected transportation zones. You will find the geographical data of the transportation zones on the [General](#) tab page of the location master record.

Mass Geocoding

You can calculate geographical coordinates (geocoordinates) for multiple locations.

Use

You use this report to calculate geocoordinates for longitude and latitude for your locations. Geocoordinates are essential for the distance and duration calculation between two or more locations. The system defines the geocoordinates for a location using the following data:

- Longitude
- Latitude
- Precision of the location (for example country, region, postal code, street, house number)
- Match (percentage to which the geocoordinates match with the address defined in the location)

The report allows you to display existing geocoordinates from the database, to simulate geocoordinates that the geocoding program provides, or to directly overwrite existing geocoordinates for locations. You determine geocoordinates by defining a selection from the following:

- Location
- Location type
- Transportation zone
- Validity end of geographical data

You can indicate the validity of geocoordinates in the Customizing activity [Define GIS Providers](#). For more information, see [Geocoding](#).

Once you have a selection, you define if you want to calculate geocoordinates for all selected locations or only for locations that do not yet have geocoordinates as follows:

- If you want the system to calculate geocoordinates for all of your selected locations, select the [All Selected Locations](#) radio button.

i Note

If you have defined geocoding levels in Customizing, the system calculates the geocoordinates on the basis of the geocoding levels.

- If you want the system to only calculate geocoordinates for those locations that have initial geocoordinates, select the [Only Locations Without Existing Geocoordinates](#) radio button. You select the radio button for locations where no geocoordinates were calculated upon the initial maintenance, or for locations that were transmitted from a back-end

system by means of the Core Interface. If you want the system to only calculate geocoordinates for those locations that have initial geocoordinates on that geocoding level, additionally select a specific geocoding level.

- If you want to calculate geocoordinates based on a specific geocoding level, select the **Only Specific Geocoding Level** radio button.

Furthermore, you can execute geocoding as follows:

- If you want the system to display existing geocoordinates from the database for selected locations, select the **Show Existing Geocoordinates** radio button.
- If you want the system to simulate the geocoordinate determination, select the **Simulate Geocoordinates** radio button.
- If you want the system to overwrite any existing geocoordinates with the geocoordinates that result from this calculation, select the **Overwrite Geocoordinates** radio button.

Prerequisites

- The system calculates initial geocoordinates for a location on the basis of existing address data. You must therefore ensure that you enter all relevant information (at least country or region) in the address data fields for your locations in the location master.
- You have defined geocoding levels in Customizing. For more information, see Customizing for **Transportation Management** under ► **Master Data** ► **Transportation Network** ► **Location** ► **Geocoding** ► **Define Geocoding Levels** ►.

Activities

To access the **Mass Geocoding** report, on the SAP Easy Access, screen choose ► **Logistics** ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Geocoding** ► **Mass Geocoding** ►.

To access the **Mass Geocoding in Batch** report, on the SAP Easy Access, screen choose ► **Logistics** ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Geocoding** ► **Mass Geocoding in Batch** ►.

Deletion of Obsolete Geocoordinates

You can delete geographical coordinates (geocoordinates) after deleting geocoding levels.

Use

You use this report to delete obsolete geocoordinates after one or more geocoding levels have been deleted.

Prerequisites

You have defined geocoding levels in Customizing. For more information, see Customizing for **Transportation Management** under ► **Master Data** ► **Transportation Network** ► **Location** ► **Geocoding** ► **Define Geocoding Levels** ►.

Features

You can simulate the deletion of obsolete geocoordinates by deselecting the **Delete Obsolete Geocoordinates** checkbox. You can use the **Delete Geocoordinates in Batch** report to schedule background processing for the deletion of obsolete geocoordinates after one or more geocoding levels have been deleted.

Activities

To access the [Delete Geocoordinates](#) report, on the SAP Easy Access, screen choose ►► [Logistics](#) ► [Transportation Management](#) ► [Master Data](#) ► [Transportation Network](#) ► [Geocoding](#) ► [Delete Geocoordinates](#) ►.

To access the [Delete Geocoordinates in Batch](#) report, on the SAP Easy Access, screen choose ►► [Logistics](#) ► [Transportation Management](#) ► [Master Data](#) ► [Transportation Network](#) ► [Geocoding](#) ► [Delete Geocoordinates in Batch](#) ►.

Transportation Zone

Definition

An object that groups a number of locations.

Use

You use this business object to group locations into transportation zones. The system can partly transfer the properties that you assign to a transportation zone to all its locations. This function reduces the volume of master data stored in the system.

❖ Example

A number of locations are supplied by the same warehouse. You group these in a transportation zone.

Structure

A transportation zone is identified by a name. To define a transportation zone, you specify its name and type. In Transportation Management (TM), the following types of transportation zones exist:

- Direct zone
- Postal code zone
- Region zone
- Mixed zone

i Note

A mixed zone is a combination of any of the other zone types.

Integration

This business object is used in the definition of more complex master data objects that are part of the transportation network such as transportation lanes (see [Transportation Lane](#)).

Definition of Transportation Zones

Use

You use this process to define transportation zones in Transportation Management (TM).

Prerequisites

- You have defined locations (see [Definition of Locations](#)).

Process

1. You define a transportation zone in the **Transportation Network Cockpit** app, which you open from the launchpad.
2. You create a transportation zone by entering its name and a description for the zone.

i Note

The system automatically sets the zone type to **M** (mixed). If you require a different zone type, you can change the zone type using the input help (F4).

3. Depending on the type of transportation zone, you assign or exclude locations by creating entries on **one** of the following tab pages:

- **Zone – Location**

To define a **direct zone**, you create entries on this tab page. You assign locations to the transportation zone directly by entering the name of each location. You exclude locations by selecting the **Exclude** checkbox.

- **Zone – Postal Code**

To define a **postal code zone**, you create entries on this tab page. You assign locations to the transportation zone by specifying a postal code range that is valid for a particular country or region.

- **Zone – Region**

To define a **region zone**, you create entries on this tab page. You assign locations to the transportation zone by specifying a country/region and a region. If you only enter a country/region, the system automatically chooses all regions of this country/region.

i Note

If you want to define a **mixed zone**, you can create entries on more than one tab page.

4. You save your entries.

Caution

If you have defined a direct zone, a postal code zone, or a region zone, and you enter information on any other tab page, the system automatically changes the type of transportation zone to a mixed zone. This change is irreversible.

Transportation Zone Hierarchy

Definition

A hierarchy that can be defined for transportation zones.

Use

You use this object to structure transportation zones hierarchically by assigning one transportation zone to another. All locations included in a zone that is assigned to another zone, are automatically also part of that other zone.

Transportation zones that are included in a transportation zone hierarchy inherit properties of the superordinate zone.

The following conditions apply when you define a transportation zone hierarchy:

- You cannot assign a transportation zone to itself.
- You cannot assign a transportation zone to another zone if the former already contains the other zone.

Structure

A transportation zone hierarchy is defined by its name and its hierarchy structure.

Integration

- The **VSR optimizer** takes the transportation zone hierarchy into account during the optimization run.
- The lane determination function takes the transportation zone hierarchy into account when determining a specific transportation lane (see [Lane, Distance, and Duration Determination \(LDDD\)](#)).
- The transshipment location determination function takes the transportation zone hierarchy into account when determining all valid transshipment locations for a location.

Defining a Transportation Zone Hierarchy

Prerequisites

- You have defined transportation zones (see [Definition of Transportation Zones](#)).

Context

You use this procedure to define a transportation zone hierarchy in Transportation Management (TM).

Procedure

1. You define a transportation zone hierarchy in the **Transportation Network Cockpit** app, which you open from the launchpad.
2. Enter RELH_ZONE as the name of the transportation zone hierarchy.
3. Enter RELHS_ZONE as the name of the hierarchy structure.

i Note

TM delivers the hierarchy structure as standard.

4. Choose **Create**.
5. Assign transportation zones to the transportation zone hierarchy by using the **Hierarchy Fast Entry** screen area or by assigning them directly in the tree structure.
6. Save your entries.

Calculation of Geographical Data for a Transportation Zone

Use

You use this process to calculate the geographical data of the geographical center of a transportation zone. The calculation is based on the geographical data of all locations that are part of the transportation zone. The system calculates the following geographical data:

- Latitude
- Longitude
- Time zone
- Precision (continent, country, region, and postal code)

i Note

In Transportation Management the precision is calculated based on the country and region of the locations.

The system uses the calculated data, for example, to increase the precision of a transportation lane when calculating the distance between locations or zones.

Prerequisites

You have entered the address data of all locations (see [Definition of Locations](#)). The system calculates the geographical data for the location on the basis of the address data.

Process

1. From the launchpad, you open the [Transportation Network Cockpit](#) app.
2. You select the transportation zone for which you want to calculate the coordinates and choose [Calculate Coordinates](#).

i Note

The system does not consider locations for which you have set the [Deletion Flag](#) (see [Deleting Locations](#)) or locations that have initial coordinates.

3. The system calculates the geographical data for the geographical center of the selected transportation zone and stores the data in the background.

More Information

[Transportation Lane](#)

[Transportation Zone](#)

Determination of Locations Within a Transportation Zone

Use

You can use this report to determine all locations contained within a transportation zone.

Prerequisites

Both locations (see [Location](#)) and transportation zones (see [Transportation Zone](#)) exist.

Features

The report offers the following options:

- Determination of all locations that are directly contained in the transportation zone

In this case, the system determines all locations that are contained in this transportation zone.

- Determination of all locations (that is, locations that are indirectly contained in the transportation zone)

In this case, the system determines all locations that are contained in both this transportation zone and in the transportation zones of the corresponding transportation zone hierarchy.

The following applies when defining locations:

- A location is directly contained within a transportation zone if one of the following criteria is met:
 - It is assigned directly to the transportation zone (zone type D or M).
 - It matches the transportation zone definition (zone type R or M) with regard to the region and country from the location address.
 - It matches the transportation zone definition (zone type P or M) with regard to the postal code and country from the location address.
- A location is indirectly contained within a transportation zone if it is directly assigned to at least one transportation zone that meets the following condition: It is a lower-level transportation zone within the transportation zone hierarchy of the transportation zone that you entered in this report.

❖ Example

You have defined three transportation zones:

- Transportation zone EU (Europe)
- Transportation zone DE (Germany)
- Transportation zone FR (France)

The transportation zones DE and FR are subordinate to the transportation zone EU.

If you run the report for transportation zone DE and specify that the system is to determine all locations that are directly contained within this zone, the system displays all locations of transportation zone DE only. This result is irrespective of whether you have manually assigned the locations to the transportation zone or whether the system has assigned them using the address field.

If you specify that the system is to determine all locations, it displays the locations from the transportation zones EU and FR as well.

Activities

To execute the report, on the [SAP Easy Access](#) screen, choose [Transportation Management](#) [Master Data](#) [Transportation Network](#) [Determine Locations Within a Transportation Zone](#).

Transportation Lane

Definition

A relationship between two locations, two transportation zones, or a combination of locations and zones that expresses the direct reachability of the locations or of all locations within the zones for a specific means of transport.

Use

You use this business object when planning distribution and procurement between different locations.

Structure

A transportation lane is defined by a source location or zone, a destination location or zone, and a means of transport (MTr). You have the option of assigning carriers to the transportation lane.

Integration

This business object is used by several processes in Transportation Management (TM). For more information, see:

- [Planning](#)
- [Creation of Business Shares and Transportation Allocations](#)
- [Carrier Selection](#)

Definition of Transportation Lanes

Use

You use this process to define transportation lanes in Transportation Management (TM).

Prerequisites

- You have defined means of transport (MTr) in Customizing for [Transportation Management](#) under [Master Data](#) [Resources](#) [Define Means of Transport](#) .
- You have defined at least two locations or transportation zones. For more information, see [Definition of Locations](#) or [Definition of Transportation Zones](#).
- If you want to assign a carrier to a means of transport, you have assigned the business partner role [Carrier](#) to the business partner that functions as the carrier. For more information, see [Definition of Business Partners](#).

Process

1. You define a transportation lane on the launchpad by opening the [Define Transportation Lane](#) app.
2. You create a transportation lane by entering information on either of the following tab page pages:
 - [Tr.Lane](#)

You create the transportation lane by entering or choosing the source location or zone and the destination location or zone.

- o **Intra-Zone Lane/Loc. Transp. Lane**

You use this tab page to define an intrazone lane or a location transportation lane.

- You create an intrazone lane by entering or choosing a transportation zone.
- You create a location transportation lane by entering or choosing a location.

i Note

Defining a location transportation lane has no impact on the reachability of locations. It can, however, be useful in specific contexts such as the initialization of transportation allocations (see [Transportation Allocation](#)).

For example, you might create a location transportation lane for a location A, define a planning period for one or more means of transport, assign one or more carriers to the transportation lane, and define a maximum freight order quantity for each carrier. This information can be evaluated during the creation of transportation allocations, thereby limiting the number of transportation activities from or to location A that are assigned to a specific carrier within the planning period.

- o **Mass Maint. (Create)**

You create multiple transportation lanes by copying an existing transportation lane.

i Note

On all tab pages, when you choose a transportation zone via the input help you can identify the transportation zones by the location type **1005**. This location type is used as a marker for transportation zones in the system. It cannot be used to define transportation zones in the **Define Location** transaction (see [Definition of Locations](#)). To define transportation zones, see [Definition of Transportation Zones](#).

3. You assign a means of transport to the transportation lane.

Means of transport are used to move goods between the locations or zones of a transportation lane. You specify attributes for the means of transport that apply to transportation activities between the source and destination of the goods to be transported, such as the transportation duration, the transportation distance, and the transportation costs.

i Note

You can only create a transportation lane if you assign at least one means of transport to it.

4. You have the option of assigning carriers to the transportation lane.

A carrier offers services along a transportation lane for a specific means of transport. You, therefore, assign carriers to the transportation lane by assigning them to the means of transport. Once you have assigned a carrier to the means of transport, you can specify the attributes that are specific to this combination of carrier and MTr. For example, the transportation costs, the priority, the share of business, or the capacity for the transportation allocation.

i Note

If you want to use automatic carrier selection, you have to assign carriers to means of transport.

Result

You have created transportation lanes. You can change or display transportation lanes that you have defined on the launchpad by opening the **Define Transportation Lane** app.

You can change or display multiple transportation lanes on the [Mass Maint. \(Display/Change\)](#) tab page.

You can use transportation lanes in transportation processes such as dispatching or subcontracting.

Use of Change Documents for Transportation Lanes

Use

You use this process to trace changes that are made to transportation lanes in Transportation Management.

The changes are logged in change documents, which allow you to find out at any time, what was changed, when the changes occurred, and how the changes were made. Change documents facilitate the analysis of errors. To use change documents for transportation lanes, you have to define the business object as a change object in the system.

Process

1. You activate change documents in Customizing. For more information, see Customizing for [Transportation Management](#) under [►Master Data►Transportation Network►Transportation Lane►Activate Change Documents►](#).
2. You display the change documents for transportation lanes in the [Display Change Documents](#) app, which you open from the launchpad.

Distance and Duration Determination (DDD)

Use

You can use this function in Transportation Management (TM) to determine the distance of a transportation lane and the duration it takes the means of transport to travel that distance. The system uses the distance and duration of the transportation lane in planning processes. The [VSR optimizer](#), for example, determines the most suitable means of transport for the transportation processes on the basis of the distance and duration.

Further, you can use this function to determine the distance and duration between two locations in a business document of the freight order management area and in planning. For more information, see [Transportation Distance and Transportation Duration](#) and [Determination of Transportation Distance and Transportation Duration](#).

Distance and duration determination is also used in the charge management area.

Prerequisites

To ensure a more flexible approach to the determination of distances and durations, you can use the process controller as a flexible framework to create different distance and duration determination strategies so you can connect to external geographical information systems (GIS) that offer different calculations than the straight line distance and the calculation offered by the standard geocoding program. You configure the process controller for the distance and duration determination in Customizing for Transportation Management. For more information, see Customizing for [Transportation Management](#) under [►Master Data►Transportation Network►Transportation Lane►Distance and Duration DeterminationGeocoding►Configure Process Controller for Distance and Duration Determination►](#).

For more information about further prerequisites, see [Settings for Distance and Duration Determination](#).

Activities

In TM, you can implement a business add-in (BAI) for distance and duration determination that can manipulate the calculated values for the distance and duration. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) [Master Data](#) [Transportation Network](#) [Reuse Component](#) [Enhancement of Basic Distance and Duration Determination](#) [BAI: RC DDD - Basic Distance and Duration Determination](#).

More Information

[Distance and Duration Determination](#).

Lane, Distance, and Duration Determination (LDDD)

Use

The system uses this function to determine the most specific transportation lane for your transportation process. The lane, distance, and duration determination (LDDD) considers the hierarchy of the means of transport and the hierarchy of the transportation zone at the source and destination location. In addition to determining the transportation lane, the system determines the distance of the lane and the duration it takes the means of transport to travel that distance.

LDDD is used in planning processes in Transportation Management (TM). The [VSR optimizer](#), for example, determines the most specific transportation lane for transportation processes on the basis of the lane, distance, and duration determination.

Prerequisites

- In Customizing, you have defined means of transport to which you have assigned superordinate means of transport. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) [Resources](#) [Define Means of Transport](#).
- You have defined hierarchies for the source location or zone and the destination location or zone (see [Defining a Transportation Zone Hierarchy](#)).
- You have defined transportation lanes (see [Definition of Transportation Lanes](#)).
- You have defined the hierarchy priorities for lane determination in Customizing. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) [Transportation Network](#) [General Settings for Transportation Network Determination](#).

i Note

In the standard system, the hierarchy of the means of transport is considered first during determination.

Activities

In TM, you can implement a business add-in (BAI) for lane, distance, and duration determination that enables you to manipulate the values for the distance and duration and the transportation lane calculated by the system. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) [Master Data](#) [Transportation Network](#) [Reuse Component](#) [Enhancement of Lane, Distance and Duration Determination](#) [BAI: RC LDDD - Lane, Distance and Duration Determination](#).

More Information

[Distance and Duration Determination](#)

Distance and Duration Determination

Use

You can use this function to determine the distance of a transportation lane and the duration it takes the means of transport to travel that distance. The system determines the distance and duration in the following ways:

- The system calculates the shortest distance (line of flight) between locations.
- The system calculates the precise distance between locations based on the means of transport by using a third-party geographic information system (GIS).

The distance and duration determination (DDD) uses geocoordinates that are stored in the system for each location. For a more precise determination of distances and durations, you can use geocoding levels and DDD strategies to define on which level of detail you want to determine the distance and duration. For example you can determine the distance and duration based on the house number, the postal code or the region of the location's address. If the distance between a start and a destination location is very large, you may not need to use the most detailed geocoordinates to determine the distances and durations as it does not affect the business process significantly.

To use this function, you must have activated the business function Enhanced Services for Transportation Network (SCM_SCMB_TR_NETWORK).

Prerequisites

For information, see [Settings for Distance and Duration Determination](#).

Features

Determining Distances and Durations for the DDD Buffer

The retrieval of data from a GIS can be time consuming and it can affect the performance of the system. For this reason, you can store retrieved data in the DDD buffering table. You can use the [Determine Distances and Durations for DDD Buffer \(Selected Geography\)](#) report to precalculate the distance and duration for a set of start and destination locations and save the result in the distance and duration determination (DDD) buffer. For large volumes of data you can also run background jobs for this function. For more information, see [Determination of Distances and Durations Based on Selected Geography](#).

Whereas determination with this report is based on the selected geography, the [Determine Distances and Durations for DDD Buffer \(Selected Documents\)](#) report determines distances and durations based on selected business documents. For more information, see [Determination of Distances and Durations Based on Selected Documents](#).

Adjusting Distances and Durations for the DDD Buffer

You can use the [Adjust Distances and Durations for the DDD Buffer](#) report to manually adjust distances and durations for the distance and duration determination (DDD) buffer. For more information, see [Adjustment of Distances and Durations for the DDD Buffer](#).

Deleting Buffered Distances and Durations

You can use the [Deletion of Buffered Distances and Durations](#) report to delete obsolete entries from the distance and duration determination (DDD) buffer. For more information, see [Deletion of Buffered Distances and Durations](#).

Determining the Distance and Duration for a Single Transportation Lane

You can determine distances and durations for a single transportation lane. For more information, see [Determining the Distance and Duration for a Single Transportation Lane](#).

Determining the Distance and Duration for Multiple Transportation Lanes

You can determine distances and durations for multiple transportation lanes. For more information, see [Determining the Distance and Duration for Multiple Transportation Lanes](#).

Settings for Distance and Duration Determination

Use

Determining Distance and Duration Based on the Shortest Distance

If you want the system to determine the distance and duration for the shortest distance (line of flight) between two locations, you have done the following:

- In the Customizing activity [Define Means of Transport](#), you have made the following settings:
 - You have defined means of transport and specified the average speed and the distance factor for them.
 - You have deactivated the [GIS Qlty](#) (GIS Quality) checkbox.
- You have defined locations in the location master. If you have entered address data for the location, the system calculates the geographical data for the location as well as any additional geocoordinates based on geocoordinate determination strategies you have maintained. The system uses the geographical data of a location as a basis for the DDD.

Determining Distance and Duration as well as Routes by Means of a GIS

If you want the system to determine the distance and the duration as well as routes by means of a geographic information system (GIS), you have done the following:

- You have set up the standard integration with SAP HANA spatial services (see [Standard Integration with SAP HANA spatial services](#)) or you have used the process controller to create your own DDD or georouting strategy and method.
- In the Customizing activity [Set Usage of GIS Tool](#), you have set the usage of the GIS tool.
- You have maintained means of transport for which you have specified the low, medium, and high speeds, and you have activated the [GIS Qlty](#) (GIS Quality) checkbox in the Customizing activity [Define Means of Transport](#).

i Note

If you do not make any entries for the speeds, the system can determine the distance but not the duration. In this case, the system uses a standard value for the duration.

- If you want to determine distances and durations on the basis of specified geocoordinates, you have done the following in Customizing:
 - You have defined geocoding levels in the Customizing activity [Define Geocoding Levels](#).
 - You have defined GIS zones for use in the DDD or georouting strategies in the Customizing activity [Define GIS Zone](#).
 - You have assigned countries or regions to the GIS zones for use in the DDD or georouting strategies in the Customizing activity [Assign Country/Region to GIS Zone](#).
 - You have defined DDD or georouting strategies in the Customizing activity [Define Strategy for Distance and Duration Determination and Georouting](#).

- o You have defined global settings for the distance and duration determination in the Customizing activity [Define Global Settings for Distance and Duration Determination](#).
- o You have defined number range intervals for speed profiles in the Customizing activity [Define Number Range Intervals for Speed Profiles](#).
- o Optionally, you have defined GIS providers in the Customizing activity [Define GIS Providers](#) and indicated the validity of geographical data based on the GIS strategies for a given GIS provider. The validity specifies the number of days or hours that geographical data from a GIS provider is stored in the system before it is automatically deleted and redetermined. You can also indicate whether and how GIS data is to be stored. It can be stored based on the validity of the GIS provider (in days) or the validity of the GIS strategy (in hours).

Determining the Distance and Duration for a Single Transportation Lane

Procedure

1. Define a transportation lane.
2. Display the detail screen for the means of transport and switch to change mode.
3. On the detail screen, in the [Parameters](#) frame, choose [Generate Proposal](#).

Results

The system determines the distance and duration on the basis of either a geographic information system or the shortest distance (depending on the assigned means of transport) and enters values in the [Transptn Distance](#) (Transportation Distance) and [Trsp. Duration](#) (Transportation Duration) input fields.

i Note

You can enter values for the distance and duration manually when defining a transportation lane.

Next Steps

For more information about distance and duration determination, see [Distance and Duration Determination](#).

For more information about creating transportation lanes, see [Create Transportation Lanes](#).

Determining the Distance and Duration for Multiple Transportation Lanes

Procedure

1. Define multiple transportation lanes in the [Mass Creation of Transportation Lanes](#) report.
2. In the Transport Duration and Distances frame, choose either of the following:
 - o [Calculate transport durations and distances](#)

The system redetermines the distances and duration for the transportation lanes created. It determines the values on the basis of either a GIS or the shortest distance and enters values in the [Transptn Distance](#) and [Trsp. Duration](#)

input fields of the means of transport.

- **Adopt Transport. Duration and Distances from Template**

The system reuses the distances and duration of the transportation lane that is used as a template and enters values in the **Transptn Distance** and **Trsp. Duration** input fields of the means of transport.

When you run mass maintenance, you need to run the **Determine Distances and Durations for DDD Buffer (Selected Geography)** report. The system determines the distances and durations for all the transportation lanes created.

To run the report, on the Easy Access screen, search for **Determine Distances and Durations for DDD Buffer (Selected Geography)** (/SCMB/DDD_PRECALC).

More Information

For more information about distance and duration determination, see [Distance and Duration Determination](#).

For more information about creating transportation lanes, see [Create Transportation Lanes](#).

For more information about the precalculation of distances and durations, see [Determination of Distances and Durations Based on Selected Geography](#).

Determination of Distances and Durations Based on Selected Geography

Use

You can use this function to precalculate the distance and duration for a set of start and destination locations and save the result in the distance and duration determination (DDD) buffer. For large volumes of data you can also run background jobs for this function.

The precalculation of distances and durations speeds up the distance and duration determination by storing already calculated results for later reuse. You can either add new entries only to the buffer or you can overwrite already existing buffer entries. You can also simulate the selection. By simulating, the report displays the selected entries that the system potentially adds to the buffer. The system does not make any changes to the buffer. You set up your selection based on the following data:

- DDD strategy

You select a DDD strategy and a geocoding level that you want to use for the distance and duration determination. If you enter a DDD strategy and leave the **Geocoding Level** field empty, the system precalculates the distance and duration for all geocoding levels of the locations. If you want the system to set the DDD strategy and the geocoding level dependent on the settings in the DDD strategy Customizing, you can also select the **Strategy Determination** checkbox.

- Means of transport

You can select one or several means of transport. The system uses the means of transport for the distance and duration calculation. If you leave the **Means of Transport** field empty, the system uses all means of transports maintained in the system. The system calculates the distance and duration for all selected start and destination location pairs in combination with all selected means of transport.

- Precision of DDD

You can select the precision of the determined durations and distances. By selecting **Undefined**, for example, you can select all distances and durations that could not be determined.

- Validity end of buffer entry

You can select a range of validity end dates. You can indicate the validity of distances and durations in the Customizing activity [Define GIS Providers](#). For more information, see [Settings for Distance and Duration Determination](#).

- Start and destination location

You can select a set of start and destination locations for the distance and duration calculation.

By selecting the [Use Only Existing Transportation Lanes](#) checkbox, the system only selects combinations that have start and destination locations with a corresponding transportation lane maintained. If you do not select the [Use Only Existing Transportation Lanes](#) checkbox, the system considers all combinations between the start and destination locations. Combinations with identical start and destination are ignored.

To use this report, you must have activated the business function Enhanced Services for Transportation Network (SCM_SCMB_TR_NETWORK).

Prerequisites

- You have defined strategies for distance and duration determination in the process controller. The strategies provide the connection to your GIS provider. You can define your own strategy or use the standard strategy DDD_STD.

You define strategies in the Customizing [Define Strategy](#). For more information, see Process Controller.

i Note

The system updates the DDD buffer by means of the DDD strategies, so the strategies you use must contain the standard method for the update of the buffer.

- You have determined geocoordinates for all start and destination locations. You maintain geocoordinates with the [Mass Geocoding](#) report. To run the [Mass Geocoding](#) report, on the [SAP Easy Access](#) screen search for [Mass Geocoding](#) (/SAPAPO/MASSGEOCODE).
- You have indicated in the Customizing activity [Define Global Settings for Distance and Duration Determination](#) whether a failed distance and duration determination is to be buffered with a precision value of [Undefined](#).

Activities

To precalculate the distances and durations, on the [SAP Easy Access](#) screen search for [Precalculate Distances and Durations for the DDD Buffer](#) (/SCMB/DDD_PRECALC).

To precalculate the distances and durations as a background job, on the [SAP Easy Access](#) screen search for [Precalculate Distances and Durations for the DDD Buffer in Batch](#) (/SCMB/DDD_B_PRECALC).

Determination of Distances and Durations Based on Selected Documents

You can use this function to determine distances and durations for start and destination locations based on selected business documents.

Use

This function determines distances and durations for a set of start and destination locations based on a number of selected business documents and saves the result in the distance and duration determination (DDD) buffer.

You can start this function in simultaneous parallel sessions only if you have indicated a parallel processing profile in your planning profile. We recommend that you use background processing if you want to process business documents simultaneously.

Integration

For more information about the integration of this function into overall background processing, see [Background Processing](#).

Prerequisites

- You have created selection profiles and a planning profile. For more information, see [Selection Profile](#) and [Planning Profile](#).
- You have set up distance and duration determination with the DDD buffer. For more information, see [Settings for Distance and Duration Determination](#).

Features

This function determines distances and durations for all pairs of locations that are relevant for a given combination of selection profiles and planning profiles.

You can run this report as a preparation task so that distances and durations are already available when you start vehicle and scheduling (VSR) routing. This will reduce runtime in the VSR optimizer preprocessing.

Activities

To use this function, you must run program /SCMTMS/PLN_DDD. You must specify the following settings:

- Selection profiles
- Planning profile
- Capacity selection settings
- Checkbox to indicate whether all messages are to be displayed
- Checkbox to indicate whether messages should be logged in the application log

As a result, the DDD buffer is filled.

The system logs all messages that it issues when executing the function in the application log under object /SCMTMS/TMS. If you run the function in the background, the system uses the subobject BP for logging. If you run the function manually, the system logs the messages under subobjects according to the business object that created the message.

Related Information

[Planning Run](#)

Adjustment of Distances and Durations for the DDD Buffer

You can use this report to manually adjust distances and durations for the distance and duration determination (DDD) buffer.

Use

If distances and durations could not be determined for some locations, for example, you can use this report to filter for these distances and durations using the **Precision of DDD** field and then enter distances and durations manually.

You set up your selection based on the following data:

- Strategy selection

You can set up the strategy selection as follows:

- If you want to delete buffer entries with a specific strategy and a specific geocoding level only, you select a DDD strategy and a geocoding level.
- If you want to delete all entries with a certain strategy regardless of the geocoding level, you select a DDD strategy and leave the **Geocoding Level** field empty.
- To delete all entries with a specific geocoding level, regardless of the strategy, you can leave the **Strategy** field empty and select a specific geocoding level.
- If you leave both fields empty, the system deletes all combinations of strategy and geocoding level in the selection.

- Means of transport

You can select one or several means of transport if you want to delete buffer entries with specific means of transport only. If you leave the **Means of Transport** field empty, the report uses all means of transports maintained in the system.

- Precision of DDD

You can select the precision of the determined durations and distances. If you select **Undefined**, for example, the system displays all distances and durations that could not be determined. You can then enter distances and durations manually.

- Validity end of buffer entry

You can select a range of validity end dates. You can indicate the validity of distances and durations in the Customizing activity **Define GIS Providers**. For more information, see [Settings for Distance and Duration Determination](#).

- Time-dependent selection

You can select buffer entries for the deletion dependent on the date they were last changed or accessed.

- Start and destination location selection

You can select a set of start and destination locations to delete only buffer entries with selected locations.

Prerequisites

- You specify the interval in which the system updates the last access date for the time-dependent selection in the Customizing activity **Define Global Settings for Distance and Duration Determination**. In this Customizing activity, you also indicate whether a failed distance and duration determination is to be buffered with a precision value of **Undefined**.

Activities

To adjust distances and durations, on the **SAP Easy Access** screen search for **Adjust Distances and Durations for the DDD Buffer (/SCMB/DDD_MAINTAIN)**.

Deletion of Buffered Distances and Durations

Use

You can use this function to delete obsolete entries from the distance and duration determination (DDD) buffer. For large volumes of data you can also run background jobs for this function.

You delete entries that are no longer needed or entries that are invalid because of a change in the system setup, for example a new GIS Provider has been defined with a new strategy or a geocoding level has been removed. It is also possible to delete old entries dependent on their last change or access date.

You set up your selection based on the following data:

- Strategy selection

You can set up the strategy selection as follows:

- If you want to delete buffer entries with a specific strategy and a specific geocoding level only, you select a DDD strategy and a geocoding level.
- If you want to delete all entries with a certain strategy regardless of the geocoding level, you select a DDD strategy and leave the **Geocoding Level** field empty.
- To delete all entries with a specific geocoding level, regardless of the strategy, you can leave the **Strategy** field empty and select a specific geocoding level.
- If you leave both fields empty, the system deletes all combinations of strategy and geocoding level in the selection.

- Means of transport

You can select one or several means of transport if you want to delete buffer entries with specific means of transport only. If you leave the **Means of Transport** field empty, the report uses all means of transports maintained in the system.

- Precision of DDD

You can select the precision of the determined durations and distances. By selecting **Undefined**, for example, you can select all distances and durations that could not be determined and then delete them.

- Validity period selection

You can specify that only invalid entries are to be considered. Further, you can select a range of validity end dates for your buffer entries. You can indicate the validity of distances and durations in the Customizing activity **Define GIS Providers**. For more information, see [Settings for Distance and Duration Determination](#).

- Time-dependent selection

You can select buffer entries for the deletion dependent on the date they were last changed or accessed.

- Start and destination location selection

You can select a set of start and destination locations to delete only buffer entries with selected locations.

To use this report, you must have activated the business function Enhanced Services for Transportation Network (SCM_SCMB_TR_NETWORK).

Prerequisites

- You specify the interval in which the system updates the last access date for the time-dependent selection in the Customizing activity **Define Global Settings for Distance and Duration Determination**. In this Customizing activity, you also indicate whether a failed distance and duration determination is to be buffered with a precision value of **Undefined**.

Activities

To delete distances and durations, on the [SAP Easy Access](#) screen search for [Delete Buffered Distances and Durations](#) (/SCMB/DDD_DELETION).

To delete distances and durations as a background job, on the [SAP Easy Access](#) screen search for [Delete Buffered Distances and Durations in Batch](#) (/SCMB/DDD_B_DELETION).

Checking and Repairing the DDD Buffer Table

Use

You use the [Check and Repair Buffering Table](#) report to check and repair any inconsistencies that might occur in the distance and duration determination (DDD) buffering table. You can do the following with the report:

- Delete all entries in the buffering table
- Check the following for consistency:
 - Means of transport
 - DDD strategies
 - Locations

To use this report, you must have activated the business function Enhanced Services for Transportation Network (SCM_SCMB_TR_NETWORK).

Activities

To access this report, on the SAP Easy Access screen, search for [Check and Repair Buffering Table](#) (/SCMB/DDD_CLEANUP).

More Information

For more information about DDD, see [Distance and Duration Determination](#).

Checking Buffered Distances and Durations for Inconsistencies

Use

You use the [Check Buffered Distances and Durations for Inconsistencies](#) report to check the distance and duration determination (DDD) buffer for inconsistencies which may arise from a triangle inequality. This check considers entries to be inconsistent if the direct connection between two geocoordinates is longer or slower than the connection via intermediate geocoordinates. The check returns information on the type of inconsistency, the involved locations for the geocoordinates and the alternative paths that are shorter or faster to connect the geocoordinates from a DDD entry.

You can filter the displayed entries using the selection fields. Further, you can specify that the system is to skip consistent entries.

For more information, see the system documentation.

i Note

Currently, the report can handle up to 16 million DDD buffer entries.

Activities

To access this report, on the SAP Easy Access screen, search for [Check Buffered Distances and Durations for Inconsistencies \(/SCMB/DDD_CHECK_TRIA\)](#).

More Information

For more information about DDD, see [Distance and Duration Determination](#).

Schedule

Use

You use this function to define a sequence of stops such as ports, airports, or gateways that is valid for a specific period of time. Ships, trucks, or airplanes can move goods at recurring times along the whole sequence or any part of it. The movement of goods depends on the transit duration between each of the stops as well as the cut-off times and the length of stay at each stop.

As part of the new schedule concept, you can use different carrier schedules and gateway schedules to define information about your stop sequence and departure rules. For more information about departure rules, see [Use of Departure Rules](#).

Features

There are up to three categories of schedules for each mode of transport available to you in Transportation Management (TM).

- Direct and indirect gateway schedules
- Gateway schedules with reference
- Carrier schedules

Gateway Schedules

Gateway schedules are schedules whose start stops and destination stops are transportation hubs (gateways) such as container freight stations (CFS) where freight is consolidated and deconsolidated. When you create a gateway schedule, you can enter a reference to a carrier schedule for each transportation stage and so transfer departure rules and voyages from the carrier schedule to the gateway schedule. This function is based on the transportation mode and is available for the sea and air modes only.

For more information, see [Direct Creation of Freight Documents and Selection of Schedules](#).

i Note

You can enter only one source gateway and one target gateway for each gateway schedule.

The following gateway schedules are supplied in the standard SAP delivery:

- Road Gateway Schedule

As a logistics service provider, you use a road gateway schedule to define a **direct** connection between two gateways.

Consequently this involves a direct gateway schedule. Freight is transported between the two gateways in the road gateway

schedule by road.

- Sailing Schedule

You use sailing schedules to define a schedule for the customer in an ocean freight scenario. Sailing schedules denote a connection between two gateways that are designated as container freight stations in the sea transportation mode, and include information about the port of loading and port of discharge. Unlike the road gateway schedule, this involves an **indirect** gateway schedule for the sailing schedule. However, they do not include information about any of the other stops in the sequence.

- Master Flight Schedule

As a logistics service provider, you use the master flight schedule to define a connection between the gateways at two airports. As the master flight schedule contains further information about the airport of departure and the airport of destination, this involves an **indirect** gateway schedule, similar to the sailing schedule.

You can reference from one master flight schedule to several carrier flight schedules, and therefore map connecting flights in your schedule. You can define the individual references for each transportation stage of the main carriage.

If you define a restriction to certain organizational units for a master flight schedule, this restriction also applies to any air freight bookings that you generate from this master flight schedule. This means that only transportation planners from the specified organizational unit can plan the air freight bookings in question. For more information, see [Publication and Locking of Air Freight Bookings](#).

Gateway Schedules with Reference

In the sea and air transportation modes, you can also reference one or more carrier schedules from a gateway schedule and as a result map for instance connecting flights in your schedule in air transportation mode. You can define the individual references for each transportation stage of the main carriage.

i Note

If there are changes in one of the referenced carrier schedules, and you have already entered these in the carrier schedule yourself, then the status of referenced data in your gateway schedule is changed from **Data Is Up-to-date** to **Data Is Not Up-to-date**. For each transportation stage, the system shows you whether the data in the relevant referenced carrier schedule has changed.

❖ Example

You open your master flight schedule from Frankfurt to Los Angeles via New York. You note that the status of the referenced data is **Data Is Not Up-to-date**. You now check in the departure rules of your schedule which transportation stage is affected by this change. The status of the referenced data for the transportation stage from Frankfurt to New York is still **Data Is Up-to-date**. However, the status of the referenced data for the transportation stage from New York to Los Angeles is **Data Is Not Up-to-date**.

You assign the updated carrier flight schedule to the no longer up-to-date transportation stage of the master flight schedule again. The system then changes the status of the referenced data for both the transportation stage from New York to Los Angeles and for the entire master flight schedule to **Data Is Up-to-date**.

If these changes involve a postponement of the departure dates and one of the two flight schedules is assigned to an air freight booking, the system changes the status of the referenced data in the air freight booking to **Data Is Not Up-to-date**. For additional information about using schedules in freight documents, see [Use of Schedules](#).

⚠ Caution

Once you have assigned a new or updated carrier schedule to the gateway schedule, you have to regenerate the departures in the gateway schedule.

i Note

- You can change the transit duration of a gateway schedule even after departure rules and voyages have been created.

Example: For a sailing schedule with a source container freight station (CFS) and a destination CFS, you can change the pick-up transit duration (that is, the duration from the source CFS to the first port of loading) and the delivery transit duration (that is, the duration from the last port of discharge to the destination CFS). The system considers these transit durations for new departure rules or voyages in the corresponding schedule. Changing the transit durations does not have an impact on existing departure rules and voyages.

- Further, you can change the transit duration from the last port of discharge to the destination gateway after you have created a voyage. The arrival time is derived from the transit duration. If you then copy time changes from the underlying carrier schedule's voyage into the gateway schedule voyage, the system keeps the manually changed transit duration and uses it to recalculate the arrival time.

Gateway Schedules Without Consolidation

As in the freight booking, you can also specify whether a consolidation takes place for a gateway schedule. For example, you can define a gateway schedule with a source gateway but not a destination gateway if you want a consolidation to take place only at the source gateway. You define this in Customizing in the schedule type.

Carrier Schedules

Carrier schedules define a sequence of stops for which the start stop and target stop are not gateways. By specifying departure rules, you can generate departures automatically. The system calculates the arrival and departure dates and times at each stop in the sequence, taking into account the transit duration, cut-off times, and availability of the goods for each stop. You can then modify each voyage individually.

In the sea and air transportation modes, you can also reference carrier schedules from a gateway schedule. For more information, see [Direct Creation of Freight Documents and Selection of Schedules](#).

The following carrier schedules are supplied in the standard SAP delivery:

- Ocean Carrier Schedule

You use the ocean carrier schedule to define a standard sequence of stops that are served by a ship. By specifying departure rules, you can generate voyages automatically.

- Carrier Flight Schedule

You use the carrier flight schedule to define a standard sequence of stops that are typically served by **one** plane. Alternatively, you can also define carrier flight schedules for connection flights with multiple transportation stages, for which various carriers with different aircraft types and flight numbers execute each of the transportation stages. If loading and unloading activities do **not** take place at one of these stops but other activities such as refueling do take place, you can classify this as a service stop.

Finally, by specifying departure rules, you can generate flights automatically.

Moreover, you can specify that the carrier entered in the carrier flight schedule is the operating (executing) carrier of the schedule. Consequently, you can map codeshare flights in your schedule.

❖ Example

Airline XX offers a flight under their own flight number XX123. A different airline, ZZ, by arrangement with airline XX sells the flight under flight number ZZ456, even though the other airline does not operate the flight itself. Airline XX is therefore the operating carrier, and airline ZZ is the marketing carrier.

- Road Schedule

You use the road schedule to define a standard sequence of stops that are served by a truck.

For the ad hoc loading process, you can define warehouse loading and warehouse unloading stops. For more information, see [Ad Hoc Loading](#).

Additional Functions

You can also create schedule-based freight documents by executing the report /SCMTMS/MP_SCHED_CREATE_TOR. For more information, see the system documentation.

You can also use transaction WUF to display the documents that refer to a specific schedule. This function is available for the following document types:

- Schedules
- Freight bookings
- Freight orders
- Schedule-based allocations

More Information

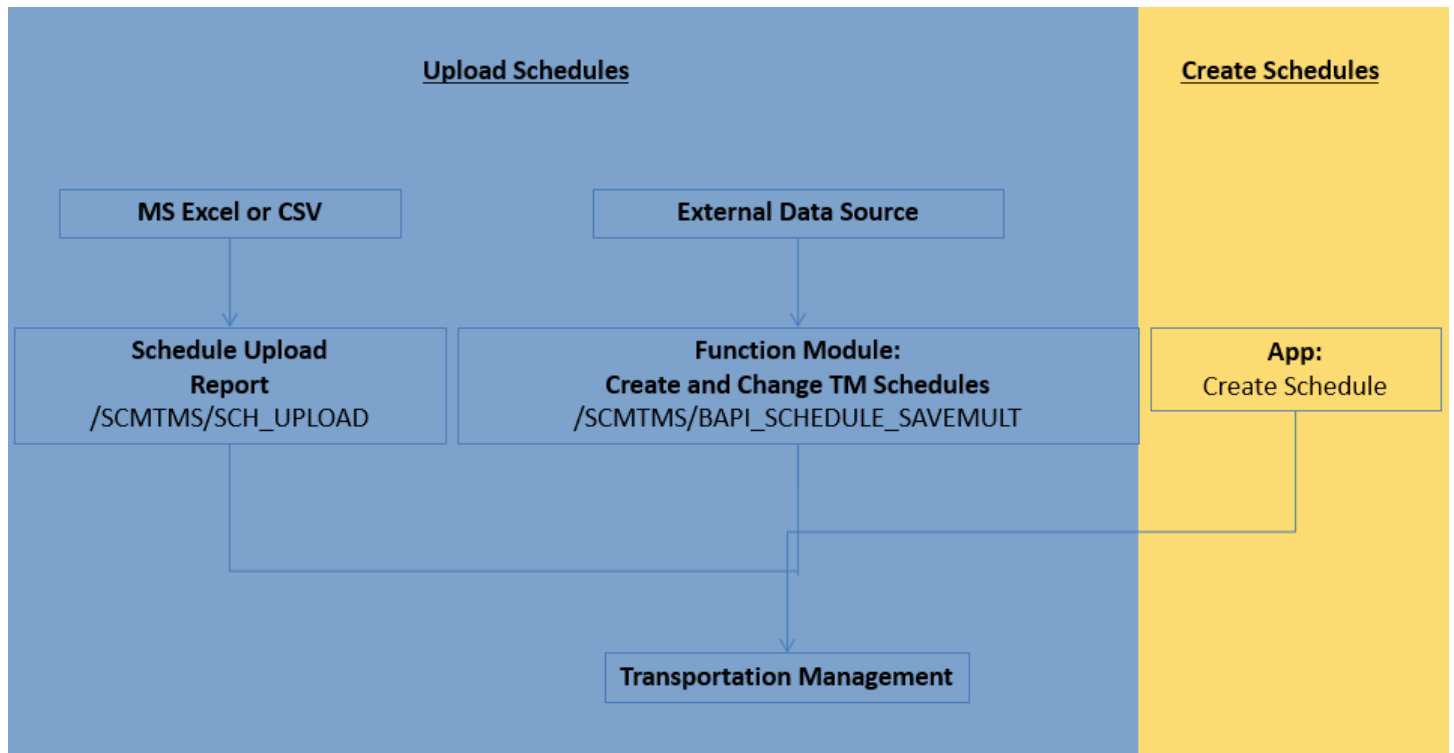
[Creating Schedules](#)

[Use of Schedules](#)

Creating Schedules

Use

You can create various carrier schedules and gateway schedules that you want to use as part of your transportation planning activities. You can create individual schedules manually or upload one or more schedules from an external data source to Transportation Management (TM). For more information, see [Manual Creation of Schedules](#) and [Uploading of Schedules](#).



Creation of schedules in TM

Prerequisites

- You have defined locations (see [Location](#)).
- You have defined business partners and assigned them to the business partner role **Carrier**. For more information, see [Definition of Business Partners](#).
- You have defined the modes of transportation in Customizing for Transportation Management under ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Transportation Lane** ► **Define Transportation Mode** ►.
- You have defined number range intervals for schedules and departures in Customizing for Transportation Management under ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Schedule** ►.
- You have defined a schedule type in Customizing for Transportation Management under ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Schedule** ► **Define Schedule Types** ►.

More Information

[Interaction Between Organizational Units](#)

[Schedule](#)

[Default Route](#)

Manual Creation of Schedules

Use

You use this process to create various carrier schedules and gateway schedules that you want to use as part of your transportation planning activities. To create schedules, open the **Create Schedule** app on the launchpad.

Process

1. Entering the Schedule Type

To specify the type of schedule you want to create, you enter the schedule type that you defined before in Customizing. TM provides the following standard schedule types:

- Ocean carrier schedule: 1000
- Carrier flight schedule: 1100
- Road schedule: 1200
- Sailing schedule: 2000
- Sailing schedule with reference to ocean carrier schedules: 2100
- Road gateway schedule: 2200
- Master flight schedule: 2500
- Master flight schedule with reference to carrier flight schedules: 2600

Depending on the schedule type you entered, the system dynamically generates a user interface for creating schedules.

2. Entering the General Schedule Data

First you enter the general data, such as the carrier, transportation mode, and the transportation group for your schedule.

You can also specify a shipping type. If you specify a shipping type, the system only takes the schedule into account for forwarding orders with the same shipping type during automatic planning. For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Type Codes and Role Codes](#) > [General Type Codes and Role Codes](#) > [Define Shipping Type](#).

If you create a sailing schedule with reference, you can reference to existing ocean carrier schedules. If you create a master flight schedule with reference, you can reference to carrier flight schedules.

If you create a master flight schedule, you can define an air cargo security status for this schedule. For more information, see [Air Cargo Security in Business Document Processing](#) and [Air Freight-Specific Consolidation Processes](#).

3. Entering Location Data

You enter the standard sequence of stops along with the transit duration, the cutoff times, the length of stay, and so on. You can also enter a start location and end location.

4. Creating Departure Rules

You enter departure rules to specify the frequency of voyages, departures, or flights from a stop. You also define a time period during which the departure rule is valid. For example, you can specify that a ship sails from the port of Newark every Thursday at 3:00 pm between January 1 and June 30. For more information about creating departure rules, see [Example of the Use of Departure Rules](#).

If you are working with a gateway schedule with reference and have referenced carrier schedules, you can copy the departure rules from the relevant carrier schedules at transportation stage level.

5. Creating Voyages, Departures, and Flights

After you have created your departure rules, generate voyages, departures, or flights to calculate the actual departure dates/times automatically. The system displays the actual departure dates/times on the appropriate tab.

6. Changing Departure Dates/Times

You select a departure date/time that the system created automatically, and adjust the relevant data as needed. You can also add a departure date/time that is not based on a departure rule.

❖ Example

You created an ocean carrier schedule, in which your ship sails from port A via port B to port C and puts to sea every Thursday at 3:00 pm. You receive a notification that port B will be closed next Thursday; therefore, for the trip next week, you enter that your ship should sail directly from port A to port C.

If you have referenced one or more carrier schedules from a gateway schedule, you cannot change the data copied from the carrier schedules. If you want to change this data, you have to change it directly in the relevant carrier schedule. As an alternative, you can assign a different carrier schedule to the gateway schedule, and create new departures.

7. Deleting Schedules

You can delete entire schedules or departures, if they lie in the past, using report `/SCMTMS/SCH_DELETION`. For more information, see the system documentation.

Uploading of Schedules

Use

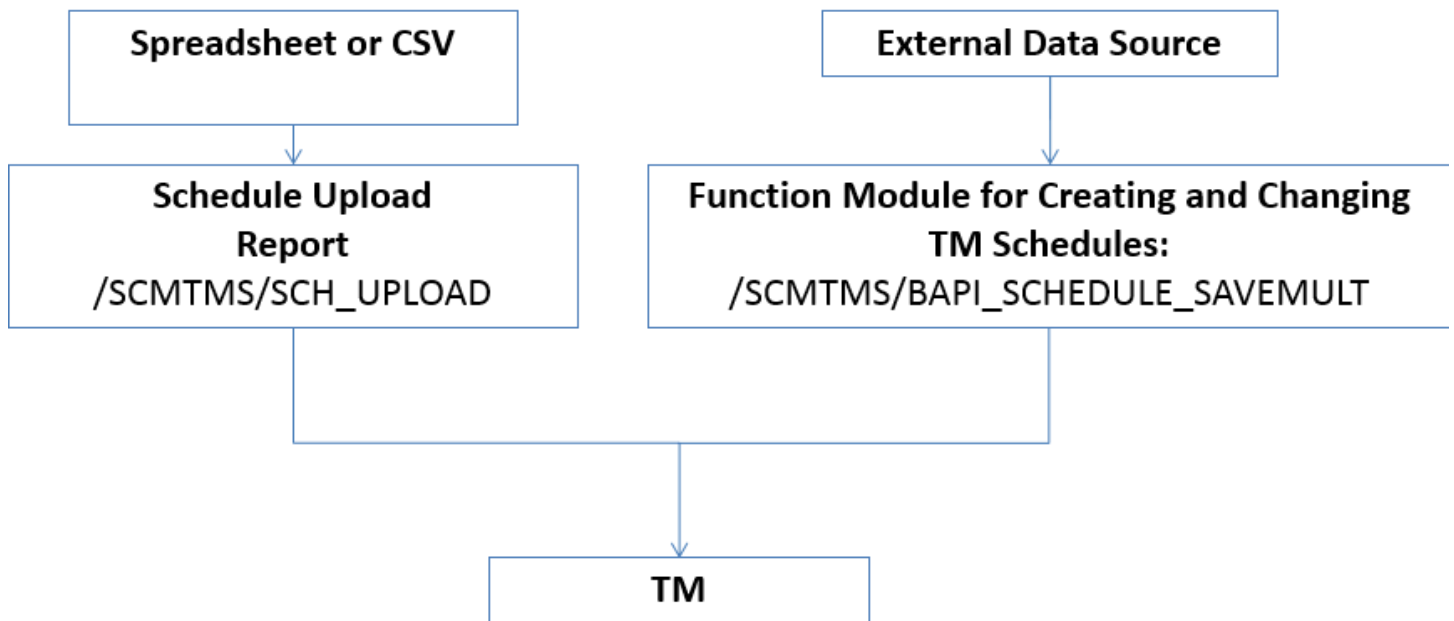
You can upload schedules from an external data source to Transportation Management (TM) by calling function module `/SCMTMS/BAPI_SCHEDULE_SAVEMULT`. This function is available for the sea, air, and road transportation modes. For more information, see the system documentation.

Alternatively, you can upload schedules in a format used by a spreadsheet program or in CSV format to TM by executing report `/SCMTMS/SCH_UPLOAD`. This function is available for the sea and air transportation modes. For more information, see the system documentation.

i Note

You can delete schedules created manually and uploaded by calling function module `/SCMTMS/BAPI_SCHEDULE_DELMULTI` or by executing report `/SCMTMS/SCH_DELETION`.

Upload Schedules



Uploading of Schedules

More Information

SAP Note [1743069](#)  for uploading schedules for the air transportation mode

SAP Note [2136548](#)  for uploading schedules for the sea transportation mode

Transportation Costs

Use

You can define transportation costs in your schedules based on the quantity of goods that are to be transported. These costs are then used by the optimizer to calculate the most cost-effective route; they do not represent actual costs.

i Note

You can also define transportation costs based on other criteria such as distance or duration in the planning profile. For more information, see [Constraints and Costs Settings](#).

Prerequisites

In Customizing for [Transportation Management](#), you have completed the activities under [►Master Data►Transportation Network►Schedules](#) .

Features

You can calculate transportation costs in gateway schedules and carrier schedules based on the costs per quantity.

Gateway Schedules

When you enter quantity costs in a gateway schedule such as a sailing schedule or an LCL overland schedule, the optimizer calculates the transportation costs by multiplying the costs per quantity by the actual quantity to be transported. The system only considers the stage between the port of loading and the port of discharge.

❖ Example

If you enter a quantity cost of 200 per ton and want to calculate the cost of transporting 1.5 tons, the system multiplies 200 x 1.5 and returns a transportation cost of 300.

Carrier Schedules

When you enter quantity costs in a carrier schedule such as an ocean carrier schedule, the optimizer calculates the transportation costs based on the number of stages. It therefore multiplies the costs per quantity by the actual quantity to be transported, and then multiplies the result by the number of stages over which goods are transported.

❖ Example

If you enter a quantity cost of 200 per ton and want to calculate the cost of transporting 1.5 tons over 5 stages, the system multiplies 200 x 1.5 x 5 and returns a transportation cost of 1500.

Activities

To define quantity costs in schedules, open the [Create Schedule](#) app from the launchpad.

Example: Schedules for Sea Traffic

Use

As a logistics service provider (LSP), you receive an order to transport freight from your container freight station (CFS) in Nuremberg, Germany (CFS_NUR) to a CFS in Chicago, United States (CFS_CHI). You want to specify that all freight transported between Nuremberg and Chicago is to be routed by default via the ports in Hamburg (PORT_HAM) and New York (PORT_NYK) and their respective CFSs (CFS_HAM and CFS_NYK).

Process

1. You create a sailing schedule from the CFS in Hamburg (CFS_HAM) to the CFS in New York (CFS_NYK), and specify the corresponding ports as Hamburg (PORT_HAM) and New York (PORT_NYK).
You specify LCL (less than container load) as the shipping type.
2. You create a road gateway schedule with which goods are transported from the CFS in Nuremberg (CFS_NUR) to the CFS in Hamburg (CFS_HAM), and a second road gateway schedule with which goods are transported from the CFS in New York (CFS_NYK) to the CFS in Chicago (CFS_CHI).
3. You create a default route with the following sequence of locations:
 - a. CFS in Nuremberg (CFS_NUR)
 - b. CFS in Hamburg (CFS_HAM)
 - c. CFS in New York (CFS_NYK)
 - d. CFS in Chicago (CFS_CHI)

- On the **Stages** tab page of the forwarding order user interface under **Actual Route**, you select the stage from Nuremberg to Chicago and choose **Schedule** > **Select**.

The system lists a number of proposals.

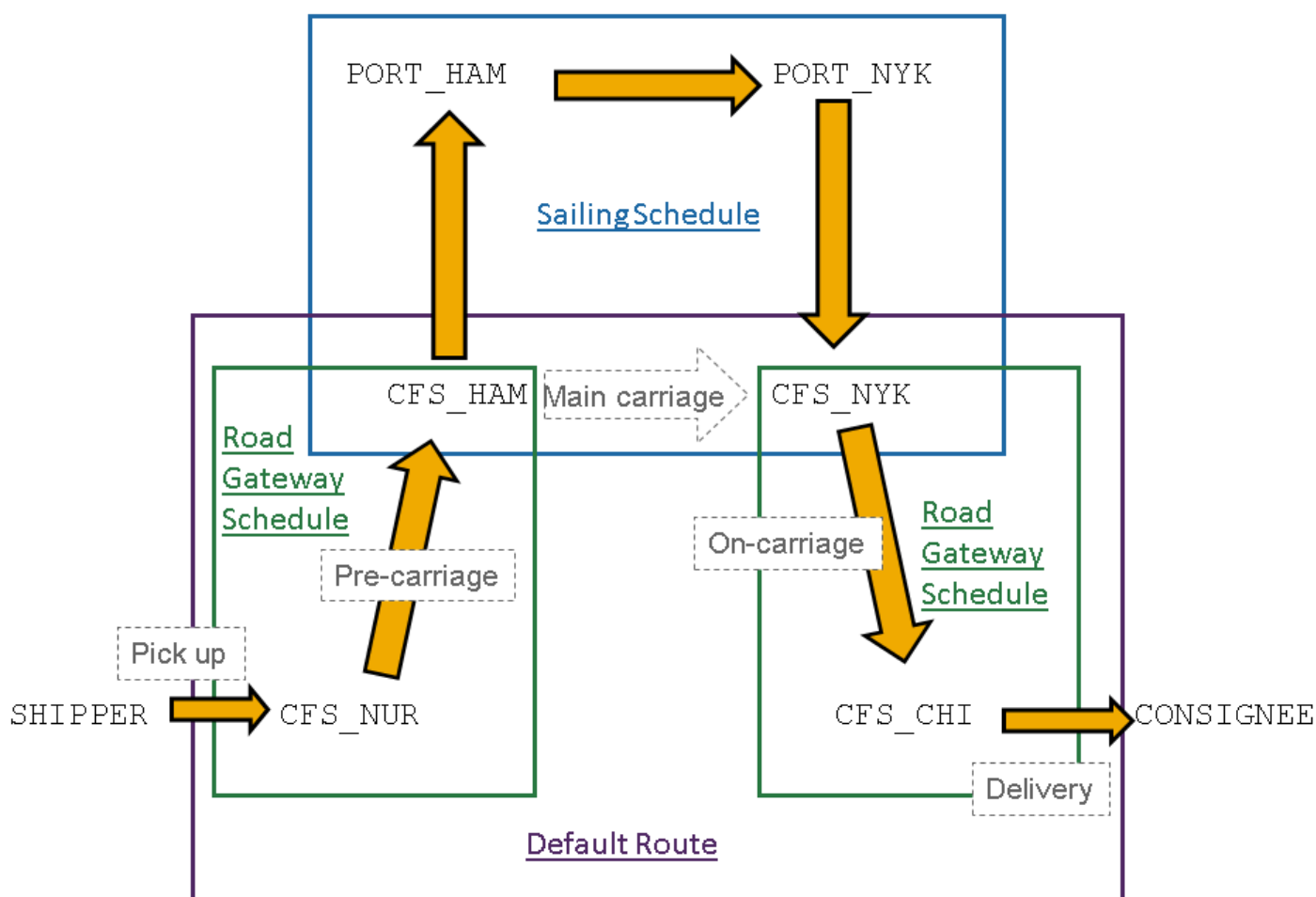
i Note

Note that the system only takes into account schedules with shipping type LCL or schedules that do not have a defined shipping type.

- You select the transportation proposal that you require and choose **Done**.

The system automatically adds the pre-carriage and on-carriage stages along with additional data such as the port of loading and port of discharge, voyage number, and the name of the vessel.

This process is shown in the following graphic:



Example: Using sailing schedules, road gateway schedules, and default routes

Result

When you select the appropriate sailing schedule from the forwarding order, the system automatically enhances the data on the **Stages** tab page by adding the ports of loading and discharge, as well as the pre-carriage and on-carriage stages.

Use of Departure Rules

Use

By specifying departure rules, you can generate departures automatically. The system calculates the arrival and departure dates and times at each stop in a stop sequence, taking into account the following durations and times for each stop:

- Transit duration
- Length of stay
- Cargo cut-off time
- Availability time

You can generate concrete departures from your departure rules and adjust them individually for each departure if required.

You can define departure rules at schedule or transportation stage level for carrier schedules and gateway schedules.

You can also apply departure rules from a referenced carrier schedule to a gateway schedule.

Prerequisites

You have created the required locations in the master data (see [Location](#)).

Features

Creation of Departure Rules at Schedule Level

If your standard stop sequence has just one source location and one destination location, your departure rule encompasses just one transportation stage. The departure rule that you define therefore applies at schedule level.

Creation of Departure Rules at Transportation Stage Level

To be able to define departure rules at transportation stage level, your standard stop sequence must contain more than two locations.

When you create departure rules at transportation stage level, you can define multiple departure rules with diverging departure times and arrival times for the various days of the week for which your schedule applies. This allows you to define different transit durations, lengths of stay, cargo cut-off times, and availability times for each day of the week.

Referencing Schedules with Departure Rules

In the gateway schedule with reference, you can reference a departure rule for a transportation stage from a carrier schedule for each transportation stage of the main carriage. The referenced departure rules can come from different carrier schedules. Alternatively, you can copy the departures for the entire main-carriage stage directly from a referenced carrier schedule.

If times are changed in the referenced carrier schedule, the system changes the status of the referenced data in the referencing gateway schedule from **Data Is Up-to-date** to **Data Is Not Up-to-date**. In the departure rule of the gateway schedule with reference, the system enters the status of the referenced data at transportation stage level. From here you can navigate via a link to the relevant referenced carrier schedule, and there check exactly which times have changed. If you transfer these changes to your gateway schedule with reference either by assigning the same carrier schedule with the up-to-date data for the transportation stage concerned again, or by assigning a different carrier schedule for the first time, the system changes the status of the referenced data from **Data Is Not Up-to-date** back to **Data Is Up-to-date**.

Automatic Calculation of Times

If, in the **Departure Rules** screen area of your schedule, you select the **Auto Fill Times** checkbox and specify particular times for various transportation stages, the system calculates the missing times for the various transportation stages of the departure rule automatically, taking the different times zones of each of the locations into account. You can choose between the following modes for the automatic calculation of times:

- Duration-based mode

You use the duration-based mode to calculate the arrival time of a transportation stage and the departure times and arrival times of later transportation stages based on the transit duration and the length of stay of a transportation stage.

- Time-based mode

You use the time-based mode to calculate the transit duration and the length of stay of transportation stages based on the departure times and arrival times of these transportation stages.

In carrier schedules, for example, in time-based mode the system uses the arrival time at the destination location of the first transportation stage and the length of stay at the destination location of the first destination stage to calculate the departure time at the source location of the second transportation stage.

In the gateway schedule, you can also define the following durations and times for the pre-carriage and on-carriage as separate transportation stages in the departure rule:

- Transit duration
- Length of stay
- Cargo cut-off time
- Availability time

You can change the cargo cut-off times in the transportation stage for the pre-carriage only and the availability time in the transportation stage for the on-carriage.

If you create a gateway schedule with reference in the transportation modes sea and air, you can change the departure time from the source location of the first transportation stage and the arrival time at the destination location of the last transportation stage in the gateway schedule. If you want to change any other times in the departure rule from the carrier schedule, you can only do so in the actual carrier schedule.

Activities

You first define in your schedule the locations of the standard stops for your schedule in the **Standard Stop Sequence** screen area and specify the cut-off time and length of stay for each location. Based on this standard stop sequence, you create the departure rules for the different departure times on the various days of the week.

After you have created your departure rules, choose **Create Trips** to calculate the actual departure times automatically. If you want to change the dates of the departures generated automatically from the departure rule, select the trip in question and enter the deviating dates. You can also add a trip that is not based on a departure rule. If you have assigned a new or updated carrier schedule to a transportation stage of your departure rule, you have to regenerate the departures with the changed data.

Example

[Example of the Use of Departure Rules](#)

Example of the Use of Departure Rules

You want to define several departure rules for your carrier flight schedule from Frankfurt airport to John F. Kennedy airport in New York via London Heathrow with differing departure times and transit durations on the various days of the week. One of these departure rules applies every Monday.

In this departure rule, you specify the departure time from Frankfurt airport as 05:00 CET and the transit duration for the flight to London Heathrow, which is one hour. The system calculates the arrival time in London Heathrow as 05:00 GMT UK.

Based on your cut-off times and the length of stay, which you specified as three hours, the system calculates the departure time for the flight to John F. Kennedy airport in New York. The time determined is 08:00 GMT UK. Since you specified eight hours as the transit duration, the system calculates the arrival time at John F. Kennedy airport in New York as 11:00 EST.

If the departure time from Frankfurt airport is moved back by two hours to 07:00 CET, the system adjusts the departure times and arrival times in London and New York accordingly. The new arrival time in New York is therefore 13:00 EST.

You can use this departure rule as a transportation stage in the departure rule for a master flight schedule by first entering your airport of departure and your airport of destination in the standard airport sequence of your master flight schedule, and assigning a corresponding carrier flight schedule to it. If you then create departure rules for your master flight schedule, you can select a suitable departure rule from the referenced carrier flight schedule. The departure rule from the carrier flight schedule appears as the only transportation stage in the departure rule for the master flight schedule.

The pre-carriage for this master flight schedule is from a gateway at Frankfurt airport to the airport. In the on-carriage, the goods are transported from the John F. Kennedy airport in New York to a gateway at the airport. The source location of the main carriage is Frankfurt airport and the destination location is John F. Kennedy airport in New York. The departure time from Frankfurt airport is 07:00 CET and the arrival time in New York is 13:00 EST.

In the gateway schedule, you can now define the transportation stages of the pre-carriage and on-carriage as parts of the departure rule. The departure time from your gateway at Frankfurt airport is 05:00:00 AM CET. The transit duration from the gateway to the airport amounts to two hours. This means the goods arrive at Frankfurt airport on time for the departure at 07:00.

You specify the length of stay at John F. Kennedy airport in New York as two hours. The availability time at the gateway is therefore 15:00 EST. The customer can pick the goods up here at 15:00 EST.

If, in the referenced carrier flight schedule, the departure time at Frankfurt airport is moved from 5:00 EST to 7:00 CET so that the arrival time at John F. Kennedy airport in New York is no longer 13:00 EST but 15:00 EST, the system changes the status of the referenced data in the master flight schedule from **Data Is Up-to-date** to **Data Is Not Up-to-date**. In the departure rule of the master flight schedule, you can navigate via a link to the referenced carrier flight schedule, and there check exactly which times have changed. If you transfer the changed departure time at Frankfurt airport and the changed arrival time at John F. Kennedy airport in New York to your master flight schedule, the system changes the status of the referenced data from **Data Is Not Up-to-date** back to **Data Is Up-to-date**.

Trade Lane

Definition

A trade lane is a classification of any transport-related activities representing a combination of zones or locations combined with a transportation mode or means of transport and providing an orientation between the source and destination.

Examples of trade lanes include:

- All transports from Hamburg to New York
- All transports within Germany
- All transports that leave Bavaria

Trade lanes are not a substitute for transportation lanes. For planning purposes transportation lanes must be maintained even if a suitable trade lane exists.

Trade lanes can form a hierarchy. This means that more generic trade lanes can contain more specific trade lanes. For example, “Within Germany Using Means of Transport Truck” contains “Along the Way from Frankfurt to Hamburg with Means of Transport Refrigerated Truck” and “Within Bavaria with Means of Transport Truck for Liquids”.

Trade lanes may overlap, for example, “From Germany Using Airplane” and “Within Europe Using Airplane”.

Trade lane hierarchies contain the following:

- Transportation zone hierarchies, for example, “From Germany” contains “From Bavaria”, and this contains “From Plant_MÜNCHEN”
- Hierarchical relationships between means of transport, for example, “Within Germany Using Means of Transport Truck” contains “Within Germany Using Means of Transport Refrigerated Truck”
- Means of transport/transportation mode relationship, for example, “Within Germany Using Transportation Mode Street” contains “Within Germany Using Means of Transport Refrigerated Truck”

In previous releases, trade lanes were created from, and associated with other business objects such as Business Shares, Allocations, and Freight Agreements. You can now use trade lanes as full master data objects, creating, copying, and reusing them according to the needs of your business. You can view all available trade lanes in a worklist and, view information such as a where-used list.

You can configure the types of trade lanes that can be used to create trade lanes. For more information see Customizing for Transportation Management under ► [Transportation Management](#) ► [Master Data](#) ► [Transportation Network](#) ► [Trade Lane](#) ►

Structure

You define the following settings for a trade lane:

- Description
- Transportation Mode
- Means of Transport
- Orientation

Possible settings:

- Inbound
 - Outbound
 - From
 - To
 - Along
 - Within
- Source
- Source Type
- Destination

- Destination Type

Integration

This business object is used by several processes in Transportation Management (TM). For more information, see:

- [Creation of Business Shares and Transportation Allocations](#)
- [Carrier Selection](#)
- [Agreement Maintenance](#)

Trade Lane Upload

Use

You use this function to automatically create and delete trade lanes in TM by uploading a list of trade lanes.

Integration

For more information about the integration of Trade Lanes into TM, see [Trade Lane](#).

Prerequisites

You have defined number range intervals for trade lanes Customizing for Transportation Management. For more information see Customizing for Transportation Management under ► [Transportation Management](#) ► [Master Data](#) ► [Transportation Network](#) ► [Trade Lane](#) ► [Define Number Range Intervals for Trade Lanes](#) ►.

i Note

In order to use a specific Trade Lane ID, you have to define an external number range. If you do not define an external number range, the report assigns automatically a number from the internal number range to the trade lane.

You have defined types of trade lanes in Customizing for Transportation Management. For more information see Customizing for Transportation Management under ► [Transportation Management](#) ► [Master Data](#) ► [Transportation Network](#) ► [Trade Lane](#) ► [Define Trade Lane Types](#) ►.

You have created an XLS or XLSX file containing a list of trade lanes that you want to create or delete in TM.

- The list contains the following columns:
 - Excel ID
 - Trade Lane ID
 - Trade Lane Type
 - Transportation Mode
 - Means of Transport
 - Orientation
 - Source Type

- Source
- Destination Type
- Destination
- Language
- Trade Lane Description
- You enter the details for each trade lane in a separate line.
- You can enter multiple descriptions of a single trade lane in different languages. In order to add a description in an additional language, enter a line with the same **Excel ID**. Leave all fields blank, except **Excel ID**, **Language** and **Trade Lane Description**. When you use the same **Excel ID** repeatedly, the system adds multiple descriptions for the same trade lane.
- The following restrictions apply:
 - You cannot enter identical trade lanes with different Trade Lane IDs.
 - You must assign a unique Excel ID to each trade lane.

i Note

You cannot enter both a Transportation Mode and Means of Transport at the same time.

Activities

After you have selected a file that contains the list of trade lanes that you want to upload, you can choose whether you want to create or to delete these trade lanes in TM.

Furthermore, you can specify which lines of the XLS or XLSX file shall be uploaded, in order to exclude explanatory header lines or similar irrelevant lines.

Output

After the list of trade lanes has been uploaded to TM, the system runs several checks on the entered data, for example, whether the data corresponds to predefined requirements or whether entered data is a duplicate of already existing trade lanes.

After checking the data the system displays a log message and writes the data to TM. If you choose to run the upload in test mode, the system runs the background checks and creates a log message, but does not write the data to TM.

Default Route

Use

You use this function to define a sequence of transshipment locations. For each default route you can specify the transportation mode, carrier, schedule, and so on, for each transportation stage. The system considers default routes as possible routes for a shipment between two locations.

In the transportation cockpit, you can find default routes for transportation stages (freight unit and transportation unit stages) using the **Apply Default Route** button.

You can also find default routes on the user interfaces of these business documents.

You can also use default routes in VSR optimization (see [Use of Default Routes in VSR Optimization](#)).

Prerequisites

- You have defined [locations](#).
- You have configured the settings in the Customizing activity [Define Default Route Types](#) under [Transportation Management > Master Data > Transportation Network](#).

Activities

To create a default route, open the [Create Default Route](#) app on the launchpad.

Transportation Management is shipped with default route type 3000.

More Information

For more information about how default routes are used in conjunction with forwarding orders, see [Direct Creation of Freight Documents and Selection of Schedules](#).

For an example of how default routes can be used together with schedules, see [Example: Schedules for Sea Traffic](#).

For more information about the interaction of the default route and the determination of transportation distance and transportation duration, see [Determination of Transportation Distance and Transportation Duration](#).

Use of Default Routes in VSR Optimization

Use

You use this function to consider default routes when you generate routes for the transport of freight units and transportation units in VSR optimization. In your optimizer settings, you can specify how default routes are to be used when you process freight units and transportation units in VSR optimization.

i Note

This function is not supported for trailer units.

Prerequisites

You have defined default routes (see [Default Route](#)).

You have selected the required checkboxes in Customizing for Transportation Management under [Transportation Management > Master Data > Transportation Network > Define Default Route Types](#) under [Optimizer Settings](#).

In the optimizer settings of your planning profile, you have selected an option to consider default routes in VSR optimization. To do this, open the [Edit Optimizer Settings](#) app on the launchpad.

Features

When you use default routes in VSR optimization, you have the following options:

- Only consider default routes

The VSR optimizer searches only for default routes as possible routes for the transport of your freight units and transportation units.

- Prefer default routes

The VSR optimizer searches for default routes but also generates alternative routes as possible routes for the transport of your freight units and transportation units. If the VSR optimizer identifies a default route as a possible route, it automatically selects the default route without considering the planning costs of the alternative routes.

- Also consider default routes

The VSR optimizer searches for default routes but also generates alternative routes as possible routes for the transport of your freight units and transportation units. The VSR optimizer then selects the route with the lowest planning costs.

- Do not consider default routes

The VSR optimizer does not search for default routes as possible routes for the transport of your freight units and transportation units. If the VSR optimizer generates only routes based on transshipment locations, these routes may by chance correspond to one of your default routes.

Standard Integration with SAP HANA spatial services

Transportation Management is capable of using information determined by external geographical information system (GIS) applications for geocoding, distances and durations, and georouting.

Using external GIS services requires custom code since the used Supply Chain Execution (SCE) GIS integration framework is a generic framework and does not provide any vendor-specific connectors. With the introduction of SAP HANA spatial services, SAP now offers a corresponding integration for this framework. The use of this integration enables the consumption of SAP HANA spatial services without writing custom coding.

For more information about setting up this standard integration, see SAP Note [2751622](#)  (Setup Guide for SAP HANA Spatial Services in SCE Applications).

For more information about SAP HANA spatial services, see [SAP HANA spatial services](#).

Related Information

[Using GIS Metrics](#)

Using GIS Metrics

Use

You can use GIS (Geographical Information System) metrics for monitoring and reporting purposes. GIS metrics are created in the GIS Integration Framework when GIS requests are processed. In Transportation Management, you can display GIS metrics and delete them.

Prerequisites

For more information about the prerequisites for using the GIS Integration Framework, see SAP Note [2751622](#)  (Setup Guide for SAP HANA Spatial Services in SCE Applications).

Activities

To display GIS metrics, on the SAP Easy Access menu choose ► **Logistics** ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Display GIS Metrics** ►.

To delete GIS metrics, on the SAP Easy Access menu choose ► **Logistics** ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Delete GIS Metrics** ►.

Related Information

[Display of GIS Metrics](#)

[Deletion of GIS Metrics](#)

[Process Controller Configuration](#)

Display of GIS Metrics

Use

You can use this report to display GIS (Geographical Information System) metrics. GIS metrics are created in the GIS Integration Framework when GIS requests are processed. They can be used for monitoring and reporting purposes.

Features

Selection

You need to indicate the following:

- A process controller strategy that is relevant for the GIS Integration Framework
- A horizon or a time window to select the relevant entries

Outputs

On the **GIS Metrics Display Summary** screen, you can see the following information, for example:

- The GIS service that was used (geocoding, georouting, or distance and duration determination)
- The GIS provider who processed the GIS requests
- The number of successful and failed calls to SAP HANA spatial services
- The number of successful GIS requests, such as the number of routes for georouting
- The number of failed GIS requests
- The number of transaction objects, such as the number of source locations for distance and duration determination

Activities

To start the report, enter transaction code /SCMB/GIS_METR_DISPL in the command field.

Related Information

Deletion of GIS Metrics

Use

You can use this report to delete GIS (Geographical Information System) metrics to clean up your database. GIS metrics are created in the GIS Integration Framework when GIS requests are processed.

Features

Selection

You need to indicate the following:

- A process controller strategy that is relevant for the GIS Integration Framework
- A horizon or a time window to select the relevant entries

Outputs

On the [GIS Metrics Deletion Summary](#) screen, you can see the following information, for example:

- The GIS service that was used (geocoding, georouting, or distance and duration determination)
- The GIS provider who processed the GIS requests
- The number of successful and failed calls to SAP HANA spatial services
- The number of successful GIS requests, such as the number of routes for georouting
- The number of failed GIS requests
- The number of transaction objects, such as the number of source locations for distance and duration determination

Activities

To start the report, enter transaction code /SCMB/GIS_METR_DEL in the command field.

Related Information

[Process Controller](#)

Transportation Mode

Definition

You can use these features to define master data that is specific to a particular transportation mode, for example, air freight or ocean freight. You can specify the following settings:

- Air freight and ocean freight codes (see [Transportation-Mode-Specific Codes](#))

- Air waybill stock numbers (see [Waybill Stock Definition](#))

More Information

For more information about transportation modes, see Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Transportation Lane](#) > [Define Transportation Mode](#).

Transportation-Mode-Specific Codes

Definition

Transportation Management allows you to define codes that are specific to various transportation modes. These codes are explained in detail below.

Air-Freight-Specific Codes

- Location codes

You can define air-freight-specific locations such as airports or cities in the vicinity of airports. When you create an air-freight-specific document such as an air forwarding order or air freight booking, you can enter the airport of departure and the airport of destination on the user interface. The system then automatically determines the correct location code from the master data and displays it in read-only mode.

If you enter a one-time location (see [One-Time Location](#)) in your document, you can specify the location code directly in the document.

The system also takes location codes into account when calculating charges.

You define locations codes in Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Location](#) > [Define IATA Location Codes](#).

- Aircraft type codes

You can define aircraft type codes and assign a category to them to specify, for example, whether the aircraft is a freighter or passenger aircraft.

You define aircraft type codes in Customizing for [Transportation Management](#) under [Master Data](#) > [Resources](#) > [Define IATA Aircraft Type Codes](#).

- Airline codes

You can define airline codes for each of your air-freight carriers and assign an air waybill prefix to them. You can also specify whether the carrier is a member of the International Air Transport Association (IATA).

When you then create an air-freight document such as an air forwarding order, you can enter your airline code on the user interface. The system then automatically identifies the associated air waybill prefix. If there is a one-to-one relationship between the airline code and the associated carrier, the system can also determine the relevant carrier when you enter the airline code.

You define airline codes in Customizing for [Transportation Management](#) under [Master Data](#) > [Business Partners](#) > [Define IATA Airline Codes](#).

- Handling codes

You can define handling codes to specify how certain types of freight are to be transported. For example, you can specify that goods are perishable and must, therefore, be transported in a refrigeration unit.

You define handling codes in Customizing for **Transportation Management** under **Basic Functions** **General Settings** **Define Handling Codes**

- Traffic conference areas and region codes

You can define traffic conference areas, which designate a part of the world, and assign them to regions. You can also assign a country or region to the different regions.

You define traffic conference areas and region codes in Customizing for [Transportation Management](#) under [Master Data](#) [Transportation Network](#) [Define IATA Traffic Conference Areas](#) and [Master Data](#) [Transportation Network](#) [Define IATA Region Codes](#) respectively. You can also assign countries or regions to region codes by choosing [Master Data](#) [Transportation Network](#) [Assign Countries/Regions to IATA Regions](#).

- Other charge codes and notes

You can define charge codes that are defined by the International Air Transport Association (IATA) as **Other Charges**. You can also enter IATA notes, which the system takes into account when calculating charges.

You define other charge codes and notes in Customizing for **Transportation Management** under **Basic Functions** **Charge Calculation** **Basic Settings** **Maintain IATA Other Charge Codes** and **Basic Functions** **Charge Calculation** **Air Freight Settings** **Maintain IATA Notes**.

Ocean-Freight-Specific Codes

- Standard carrier alpha code (SCAC)

You define SCACs in Customizing and specify the period for which they are to be valid. When you assign SCACs to carriers, the values that you have defined are proposed in the SCAC input help. You can, therefore, enter a SCAC to find the associated carrier and vice versa.

You define SCACs in Customizing for **Transportation Management** under **Master Data** **Business Partners** **Define Standard Carrier Alpha Codes**.

- United Nations Code for Trade and Transport Locations (UN/LOCODE)

You define UN/LOCODEs in the location master data and assign them to a container freight station (CFS) or a port, for example. When you enter a location in your ocean forwarding order or ocean booking, the system automatically determines the correct UN/LOCODE from the master data and displays it in the document in read-only mode.

If you enter a one-time location in your document, you can specify the UN/LOCODE directly in your document.

You define UN/LOCODEs in the [Define Location](#) app which you can open from the launchpad.

More Information

For more information about defining business partners (airlines), see [Definition of Business Partners](#).

For more information about locations, see [Location](#).

For more information about SCACs and airline codes in Freight Order Management, see [Carrier Categorization](#).

Product

Definition

A tangible or intangible good that is part of the business activities of a company. It can be traded and it contributes directly or indirectly to the value chain.

Use

You use this object to manage global product master data.

i Note

Transportation Management (TM) does not use location-dependent products. TM uses product information which is maintained under

- Basic Data 1
- Basic Data 2
- Additional Data
 - Material description in different languages
 - Units of Measure

Structure

The product master data is maintained on a number of tab pages.

i Note

Some of the tab pages contain data that is not relevant for TM.

You identify each product with a unique product number, and an appropriate description.

You define whether the product is a packaging material by choosing the material group for packaging materials.

More Information

[Product Master](#)

Definition of Products

Use

You use this process to define products that are transportable goods.

Prerequisites

In Customizing for **Transportation Management**, you have defined the following:

- You have defined transportation groups under ► **Transportation Management** ► **Master Data** ► **Product** ► **Maintain Transportation Group** .
- You have defined the packaging material types under ► **Transportation Management** ► **Master Data** ► **Product** ► **Define Packaging Material Types** .

Process

This is custom documentation. For more information, please visit [SAP Help Portal](#).

You define a product in the [Create Material](#) app, which you can open from the launchpad.

Use of Change Documents for Products

Use

You use this process to trace changes that are made to products in Transportation Management (TM).

The changes are logged in change documents, which allow you to find out at any time, what was changed, when the changes occurred, and how the changes were made.

Process

You display the change documents for products in the [Display Change Documents](#) app, which you open from the launchpad.

Classification of Goods

Commodity codes are a set of classification codes used in the global marketplace to identify goods and services. Commodity codes group together goods and services and can differ for each country.

Use

You can classify and evaluate your goods to be transported (commodities) according to characteristics of your choice. Each commodity has its own commodity code. You can group commodity codes using a commodity code type. The classification schemas (commodity code types) used for this purpose may depend on the transportation mode or country, for example.

When classifying goods, you can take into account transportation-relevant product characteristics, such as weight, density, loading properties, and handling (of dangerous goods, for example).

You can use this function in Transportation Management (TM) for the following purposes:

- Charge calculation for incoming invoices
- Printing

Commodity codes can be used when printing business documents. For more information, see [Printing](#).

You have the following two options for determining commodity codes in TM:

- Using commodity codes assigned to the material master
- Using commodity codes assigned to items locally

Related Information

[Commodity Codes Assigned to Material Master](#)

[Commodity Codes Assigned to Items Locally](#)

Commodity Codes Assigned to Material Master

You can use a source based on SAP Governance, Risk and Compliance (GRC), that is, a source in the master data master, to create commodity codes in Transportation Management.

Prerequisites

In Customizing for **Transportation Management**, under **Freight Order Management > Freight Order > Define Freight Order Types**, you have chosen **Use Commodity Codes Assigned to Material Master**.

You have made settings in the following activities in Customizing for **Transportation Management**, under **Master Data > Classification of Goods**:

- **Define Numbering Scheme for Commodity Codes**
- **Define Numbering Scheme Content for Commodity Codes**
- **Activate Numbering Scheme Content for Commodity Codes**
- **Assign Numbering Scheme Content for Commodity Codes to Country/Region**

Use

You can use numbering schemes to create entries for commodity codes. You can maintain all the different types of numbering schemes, their respective content, the link between the numbering scheme and the content, and also the country assignment by making settings in Customizing.

You then assign the commodity code to a product and classify your products. To do so, you use the following apps on the launchpad:

- **Manage Commodity Codes for Transportation**

You use this app to create entries for commodity codes for your chosen numbering scheme content.

You select the numbering scheme content, enter the commodity code for that numbering scheme content, and enter the validity period. You can enter a validity period within the validity period you chose in the Customizing activity **Activate Numbering Scheme Content for Commodity Codes**. You can also enter a description for the commodity code in a language that you choose.

- **Classify Products - Commodity Codes for Transportation**

The new entry for the commodity code is then visible in this app. You can search for the new entry using the numbering scheme and assign the commodity code to a product and enter the validity period.

You can display the commodity codes assigned to products by navigating to the detailed view of a freight order, freight booking, freight unit, or consignment order. In the **Items** tab, select a product, and then **Commodity Codes**.

Commodity Codes Assigned to Items Locally

You can use a local source in Transportation Management to create commodity codes.

Prerequisites

In Customizing for **Transportation Management**, under **Freight Order Management > Freight Order > Define Freight Order Types**, you have chosen **Use Commodity Codes Assigned to Items Locally**.

You have defined commodity codes and commodity code types in Customizing for **Transportation Management** under **Master Data > Classification of Goods > Define Commodity Codes**.

Features

Commodity codes classify goods (commodities) based on various classification schemas (commodity code types). You can enter commodity codes that belong to various classification schemas as value lists in Customizing.

You can make a selection on the following UIs:

- Forwarding Order

You can assign the commodity codes to cargo items when you create a forwarding order.

- Freight Booking

The system copies the commodity codes from the forwarding order. You can adjust the commodity codes in the freight booking as required. For more information, see [Freight Booking](#) and [Management of Air Freight Bookings for Airlines](#).

- Freight Order

The system copies the commodity codes from the forwarding order. You can adjust the commodity codes in the freight order as required. For more information, see [Freight Order](#).

Example

The commodity code type **Cereal** has the abbreviation **CR** and the following commodity codes:

Commodity Code	Description
100	Wheat
101	Organic wheat
200	Barley
201	Organic barley
300	Rye
301	Organic rye

Example: Classification of Goods for Commodity Codes Assigned to Items Locally

Use

You use this process to classify and evaluate your goods to be transported according to characteristics of your choice.

Prerequisites

In Customizing for **Transportation Management**, under **Freight Order Management > Freight Order > Define Freight Order Types**, you have chosen **Use Commodity Codes Assigned to Items Locally**.

You have defined commodity codes and commodity code types in Customizing for [Transportation Management](#) under [Master Data](#) > [Classification of Goods](#) > [Define Commodity Codes](#).

Process

The following example outlines how to classify a cargo:

1. Changing the forwarding order

Open the [Change Forwarding Order](#) app on the launchpad. You display the detail data for a main cargo item and open the [Commodity Codes](#) tab page.

You define the cargo as commodity code type [Cereal](#) with commodity code [300](#) (rye) and save your entries. When you save the data, the system copies the classification to the corresponding freight order or freight booking.

2. Displaying or checking the freight order or freight booking for the forwarding order

Open the [Change Air Freight Booking](#) or [Change Ocean Freight Booking](#) app on the launchpad. You display the detail data for a main cargo item and open the [Commodity Codes](#) tab page. You now check whether the system has copied the commodity code type and commodity code correctly from the forwarding order.

→ Recommendation

From the forwarding order, you can also navigate to the follow-on documents [Freight Order](#) or [Freight Booking](#) using the [Document Flow](#) tab page, and to the [Commodity Codes](#) tab page using the [Cargo Management](#) tab page.

3. Changing the freight order or freight booking for the forwarding order

In the freight order or freight booking, you display the detail data for a main cargo item and open the [Commodity Codes](#) tab page. You select the [Deviating Commodity Codes](#) checkbox, change the commodity code to [301](#) (organic rye), and save your entries. When you save your data, the commodity codes in the freight order or freight booking are adjusted. The commodity codes in the forwarding order remain unchanged. Once the data is saved, no further data is copied from the forwarding order to the freight order or freight booking.

Result

Goods have been classified in the forwarding order.

Goods in the freight order or freight booking are classified either from the forwarding order or locally in the freight order or freight booking using the [Deviating Commodity Codes](#) checkbox.

Equipment Groups and Equipment Types

Use

You can use equipment types to group resources that have the same attributes. You can use equipment groups to group equipment types into categories. If you work with containers, you need to assign the container groups to the relevant transportation modes.

You can perform planning based on equipment groups and types.

Features

Grouping of Resources

This is custom documentation. For more information, please visit [SAP Help Portal](#).

When you define a resource in the resource master data transaction, you can assign an equipment group and type. By doing so, this resource belongs to this equipment type and shares its attributes. You can also copy equipment group and type data into your resource and use it as a template.

You can use an equipment type in various profiles, such as the capacity availability settings, to control the behavior of all resources assigned to this equipment type.

Planning Based on Equipment Types

For scenarios, in which you regularly subcontract your transportation demands to a carrier and prefer to work with vehicle or container types, we recommend that you define equipment types. An equipment type can be assigned to an unlimited number of capacity documents at the same point in time. Note that if you want to use planning based on equipment types, the name of the equipment type needs to be unique across all equipment groups.

For more information, see [Planning Based on Equipment Types](#).

Freight Unit Building

When you create freight units, you can take into account equipment types and their dimensions instead of the planning-relevant quantities.

For more information, see [Consideration of Equipment Data During FU Creation](#).

Container

You can use equipment groups and types, for example, when creating forwarding order or forwarding quotation items of the category **Container**. The physical property data of the container is then available in the forwarding order and subsequent business documents as information for the planner. In addition, after you have entered the equipment group, the equipment type, and the number of containers in the forwarding order, the system calculates the tare weight and the capacity of the container in twenty-foot equivalents units (TEUs) based on the Customizing settings. If you have assigned an equipment group to the item type in Customizing, the equipment group is entered automatically for an item with this item type, and equipment types of this group are displayed for selection.

For more information, see [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

Carrier profiles

When you define carrier profiles, you can use equipment types to describe the kind of equipment that is owned by the carrier.

For more information, see [Definition of Carrier Profiles](#).

i Note

- Equipment group and equipment type are generic terms. In Transportation Management (TM), the most specific term is used on the user interface where possible. If this is not possible, the generic term equipment type or vehicle type is used.

The following equipment types are supported:

- Vehicle type
 - Truck type
 - Trailer type
 - Locomotive type

- Railcar type
- Container type
- Handling resource type

Activities

You define equipment groups and types in Customizing.

1. To define a new equipment group, for example, a vehicle group, select the **Vehicle Groups** view and enter a name and a description and select a resource class.
2. To define a vehicle type for your new vehicle group, select your vehicle group and then the **Vehicle Types** view and enter a name and a description. By double-clicking on the vehicle type, the details screen that lets you enter the attributes of the vehicle type is displayed.

For example, you can define capacity and weight for your container types. For your truck types, you can define parameters, such as payload weight or CO₂ emissions. For more information, see [Customizing for CO₂ Emissions Calculation](#).

For more information, see Customizing for **Transportation Management** under ►► **Master Data** ► **Resources** ► **Define Equipment Groups and Types** ►.

If required, you can implement a consistency check. For a possible implementation, see Customizing for **Transportation Management**, under ►► **Business Add-Ins (BAdIs) for Transportation Management** ► **Master Data** ► **General Settings** ► **Equipment Types and Equipment Groups** ► **BAdI: Consistency Check for Equipment Types** ►. This implementation checks whether an equipment type has been assigned to exactly one equipment group.

Customizing for CO₂ Emissions Calculation

You can set the parameters by which carbon dioxide (CO₂) emissions for a vehicle type are calculated. You must specify emissions for a laden and unladen vehicle.

Greenhouse Gas Emissions

Transportation Management (TM) measure CO₂ emissions for a road freight order based on the parameters you set for a vehicle type. You indicate the value in the standardized CO₂ equivalents (CO₂e). In addition to the parameters you indicate below, you must specify the payload weight and its unit of measure to enable the calculation of CO₂ emissions.

CO₂ Emissions (Empty)

You indicate an unladen vehicle's average CO₂ emissions.

CO₂ Emissions (Full)

You indicate a laden vehicle's average CO₂ emissions. A laden vehicle refers to a vehicle's gross combination weight, that is the weight of the vehicle and the maximum permissible load weight.

Units of Measurement

You must also indicate the unit of weight, for example, kilograms, and unit of distance, for example, kilometers.

Road Freight Order

CO₂ emissions are reflected in the [Edit Road Freight Order](#) app and the [Transportation Cockpit](#) app. For more information, see [Greenhouse Gas Emissions](#).

Activities

You set greenhouse gas emissions parameters in Customizing.

1. Select the [Vehicle Groups](#).
2. Double-click on [Vehicle Types](#).
3. For groups with multiple vehicle types, double-click on the type you wish to configure.
4. Set the parameters under the section [Greenhouse Gas Emissions](#).

For more information, see Customizing for [Transportation Management](#) under [Master Data](#) > [Resources](#) > [Define Equipment Groups and Types](#) .

Definition of Resources

Use

Resources play a central role in planning and execution within Transportation Management. Resource data is relevant to the planning of order dates taking into account working times and the available capacities of the resources. Resources, in conjunction with booking, offer the capacity needed to perform transportation activities on freight units, such as transportation, loading, and unloading activities.

Features

The following resource types exist:

- Calendar resource (see [Calendar Resource](#))
- Vehicle resource (see [Vehicle Resource](#))
- Handling resource (see [Handling Resource](#))
- Drivers (see [Driver](#))
- Transportation unit resource (see [Transportation Unit Resource](#))

i Note

Drivers are defined in Transportation Management as business partners. However, they are often handled in the same way as resources and are, therefore, described here.

Activities

To create, change, or display resources, you have the following options:

- On the [SAP Easy Access](#) screen, choose [Transportation Management](#) > [Master Data](#) > [Resource](#) > [Resource](#) .
- In SAP launchpad, open the [Define Resource](#) app.

For more information, see the following sections:

- Creating handling, vehicle, and calendar resources (see [Creating Handling, Vehicle, and Calendar Resources](#))
- Creating drivers (see [Creating Drivers](#))
- Copying resources (see [Copying Resources](#))
- Copying resources from template (see [Copying Resources from Templates](#))
- Deleting resources (see [Deleting Resources](#))

Classification of Resources

Use

You can use resource classes to group resources that have the same characteristics or types (for example, vehicle resources or handling resources). This enables you to see at a glance exactly what kinds of resources are available for transportation planning.

You assign the resource class to a means of transport or an equipment group. The system then automatically assigns the correct resource class to the resource when you specify the means of transport or equipment group.

Prerequisites

- You have defined resource classes in Customizing for **Transportation Management** under **Master Data > Resources > Define Resource Class**.
- You have defined means of transport in Customizing for **Transportation Management** under **Master Data > Resources > Define Means of Transport**.
- You have defined equipment groups in Customizing for **Transportation Management** under **Master Data > Resources > Define Equipment Groups and Equipment Types**.

Features

Vehicle Resources

You can use resource classes to specify the kind of vehicle that is to be used for your transportation activities. When you enter the means of transport in the resource master data, the system automatically determines the resource class based on the means of transport.

For example, if you have assigned the resource class **Trailer** to means of transport MTR_1 and subsequently create a resource with means of transport MTR_1, the system automatically determines that your resource is a trailer.

If you do not assign a resource class to the means of transport, the system determines the resource class based on the transportation mode and whether the vehicle resource is passive. The default values in this case are shown in the following table:

Transportation Mode	Passive	Default Resource Class
Road	No	Truck
Road	Yes	Trailer
Rail	No	Locomotive

Transportation Mode	Passive	Default Resource Class
Rail	Yes	Rail Car
Sea	Not applicable	Vessel
Air	Not applicable	Airplane

Handling Resources

You can use resource classes to specify the kind of equipment that is used to handle goods at a specific location (for example, whether the piece of equipment is a dolly or a forklift truck). When you enter the equipment group in the handling resource master data, the system automatically determines the resource class based on the equipment group.

For example, if you have assigned the resource class **Forklift Truck** to the handling resource group (that is, the equipment group) **EQ1**, the system automatically determines that the resource class is to be **Forklift Truck** when you enter **EQ1** in the resource master data.

If you do not assign a resource class to the handling resource group, the resource class defaults to **Doors**.

Transportation Unit Resources

You can use resource classes to specify the kind of transportation unit resource that is to be used for your transportation activities. When you enter the equipment group in the transportation unit resource master data, the system automatically determines the resource class based on the equipment group.

For example, if you have assigned the resource class **Container** to the transportation unit group (that is, the equipment group) **CN**, the system automatically determines that the resource class is to be **Container** when you enter **CN** in the resource master data. If you do not assign a resource class to the transportation unit group, the resource class defaults to **Container**.

More Information

[Definition of Means of Transport](#)

Related Information

[Equipment Groups and Equipment Types](#)

Resource Types

In the following sections, you can find information about the resource types:

- Calendar resource (see [Calendar Resource](#))
- Vehicle resource (see [Vehicle Resource](#))
- Handling resource (see [Handling Resource](#))
- Driver (see [Driver](#))
- Transportation Unit resource (see [Transportation Unit Resource](#))

Calendar Resource

Definition

Resource that you can assign to a calendar and optionally to a shift. It is used particularly for scheduling the goods receipt and goods issue processing time.

Use

You use calendar resources to specify operating times for locations. For more information, see [Operating Times](#).

i Note

You cannot define a capacity profile for calendar resources. For more information, see [Definition of Capacity Profiles](#).

Structure

You can define, among others, the following parameters for calendar resources:

- Shifts

For more information about shifts, see [Definition of Capacity Variants](#).

- Downtimes (see [Definition of Downtimes](#))

Vehicle Resource

Definition

An instance of a particular means of transport, or a group of identical instances of means of transport that can provide transportation services.

Use

You use this business object to map the capacity and availability of vehicles that you want to use for transportation.

You can choose up to eight dimensions and units of measurement to describe the capacity. Note that the mass and volume are predefined by default. Planning can only take the vehicle capacity into account if these correspond to the dimensions and units of measurement that you have defined in a freight unit building rule.

i Note

You cannot define the available capacity for this resource type dependent on time. However, you can define a time-dependent availability using shift sequences. For more information about defining shift sequences, see [Specification of Definitions](#) and [Creating Definitions](#).

In the planning profile, you can specify whether the system can consider a manual overload of the resource capacity.

Vehicle Group and Vehicle Type

The vehicle and vehicle group types specify attributes such as the physical properties of a certain kind of vehicle. When you have created a vehicle type you can:

- use the type as a template to create vehicle resources, copying the attributes of the type to new resource

- request a special type of vehicle resource in a forwarding order or in booking order. (This is similar to the function already available for container groups and types).

For more information see Customizing for Transport Management under ► [Transportation Management](#) ► [Master Data](#) ► [Resources](#) ► [Define Equipment Groups and Equipment Types](#) ►.

Structure

You can define, among others, the following parameters for vehicle resources:

- Passive means of transport

In Customizing, you can define vehicle resources that cannot move by themselves, by selecting the [Passive Means of Transport](#) checkbox for the means of transport assigned to the vehicle resource.

For more information, see [Definition of Means of Transport](#).

- Combination vehicle

You can combine vehicle resources as follows:

- Vehicle-resource combination

If you have your own fleet and want to dynamically couple or uncouple a tractor and trailer, we recommend that you work with vehicle-resource combinations. VSR optimization can dynamically combine the individual components for vehicle-resource combinations. This is based on the prerequisite that you have defined means-of-transport combinations in Customizing under ► [Master Data](#) ► [Resources](#) ► [Means of Transport and Compartment](#) ► [Define Means-of-Transport Combination](#) ►. For more information, see [Definition of Means-of-Transport Combinations](#).

- Combination resource

If you do not have your own fleet you can perform VSR optimization on the level of the combination resource. This means that VSR optimization does not have to deal with the coupling and uncoupling of the individual components of the combination resource. For more information, see [Definition of Combination Resources](#).

- Loading requirements.

You can specify for vehicle resources and locations whether they can be loaded or unloaded from the left, right, back, or top. Further, you can specify the equipment that is attached to a vehicle resource and the equipment that is required for loading and unloading at a location, for example, a forklift. A vehicle resource can only be loaded or unloaded at a location if it fulfills at least one of the loading direction requirements and all of the attached equipment requirements of this location. For more information see [Loading Requirements for Vehicle Resources](#).

- Compartment

You can divide the loading space of a vehicle resource into compartments.

For more information, see [Definition of Compartments](#).

- Capacity (see [Definition of Capacity](#))
- Availability (see [Definition of Availability](#))
- Minimum number of seals

You can specify the minimum number of seals of the vehicle resource. You can use this information for the definition of incompatibilities.

- Resource template (see [Copying Resources from Templates](#))
- Physical properties (see [Definition of Physical Properties](#))

- Downtimes (see [Definition of Downtimes](#))
- Qualifications (see [Definition of Qualifications](#))
- Attached equipment (see [Definition of Attached Equipment](#))
- Grouping attributes (see [Definition of Grouping Attributes](#))
- Alternative names (see [Definition of Alternative Names](#))

Integration

To define vehicle resources, you must have defined means of transport. For more information, see [Definition of Means of Transport](#).

More Information

[Freight Unit Building Rule](#)

[Definition of Dimensions and Units of Measurement](#)

[Planning Profile](#)

[Incompatibilities](#)

Loading Requirements for Vehicle Resources

You can define loading requirements for vehicle resources.

These requirements can be taken into account in VSR optimization, load consolidation, and for the generation of transportation proposals. The system considers the requirements specified for locations and resources and can identify incompatibilities that would, for example, make a resource unsuitable for use at a specific location. For manual planning, you can specify how the system should handle, for example, a plan that includes a resource that is not compatible with the specified location.

Features

Loading Direction Profiles

You can use loading direction profiles to specify:

- the direction from which loading or unloading is possible
- a preference for loading direction, if required (not yet supported)

For more information see Customizing for **Transportation Management** under ► **Master Data** ► **Resources** ► **General Settings** ► **Define Loading Direction Profile** ►.

You can also specify loading direction profiles for locations. For more information see [Loading Requirements for Locations](#).

Attached Equipment Profiles

You can define attached equipment profiles that describe the equipment provided by the resource for loading/unloading. For more information see Customizing for **Transportation Management** under ► **Master Data** ► **Resources** ► **General Settings** ► **Define Attached Equipment Profile** ►.

You can also define attached equipment profiles to describe the equipment provided by a location for loading/unloading. For more information see [Loading Requirements for Locations](#).

In assessing compatibility, the system compares the location's attached equipment profile with the resource's attached equipment profile. Other equipment that is assigned to the resource, for example, loading equipment in a vehicle loading profile, is not relevant in this context.

Vehicle Loading Profiles

You can use vehicle loading profiles to define specific loading constraints of a vehicle as follows:

- Loading Direction Profile
- Loading Equipment Group/Type (not yet supported)

Models a loading equipment, for example, a forklift, the dimensions and weight of which can then be considered during planning

- Loading Equipment Fixed (not yet supported)

Defines whether the loading equipment is attached to the vehicle for every transport.

- If it is fixed, planning can rely on the availability of the loading equipment without further checks.
- If the loading equipment is not fixed, it could be removed in order to gain additional loading space and/or loading weight capacity.

- Loading Equipment Outside Cargo Space (not yet supported)

Defines whether the vehicle's loading equipment is placed outside the cargo space.

For more information see Customizing for [Transportation Management](#) under [Master Data](#) > [Resources](#) > [General Settings](#) > [Define Vehicle Loading Profile](#) ➤

Planning Profile

In the incompatibility settings of your planning profile, you can define constraints for loading direction and attached equipment. Note that once you define these constraints, we recommend that you check and adapt your master data. That means, add the loading direction profile and the attached equipment profile to the locations and vehicle resources that should consider these constraints.

Example: If a location has no loading direction profile defined, no restrictions with regards to this constraint are applied. That would mean, vehicle resources can be loaded or unloaded from all sides at this location.

For more information, see [Planning Profile](#).

Related Information

[Loading Requirements for Locations](#)

[Planning Profile](#)

Definition of Means of Transport

Use

You use means of transport to define groups of vehicle resources.

Features

Means of transport are groups of vehicle resources, for example, container ships or cargo ships. You assign a means of transport to each vehicle resource.

You can define the following attributes, for example:

- Your Own Means of Transport

You can specify that the means of transport is part of your own fleet.

- Passive Means of Transport (Trailers)

You can specify that a means of transport cannot move by itself and that it therefore needs to be coupled to a vehicle that can move by itself. The incompatibilities (see [Incompatibilities](#)) as well as the means-of-transport combinations that you have defined determine whether or not a trailer can be coupled to a vehicle.

You can define a coupling duration and an uncoupling duration for a trailer.

The system schedules the uncoupling activity for a trailer when a different vehicle has to move it in the next stage. If the system has assigned an empty trailer to a stage, it schedules the coupling activity directly before the transportation activity and the uncoupling activity directly after the transportation activity.

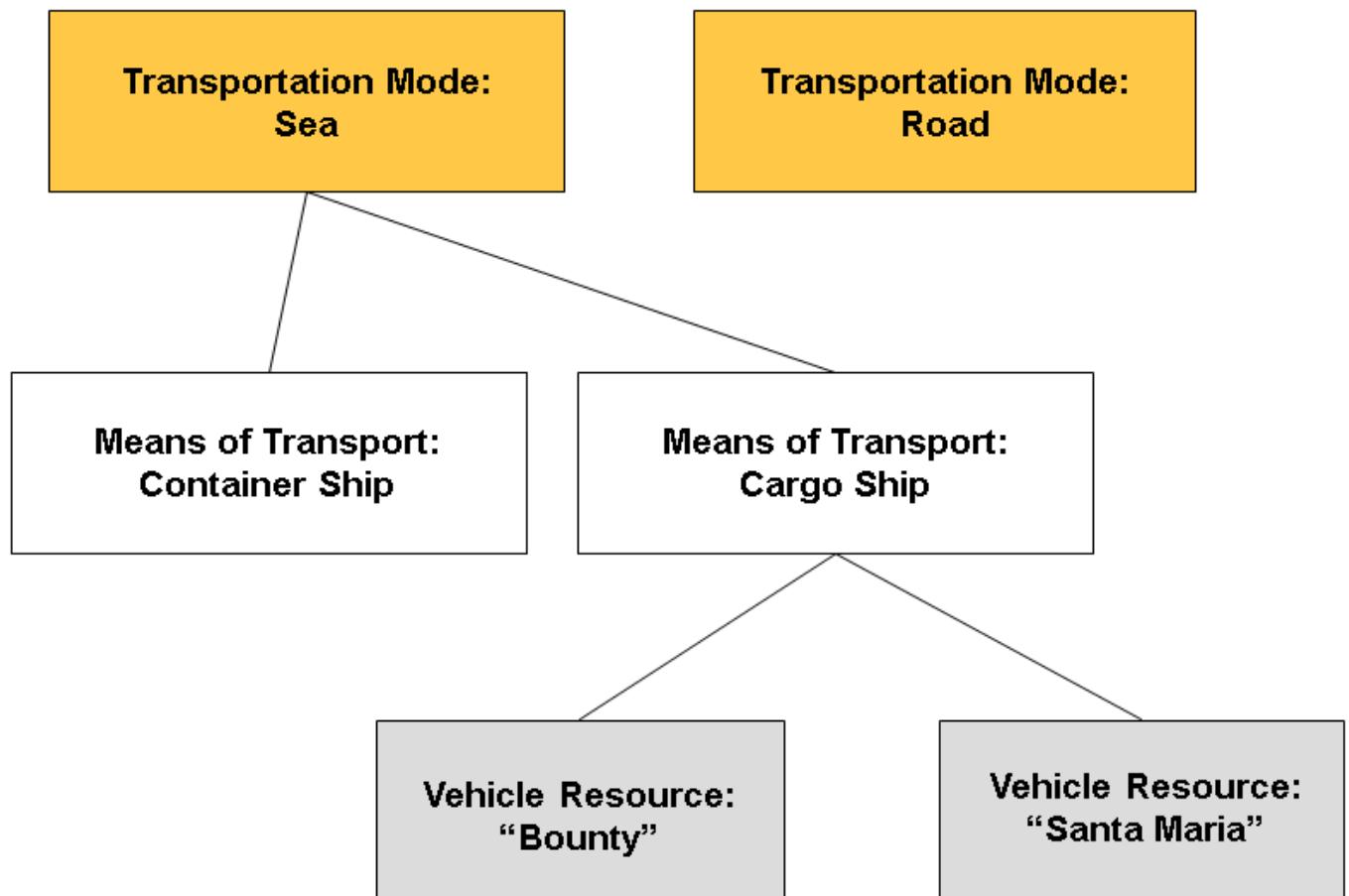
i Note

- You cannot couple trailers to schedules (see [Schedule](#)).
- Trailers must have capacity.
- You cannot define schedules for trailers.

- Transportation Mode

The transportation mode specifies how the goods are transported, for example, as a sea shipment.

The following figure represents the relationship between vehicle resources, means of transport, and transportation modes:



Relationship Between Vehicle Resources, Means of Transport, and Transportation Modes

The vehicle resource is a particular means of transport, for example, the ship “Santa Maria”.

- Superordinate Means of Transport

You can specify a means of transport that is superordinate to a means of transport.

A means of transport A that is superordinate to another superordinate means of transport B is also superordinate to any subordinate means of transport of the means of transport B.

- Multiresource

You can specify that the resource assigned to this means of transport is a multiresource. You can use this attribute if an unlimited number of resources are available for a means of transport but you want to define the settings for these resources only once. For more information, see [Use of Multiresources](#).

- No Transportation Lanes Required

You can activate planning with an unlimited transportation network. That means, a vehicle resource with this means of transport can travel between all locations. Note that this setting is relevant for vehicle scheduling and routing (VSR) optimization and for scheduling.

Activities

You create means of transport in Customizing for **Transportation Management** under ► **Master Data** ► **Resources** ► **Define Means of Transport** ►.

You define coupling and uncoupling durations for passive means of transport in Customizing for **Transportation Management** under ► **Master Data** ► **Resources** ► **Define Coupling/Uncoupling Duration** ►.

Definition of Means-of-Transport Combinations

Use

You use a means-of-transport combination (MTr combination) to model vehicle combinations. An MTr combination consists of one means of transport that can move independently and can move at least one passive means of transport (passive MTr).

Prerequisites

You have selected the **Passive** checkbox for the corresponding means of transport for each passive MTr. For more information, see Customizing for **Transportation Management** under **►►Master Data ► Resources ► Define Means of Transport ►**.

Features

You can define a total capacity for MTr combinations. The total capacity is the maximum loading capacity without the tare weight of the vehicle and passive MTr.

You can define incompatibilities for MTr combinations by using attributes (see [Incompatibilities](#)).

Activities

You create MTr combinations in Customizing for **Transportation Management** under **►►Master Data ► Resources ► Define Means-of-Transport Combination ►**.

You define attributes for MTr combinations in Customizing for **Transportation Management** under **►►Master Data ► Resources ► Define Attributes of MTr Combination/Compartment ►**.

Example

The following table shows the vehicle resources and means of transport you have defined, and the assignments you have made:

Vehicle Resources	Means of Transport
TRUCK1	MTRTRUCK1
TRUCK2	MTRTRUCK2
PASSIVEVEH1	MTRPASSIVE1
PASSIVEVEH2	MTRPASSIVE2

According to your business needs, you model the following MTr combinations:

- MTRTRUCK1 + MTRPASSIVE1
- MTRTRUCK1 + MTRPASSIVE2
- MTRTRUCK2 + MTRPASSIVE1

Definition of Combination Resources

You use combination resources to define groups of vehicle resources.

Features

A combination resource is the fixed amalgamation of one vehicle resource and one or more passive vehicle resources, for example, one tractor and two trailers. These component resources are coupled in a fixed sequence and are always used together.

Unlike means of transport combinations, once you define a combination resource, the components are fixed and cannot be changed, for example, as part of VSR optimization. For more information see about the difference between combination resources and vehicle-resource combinations, see [Vehicle Resource](#).

Combination resources simplify the handling of fixed truck-trailer combinations in Transportation Management (TM), by using a single road freight order that includes the assigned trailers as local items instead of separate trailer unit documents.

You can perform VSR optimization on the level of the combination resource. This means that VSR optimization does not have to deal with the coupling and uncoupling of the individual components of the combination resource but can still do load planning on the truck-trailer level. For more information about using combination resources in planning, see [Use of Combination Resources](#).

Activities

Configuring Combination Resources

You configure a combination resource in Customizing for [Transportation Management](#) under [Master Data > Resources > General Settings > Define Equipment Groups and Equipment Types](#).

When you configure a vehicle group as a combination resource you must:

- Specify one truck vehicle resource.
- Define a number sequence and assign the lowest sequence number to that truck.

Creating Combination Resources

In master data for resources, you create a resource with the following settings:

Setting	Value	Description
Resource Type	09	Vehicle Resource
Resource Class	01	Truck

You specify a vehicle group and vehicle type that have been configured for a combination resource in Customizing. This vehicle group and vehicle type combination identifies the resource as a combination resource.

You can work with the combination resource on the [Combination Resource](#) tab and, for example, add an additional passive resource such as a trailer.

Restrictions

The following restrictions apply:

- You must specify one truck resource.
- You must define a number sequence, the lowest number of which you assign to the truck resource.
- If you have previously defined a vehicle type as part of a means-of-transport combination, you cannot use that resource as part of a combination resource. For more information see [Definition of Means-of-Transport Combinations](#).

- If you have defined a compartment profile for a vehicle resource, you cannot use that vehicle resource in a combination resource. For more information see [Definition of Compartments](#).
- If a vehicle resource is defined as a combination resource, it cannot be used as a component in another combination resource.
- If a vehicle resource is used as a component of a combination resource, it cannot be used standalone during planning. If you need the same vehicle to be planned both as part of a combination resource and standalone, you need to create two separate vehicle resources.

Definition of Compartments

Use

You use compartments to map the loading space of a vehicle.

Features

You define compartments by using compartment types. For example, you use the compartment type to define the capacity of the compartment, possible steps, and a compartment profile. In the compartment profile, you specify the number and type of compartments that a means of transport can have. You then assign this compartment profile to the means of transport.

You can define a fixed number of compartments for each resource.

For each compartment, you can define a capacity for each dimension. The capacity consumption within a compartment is linear. By defining steps, you can model a capacity consumption on the basis of the step. This is recommended if the compartment has a door or if you want to work with movable partitions.

The relationship between the total capacity of the vehicle resource and the total capacity of all assigned compartments determines whether the individual compartments are fixed or flexible. This is not an attribute of the compartment. A resource with fixed compartments means that the total capacity of all assigned compartments is the same as the total capacity of the resource. A resource with flexible compartments means that the total capacity of all assigned compartments is greater than the total capacity of the resource. For more information, see [Examples of Compartments](#).

You can define incompatibilities for compartments based on attributes.

Activities

You define compartment types in Customizing for **Transportation Management** under **Master Data > Resources > Define Compartment Type**.

Example

You use compartments for the following:

- You have to separate two freight units on a truck because they contain refrigerated goods and fruit.
- You want to transport different types of gasoline in one tanker.

More Information

[Definition of Dimensions and Units of Measurement](#)

Compartment

Definition

Part of the loading space of a vehicle resource (vehicle or trailer).

Use

You use compartments for the following, for example:

- You have to separate two freight units on a truck because they contain refrigerated goods and fruit.
- You want to transport diesel and super fuel in one tanker.

Structure

You define compartments by using compartment types. You can define a fixed number of compartments per vehicle resource (vehicle or trailer).

For each compartment, you can define a capacity per dimension. The capacity consumption within a compartment is linear. By defining steps, you can model a capacity consumption on the basis of the step. This is recommended if the compartment has a door, or if you want to work with movable partitions, for example.

The relationship between the total capacity of the vehicle resource and the total capacity of all assigned compartments determines whether the individual compartments are fixed or flexible. This is not an attribute of the compartment. A vehicle resource with fixed compartments means that the total capacity of all assigned compartments is the same as the total capacity of the vehicle resource. A vehicle resource with flexible compartments means that the total capacity of all assigned compartments is greater than the total capacity of the vehicle resource. For more information, see [Examples of Compartments](#).

The system takes into account compartments in VSR optimization and in manual planning in the Transportation Planning/Vehicle Scheduling (TP/VS) component, as well as in SAP Transportation Management (SAP TM).

Integration

- You define compartment types in Customizing for SCM Basis. You use the compartment type to define the capacity of the compartment, possible steps, and a compartment profile, for example. In the compartment profile, you specify the number and type of compartments that a means of transport is to have. You then assign this compartment profile to the means of transport.

For more information, see the Implementation Guide (IMG) for SCM Basis under [▶▶ Master Data ▶▶ Trailers and Compartments ▶ Define Compartment Type. ▶](#)

- To be able to work with compartments in TP/VS, you must set the [Activate Comps](#) indicator in Customizing for Transportation Planning/Vehicle Scheduling (TP/VS).

For more information, see the IMG for TP/VS under [▶▶ Trailers and Compartments ▶ Activate Trailers and Compartments ▶](#).

- You have defined a number range for the handling unit [Compartment Type](#) in TP/VS.

For more information, see the IMG for TP/VS under ► [Basic Settings](#) ► [Define Number Range for HU Compartment Type](#) ►

- You have assigned packaging materials to means of transport and compartment types in TP/VS.

For more information, see the IMG for TP/VS under ► [Interfaces](#) ► [Maintain Mapping Table for Vehicles/Compartment Type](#) ►.

- You can define compatibilities and incompatibilities for compartments.
- Dynamic route determination in TP/VS does not support compartments. For more information, see SAP Library on SAP Help Portal at ► <http://help.sap.com> ► [SAP Supply Chain Management \(SAP SCM\)](#) ► [SAP Advanced Planning and Optimization \(SAP APO\)](#) ► [Transportation Management](#) ► [Transportation Planning/Vehicle Scheduling \(TP/VS\)](#) ► [Dynamic Route Determination](#) ►.

Examples of Compartments

Fixed Compartments

Vehicle Capacity: 15 m³



1: CT X

Max. Cap.: 5 m³

2: CT X

Max. Cap.: 5 m³

3: CT X

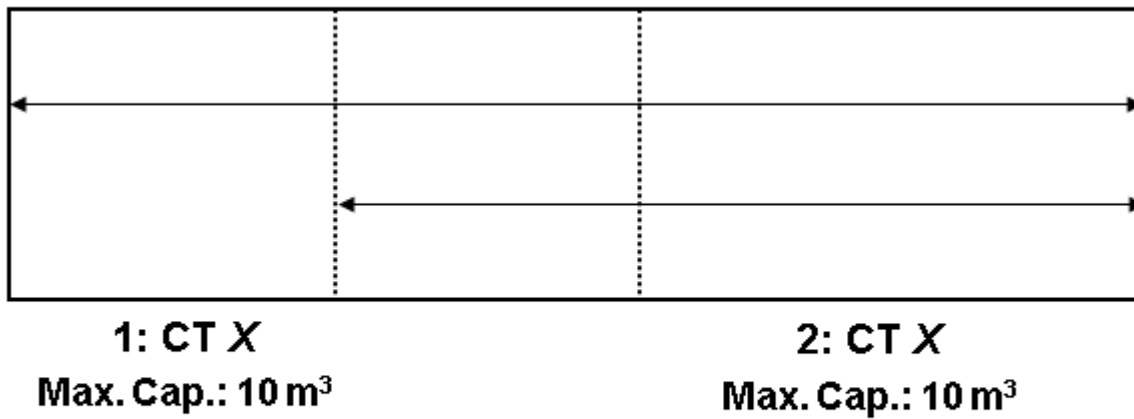
Max. Cap.: 5 m³

Legend:

CT = Compartment Type

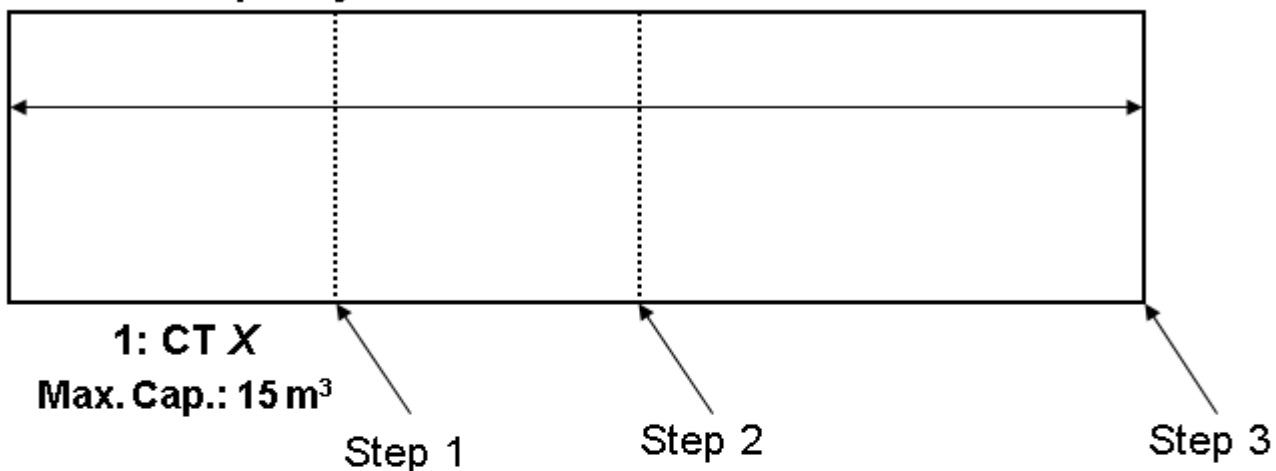
In this example, the vehicle resource comprises three compartments with the compartment type X. Each compartment has a maximum capacity of 5 m³. The total capacity of the vehicle resource is 15m³.

Flexible Compartments (Without Steps)

Vehicle Capacity: 15 m³**Legend:**

CT = Compartment Type

In this example, the vehicle resource comprises two compartments with the compartment type X. Each compartment has a maximum capacity of 10m³. The total capacity of the vehicle resource is 15m³, in other words, you are not allowed to load more than 15m³ onto the vehicle resource.

Flexible Compartments (with Steps)**Vehicle Capacity: 15 m³****Settings for Compartment Type X**

Step	Capacity	Linear
1	5 m ³	No
2	10 m ³	No
3	15 m ³	No

Legend:

CT= Compartment Type

In this example, the vehicle resource comprises one compartment with the compartment type X. The compartment has a maximum capacity of 15 m³. The total capacity of the vehicle resource is 15m³. The compartment has three steps (5 m³, 10 m³, and 15m³).

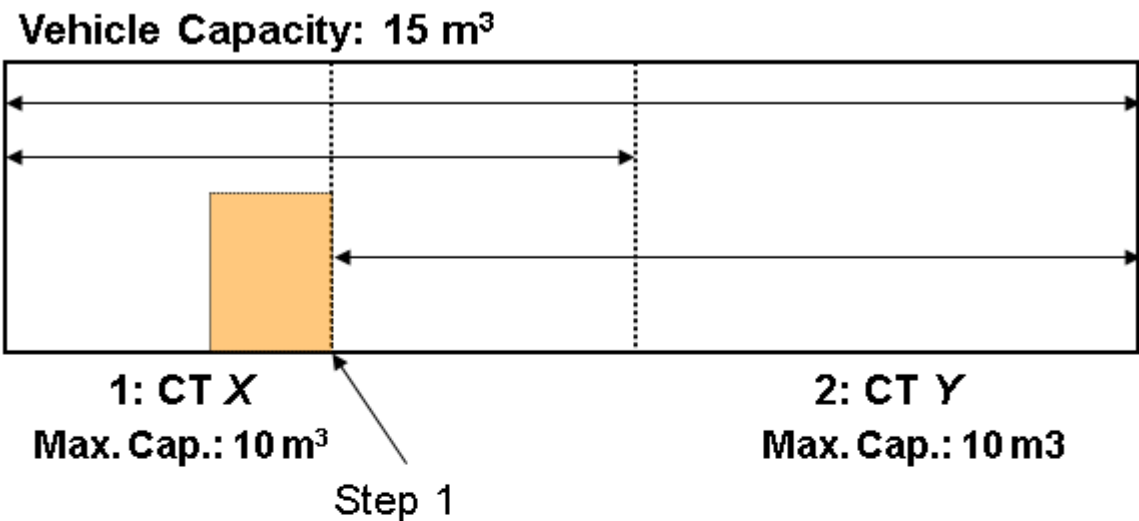
This means the following for the compartment:

- Zero consumption is not possible for the compartment. That means if you do not load this compartment, it still consumes a capacity of 5m³.
- Each load that is greater than or equal to 0 m³ but less than or equal to 5m³ results in a capacity consumption of 5m³. That means if you load this compartment with 3m³, for example, it still consumes a capacity of 5m³.
- Each load that is greater than 5 m³ but less than or equal to 10m³ results in a capacity consumption of 10m³.
- Each load that is greater than 10 m³ but less than or equal to 15m³ results in a capacity consumption of 15m³.

i Note

In a realistic scenario, you would define a second compartment, for example, a flexible compartment without steps.

Flexible Compartments (Zero Consumption Is Not Possible)



Settings for Compartment Type X

Step	Capacity	Linear
1	5 m ³	No

Legend:
CT = Compartment Type

In this example, the vehicle resource comprises two compartments. Compartment 1 has compartment type X, compartment 2 has compartment type Y. Each compartment has a maximum capacity of 10m^3 . The total capacity of the vehicle resource is 15m^3 . Compartment 1 has a door and a step (5m^3). Compartment 2 does not have any steps.

This means the following for the two compartments:

Compartment 1:

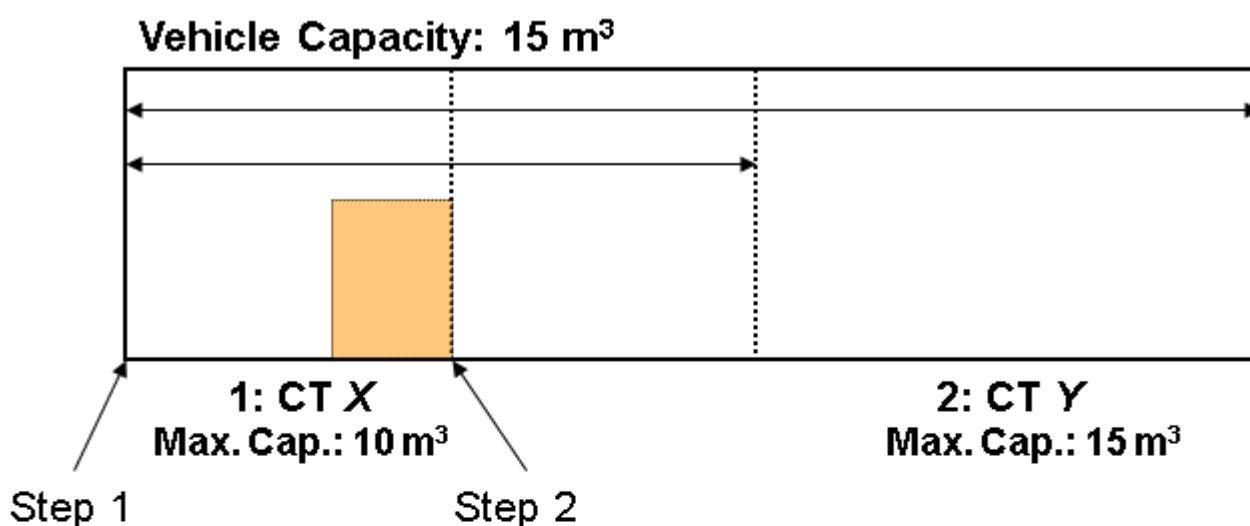
- Zero consumption is not possible for the compartment. That means if you do not load this compartment, it still consumes a capacity of 5m^3 .
- Each load that is greater than or equal to 0m^3 but less than or equal to 5m^3 results in a capacity consumption of 5m^3 .
- Each load x that is greater than 5m^3 but less than or equal to 10m^3 results in a capacity consumption of $x\text{m}^3$.

Compartment 2:

Each load $x\text{m}^3$ results in a capacity consumption of $x\text{m}^3$ (linear capacity consumption).

Flexible Compartments (Zero Consumption Is Possible)

Example 1



Settings for Compartment Type X

Step	Capacity	Linear
1	0m^3	No
2	5m^3	No

Legend:

CT = Compartment Type

In this example, the vehicle resource comprises two compartments. Compartment 1 has compartment type X and a maximum capacity of 10m^3 . Compartment 2 has compartment type Y and a maximum capacity of 15m^3 . The total capacity of the vehicle resource is 15m^3 . Compartment 1 has two steps (0m^3 and 5m^3). Compartment 2 does not have any steps.

This means the following for the two compartments:

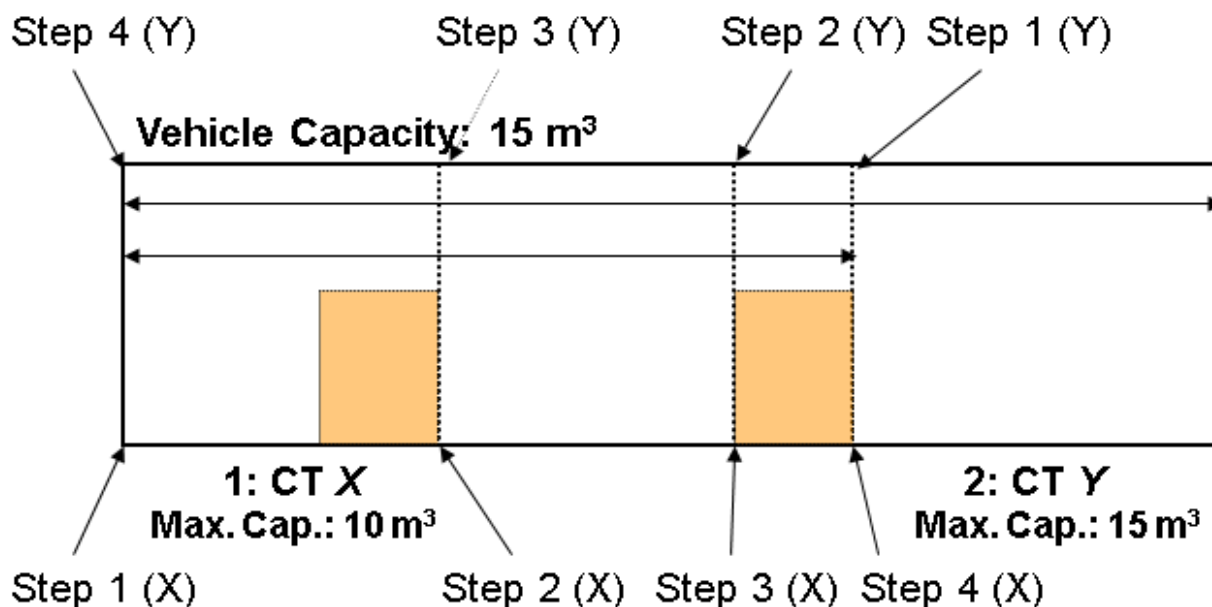
Compartment 1:

- Zero consumption is possible for the compartment. That means if you leave this compartment empty, it does not consume any capacity.
- Each load that is greater than 0m^3 but less than or equal to 5m^3 results in a capacity consumption of 5m^3 . This compartment might be a refrigerated part with a door, for example. As soon as you load one frozen pizza, you have to turn on the refrigerated part. The second compartment has to start behind the door.
- Each load x that is greater than 5m^3 but less than or equal to 10m^3 results in a capacity consumption of $x\text{m}^3$ (due to the maximum capacity).

Compartment 2:

Each load $x\text{m}^3$ results in a capacity consumption of $x\text{m}^3$ (linear capacity consumption).

Example 2



Settings for Compartment Type X

Step	Capacity	Linear
1	0 m ³	No
2	5 m ³	No
3	8 m ³	Yes
4	10 m ³	No

Settings for Compartment Type Y

Step	Capacity	Linear
1	5 m ³	Yes
2	7 m ³	No
3	10 m ³	Yes
4	15 m ³	No

Legend:

CT = Compartment Type

In this example, the vehicle resource comprises two compartments. Compartment 1 has compartment type X and a maximum capacity of 10m³. Compartment 2 has compartment type Y and a maximum capacity of 15m³. The total capacity of the vehicle resource is 15m³. Compartment1 has four steps (0m³, 5m³, 8m³, and 10m³). Compartment2 also has four steps (5m³, 7m³, 10m³, and 15m³). There are two doors.

This means the following for the two compartments:

Compartment 1:

- Zero consumption is possible for the compartment.
- Each load that is greater than 0 m³ but less than or equal to 5m³ results in a capacity consumption of 5m³.
- Each load x that is greater than 5 m³ but less than or equal to 8 m³ results in a capacity consumption of x m³ (linear capacity consumption since you have set the **Linear** indicator for this step).
- Each load that is greater than 8 m³ but less than or equal to 10m³ results in a capacity consumption of 10m³.

Compartment 2:

- Each load x that is greater than 0 m³ but less than or equal to 5 m³ results in a capacity consumption of x m³ (linear capacity consumption since you have set the **Linear** indicator for this step).

- Each load that is greater than 5 m³ but less than or equal to 7m³ results in a capacity consumption of 7m³.
- Each load x that is greater than 7 m³ but less than or equal to 10 m³ results in a capacity consumption of x m³ (linear capacity consumption since you have set the **Linear** indicator for this step).
- Each load that is greater than 10 m³ but less than or equal to 15m³ results in a capacity consumption of 15m³.

Definition of Dimensions and Units of Measurement

Use

You use dimensions to group units of measurement. You need units of measurement to display product quantities, for example, for freight unit building rule (FUB rules), and to describe the capacity of vehicle resources (see [Freight Unit Building Rule](#) and [Vehicle Resource](#)).

i Note

The system might display quantities with 14 decimal digits. If you want the system to round quantities to a specific number of decimal digits, follow the instructions described in SAP Note [1132219](#) .

Prerequisites

- You have checked in Customizing under **General Settings** > **Check Units of Measurement**  that the dimensions delivered as a standard and the associated units of measurement are complete.
- When using FUB rules, the system checks if the forwarding order contains the relevant quantities based on the FUB rules. If the forwarding order does not contain the relevant quantities, the system uses the product data. For this purpose, you have defined the following in the product master data:
 - Base unit of measure (BUoM)
 - Conversion factors

You need conversion factors to convert the base units of measure into the required units of measurement. You define these in the product master on the **Units of Meas.** tab page. For more information, see [Definition of Products](#).

If you have not defined a conversion factor, the system cannot take split quantities or rounding values into account when building freight units from business documents (forwarding orders, for example). In this case, it sets the quantities to zero.

Example

Examples for Dimensions and Units of Measurement

The following table provides you with examples of dimensions and of some units of measurement you can assign to them.

Dimension	Units of Measurement
Mass	Kilogram; milligram
Volume	Cubic centimeter; cubic decimeter

Example for the Conversion of Units of Measurement

The following units of measurement are predefined as a default:

- **Kilogram** (dimension **mass**)
- **Cubic decimeter** (dimension **volume**)

You also define the following unit of measurement:

- **Pallet** (no dimension)

Your forwarding order contains product A and product B each of which have the quantity "1 piece". You have defined **piece** as BUoM for both products.

You have defined the following conversion factors for product A:

- 1 piece = 2 kg
- 1 piece = 3 dm³

You have defined the following conversion factor for product B:

- 1 piece = 0.5 pal

In this case, product A is displayed with quantity 2 kg or 3 dm³. In the unit of measurement **pallet** (pal), the quantity 0 is displayed because you have not defined a conversion factor.

Product B is displayed with the quantity 0.5 pal. Since you have not defined conversion factors for **kilogram** or **cubic decimeter**, no conversion occurs. The system displays the value 0.

Handling Resource

Definition

An installation or device that is used to handle the goods at one specific location. The handling resource offers handling capacity that allows the goods to be loaded onto or unloaded from vehicle resources.

Use

You use this business object to handle goods.

You can specify the capacity of handling resources based on shifts using the capacity profile.

❖ Example

You have 50 doors for loading and unloading freight units. In the morning and in the evening the full capacity is needed, in other words, all doors are open. During the day, only 20 doors are needed, in other words, the capacity is reduced.

Structure

You can define, among others, the following parameters for handling resources:

- Resource template (see [Copying Resources from Templates](#))

- Capacity profile (see [Definition of Capacity Profiles](#))
- Availability (see [Definition of Availability](#))
- Downtimes (see [Definition of Downtimes](#))
- Qualifications (see [Definition of Qualifications](#))
- Attached equipments (see [Definition of Attached Equipment](#))
- Grouping attributes (see [Definition of Grouping Attributes](#))
- Alternative names (see [Definition of Alternative Names](#))

Example

- Forklift
- Door

More Information

[Vehicle Resource](#)

Driver

Definition

A driver is any person who can operate vehicles and perform transportation-related tasks as a result of certain qualifications.

You specify a driver in TM business documents.

You can assign a default driver for a truck. If you have defined required qualifications for a vehicle resource, the system checks the qualifications of the driver to ensure that the driver meets the required qualifications of the vehicle.

When you assign a vehicle to a freight order, the system checks the driver assignment. If no driver has previously been assigned to the freight order the system assigns the vehicle's default driver to the freight order.

Structure

You can define the following parameters for drivers:

- Availability (see [Definition of Availability](#))
- Absences (see [Definition of Downtimes](#))
- Qualifications (see [Definition of Qualifications](#))
- Supply Chain Unit

Represents the driver's home location.

- Driver Type

You can define driver types to categorize drivers depending on the duties they perform and based on certain parameters such as their qualification. You can define this parameter in the Customizing for [Transportation Management](#) under ►

Transportation Unit Resource

Definition

An instance of a particular transportation unit like a container that you can use to map the capacity and availability that you want to use for transportation.

You can choose up to eight dimensions and units of measurement to describe the capacity. Note that the mass and volume are predefined by default. Planning can only take the transportation unit resource capacity into account if these correspond to the dimensions and units of measurement that you have defined in a freight unit building rule.

i Note

You cannot define the available capacity for this resource type according to time. However, you can define a time-dependent availability using shift sequences. For more information about defining shift sequences, see [Specification of Definitions](#) and [Creating Definitions](#).

In the planning profile, you can specify whether the system can consider a manual overload of the resource capacity.

Equipment Group and Equipment Type

The equipment and equipment group types specify attributes such as the physical properties of a certain kind of transportation unit resource. When you have created a transportation unit resource type you can:

- Use the type as a template to create transportation unit resources, copying the attributes of the type to a new resource
- Request a special type of transportation unit resource in a forwarding order or in a booking order

For more information see Customizing for Transport Management under ► [Transport Management](#) > [Master Data](#) > [Resources](#) > [Define Equipment Groups and Equipment Types](#) ►

Structure

You can define, among others, the following parameters for transportation unit resource resources:

- Capacity (see [Definition of Capacity](#))
- Availability (see [Definition of Availability](#))
- Multiresource

For container resources, you can specify whether the resource is a single resource or a multiresource (see [Use of Multiresources](#)).

- Minimum number of seals

You can specify the minimum number of seals of the transportation unit resource. You can use this information for the definition of incompatibilities.

- Resource template (see [Copying Resources from Templates](#))
- Physical properties (see [Definition of Physical Properties](#))
- Downtimes (see [Definition of Downtimes](#))

- Qualifications (see [Definition of Qualifications](#))
- Attached equipment (see [Definition of Attached Equipment](#))
- Grouping attributes (see [Definition of Grouping Attributes](#))
- Alternative names (see [Definition of Alternative Names](#))

More Information

[Freight Unit Building Rule](#)

[Definition of Dimensions and Units of Measurement](#)

[Planning Profile](#)

[Incompatibilities](#)

Creating, Copying, and Deleting Resources

In the following sections you can find information about:

- Creating handling, vehicle, calendar, and transportation unit resources (see [Creating Handling, Vehicle, and Calendar Resources](#))
- Creating drivers (see [Creating Drivers](#))
- Copying resources (see [Copying Resources](#))
- Creating resources based on resource templates (see [Copying Resources from Templates](#))
- Deleting resources (see [Deleting Resources](#))

Creating Handling, Vehicle, Calendar, and Transp. Unit Resources

Prerequisites

- You have specified a factory calendar in Customizing for **Transportation Management** under ► **Master Data** ► **Calendar** ► **Maintain Factory Calendar** .
- If you want to create a vehicle resource, you have to have specified a means of transport in Customizing for TM under ► **Transportation Management** ► **Master Data** ► **Resources** ► **Define Means of Transport** .

Context

This procedure describes how to create the following resource types:

- Handling resource (see [Handling Resource](#))
- Vehicle resource (see [Vehicle Resource](#))
- Calendar resource (see [Calendar Resource](#))
- Transportation Unit resource (see [Transportation Unit Resource](#))

Procedure

1. You have the following options:
 - On the **SAP Easy Access** screen, choose ► **Transportation Management** ► **Master Data** ► **Resources** ► **Define Resource** ►.
 - In the SAP launchpad, open the **Define Resources** app.
2. Choose the **Create Resources** pushbutton.
3. Select the tab page for the resource type for which you want to create a new resource.

i Note

You cannot change the resource type later.

4. For **handling resources**, specify the resource class, the time zone or location, and the factory calendar.

For **calendar resources**, specify the time zone or location, and the factory calendar.

For **vehicle resources**, specify the resource class, the time zone or depot location, the factory calendar, the means of transport, the dimension, and the capacity.

For **transportation unit** resources, specify the resource class, the time zone or depot location, the factory calendar, the dimension, and the capacity.

i Note

You can specify the depot location in the **Location** column.

For more information about resource classes, see [Classification of Resources](#).

5. In addition, you can define other resource-specific information, such as downtimes, attached equipment, time-continuous capacity, or qualifications.

Next Steps

[Definition of Means of Transport](#)

[Definition of Dimensions and Units of Measurement](#)

[Definition of Capacity](#)

[Definition of Downtimes](#)

[Definition of Attached Equipment](#)

[Definition of Qualifications](#)

Creating Drivers

Prerequisites

- You have defined a factory calendar in Customizing for **Transportation Management** under ► **Master Data** ► **Calendar** ► **Maintain Factory Calendar** ►.

- You have defined qualifications in Customizing for [Transportation Management](#) under [Master Data](#) > [Resources](#) > [Maintain Settings for Qualifications](#) .

Context

You use this procedure to create the driver business partner.

Procedure

1. In transaction BP, you create a new business partner of the type **Person** and specify the BP role as **Driver**. The entry screen changes and a new **Driver** tab page is displayed.
2. On the **Driver** tab page, you enter the attributes of the driver such as the labor factor and the supply chain unit (the home location of the driver).
3. In the **Qualifications** screen area, you enter the qualifications that are held by the driver. This enables you to specify, for example, the type of license that the driver has or whether the driver has any experience of transporting dangerous goods. For more information, see [Definition of Qualifications](#).
4. In the **Availability** screen area, you enter the availability of the driver by specifying the valid from and valid to dates and times. You must also enter a factory calendar.

You can define a number of capacity variants and specify which of these variants is to be used as the active variant. Note that TM does not evaluate the type of capacity (minimum, mean, or maximum capacity). For more information, see [Definition of Capacity Variants](#).

You can also enter planned absences such as vacation periods and unplanned absences such as sick leave in this area. For more information, see [Definition of Downtimes](#).

Next Steps

For more information about business partners, see [Definition of Business Partners](#).

Copying Resources

Use

You use this procedure to create a copy of one or more resources of the following type:

- Handling resource (see [Handling Resource](#))
- Calendar resource (see [Calendar Resource](#))
- Transportation Unit resource (see [Transportation Unit Resource](#))

Prerequisites

You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).

Procedure

1. In the **SAP Easy Access** screen, choose [Master Data](#) > [Resources](#) > [Define Resource](#) , or from the launchpad open the **Define Resource** app. Enter the resource you want to copy and choose the **Change** pushbutton. Select the resource you want to copy, and choose the **Copy Resources** pushbutton.

2. Choose the **Copy Resources** pushbutton.

3. In the **Copy Resources** dialog box, you can create one or more copies of a resource.

- **Single copy**

Choose the copy mode **Copy Resources** and in the **To** field group maintain the name of the copy, its location, and a short description.

- **Multiple copies**

Choose the copy mode **Make Multiple Copies of Resources**. Enter the number with which you want the copied resources to begin and the number of resources that you want to create. This number is added to the end of the resource name as a value for counting.

❖ Example

Assume that the name of the resource that you want to copy is Res_00. If you choose 1 as the start number and 10 as the number of resources, the system creates the resource copies Res_001 to Res_0010.

If you have selected more than one resource in the **Change Resources: Header Data** screen, the system calls up the selected resources consecutively in the **Copy Resources** screen.

Result

You have created a copy of one or more resources. Each copy has its own name.

Copying Resources from Templates

Prerequisites

- You have created a resource for which you have selected the **Resource Template** checkbox on the **Transportation** tab page.

For more information about creating resources, see [Creating Handling, Vehicle, and Calendar Resources](#).

- For handling resources that you want to mark as resource templates, you have specified template types in Customizing for **Transportation Management** under **Master Data** > **Resources** > **Specify Resource Template Types**.

i Note

For vehicle resources, the template type is automatically **Means of Transport**.

Context

This procedure describes how to create a copy of one or more resources of the following type from resource templates:

- Handling resource (see [Handling Resource](#))
- Vehicle resource (see [Vehicle Resource](#))

Procedure

1. You have the following options:

- On the [SAP Easy Access](#) screen, choose [Logistics](#) [Transportation Management](#) [Master Data](#) [Resources](#) [Define Resource](#) and choose the [Resources \(Copy Resources from Template\)](#) pushbutton.
 - On the [SAP Easy Access](#) screen, choose [Logistics](#) [Transportation Management](#) [Master Data](#) [Resources](#) [Copy Resource from Template](#).
 - Open the [Copy Resource from Template](#) app from the launchpad.
2. In the [Name of New Resource](#) group box, you specify the data for the new resource that you want to create from the resource template.
 3. In the [Choose Resource Template](#) group box, you specify whether the resource template is one of the following:
 - [Vehicle Resource](#)

By specifying a means of transport, the system displays a list with all vehicle resources marked as resource templates that you have assigned this means of transport to.
 - [Handling Resource](#)

By specifying a template type, the system displays a list with all handling resources marked as resource templates that you have assigned this template type to.
- If you want to specify the name of the resource to be copied directly, select the [From other Resource](#) checkbox, and specify the name in the [Resource Name](#) field.
- i Note**
- By selecting this checkbox, you can copy resources that have not been marked as templates or resources that cannot be marked as templates (for example, calendar resources).
4. To copy the data from the existing resource to the new resource, choose the [Resource \(Copy Resource\)](#) pushbutton.

Deleting Resources

Prerequisites

You can only delete resources that are not planned or used in other master data. For this reason, check for existing dependencies on the [SAP Easy Access](#) screen by choosing [Logistics](#) [Transportation Management](#) [Master Data](#) [Where-used List](#).

Context

This procedure describes how to delete the following resource types:

- Vehicle resources (see [Vehicle Resources](#))
- Handling resources (see [Handling Resources](#))
- Calendar resources (see [Calendar Resources](#))
- Transportation Unit resources (see [Transportation Unit Resource](#))

Procedure

1. In the [SAP Easy Access](#) screen or in the [Define Resource](#) app, which you open from the launchpad, choose [Master Data](#) [Resources](#) [Define Resource](#). Enter the resource you want to delete and choose the [Change](#) pushbutton.
2. Select the resource you want to delete and choose [Delete Resource](#).

If there are no dependencies, the system deletes the resource.

If the resource has dependencies, the system displays an error message. You must remove all of these dependencies before you can delete the resource.

3. Choose **Save** to confirm deletion.

Definition of Capacity

Use

You use this function to specify the capacity that is available at specific points in time for a resource.

Features

You can define the following capacity categories:

- **Standard available capacity**

You specify this capacity in the header data when creating a new resource. This capacity does not take a multiple-shift operation into account, in other words, this capacity is always the same.

The standard available capacity is limited only by the resource availability (see [Definition of Availability](#)).

- **Time-continuous capacity**

The time-continuous capacity takes a multiple-shift operation into account. A capacity variant is used to define this capacity based on different shift programs. For more information, see [Definition of Capacity Variants](#).

You can define data for the time-continuous capacity for vehicle resources, drivers, and calendar resources. For handling resources, you define a capacity profile.

- **Capacity profile**

The capacity profile corresponds to the time-continuous capacity, but is only available for handling resources. For more information, see [Definition of Capacity Profiles](#).

Moreover, the capacity profile provides you with an overview of the available capacity based on the standard available capacity, and the capacity derived from capacity variants.

i Note

In downtimes, only reduced capacity is available for handling resources, and no capacity is available for all other resources. For more information, see [Definition of Downtimes](#).

Definition of Capacity Variants

Use

If you work in a multiple-shift operation and want to plan using shifts, you can use capacity variants to define different capacities for one or more resources.

Prerequisites

You have specified capacity variants in Customizing for [Transportation Management](#) under [Master Data](#) [Resources](#) [Specify Capacity Variants](#). You can also specify these capacity variants on the [Transportation Resource Master](#) screen by choosing [Current Settings](#) [Capacity Variants](#).

Features

You can define shift sequences with shifts, breaks, and shift factors in a capacity variant. You define these data based on definitions (see [Specification of Definitions](#)).

If you define a capacity variant as the **active variant** on the [General Data](#) tab page of a resource, the system uses the capacity information from the capacity variant instead of the standard available capacity.

i Note

The system considers the data of a capacity variant only if you have defined it as an active variant.

You can use the capacity profile to gain an overview of the available capacity that is determined from the active capacity variant for the validity period of the resource. For more information, see [Definition of Capacity Profiles](#).

Specification of Definitions

Use

You use this function to specify resource-independent and date-independent templates for defining the available capacities of the capacity variants.

Features

The definitions, their use, and their data are summarized in the following table.

Definition	Use	Data
Shift sequence	In the shift sequence, you define the daily shifts for any number of consecutive days, for example, for a week. For each shift, you define working and break times, the utilization rate, and the capacity (in the case of handling resources). You assign shift sequences to the time intervals of a capacity variant. In each interval, the shift sequence is repeated periodically (for example, every seven days). You also specify on which day of a shift sequence an interval starts.	• Day numbers • Zero to nine shifts per day • Processing of shifts that start or end on non-workdays
Shift	You define shift data in a shift definition.	• Shift • Start and end of the shift • Break pattern • Shift factors
Breaks	In a break pattern, you define the breaks for a shift.	• Break pattern

Definition	Use	Data
		• Break number • Start and end of the breaks • Relative start of the break, in relation to the start of the shift, and the total break duration
Shift factors	You define a shift factor to define the rate of resource utilization and the capacity of a shift.	• Shift factor definition • Validity • Rate of resource utilization • Capacity for handling resources

Activities

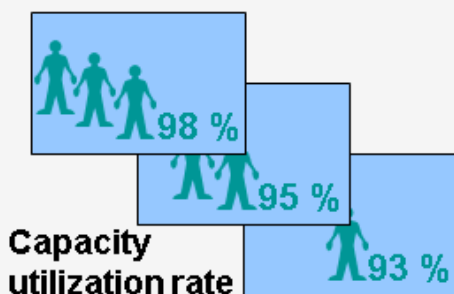
For more information about creating definitions, see [Creating Definitions](#).

To check whether a definition is used in resources, on the [Display Definitions](#) screen, choose .

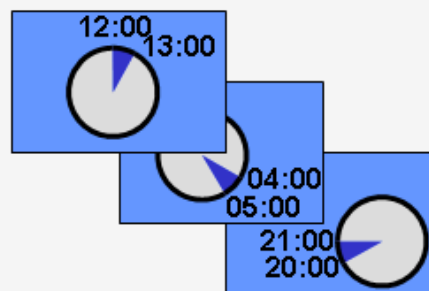
Example

The following figure is an example of how you can use definitions to define the capacity of capacity variants for a resource.

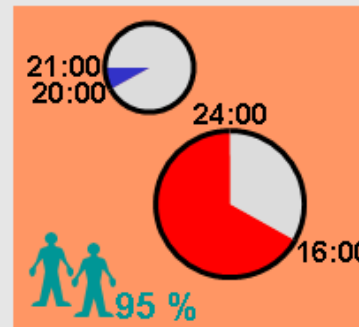
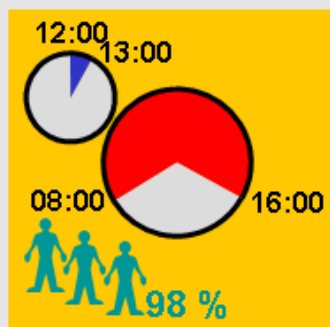
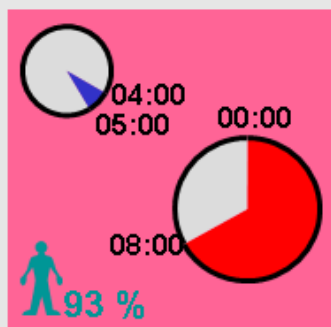
Shift factor definitions



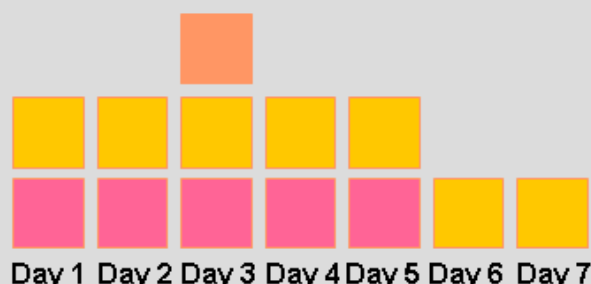
Break patterns



Shift definitions



Shift sequences

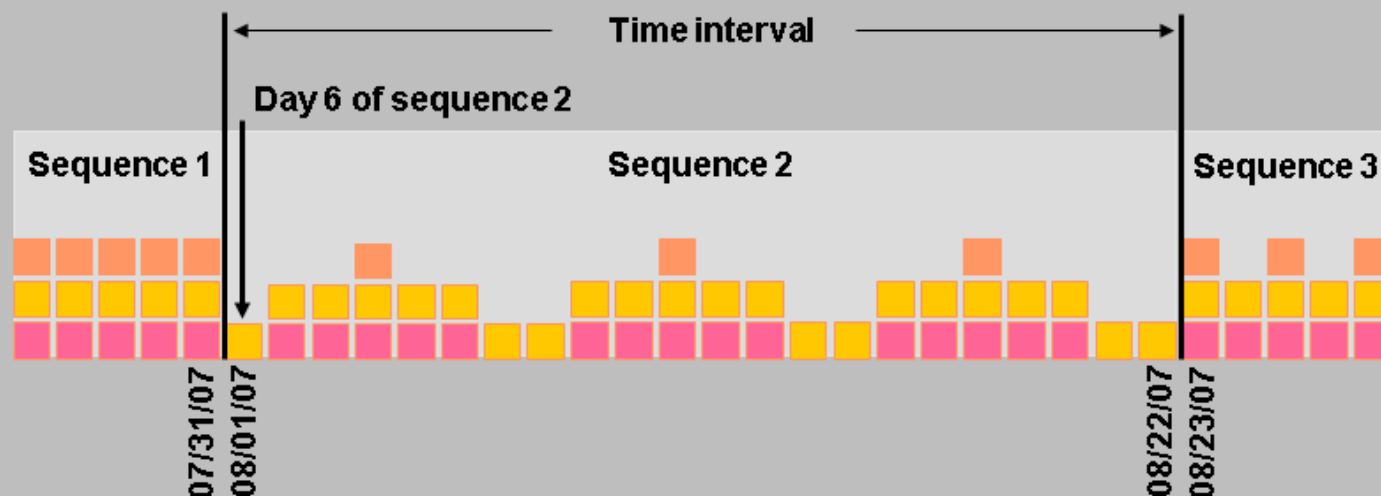


Sequence 1

Sequence 2

Sequence 3

Time intervals of capacity variants



Capacity Variants for a Resource

Creating Definitions

Prerequisites

You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).

Procedure

1. On the **SAP Easy Access** screen, choose **►Transportation Management► Master Data► Resources► Define Resource ►** or in the launchpad, open the **Define Resource** app.
2. You have the following options:
 - On the **Resources** screen, choose the **Definitions** pushbutton.
 - Choose the **Change** pushbutton. On the **Change Resources: Header Data** screen, choose the **Definitions** pushbutton.
3. Choose the **Breaks** tab page.
4. Create the new break pattern, for example, lunch, and assign a break number (dependent on how many breaks you want to define for the day).

There are two ways to define breaks:

- You specify the start and end of the break.
- You specify the number of hours after which you want the break to take place and enter the duration of the break. The reference point is always the beginning of the shift.

If you want to define a recurrent break, for example, a break for every 2 hours, you must define a break for 2 hours, 4 hours, 6 hours, and so on after the shift begins (taking into account the duration of any previous breaks).

5. Choose the **Shifts** tab page.
6. Enter a description for the shift, a validity end, and a start and end for the shift. Assign your break pattern to this shift.

i Note

The system takes into account shift factors only for handling resources.

7. Choose the **Shift Sequences** tab page.
8. Enter a description, the day number, and the validity end. In the **Non-Workdays** field, you can choose from the following options to define the start and end of the shift:
 - **Can Finish on a Non-Workday**
 - **Can Start on a Non-Workday**
 - **Can Only Start or Finish on a Workday**
 - **Can Start or Finish on a Non-Workday**

Creating Capacity Variants

Prerequisites

- You have created definitions (see [Creating Definitions](#)).
- You have specified capacity variants in Customizing for **Transportation Management** under **►Master Data► Resources► Specify Capacity Variants ►**.

Context

You can use this procedure to combine capacity-relevant data such as different shifts and breaks in a capacity variant.

Procedure

1. On the [SAP Easy Access](#) screen or on the launchpad, open the resource for which you have created definitions.
2. On the [Change Resources: Header Data](#) screen, choose the [Capacity Variants](#) pushbutton. To create a new variant, choose [Interval](#).
3. Enter a capacity variant and a validity period. Assign the shift sequence to the capacity variant. Select the first day so that the first day of the time interval is the first day of the shift sequence.

❖ Example

You have defined a four-day shift sequence. If you enter 2 as the first day, the first day of the time interval has the shifts from the second day of the shift sequence.

The following ways exist for defining workdays:

- Workdays according to the factory calendar
 - No workdays (overwrites the factory calendar)
 - Workdays (overwrites the factory calendar)
4. Save the interval and go back to the resource master data.
 5. On the [General Data](#) tab page, assign the capacity variant to the resource by specifying its name in the [Act Var.](#) (Active Variant) field.

Definition of Capacity Profiles

Use

The capacity profile provides you with an overview of the available capacity of a handling resource per day from the current date to the end of the resource's validity period for the following capacities:

- Standard available capacity
- Capacity variants

Prerequisites

If you want the capacities from capacity variants to be displayed in the capacity profile, you have to specify capacity variants for handling resources. For more information, see [Creating Capacity Variants](#).

Features

You can manually change the standard available capacity for selected resources without having to define a new shift or a new interval in the capacity variant if, for example, the working time or the rate of resource utilization varies for individual days. The system saves these changes to the database in a capacity profile for the standard available capacity or for the selected capacity variant.

If, for example, you change the standard available capacity by changing the shift the system adjusts all unchanged data in the capacity profile. All variants entered manually are retained.

As soon as a capacity profile for the standard capacity or for the capacity variant exists in the database, the system takes the variants defined there into consideration - instead of the corresponding capacity – when determining the valid capacity.

i Note

Depending on the validity period defined for the resource and the number of shifts, the overview may contain a large number of entries.

Activities

Displaying the Capacity Profile

By choosing **Capacity Profile** on the **Display Resources: Header Data** screen, you can call an overview of the standard available capacity, and the available capacity generated from the capacity variants.

Changing the Available Capacity of a Shift

If you want to change the available capacity of a shift, proceed as follows:

1. In the capacity profile, choose the  **Shifts (Change Shifts)** pushbutton.

A dialog box appears in which you can specify the period and shift for which you want to change the available capacity.

i Note

You can also mark the period on the calendar. The system then automatically marks the corresponding shifts in the overview. The marked period is also shown in the dialog box.

2. Enter the changed capacity in the **Available Capacity** group box.

Determination of the Valid Available Capacity

Use

This process describes how the system determines the valid available capacity of a resource.

For planning purposes, the system takes the following capacity categories into consideration in a hierarchical sequence to determine the capacity that is available at a specific point in time at a resource:

1. Standard available capacity (taking the factory calendar into account)
2. Capacity profile (for handling resources) or time-continuous capacity (for all other resources)

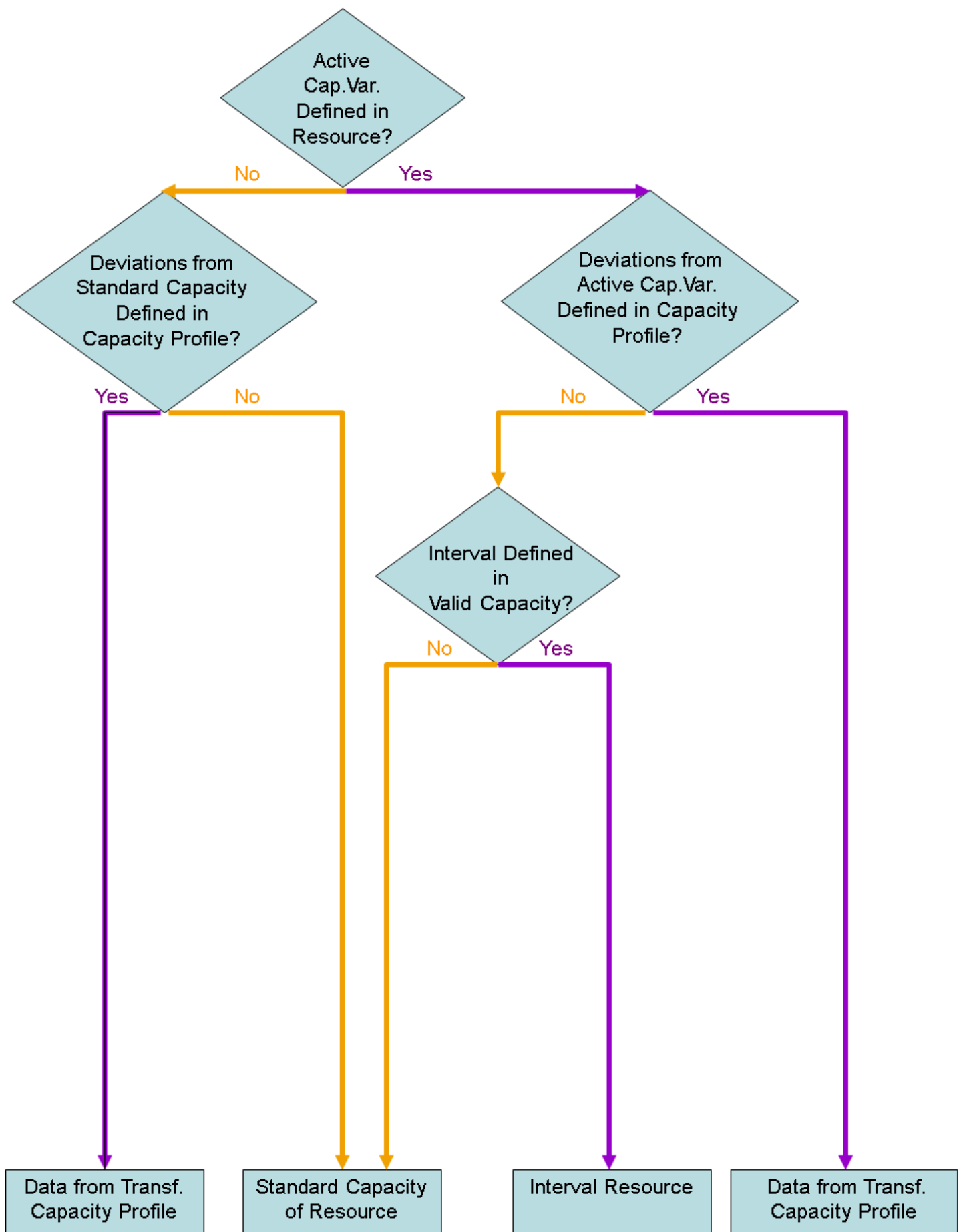
Process

The system checks the resource at the current point in time in the following sequence. If the result of the check is negative, the system moves on to the next step:

1. If you have defined a **downtime** for the resource for the current point in time, only reduced capacity is available for handling resources and no capacity is available for all other resources.
2. If you have specified an **active capacity variant** in the resource **and** different data is defined in a **capacity profile** for the current date, the system uses the capacity defined in the capacity profile.

3. If you have specified an **active capacity variant** for the resource, for which a **capacity profile does not exist**, and an **interval** with a capacity that is valid at the current time is defined in this variant, the system uses this capacity.
4. If you have also created a **capacity profile** for the **standard available capacity**, the system uses the capacity defined in the capacity profile.
5. The system only uses the **standard available capacity** if it could not determine a valid resource capacity for the current point in time during one of the previous checks.
6. If you have not defined any standard available capacity for the resource, the resource does not have any capacity.

The following figure shows the process flow:



Determination of the Valid Available Capacity

Definition of Availability

Use

You use this function to specify general availability for the following resource types:

- Vehicle resources (see [Vehicle Resources](#))
- Handling resources (see [Handling Resources](#))

Features

The actual availability of a resource within the general availability depends on the factory calendar, downtimes, and shift and break definitions.

i Note

Planning takes into account only available resources.

Activities

You can specify the general availability for vehicle resources or handling resources. On the launchpad open the [Define Resources](#) app, or on the [SAP Easy Access](#) screen, choose [Master Data](#) > [Resources](#) > [Define Resource](#) . Enter the resource you want to edit and choose [Change](#). Choose the [Transportation](#) tab page.

Definition of Downtimes

Use

You use this function to specify downtimes, that is time segments in which resources are not available.

You can define downtimes for all resource types and for the [Driver](#) business partner.

Prerequisites

You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).

Features

Downtimes override the availability of a resource. They exclude vehicle resources from planning. In the case of handling resources, downtimes can reduce the available capacity (but a remaining available capacity of 0 also excludes a handling resource from planning).

A differentiation is made between scheduled downtime (for example, for maintenance) and non-scheduled, unexpected downtime (for example, due to a machine breakdown).

Downtimes differ from a regular, scheduled non-working time based on the factory calendar or the shift sequence (for example, break or holiday).

i Note

Downtimes outside the validity period are not taken into consideration during planning. It is not useful to define downtimes that are outside the validity period of the resource, since a capacity has not been defined for the resource for these times.

For a driver, you can specify absences and non-working times. Absences can represent sickness or vacation. Non-working times, for example weekends or public holidays, are defined in the calendar. Scheduling considers both absences and non-working times.

Vehicle scheduling and routing (VSR) optimization interprets a downtime for a single resource in the following way:

- The resource is not available for planning during the downtime.
- The VSR optimizer needs to empty the resource, that is, the resource must be unloaded before a downtime.
- The VSR optimizer needs to bring the resource to the location for which the downtime has been defined.

i Note

- VSR optimization does not consider downtimes for multiresources.
- VSR optimization and scheduling do not consider downtimes and availabilities (factory calendars and shift sequences) for drivers, trailer resources, container resources, and railcar resources.

Activities

You define a downtime for the resource that you have created on the [Change Resources: Header Data](#) screen by specifying on the [Downtimes](#) tab page a start date/time and end date/time as well as the downtime type (scheduled or non-scheduled downtime). In addition, you can enter a language-dependent short text that describes the downtime in more detail.

For [Driver](#) business partners, you define downtimes in the [Absence](#) screen area of the business partner master data.

Example

A handling resource has a capacity of 5. Due to a reduced load, you specify a downtime to reduce the handling resource's capacity for a specific period of time to 2.

Definition of Qualifications

Use

You use this function to specify qualifications, such as skills, capabilities, and permissions needed to operate vehicle resources and handling resources.

Prerequisites

- You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).
- You have defined qualification types and qualifications in Customizing for [Transportation Management](#) under [Master Data > Resources > Maintain Settings for Qualifications](#).

Features

You can specify a validity period for a qualification, defining a start and end date within which the qualification is valid.

Activities

You assign qualification types and qualifications in the resource master on the [SAP Easy Access](#) screen, or in the [Resources](#) app, which you can open from the launchpad.

Example

You have maintained the qualification type [Driving License](#) with qualification [A \(Heavy Vehicle\)](#).

If you assign qualification [A \(Heavy Vehicle\)](#) of type [Driving License](#) to a vehicle resource, it means that this resource can only be operated with such a driving license.

Definition of Attached Equipment

Use

You use this function to specify equipment that is attached (attached equipment) to the following resource types:

- Vehicle resource (see [Vehicle Resource](#))
- Handling resource (see [Handling Resource](#))
- Transportation Unit resource (see [Transportation Unit Resource](#))

Prerequisites

- You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).
- You have specified categories of attached equipment in Customizing for [Transportation Management](#) under [Master Data](#) ► [Resources](#) ► [Maintain Settings for Attached Equipment](#) ►.

Activities

On the [SAP Easy Access](#) screen or in the [Resources](#) app, which you can open from the launchpad, you assign attached equipment to the categories you have specified in Customizing on the [Change Resources: Header Data](#) screen by choosing the [Attached Equip.](#) (Attached Equipment) tab page.

Example

You have specified the following categories of attached equipment in Customizing:

- FIREEXTINGUISHER
- ON_BOARD_UNIT
- NAVIGATIONSYSTEM

A truck has the following attached equipment:

- On-board unit (serial number 001-001-001)

- Fire extinguisher (serial number 001-001-0815)
- Portable navigation system (serial number 010-101-010)

In the resource master, you assign the attached equipment to the categories of attached equipment:

- Attached equipment 001-001-001 to category FIREEXTINGUISHER
- Attached equipment 001-001-0815 to category ON_BOARD_UNIT
- Attached equipment 010-101-010 to category NAVIGATIONSYSTEM

Definition of Physical Properties

Use

You use this function to specify the capacity requirement of a vehicle resource or transportation unit resource during transportation.

Prerequisites

- You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).
- If you want to use packaging materials as a template for the dimensions **Volume** and **Weight**, and for a dimensionless capacity, you have to have defined packaging materials in the product master (see [Product](#)).

Features

You can specify the capacity requirement with a unit of measure for multiple dimensions. For dimensionless capacities, you can specify multiple values.

You can use a packaging material as template for the dimensions **Volume** and **Weight**, and for dimensionless capacity. If the packaging material specifies values for volume, weight, or capacity the system automatically transfers these values into the physical properties of the resource. Here, you can overwrite the values that have been transferred automatically.

Activities

You specify or change the physical properties on the [SAP Easy Access](#) screen or in the [Define Resource](#) app, which you open from the launchpad. You choose on the [Change Resources: Header Data](#) screen the **Phys. Properties** (Physical Properties) tab page.

Example

To define how much volume a container needs on a ship, you specify physical properties for the dimension **Volume**.

Definition of Grouping Attributes

Use

You use this function to group the following resource types into categories of grouping attributes:

- Vehicle resource (see [Vehicle Resource](#))
- Handling resource (see [Handling Resource](#))
- Transportation Unit resource (see [Transportation Unit Resource](#))

Prerequisites

- You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).
- You have specified categories of grouping attributes and grouping attributes in Customizing for **Transportation Management** under ► **Master Data** ► **Resources** ► **Maintain Settings for Grouping Attributes** ►.

Activities

On the **SAP Easy Access** screen or on the launchpad you assign resources to a grouping attribute category and one of its grouping attributes on the **Change Resources: Header Data** screen by choosing the **Grouping** tab page.

Example

You have specified the following:

- Grouping attribute category **Truck** with the grouping attributes **A**, **B**, and **C**
- Grouping attribute category **Container** with the grouping attributes **C**, **CC**, and **CCC**
- Grouping attribute category **Emissions Standard** with the grouping attributes **S1**, **S2**, **S3**, **S4**, and **S5**

Now you can group resources into trucks, containers, or resources with a specific emissions standard.

Definition of Alternative Names

Use

You use this function to specify alternative names for the following resource types:

- Vehicle resource (see [Vehicle Resource](#))
- Handling resource (see [Handling Resource](#))
- Transportation Unit resource (see [Transportation Unit Resource](#))

Alternative names only play a role for selecting and displaying resources.

Prerequisites

- You have created a resource. For more information, see [Creating Handling, Vehicle, and Calendar Resources](#).
- You have specified categories of alternative names in Customizing for **Transportation Management** under ► **Master Data** ► **Resources** ► **Specify Categories of Alternative Names** ►.

Activities

On the [SAP Easy Access](#) screen or in the [Resources](#) app, which you open from the launchpad, you assign categories of alternative names to resources and specify alternative names on the [Change Resources: Header Data](#) screen by choosing the [Alternative Names](#) tab page. Note that an alternative name has to be unique within an alternative name category.

❖ Example

Resource 001000815 is a vehicle resource. For this resource, you have specified the alternative name TRUCK_007 for the category OURCUSTOMERS_RES_1. No other resource can have this alternative name within this category.

Charge Management and Service Product Catalogs

Use

You use Charge Management and Service Product Catalogs to set up the master data that the system uses to calculate transportation charges, such as revenues, costs, and profits. Once the system calculates the charges, you can trigger invoice creation and verification.

You can also include service products in forwarding agreements or service product catalogs. This enables you to specify services, such as fumigation of a container and cargo insurance in a forwarding order.

Integration

Charge Management and Service Product Catalogs is a component of Transportation Management (TM). The following TM components are integrated with Charge Management and Service Product Catalogs because you can automatically calculate transportation charges in these components on the basis of the data stored in Charge Management and Service Product Catalogs:

- [Forwarding Order Management](#)
- [Freight Order Management](#)
- [Forwarding Settlement](#)
- [Freight Settlement](#)

Charge Management and Service Product Catalogs is integrated with the Agreement Management component because TM also calculates transportation charges on the basis of the data stored in agreements. For more information, see [Agreement Management](#).

Features

- The system calculates transportation charges based on agreements (see [Agreement](#)) and the following master data in Charge Management and Service Product Catalogs:
 - [Calculation Sheet](#)
 - [Rate Table](#)
 - [Scale](#)

Note that your TM system can calculate transportation charges based on an agreement maintained in an external system. For more information, see [External Agreements](#).

- You can use the master data cockpit to access an overview of the charge management master data per organizational unit, business partner, and charge type. The system also shows the relationship between the different types of charge

management master data. For more information, see [Master Data Cockpit](#).

- When resolving the calculation sheet or the local calculation sheet, the system also takes into account the following:
 - Relevant rates
 - Additional charges
 - Other surcharges
 - Discounts for the selling (debit) as well as the buying (credit) side for all transportation modes such as trucking (land transport), rail, sea, air, and inland vessel cargo (barge)

More Information

[Setup of MD for Charge Management and SP Catalogs](#)

[Service Product Catalogs](#)

Setup of MD for Charge Management and SP Catalogs

Use

Charge Management and Service Product Catalogs uses agreements and master data to calculate transportation charges for a forwarding order or freight order, and for transportation planning (see [Planning](#)).

You can also include service products in forwarding agreements or service product catalogs. This enables you to specify services, such as fumigation of a container and cargo insurance in a forwarding order.

Prerequisites

- You have configured the prerequisites for the individual charge management master data objects as mentioned in the prerequisites of the respective master data object.
- You have configured the prerequisites for agreements (see [Agreement Maintenance](#)).
- You have configured the prerequisites for service products (see [Service Product Catalogs](#)).
- If you use Transportation Management (TM) for air freight transportation, you must make certain air-freight-specific settings (see [Charge Management Settings for Air Freight](#)).

Process

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

1. Optionally, you define the following charge calculation rules on the [Maintain Charge Calculation Rules](#) screen:
 - [Agreement Determination Rules - Standalone](#)
 - [Agreement Determination Rules - Exclusion](#)
 - [Calc. Sheet Determination Rules - Standalone](#)
 - [Calc. Sheet and Calc. Sheet Item Precondition Rule](#)

- [Grouping Rules for Resolution Base](#)
- [Rules for Rate Table Determination](#)

Standalone means that the standard determination is switched off and the condition associated with the organizational unit is used for the charge calculation. If you select the **Std Determ.** checkbox, the system uses the standard determination if the standalone determination rule fails. **Exclusion** means that the standard determination is active and that constraint and filter options are available. For more information, see [Rules and Conditions for Charge Management and SP Catalogs](#).

2. Optionally, you define a condition with a condition type name starting with **Charge Calculation** (see [Condition](#)).
3. You create an agreement (see [Agreement](#)).

You can include service products in forwarding agreements or service product catalogs.

4. You create a calculation sheet (see [Calculation Sheet](#)).
5. You create a rate table definition and maintain rate values for defined validity periods (see [Rate Table](#)).

You then release the validity periods.

We recommend that you use calculation sheet templates and rate table templates to quickly define and maintain rates. We recommend that you do not create one central rate table with rates for all your business partners.

6. You create a scale (see [Scale](#)).

i Note

You can create an agreement, calculation sheet, rate table, and scale from one consolidated user interface, starting with creating the agreement. Once you create the agreement, you can create the following from within the agreement:

- A calculation sheet for each agreement item
- A rate table from the **Calculation Sheet** tab page for each charge line in the calculation sheet
- A scale from the **Rate** tab page of the **Calculation Sheet** tab page

If you copy an agreement, the system copies the calculation sheets, rate tables, and scales that you created from within the agreement or that you referenced in the agreement.

If you delete an agreement, the system deletes the calculation sheets, rate tables, and scales that you created from within the agreement. However, the system does **not** delete the calculation sheets, rate tables, and scales that you referenced in the agreement.

More Information

[Charge Management and Service Product Catalogs](#)

[Charge Calculation](#)

Charge Management Settings for Air Freight

Use

The following is an overview of the charge management master data settings you should make if using Transportation Management (TM) for air freight transportation.

Prerequisites

You have configured the prerequisites for the individual charge management master data objects as mentioned in the prerequisites of the respective master data object.

Customizing Settings

- You can maintain charges grouped as other charges by the International Air Transport Association (IATA) in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Maintain IATA Other Charge Codes**.
- You can define IATA locations and assigned them to IATA city codes in Customizing for **Transportation Management** under **Master Data > Transportation Network > Location > Define IATA Location Codes**.
- You can define IATA airline codes in Customizing for **Transportation Management** under **Master Data > Business Partner > Define IATA Airline Codes**.
- You can maintain IATA notes in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Air Freight Settings > Maintain IATA Notes**.
- You can define unit load device (ULD) rate types in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Air Freight Settings > Define ULD Rate Types**.
- You can map ULD types to ULD rate types in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Air Freight Settings > Map ULD Types to ULD Rate Types**.
- You can define IATA commodity codes in Customizing for **Transportation Management** under **Master Data > Classification of Goods**. For more information about commodity codes, see [Classification of Goods](#).
- You can define air-freight-specific charge types with the following attributes in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charge Types**:
 - Charge type classification

In air freight, the transportation charges are broadly classified as follows:

- Weight charges
- Other charges
- Commission
- Additional charges
- Valuation charges

You can classify charge types to represent this classification in the system. You can specify charge type classifications for air freight in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Freight Term Settings for Transportation Modes > Define Classifications for Transportation Modes**.

- IATA other charge code
- Transportation mode category of **Air**
- IATA charge due of **Due Agent** or **Due Carrier**

If you classify the charge type as a **Weight** or **Valuation** charge, you specify the IATA charge due as being **Due Carrier**. If you classify the charge type as an **Other** charge, you can specify the IATA charge due as being **Due Agent** or **Due Carrier**, as appropriate. For example, although the printing of the air waybill is the responsibility of the carrier, it can be carried out by the logistics service provider. To allow the system to charge the correct business

partner for the printing, you define a printing charge type, classify it as **Other**, and then specify the IATA charge due as being **Due Agent**.

- You can specify air-freight-specific freight terms in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Freight Term Settings for Transportation Modes** ► **Define Freight Terms for Transportation Modes** ►.
- You can specify air-freight-specific pay definitions in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Freight Term Settings for Transportation Modes** ► **Define Pay Definitions for Transportation Modes** ►.

TACT Rate Upload

If you want to calculate air freight transportation charges based on TACT rates, you should upload TACT rates to TM (see [TACT Rate Upload](#)). Note that you can calculate an air freight transportation charge based on TACT rates or contract rates, as appropriate.

Calculation Sheet Settings

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

For each calculation sheet line item, you can specify whether the system should use contract rates, which are available in standard rate tables, or TACT rates to calculate air freight charges. If using a contract rate, you must specify a rate table, a rate table determination rule, or an amount. If using a TACT rate, the system automatically determines the relevant TACT rate based on mandatory parameters (see [TACT Rates](#)).

You must specify an air-freight-specific charge type for each relevant calculation sheet line item. The system automatically enters the charge type classification, charge due, and IATA other charge code, depending on the charge type's Customizing settings. If these settings are not specified in the charge type's Customizing settings, you can specify them directly in the calculation sheet.

For more information about calculation sheets, see [Calculation Sheet](#).

Rate Table Settings

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

If you specify an air-freight-specific charge type for a rate table, you must specify the rate category. Once you specify the rate category, you can define the rate table as follows:

- ULD

If you define a rate table as a ULD rate category, you must define a scale with a weight-based calculation base. With ULD rate tables, you can maintain rates specific to ULD rate types or across ULD rate types for a particular airport pair.

If you define a scale with the **ULD Rate Type** calculation base, the system applies default values for the weight-based scale items based on the ULD rate type. For example, the default rate values can be the minimum pivot weight (ULD charge code /A or /B, indicating rate class U (ULD rate)) and the over-pivot weight (ULD charge code /C, indicating rate class E (extension rate)) for each ULD rate type and airport pair combination. The scale item type for the minimum pivot weight can be a **To Scale** and for the over-pivot weight can be a **Base Scale**. You define the minimum pivot weights (chargeable weights) for ULD rate types in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Air Freight Settings** ► **Define ULD Rate Types** ►. You then map ULD types to ULD rate types in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Air Freight Settings** ► **Map ULD Types to ULD Rate Types** ►.

If you do not define a scale with the **ULD Rate Type** calculation base, you can enter the minimum pivot weight as a scale item. You specify the rate values for an airport pair.

When creating the scale items for weight-based scales in a ULD rate table, you can create multiple pivot weights for the ULD rate types. You can then use different pivot weights when required, for example, for different origin and destination location pairs, days of the week, or flight codes. When creating multiple pivot weights for ULD rate types, the system creates two weight scale items with ULD charge code /A or /B and one with charge code /C. These scale items represent the pivot and the over-pivot rates respectively. The scale value of the first pair of weight scale items defaults to the pivot weight defined for the ULD rate type in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Air Freight Settings > Define ULD Rate Types**. You must manually enter the scale value of subsequent scale items.

i Note

If there is no scale in the rate table with the **ULD Rate Type** calculation base, you create multiple pivot weights using asterisk (*) for the ULD rate type.

- Specific commodity rate (SCR)

If you define a rate table as an SCR rate category, you must define a scale with the **COMMODITY_CODE** calculation base and a commodity code type, and also a scale with a weight-based calculation base. The rate class S is associated with the scale items of an SCR rate table.

- General cargo rate (GCR) – M, N, & Q

If you define a rate table as a GCR – M, N, & Q rate category, you must define a scale with a weight-based calculation base and ensure that it is mandatory to maintain a minimum value for the calculated rate.

For more information about rate tables, see [Rate](#).

Scale Settings

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A scale created from within a rate table automatically has the same transportation mode category as that of the charge type of the rate table. If the transportation mode category is **Air**, then you can specify attributes for the scale items of a scale with the **Weight** scale base (for example, ULD rate type, ULD charge code, and rate class).

In master data scale templates, the **Items** tab page displays the scale item attributes that are relevant for air freight.

For more information about scales, see [Scale](#).

More Information

[Setup of MD for Charge Management and SP Catalogs](#)

Service Product Catalogs

Use

You can use a service product catalog to capture the services that you make available to your customer. A catalog acts as a template from which your customer can pick and choose, according to their business needs.

You can specify one or more of each of the following features:

- Services in a service product
- Service products in a service product catalog
- Forwarding agreements created from one or more service products in a service product catalog
- Forwarding orders or forwarding quotations created from forwarding agreements, or from the items and service products in a forwarding agreement

In this way, individual services flow from the catalog to the agreement, and on to the order, to fulfill your customer's needs.

For example, you can create services in the system for each of cleaning, fumigation, refrigeration, and cargo insurance. You can create one service product for cleaning and fumigation, another for refrigeration, and another for cargo insurance.

You can then create a service product catalog for transporting perishable food that contains the three service products. You can create an agreement directly from the service products in the catalog, and in turn create a forwarding order from the agreement.

Prerequisites

If you want the system to automatically add services and service products when you create a catalog, you make the following settings in Customizing:

1. Define service types in Customizing for **Transportation Management** under **Basic Functions** > **General Settings** > **Define Service Types**.
2. Specify preconditions and assign service types to item types in Customizing for **Transportation Management** under **Master Data** > **Agreements and Service Products** > **Define FWA and Service Product Item Types**.
3. Specify preconditions and assign item types to catalog types in Customizing for **Transportation Management** under **Master Data** > **Agreements and Service Products** > **Define FWA and Service Product Catalog Types**.

You can also assign a calculation sheet template to a service product catalog type. When you create a service product catalog, the system uses the template to create a calculation sheet or a local calculation sheet. For more information, see [Agreement Maintenance](#).

If you do not make these settings, the system uses the default catalog type that you specify in the **Define FWA and Service Product Catalog Types** Customizing activity to create a catalog.

You have assigned charge types to service types in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Assign Charge Types to Service Types**. The system uses these settings to assign calculation sheets or local calculation sheets to service products or catalogs.

Features

You can edit and delete the service product information in the following business documents:

- Service product catalog
- Forwarding agreement
- Forwarding order

You can create an agreement from one or more service products in a catalog. You can add service products to a standard agreement that you have not created from a catalog. You can add service products from a particular catalog to a different catalog.

i Note

In a service product catalog, you can add services to a service product. In forwarding documents you can add services to an item. If you create a forwarding agreement from a service product, the system inserts the service product as an item in the agreement.

If you deselect the **Add Services** checkbox in the **Define FWA and Service Product Item Types** Customizing activity, the system creates a standard agreement line item. You cannot add services to these service products in the application. However, you can add a service type. In this case, the system treats the service product as a single-service product. For example, you can use this feature to provide insurance for an order.

The system adds the instructions that are assigned to the service product type in the Customizing activity **Define Service Types**. It displays instructions in the **Instructions** tab page, on the forwarding agreement and forwarding order screens. You can create, edit, and delete the instructions to suit your business needs.

For the service product catalog, the following features are available:

- Output management (See [Output Management](#))
- Notes
- Attachments

Notes and attachments features are also available for the individual service products in a service product catalog.

More Information

[Management of Instructions](#)

[Creation of Forwarding Order Items or Forwarding Quotation Items](#)

Condition

Use

You use conditions to determine dependent values, for example, in the following areas:

- Filtering freight units
- Determining freight unit building rules (FUB rules) (see [Determination of Freight Unit Building Rules](#))
- Incompatibilities (see [Incompatibilities](#))
- Determining the sales organization
- Determining rate tables and agreements with charge calculation rules (see [Rules and Conditions for Charge Management and SP Catalogs](#))
- Change controller (see [Change Controller](#))
- Determining means of transport and vehicle types (see [Means of Transport and Vehicle Type Determination](#))

Features

You define input values for each condition. You can use the following to do so:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

- Business object fields including user-specific fields
- Values determined in external determination classes
- Input values that you have defined in the data crawler (see [Use of Data Crawler Input Values](#))

The input values that are available each time depend on the condition type that was chosen. The condition type defines the area in which the system is to take the condition into account. SAP delivers a number of condition types in the standard system.

The output values are determined by the condition type. For example, the FUB rule results from FUB rule determination.

The system generates a BRFplus decision table (BRFplus = Business Rule Framework plus) from input and output values. The system then processes this table from top to bottom during determination. As soon as the system finds a row in the BRFplus decision table whose input values match the current input values, it copies the corresponding output values and processes them in the area that made the call. For more information about BRFplus, see [Business Rule Framework plus \(BRFplus\)](#).

Activities

Open the **Create Condition** app on the launchpad to define conditions.

You can use the **Transport** pushbutton to transport your conditions.

You can use the **Reset BRFplus** pushbutton to reset data objects that you have changed in BRFplus.

Related Information

[Definition of Conditions](#)

[Definition of a Condition for Time Windows](#)

[Definition of a Condition for MTr/VT Determination](#)

Rules and Conditions for Charge Management and SP Catalogs

Use

You can use this function to define rules and conditions for the Charge Management and Service Product Catalogs component, including agreement determination, with a single point of entry. You do this on the **SAP Easy Access** screen by choosing **Logistics** > **Transportation Management** > **Master Data** > **Maintain Charge Calculation Conditions** .

You use these rules to distinguish between the following:

- Whether the system calls a related Business Add-In (BAI) to obtain the decision from enhanced coding
- Whether the system calls BRFplus (Business Rule Framework plus) with the configured BRFplus function

Integration

Each of the available rules has either a corresponding BAI method in which you can hard code or delegate a decision or the link to BRFplus in which the business decision can be executed.

Prerequisites

You have set up the BRFplus function in the system by setting up a condition. For information about setting up conditions, see [Definition of Conditions](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

- You can define rules for agreement determination. You have the following options:
 - **Exclusion** (standard determination): The system can check both the condition and the BAdI option to see if an automatically determined agreement has to be considered for the given business context. The system checks the list of automatically determined agreements one by one to determine if they are still applicable. If more than one agreement remains after the system has applied the exclusion rule, the system considers the agreement priority.
 - **Standalone**: You have switched off standard determination, and instead, the system executes a condition that determines the agreement. If you select the **Std Determ.** checkbox, the system uses the standard determination logic if the condition execution fails or returns no result.

- You can define rules for calculation sheet determination.

Standalone: You have switched off standard determination, and instead, the system executes a condition that determines the calculation sheet. If you select the **Std Determ.** checkbox, the system uses the standard determination logic if the condition execution fails or returns no result.

- You can define a precondition rule for the calculation sheet and calculation sheet item.

You define a BRFplus condition or a BAdI to check if a certain calculation sheet or calculation sheet item is to be processed in a certain business context. The result of this rule is a true/false decision, that is, if the response from BRFplus or the BAdI is true, the calculation sheet or calculation sheet item is considered for charge calculation. For more information, see the **Precondition** tab page in the **Items** screen area of the agreement or calculation sheet.

- You can define grouping rules for resolution bases. You have the following options:

- **Condition**

The system can dynamically determine the attribute or field by which the data source is to be grouped. Both options, the direct grouping and the dynamic BRFplus-based attribute determination, work in the same way – the data source currently being processed (for example, all shipment items) is sorted by this specific attribute. The system then groups all items with the same attribute value and later handles them as one item.

- **Group By**

If you know the calculation base by which the data source is to be grouped and this value is static, you can enter the calculation base directly. The system then groups all items with the same attribute value and later handles them as one item when looking up the rate table. For more information about grouping and collective rates, see [Calculation Logic](#).

i Note

The target structures, into which the data to be grouped is cumulated, are limited in comparison to the original data source as only certain attributes can be logically cumulated. Data that cannot be grouped is handled as a separate record.

- You can define rate table determination rules.

You use this rule to flexibly determine which rate table is to be used for the current charge item. This rule provides the following options:

- **Condition**

The flexible, rule-based determination of a specific rate table. To configure this determination, the complete logistical data source is available. Only a valid rate table is expected as the return parameter.

- **Determin. By BAdI** (Rate Table Determination Using BAdI)

The rate table determination is coded inside the BAdI.

- You can define rules for charge calculation by a formula.

The system calculates the charge item without the normal rate table access. That is, either the BAdI or BRFplus directly returns the calculated amount for this charge item.

i Note

If you decide to calculate the charge item through BRFplus, you can use a mathematical formula or any other BRFplus option. In this case, no information is available in the **Calculation Base** area.

BRFplus for Charge Management and SP Catalogs

Use

You can use this function in Charge Management and Service Product Catalogs to do the following:

- Define specific business rules for various decisions within an automatic charge calculation process
- Determine agreements as follows:
 - Automatically by exclusion
 - Independently of built-in next level principle
- Influence if a certain calculation sheet, local calculation sheet, or a calculation sheet line item is calculated in a certain business context or if the calculation is only applied for certain logistical data
- Flexibly determine a rate during charge calculation
- Use a formula to calculate transportation charges

Features

Linking of Charge Management and Service Product Catalogs and BRFplus

Charge Management and Service Product Catalogs and BRFplus (Business Rule Framework plus) are linked using rules. For information, see [Rules and Conditions for Charge Management and SP Catalogs](#).

Any use of BRFplus in the Charge Management and Service Product Catalogs component is encapsulated by rules. If you can choose to use BRFplus, you can also use a Business Add-In (BAdI). With a BAdI, you can code a better-performing and simplified decision logic.

Data Source Supply for BRFplus

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

In Transportation Management (TM), the conditions maintenance manages the data source binding by naming certain fields within a certain business object. In Charge Management and Service Product Catalogs, you can either use the standard calculation bases or any data element in the forwarding order (FWO) or freight order (FO) as the basis for BRFplus conditions. You can use all available calculation bases for conditions. All available calculation bases are also available as data access definitions for conditions. The name is the same, with the suffix `_TCMC`. Charge management conditions can only use these data access definitions; other data access definitions will fail. For example, to define a condition based on source location you must use data access definition `SOURCELOC_TCMC`, which is mapped to calculation base `SOURCELOC`.

The calculation bases are in the context of the current resolution base if available, or the document header. The freight agreement and forwarding agreement determination rule and the calculation sheet precondition rule are executed on the header level. All other rules are executed on a specific resolution base, as specified in the calculation sheet. In contrast, the system evaluates calculation sheet line item precondition rules against the resolution base of the charge line. For example, a container type code would depend on the resolution base container.

You can also use any data element in the FWO or FO as the data access definition if the data element is mapped to a field in the communication structure. To do this, you create a BRFplus condition with a new criterion, for example, executing carrier, and then assign the BRFplus condition to a determination rule. You then assign the determination rule to a calculation profile of the business partner or the sales organization. During charge calculation, the system uses the determination rule in the calculation profile to determine the appropriate charge management master data.

For BRFplus data supply, you have to name all attributes that are necessary for the specific business rule decision within the conditions maintenance.

More Information

[Business Rule Framework plus \(BRFplus\)](#)

Calculation Sheet and Local Calculation Sheet

Definition

A calculation sheet or a local calculation sheet is a hierarchical table used to calculate transportation charges.

A calculation sheet or a local calculation sheet combines the charge types permitted for a document and the sequence in which the system takes these charge types into account during the calculation.

Use

The system uses calculation sheets or local calculation sheets to specify which transportation charges to calculate and how to calculate them. This helps to bill customers for transportation services and to pay suppliers or carriers for subcontracted transportation services. Calculation sheets (both kinds, including local) contain line items and each line item can result in a calculated amount.

You can create calculation sheet templates on which to base calculation sheets or local calculation sheets. For more information about master data templates, see [Calculation Sheet Maintenance](#). For more information about Customizing templates, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Basic Settings](#) > [Templates](#) > [Define Calculation Sheet Templates](#) >.

Structure

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A calculation sheet can contain the following main tab pages and screen areas:

- **General Data**

The **General Data** tab page contains general information about the calculation sheet. For example, the archiving status and whether you can use the calculation sheet to calculate charges billable to your customer, charges to be billed to you by your carrier, or both.

- **Items**

The **Items** screen area contains the details of the calculation sheet line items and you can create rate tables for each line item. You can insert blank line items one at a time or insert one or more line items with the charge type already specified. You can also control the charge type that you enter, based on your Customizing. For more information see, [Calculation Sheet and Local Calculation Sheet Maintenance](#). The system also displays certain attributes that you do not need for charge calculation, for your information.

The following columns are mandatory in the **Items** screen area:

- **Instruction Type**

If the instruction type is **Line Item Selection**, the **Operation** (Charge Item Operation) column is mandatory.

If the instruction type is **Standard**, the following columns are mandatory:

- **Currency**

If the currency is **%** and the calculation base is **not** amount-based, the **Reference-From Line Number** column is mandatory.

If the currency is **%** and the calculation base is **Customs Value**, **Goods Value**, or **Insurable Value**, you can specify a rate table for the line item. You do not need to specify a reference line item.

- **Rate Amount, Rate Table Det. Rule** (Rate Table Determination Rule) or **Rate Table**

- **Charge Type**

i Note

Each calculation sheet has an initial mandatory sum line, which you cannot delete. You can only maintain the rounding rule for this line because you cannot define a charge type.

Depending on the type of calculation sheet item you select, the **Items** screen area contains some of the following main tab pages:

- **Basic Data**

The **Basic Data** tab page contains information on each line item (charge item), including information that determines how the system processes and calculates the transportation charges for each line item. On the **Basic Data** tab page, the following important information is available for each line item:

- **Dimensional weight profile**

The dimensional weight profile the system uses to calculate the transportation charges for the line item. The system uses the dimensional weight factor in the dimensional weight profile to convert a product's gross volume into dimensional weight. The system then uses the dimensional weight to apply the correct rate.

Note that a dimensional weight profile at the line item level of a calculation sheet has priority over a dimensional weight profile in an agreement. You define dimensional weight profiles in Customizing for

Transportation Management under ► **Basic Functions** ► **Charge Calculation** ► **Data Source Binding** ► **Define Dimensional Weight Profiles** ►.

- Calculation resolution base

During charge calculation, the system determines the calculation resolution base by the origin of the data upon which the charge aspect is based. You can define the charge aspect in the resolution base objects, for example, container, product, and stage.

- Resolution base grouping rule

The rule the system uses to group selected input data, for example, destination location, weight, and volume. You use the grouping rule together with the resolution base to group charge lines by the calculation base specified in the **Group By** field of the grouping rule.

❖ Example

A calculation sheet line item contains **Package** as the resolution base and has a grouping rule assigned to it. The grouping rule is grouped by the **Destination Location** calculation base. There is a freight order with 2 packages with the same destination location. The system groups the gross weight of the 2 packages and determines a rate for the cumulative weight. This rate is lower than the rate that the system would have determined for each package.

- Calculation method

When accessing rates during charge calculation, the system applies the specified calculation method to calculate the charges for the charge type. You define customer-specific calculation methods in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Enhancements to Charge Calculation Engine** ► **Define Calculation Methods** ►.

- **Rate**

The **Rate** tab page contains details of the rate table for each standard charge item. You can also create scales for each rate table. For more information about the **Rate** tab page, see [Rate Table](#).

- **Classification**

The **Classification** tab page contains details on each charge item, such as whether the charge item is mandatory for charge calculation and its charge category.

- **Precondition**

The **Precondition** tab page contains the precondition rule or BRFplus condition for each charge item. The precondition rule determines if the system processes the charge item before charge calculation. This tab page also displays the text from the **Documentation** tab page of the condition. For information about conditions, see [Condition](#).

This tab page also contains **Trade Lanes**, **Services**, and **Partners** tab pages.

- **Cost Pull**

You can use the **Cost Pull** tab page to specify the charge types for each charge item that you want to transfer from the freight side to the forwarding side. The charge types contain the distributed costs from the freight side. If you do not enter a charge type, the system transfers the distributed costs for all charge types in the relevant freight orders. For more information, see [Cost Pull](#).

You define charge types in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Define Charge Types** ►.

- **Incoterms**

The **Incoterms** tab page contains the Incoterms for each charge item. If the Incoterm that is specified for a charge item matches the Incoterm that is specified in a forwarding order, the system adds the line item to the order. You define the Incoterms in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Charge Types** . You can change the Incoterms that you specify in Customizing, on the **Incoterms** tab page.

- o **Related Calculation Bases**

In a calculation sheet, if you enter a calculation rule based on a calculation base that has related calculation bases, you can directly enter values for the related calculation bases on the **Related Calculation Bases** tab page. You specify related calculation bases in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Data Source Binding** > **Define Calculation Bases** .

- o **Freight Terms**

The **Freight Terms** tab page contains the freight terms for each charge type, indicating that you want to capture prepaid export charges and collect import charges in a single forwarding settlement document. For more information, see [Freight Terms in Forwarding Orders](#). Note that the **Freight Terms** tab page is only visible when the calculation sheet can be used to calculate charges billable to your customer, that is, if the charge usage of the calculation sheet is **Customer**, **Customer and Service Provider**, or **All**.

You define freight terms in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Freight Term Settings for Transportation Modes** > **Define Freight Terms for Transportation Modes** .

- **Change Documents**

You can view all changes made to a calculation sheet on the **Change Documents** tab page or screen area. The changes are listed by date and time. You view the details of the changes at header level by selecting the relevant row in the **Transactions** screen area and at field level by selecting the relevant row in the **Record Level Changes** screen area. You must refresh the calculation sheet to view recently-saved changes.

Integration

A calculation sheet or a local calculation sheet is integrated with agreements, rate tables, local rate tables, scales, and local scales. For more information, see [Agreement](#), [Rate Table](#), and [Scale](#).

More Information

[Calculation Sheet Maintenance](#)

[Rules and Conditions for Charge Management and SP Catalogs](#)

[Charge Calculation](#)

[Setup of MD for Charge Management and SP Catalogs](#)

Calculation Sheet and Local Calculation Sheet Maintenance

Use

You create and maintain calculation sheets (both kinds, including local) as a step in the process to set up master data for charge calculation.

Integration

Calculation sheet maintenance is a step in the process to set up master data in the Charge Management and Service Product Catalogs component and is integrated with agreements, rate tables, local rate tables, scales, and local scales. For more information, see [Setup of MD for Charge Management and SP Catalogs](#).

You can use the master data cockpit to access an overview of the charge management master data per organizational unit, business partner, and charge type. The system also shows the relationship between the different types of charge management master data. For more information, see [Master Data Cockpit](#).

Prerequisites

You have defined number ranges in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Number Range Intervals > Assign No. Ranges to Calc. Sheets, Rate Tables, & Scales**.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

- **Templates**

You can use templates to create calculation sheets, calculation sheet line items, and preconditions.

You can define Customizing templates in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Templates > Define Calculation Sheet Templates**. You can transfer Customizing templates across different TM systems.

You can define master data templates directly in Transportation Management (TM). To define a master data template for a calculation sheet, open the **Create Calculation Sheet Template** app from the launchpad. If required, you can define a master data template based on a Customizing template.

You can view and update all calculation sheet templates in the **Calculation Sheet Templates** personal object worklist.

You can also influence the creation of a calculation sheet from a master data calculation sheet template or a Customizing calculation sheet template. For more information, see Customizing for **Transportation Management** under **Business Add-Ins (BAdIs) for Transportation Management > Master Data > Calculation Sheets > BAdI: Calculation Sheet Creation from Templates** or **BAdI: Mass Calculation Sheet Creation from Templates**.

- **Rate tables and scales**

You can create rate tables and scales from within a calculation sheet in the following ways:

- Create a rate table for each calculation sheet line item in the **Items** screen area.

You can assign a rate table to multiple line items if you allow multiple charge types in the rate table. Note that you can assign a calculation rule to each calculation sheet line item and so do not need to use a rate table. The calculation rule is applicable for the amount and not for the rate table.

- Create a scale for a rate by going to the rate table details in the **Items** screen area.

You can also copy and delete the rate tables and scales from within a calculation sheet. For more information, see [Rate Table](#) and [Scale](#).

i Note

If you copy a calculation sheet, the system copies the rate tables and scales that you created from within the calculation sheet or that you referenced in the calculation sheet. If you delete a calculation sheet, the system deletes the rate tables

and scales that you created from within the calculation sheet. However, the system does not delete the rate tables and scales that you referenced in the calculation sheet.

You can edit local rate tables and scales independently of the calculation sheet they were created within. To enable this, you select the **Edit Loc. MD Objects** (Edit Local Charge Management Master Data Objects) checkbox in Customizing for **Transportation Management** under **Basic Functions > General Settings > Define General Settings for TM**.

- Index rate tables for fuel surcharges

You can enter an index rate table when fuel surcharges change frequently, and where you have different fuel surcharges in different zones in your country or region. You can use one of the following approaches to calculate your fuel surcharges:

- Use an actual fuel surcharge rate from an index rate table that is valid on a particular date.

To enable this approach, in the line item of the calculation sheet, specify a standard rate table with a calculation base of **FSC**, and an index rate table.

- Determine an index for the fuel surcharge and apply the index to the amount or percent you specify in a line item of a calculation sheet.

To enable this approach using amounts, specify a standard rate table or amount, the index rate table, and the index base date.

To enable this approach using percentages, specify two line items in the calculation sheet. In the first line item, enter the standard rate table. In the second line item, enter the percent you want to apply, the index rate table, the index base date, and the reference-from line number for the first line item.

For both amounts and percentages, you must use the calculation method type **Fuel Surcharge Calculation** and the calculation method **FUEL_SURCHARGE**.

- Air freight charges

You can specify attributes in a calculation sheet to facilitate the calculation of air freight charges, for example:

- Rate type of **Contract Rate** or **TACT Rate**
- Rate category of **ULD**, **SCR**, or **GCR - M, N, & Q**
- Charge type classification
- IATA charge due of **Due Agent** or **Due Carrier**
- IATA other charge code

The system displays certain attributes that you do not need for charge calculation, for your information.

For more information, see [Charge Management Settings for Air Freight](#).

- Calculation rules

You can assign a calculation rule to a calculation sheet line item using the **Amount** field, instead of assigning a rate table. You can only assign one calculation rule to each calculation sheet line item and cannot assign both a calculation rule and a rate table to a calculation sheet line item.

If the calculation base has an amount-based scale base, the line currency must be % and you do not need a reference-from or reference-to line. During charge calculation, the system applies the percentage amount to the amount values, such as goods value and insurable value.

- Flat rates, relative rates, or percentages

You can specify a flat rate (amount) for a calculation sheet line item by entering an amount and a valid currency. You can specify that a calculation sheet line item is a percentage of another calculation sheet line item by entering a percentage

value, % in the **Currency** field, and a reference line number. You can also have a relative rate for a calculation sheet line item by specifying a calculation rule in the calculation sheet line item. However, you can specify only one calculation rule at the line-item level.

- Minimum and maximum rate values

You can specify a minimum rate value and a maximum rate value for each calculation sheet line item with the **Standard** instruction type. Note that in order to specify minimum and maximum rate values, you must maintain the calculation rule for the calculation sheet line item. If the system calculates the charges to be lower than the minimum rate value or higher than the maximum rate value, the system enters the minimum rate value or maximum rate value as the charge. Note that the minimum rate value must be higher than zero and you cannot specify a minimum rate value or maximum rate value if there is a rate table or rate determination rule attached to the calculation sheet line item.

- Zero rates

You can indicate that there is no charge for a calculation sheet line item by selecting the **Zero Rate** checkbox on the **Rate** tab page for the line item.

- Charge types

You can assign a charge type to calculation sheets and rate tables and specify one or more leading charge types in a calculation sheet. A charge type classifies a charge line, which plays an important role in how the system calculates the transportation charges for the charge line. For more information, see [Charge Types](#).

- Prepaid and collect charges in one settlement document

You can capture prepaid export charges and collect import charges in a single forwarding settlement document by inserting a charge type with freight terms. For more information, see [Freight Terms in Forwarding Orders](#).

- Determination preconditions

The system uses preconditions during calculation sheet determination, for example, shipping type, stage category, transportation mode, and service level.

- Cost pull

You can transfer distributed costs from the freight side to the forwarding side, and use the costs as a basis for settling with a customer or an internal organization. To transfer the result of a cost distribution from the freight side to the forwarding side, in a calculation sheet you must specify **Customer** or **Internal** in the **Charge Usage** field. You use an instruction type of **Cost** at the item level. The system automatically includes an internal charge calculation method type, and the cost pull calculation method **COST_PULL**. You must specify a cost pull strategy, and in the **Cost Pull** tab page, you can specify the charge types the system transfers to the forwarding side. For more information, see [Cost Pull](#).

- Calculation date type

You can specify the calculation date type for each calculation sheet item. Based on the calculation date type you specify, the system uses a date, such as system date, invoice date, or order date, to determine a rate from the rate table and calculates the charges for each calculation sheet item.

- Charge relevance

You can specify whether a charge line must be included for charge calculation in the invoice, order, or both.

- Validity period

You can set a validity period for each calculation sheet line item so that the charge line is relevant only during that period.

- Mass update of calculation sheet amounts

You can use the **Update Calculation Sheet Amounts** function to change the charge item amount in one or more calculation sheets at the same time. You can navigate to the function in one of the following ways:

- Open the **Charge Management** app from the launchpad. Choose the master data cockpit for the relevant agreement type and enter your search criteria. In the personal object worklist result, select the line items that contain the calculation sheets you want to update, and choose the **Update Rates/Calculation Sheet Amounts** button.
- In transaction SE38, enter /SCMTMS/TCCS_MASS_UPDATE.

For more information, see [Update Calculation Sheet Amounts](#).

- Calculation sheets in forwarding agreement quotation items

You can enable the system to calculate the contribution and the target contribution in the flat view profile of forwarding agreement quotations by selecting the **Cost** checkbox for calculation sheet items. For more information, see [Flat View](#).

Activities

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

To create a calculation sheet for an agreement item in an agreement, on the launchpad choose the **Edit Freight Agreement** app, the **Edit Forwarding Agreement** app, or the **Edit Internal Agreement** app.

To change a calculation sheet for an agreement item in an agreement, on the launchpad choose the **Edit Freight Agreement** app, the **Edit Forwarding Agreement** app, or the **Edit Internal Agreement** app. Edit the calculation sheet on the **Calculation Sheet** tab page of the **Items** screen area in the agreement.

To display or edit a calculation sheet separately from the agreement it was created within, on the launchpad choose the **Edit Calculation Sheet** app.

To create or change a calculation sheet, on the launchpad choose the **Create Calculation Sheet** app or the **Edit Calculation Sheet** app. A calculation sheet can be used in multiple agreements.

→ Recommendation

Adhere to the following guidelines when creating a calculation sheet:

- Line numbers must be greater than 10 because the system generates the first sum line as 10.
- The sum line must have reference lines greater than itself.
- A subtotal line must have reference lines less than itself.
- A subtotal line inherits the resolution base from the sum parent line.
- If you require a nested sum line, decide on the reference line numbers accordingly.

Related Information

[Calculation Sheet and Local Calculation Sheet](#)

Update Calculation Sheet Amounts

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can use the /SCMTMS/TCCS_MASS_UPDATE program to change the charge item amount in one or more calculation sheets at the same time.

You can increase or decrease the amount by a percentage or an amount. You can apply the change to all charge items contained in the calculation sheets in your selection. You can also apply the change to absolute and relative charge items. A charge item is relative if you have specified a calculation rule for the charge item in the relevant calculation sheet.

You can further refine your selection by specifying the following settings; the function updates only the calculation sheets that fit the criteria you enter:

- Validity period

You must select an existing validity period. The system displays the valid-from date in the function.

- Charge type

You must enter a charge type that is used in the relevant calculation sheets.

You can update charge item amounts for forwarding, freight, and internal agreements from the master data cockpit.

To work with the master data cockpit, open the **Charge Management** app from the launchpad. Choose the master data cockpit for the relevant agreement type and enter your search criteria. In the personal object worklist result, if you select a line item that contains a calculation sheet only and choose the **Update Rates/Calculation Sheet Amounts** button, the system opens the **Update Calculation Sheet Amounts** function. The system automatically includes the calculation sheet in the update criteria when you open the function.

If you select a line item that contains both a rate table and a calculation sheet, the system opens the **Update Rates** function. If you select a line item that contains a rate table and a different line item that contains only a calculation sheet, the system opens both the **Update Rates** and **Update Calculation Sheet Amounts** functions on the same screen. The system automatically includes the calculation sheet and rate table in the update criteria when you open the function. For more information, see [Update Rates](#).

You can only update charge items that have an instruction type of **Standard**.

Integration

The update of calculation sheet amounts is a function in calculation sheet maintenance. For more information, see [Calculation Sheet Maintenance](#).

Prerequisites

You have specified the settings required for calculation sheet maintenance. For more information, see [Calculation Sheet Maintenance](#).

Example

You want to update charge item amounts in a calculation sheet or a local calculation sheet. The following table describes the charge item settings in the calculations sheet or the local calculation sheet:

Item Hierarchy	Instruction Type	Validity	Amount (USD)	Absolute/Relative
Basic rate	Standard	01.01.2013 to 02.01.2013	100	Absolute

Item Hierarchy	Instruction Type	Validity	Amount (USD)	Absolute/Relative
Air freight charges	Standard	02.01.2013 to 03.01.2013	150	Relative
Insurance fees	Standard	03.01.2013 to 04.01.2013	250	Relative

The following table describes your inputs to the function:

Setting	Attribute
Validity period	None
Percentage for absolute charge items	+10%

The system updates the basic rate charge item from USD 100 to USD 110. The system does not include the charge items for air freight charges and insurance fees because these are relative charge items and you specified absolute charge items in the function.

More Information

[Master Data Cockpit](#)

Charge Types

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

When setting up charge management master data, you use charge types in calculation sheets and rate tables to classify charge lines. This classification plays an important role in how the system calculates the transportation charges for a charge line.

You can assign charge types to calculation sheet line items and to rate tables at header level or item level. Note that the system assigns rate tables to a line item in a calculation sheet only if the charge types match.

You define charge types in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Charge Types](#). The system validates the following when you define a charge type:

- You can define an independent charge type only with [Positive Value](#) for [Positive/Negative](#).
- When you edit an independent charge type, you can't update the value of the field [Positive/Negative](#) to [Negative Value](#) or [Positive/Negative Value](#).

i Note

The system doesn't validate the charge type when you update any other attributes of a charge type, for example calculation base or charge usage.

→ Recommendation

Group and categorize charge types by charge subcategory and charge category to help structure transportation charges. You define charge categories and charge subcategories in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Category Codes](#) and [Define Subcategory Codes](#).

Charge Type Attributes

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A charge type's attributes define how the charge type affects the calculation of transportation charges for a charge line in a calculation sheet. There are standard attributes, which apply for transportation charges for all transportation mode categories, and also air-freight-specific attributes. For more information about charge type attributes, see the Customizing activity [Define Charge Types](#).

Standard Attributes

You can specify standard attributes for a charge type, for example:

- Whether the charge line can have a positive or a negative value
- Whether the charge line can be an absolute value (an actual amount) or a percentage value of another charge line

Note that if a charge line represents a percentage, then to calculate a chargeable amount, this charge line must refer to another charge line or to a subtotal that produces an absolute value.

- Whether the charge type is a leading charge type and so must have a value for the system to be able to calculate the charges
- The transportation mode category (road, rail, sea, or air) of the charge type

You can then use the charge type only in a calculation sheet that is associated with an agreement with the same transportation mode category or with no transportation mode category.

- The classification of the charge type for transportation mode categories
- The charge usage of the charge type

You can then use the charge type only in calculation sheets or rate tables with a compatible charge usage. If you don't specify the charge usage, you can use the charge type in any calculation sheet or rate table. Note that you can use the program [Mass Update of Charge Types with Charge Usage](#) (/SCMTMS/MASS_UPD_CHRG_TYP_USAG) in transaction SE38 to update the charge usage of multiple charge types at once.

- The charge type description in multiple languages
- The Incoterm for the charge type

If a forwarding order has the same Incoterm, the system includes a line item from a calculation sheet in the forwarding order.

- The freight terms for the charge type

The freight terms indicate that you want to calculate prepaid export charges and collect import charges for a single forwarding order. If a calculation sheet has a charge type with freight terms, the system adds a [Freight Terms](#) tab page to the calculation sheet. For more information about freight terms, see [Freight Terms in Forwarding Orders](#).

Air Freight Attributes

If you want to use the charge type for calculating air freight charges, you can also specify air-freight-specific attributes, for example:

- The classification of the charge type for the transportation mode category of **Air** as one of the following:
 - **Weight**
 - **Other**
 - **Commission**
 - **Additional Charge**
 - **Valuation**
- The charge code of the charge type for charges classified as **Other Charges** by the International Air Transport Association (IATA).
- The transportation mode category as **Air**.
- Whether the charges are due to the agent or to the carrier.

For more information about air freight settings, see [Charge Management Settings for Air Freight](#).

Leading Charge Types

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can have one or more leading charge types in a calculation sheet. A leading charge type must have a value for the system to consider the calculation sheet for charge calculation. If there are multiple calculation sheets available when calculating charges, the system skips the calculation sheets without a leading charge type. The system also skips the calculation sheets if the leading charge type doesn't have a rate. If there are no calculation sheets with a leading charge type with a rate, the system chooses a calculation sheet randomly. Note that if there are multiple leading charge types in a calculation sheet, all the leading charge types must have a rate defined for the system to consider the calculation sheet for charge calculation. For more information about leading charge types and calculation sheet determination, see [Agreement and Calculation Sheet Determination](#).

i Note

For the system to take the leading charge type into account when calculating charges, you must select the **Evaluate All Agreements** checkbox in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Charges Profiles**. The system then skips the agreements that use calculation sheets without a leading charge type or with a leading charge type that doesn't have a rate.

You can specify a charge type as a leading charge type in the Customizing activity **Define Charge Types**.

Multiple Charge Types

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can allow multiple charge types in a rate table for the following reasons:

- You want to avoid defining numerous similar rate tables.
- You want to use the rate table for multiple calculation sheet line items.

In this case, the charge type of the dimension in the rate table and calculation sheet line item must match.

i Note

If you allow multiple charge types, you must adhere to the following guidelines when defining the rate table:

- One dimension must have the **Multiple Charge Type** (MULT_CHRG_TYP) calculation base.
- All charge types must have the same value type (absolute, percentage, or decided when assigning the charge type).

Related Information

[Calculation Sheet and Local Calculation Sheet](#)

[Calculation Sheet and Local Calculation Sheet Maintenance](#)

[Rate Table and Local Rate Table](#)

[Charge Calculation](#)

Validation of Charge Types in Agreement Management

When you add a new charge line to any calculation sheet or edit a charge line, the system validates the charge type of that charge line.

The validation for the charge lines is applicable for the following agreement types:

- Freight agreements
- Internal agreements
- Forwarding agreements
- Forwarding Agreement Quotation and Assignment
- Freight Agreement RFQ Master
- Service Product Catalogue

The system validates the charge types in the charge lines of calculation sheets in the following scenarios:

- Creating new agreements, agreement templates, and calculation sheet templates.
- Copying agreements, agreement templates, and calculation sheet templates from the existing ones.
- Adding a new charge type in calculation sheet items of existing agreements, agreement templates, and calculation sheet templates.

i Note

You can't manually add the charge type **Adjusted/Quoted Amount (ADJ)** to a calculation sheet or agreement.

- Modifying the existing charge type in calculation sheet items of agreements, agreement templates, and calculation sheet templates.

The validation depends on the **Disable Extended Check for Charge Types in Agrmt. Management** checkbox. You can do that from the Customizing for **Transportation Management** under **Basic Functions > General Settings > Define General Settings for TM**

- When you've not selected the **Disable Extended Check for Charge Types in Agrmt. Management** checkbox, the system validates the following when you insert a charge line item in a calculation sheet item:
 - For new calculation sheets, you can't insert charge line items that have the same charge type with the same resolution base.
 - You also can't add a charge line item with independent charge type having **Negative** or **Positive/Negative** indicator.
 - For an existing calculation sheet, the system doesn't validate the charge line items until you modify them. However, if you edit the charge types of the existing charge line items the system validates them for charge types and corresponding resolution base as mentioned earlier.

i Note

- If you edit any other attribute of existing charge line items, for example **Currency** or **Amount**, the system doesn't validate the charge line item.
- Sign indicator validation for independent charge types isn't applicable for charge lines with **Adjusted/Quoted Amount (ADJ)** charge type.

Adjusted/Quoted Amount (ADJ) is a charge type which is proposed by the system during charge calculation. This charge type is based on the proposal from the carrier and it represents the delta amount. You can't manually add it to a calculation sheet or agreement.

- When you've selected the **Disable Extended Check for Charge Types in Agrmt. Management** checkbox, the system validates the calculation sheet items based on the company code and carrier of the freight agreement where the calculation sheet is used. This validation is based on the combination of company codes and carriers in the Customizing activity **Enable Lean Services based Settlement Integration**. You can do that from the Customizing of **Transportation Management** under **Settlement > Freight Settlement > Integration for Settlement Posting > Service Procurement - Lean Services**.

You use the Customizing activity **Enable Lean Services based Settlement Integration** to enable lean service-based integration. In this case, the system only validates the calculation sheet items of those freight agreements where it finds the combination. For more information, see [Freight Settlement Integration using Lean Services](#).

i Note

If a calculation sheet isn't used in any freight agreement, the system doesn't validate the calculation sheet line item even when you add a new charge line item or edit the charge types of the existing charge lines items.

The following table shows the scenarios and their impact of using the **Disable Extended Check for Charge Types in Agrmt. Management** and company code and carrier combination in the Customizing activity **Enable Lean Services based Settlement Integration**.

Scenario	Status of Agreement	Impact
When you don't select Disable Extended Check for Charge Types in Agrmt. Management checkbox.	New	The system validates all charge types when you add them to the calculation sheet.
	Existing	The system validates only the new charge types that you add to the calculation sheet.

Scenario	Status of Agreement	Impact
When you select Disable Extended Check for Charge Types in Agrmt. Management checkbox but haven't added any carrier and company code combination in the Customizing activity Enable Lean Services based Settlement Integration .	New or Existing	The system doesn't validate any charge type.
When you select Disable Extended Check for Charge Types in Agrmt. Management checkbox and have also added carrier and company code combinations in the Customizing activity Enable Lean Services based Settlement Integration .	New	System validates all the charge types in the agreement for the carrier and purchase company code combinations that are there in the Customizing activity Enable Lean Services based Settlement Integration .
	Existing	System validates only the newly added charge types in an agreement for the carrier and purchase company code combinations that are there in the Customizing activity Enable Lean Services based Settlement Integration .

Examples

The following examples show the process of validation of calculation sheet items that are used in freight agreements, when you've selected the **Disable Extended Check for Charge Types in Agrmt. Management** checkbox:

❖ Example

You've defined the following are the settings for the two charge types **SF04** and **FB00** in the Customizing activity **Define Charge Types**:

Charge Type	Independent Charge Type	Positive/Negative
SF04	Yes	Positive
FB00	Yes	Positive/Negative

You've created a freight agreement **AGR1** with the carrier **CARR1**, purchasing organization **PUR1**, and the following local calculation sheet:

Calculation Sheet Item No.	Charge Type	Resolution Base	Amount
10	SF04	ROOT	15
20	FB00	ROOT	25

You've also defined the following combination of carrier and company code in the Customizing activity **Enable Lean Services based Settlement Integration**:

Carrier	Company Code
CARR1	*

Now, you add a new calculation sheet item 30 to the local calculation sheet for the freight agreement **AGR1**:

Calculation Sheet Item No.	Charge Type	Resolution Base	Amount
10	SF04	ROOT	15
20	FB00	ROOT	25
30	FB00	ROOT	20

As you've checked the [Disable Extended Check for Charge Types in Agrmt. Management](#) checkbox in Customizing, and also specified the combination of carrier and company code in the Customizing activity [Enable Lean Services based Settlement Integration](#), the system doesn't validate the existing calculation sheet item numbers 10 and 20, but validates only the newly added calculation sheet item number 30. As the charge type **FB00** is an independent charge with a [Positive/Negative](#) indicator, the system doesn't save the freight agreement.

❖ Example

You've defined the following the two charge types **SF04** and **FB00** in the Customizing activity [Define Charge Types](#):

Charge Type	Independent Charge Type	Positive/Negative
SF04	Yes	Positive
FB00	Yes	Positive/Negative

You've also defined the following calculation sheet **GS1**:

Calculation Sheet Item No.	Charge Type	Resolution Base	Amount
10	SF04	ROOT	15
20	FB00	ROOT	25

The calculation sheet **GS1** is used in the following two freight agreements:

Agreement	Carrier
AGR1	CARR1
AGR2	CARR2

You've also defined the following combination of carrier and company code in the Customizing activity [Enable Lean Services based Settlement Integration](#):

Carrier	Company Code
CARR1	*

You add a new calculation sheet item to the calculation sheet **GS1**:

Calculation Sheet Item No.	Charge Type	Resolution Base	Amount
10	SF04	ROOT	15
20	FB00	ROOT	25

Calculation Sheet Item No.	Charge Type	Resolution Base	Amount
30	FB00	ROOT	20

As you've selected the **Disable Extended Check for Charge Types in Agrmt. Management** checkbox in Customizing, the system searches for all the freight agreements where **GS1** is used. In this case, it's **AGR1** and **AGR2**. The system then checks the entries in the Customizing activity **Enable Lean Services based Settlement Integration** for any combination of company code and carrier (**CARR1**). It also checks the agreements where **CARR1** is used, which is **AGR1**. Then the system finally validates the charge type in **GS1** for the newly added calculation sheet item, line number 30. As the charge type **FB00** is an Independent charge with a **Positive/Negative** indicator, the system doesn't save the calculation sheet.

Related Information

[Handling Calculated Charges in Lean Services](#)

[Basic Settings for Tendering](#)

Rate Table and Local Rate Table

Definition

A rate table or a local rate table is a grouping of prices for transportation services. The prices (or rates) are listed by validity period in the rate table. You can maintain up to 14 dimensions in a rate table or a local rate table.

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can define the scales and validities of the rate table. Once you define the rate table, you can maintain the rate values in the rate table. For more information, see [Rate Definition and Maintenance](#).

You can define a rate table manually or automatically by using a rate table template with predefined scales and validity dates. For more information about master data templates, see [Rate Definition and Maintenance](#). For more information about Customizing templates, see Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Templates** > **Define Rate Table Templates**.

You can use the standard, break-weight, or clipping calculation method for a rate. The break-weight calculation method compares and selects the lower rate from either the actual rate range or the lower end of the next rate range. It assumes that one of the scales in the rate is the weight. The clipping calculation method considers all scales, even if the value lies outside of a scale. With clipping, the system works through the pricing scale level-by-level. The calculation results from each scale level are then added up to produce the overall result.

Structure

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A rate table can contain the following main tab pages and screen areas:

- **General Data**

The **General Data** tab page contains general information about the rate table, for example, whether you can use the rate table to calculate charges billable to your customer, charges to be billed to you by your carrier, or both. On this tab page, you can also indicate how the system uses the rate table to calculate transportation charges, and whether you can use an absolute value or a percentage value.

This tab page also displays scales. You can copy a scale into the rate table to create dimensions local to the rate table. However, you cannot directly reuse a scale in the rate table. You can also create new scales for the rate table, with or without a scale template. The following columns can be mandatory:

- **Calculation Base** (always mandatory)

If the calculation base is amount-based, then the **Scale Currency** column is mandatory.

If the calculation base is not amount-based, then the **Scale Unit of Measure** column is mandatory.

- **Calculation Type** (always mandatory)

- **Scale Type** (conditionally mandatory)

If the calculation type of the calculation base is **Relative**, then the **Scale Type** column is mandatory.

- **Scale Type** and **Scale Unit of Measure** (conditionally mandatory)

If the scale base is numeric, then the **Scale Type** and **Scale Unit of Measure** columns are mandatory.

For more information, see [Scale](#).

The **Details** tab page contains the scale details.

- **Dates and Values**

The **Dates and Values** tab page contains general information about each rate, such as its validity period, the status of the validity period, and the currency. The **Valid From**, **Valid To**, and **Currency** columns are mandatory on this tab page.

On the **Dates and Values** tab page, you can display the information in different ways and perform certain actions, depending on the details already in the rate table, for example:

- You can generate incremental scale items if the rate table contains a numeric scale that does not contain any scale items.
- You can choose which rates to display, that is, all rate combinations or only rates that were maintained, if the rate table contains a scale item.
- You can select which scale to use as the dimension column if the rate table contains a numeric scale and more than one scale.
- You can display the description of a rate, delete a rate, copy a rate, and freeze the columns in the **Rate** table if the rate table contains a scale item.

If you open the rate table in display mode, you can choose to show archived validity periods. Note that you cannot change to edit mode when showing archived validity periods. For more information about archived rate tables, see [Archiving Rate Tables \(TM-MD-CM-RA\)](#).

You can also filter the rates that you want the system to display.

This tab page contains **Rates**, **Calculation Rules**, and **Scale Items** tab pages. On the **Scale Items** tab page, you can add and maintain scale items and specify the actual base or factor with which the system calculates the charges for the scale. On the **Calculation Rules** tab page, you determine the values with which the single rate is to be multiplied. If there is more than one calculation rule, the system multiplies the determined rate with each of the given entries and totals the final amount during charge calculation. If the scale items are relative, then the system applies the calculation rules. On this tab page, you can also specify the level at which the system applies the calculation rule in the rate table, for example, at rate-table or scale-item level.

i Note

If you specify that the system applies the calculation rule at the **Rate Table** level, the **Price Unit** column on the **Calculation Rules** tab page is mandatory. You must specify the level as **Scale Item** if the price unit of one of the scale items is not the same as the others. In this case, you must specify the price unit on the **Scale Items** tab page for the corresponding scale. The system applies the calculation rule only when the calculation type of the scale item for the selected rate is relative.

Calculation rules and scale items are validity-dependent, meaning you can have different calculation rules and scale items for each validity period listed in the rate table.

- **Excel Integration**

On the **Excel Integration** tab page, you can upload and download Microsoft Excel sheets to speed up the maintenance of large rate tables. For more information, see [Creation and Maintenance of Rate Tables Using Microsoft Excel](#).

- **Change Documents**

You can view all changes made to a rate table on the **Change Documents** tab page or screen area. The changes are listed by date and time. You view the details of the changes at header level by selecting the relevant row in the **Transactions** screen area and at field level by selecting the relevant row in the **Record Level Changes** screen area. You must refresh the rate table to view recently-saved changes.

i Note

You can activate the **Change Documents** tab page or screen area in Customizing for **Transportation Management** under **Master Data** ► **Rate Tables** ► **Define Rate Table Types** .

Integration

A rate table or a local rate table is integrated with agreements, calculation sheets, local calculation sheets, scales, and local scales. For more information, see [Agreement](#), [Calculation Sheet](#), and [Scale](#).

More Information

[Charge Calculation](#)

[Setup of MD for Charge Management and SP Catalogs](#)

[TACT Rates](#)

Rate Definition and Maintenance

You define rate tables (both kinds, including local) and maintain rates as a prerequisite for charge calculation. If the rates for each charge item aren't fixed or are calculated using a Business Add-In (BAI) or BRFplus (Business Rule Framework plus), charge calculation requires rates for every object in Transportation Management (TM), such as a forwarding order or freight order.

Use

You can define and maintain rates directly in TM or upload and download them from Microsoft Excel or a similar XML-enabled Application Programming Interface (API). Once you define the rate table or the local rate table, you can then maintain the rate values in the rate table or the local rate table respectively. For more information, see [Rate Table](#).

→ Recommendation

Use rate table templates to quickly define and maintain rates in rate tables or local rate tables. We recommend that you don't create one central rate table with rates for all your business partners.

i Note

You can use TACT rates for calculating air freight charges (see [TACT Rates](#)).

Integration

Rate maintenance is a step in the process to set up master data in the Charge Management and Service Product Catalogs component and is integrated with agreements, calculation sheets, local calculation sheets, scales, and local scales. For more information, see [Setup of MD for Charge Management and SP Catalogs](#).

The system determines the most appropriate rate when calculating transportation charges, based on certain criteria and attributes. For more information, see [Rate Determination](#).

You can use the master data cockpit to access an overview of the charge management master data per organizational unit, business partner, and charge type. The system also shows the relationship between the different types of charge management master data. For more information, see [Master Data Cockpit](#).

Prerequisites

- You've made the following settings in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation** :
 - Data Source Binding > Define Calculation Bases** 
 - Data Source Binding > Define Resolution Bases** 
 - Basic Settings > Number Range Intervals > Assign No. Ranges to Calc. Sheets, Rate Tables, & Scales** 
- You've defined rate tables types for rate tables and local rate tables in Customizing for **Transportation Managements** under **Master Data > Rate Tables > Define Rate Table Types** .

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

- Templates

You can use templates to create rate tables with predefined scales, organizational data, calculation rules, and validity periods.

You can define Customizing templates in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Templates > Define Rate Table Templates** . You can transfer Customizing templates across different TM systems.

You can define master data templates directly in TM. To define a master data template for a rate table, open the **Create Rate Table Template** app from the launchpad. If needed, you can define a master data template based on a Customizing template.

You can view and update all rate table templates in the **Rate Table Templates** personal object workload.

- Creation and maintenance of rate tables using Microsoft Excel

You can create multiple rate tables in Microsoft Excel using rate table templates and upload them to TM.

You can maintain a large rate table, with or without scale items, using Microsoft Excel. You can enter the rates and scales into a Microsoft Excel file and upload the file to TM. Alternatively, you can define an empty rate table with validity periods in TM and download it to Microsoft Excel. You can then maintain the rate table in the Microsoft Excel file and upload it to TM.

You can also choose to download rate tables in the background and be notified by e-mail when the download to the application server path is complete (see [Background Download of Rate Tables](#)).

For more information, see [Creation and Maintenance of Rate Tables Using Microsoft Excel](#).

- Update of rates

You can use the **Update Rates** function to change the rate values in a single rate table or in multiple rate tables at the same time. You can navigate to the function in one of the following ways:

- Open the **Charge Management** app from the launchpad. Choose the master data cockpit for the relevant agreement type and enter your search criteria. In the personal object worklist result, select a line item that contains the rate tables you want to update, and choose the **Update Rates/Calculation Sheet Amounts** button.
- In the **Maintain Rate Table** app, open the rate table and choose the **Update Rates** button.
- In transaction SE38, enter /SCMTMS/RATE_MASS_UPDATE.

For more information, see [Update Rates](#).

- Approval workflow

You can enable an approval workflow when you create or change validity periods in rate tables.

You can specify that you want the system to notify the approver when you request approval for a new or changed validity period. You can also specify whether you can edit rate tables that contain the validity periods during the approval workflow. Note that the approver is the person assigned to the position that is the head of the organizational unit. Therefore, at least one organizational unit must have a position that is the head of the organizational unit. For more information, see [Approval Check](#).

The approver uses Business Workplace to approve or reject the new or changed validity period. For more information, see SAP Library for **SAP Business Workflow** on SAP Help Portal at <http://help.sap.com/netweaver>. In SAP Library, choose ► **SAP NetWeaver 7.3** ► **Application Help** ► **Function-Oriented View** ► **Application Server** ► **Application Server ABAP** ► **Other Services** ► **Services for Application Developers** ► **SAP Business Workflow** ► **Reference Documentation** ►.

i Note

You enable the approval workflow and allow the rate tables that contain the validity periods to be edited during the approval workflow when you define rate table types in Customizing for **Transportation Management** under ► **Master Data** ► **Rate Tables** ►.

- Offline approval workflow

TM can send work items for validity periods in rate tables by e-mail to the Microsoft Outlook inbox of a user, such as a customer agent. The user can accept or reject a work item and update the system from their inbox.

For more information, see [Offline Workflow for Sending and Approving Work Items](#).

- Rate table filter

On the **Date and Values** tab page, you can filter the rates that you want the system to display. In each rate table, you can also create filter variants and select a default filter variant.

When opening a rate table, the system can recommend that you use the rate table filter based on the number of rate table entries in the rate table. For larger rate tables, the use of the rate table filter can improve system performance.

You can activate the rate table filter and specify the maximum number of rate table entries that you want the system to display before recommending the use of the rate table filter. You can make these settings in Customizing for [Transportation Management](#) under [Basic Functions](#) [General Settings](#) [Define General Settings for TM](#).

- Validity periods

When adding a new validity period to a rate table, you can choose whether you want the system to copy the scale items from the existing validity period for the latest time period. The system doesn't copy the rates, just the scale items. If you choose not to copy the scale items, scale items aren't copied into the validity period.

❖ Example

You have the following validity periods in a rate table:

Valid From	Valid To
01.01.12	01.06.12
01.01.12	31.12.12
02.06.12	01.10.12

You decide to add a new validity period with scale items included. The system inserts the new validity period with the same scale items as the validity period 01.01.12 to 31.12.12.

You release validity periods so that the rate values can be used for charge calculation.

- Rate-level validity periods

You can specify validity periods at rate level within the validity period of the parent validity period. You can also change rates on a short-term basis without needing to update all the rates of the parent validity period. For more information, see [Rate-Level Validity Periods](#).

- Multiple charge types

You can allow multiple charge types in a rate table if you want to avoid defining numerous similar rate tables or if you want to use the rate table for multiple calculation sheet line items. For more information, see [Charge Types](#).

- Rate categories

You can specify the rate category of a rate table to facilitate the calculation of air freight charges. The rate categories available are:

- Unit load device (ULD)
- Specific commodity rate (SCR)
- General cargo rate (GCR) – M, N, & Q

For more information, see [Rate Categories](#) and [Charge Management Settings for Air Freight](#).

- Zero rates

You can enter a zero rate in a rate table to indicate that there's no charge. The system confirms that you entered a rate and that the rate is zero by selecting the **Zero Rate** checkbox.

- Scales

You can create scales on the [Scales](#) tab page of a rate table, for example:

- You can enter a calculation base, scale type, and unit of measure.
- You can enter a scale template.
- You can specify different currencies at the level of individual rate values by selecting the [Scale Rel. for CrCY Maint.](#) (Scale Relevant for Currency Maintenance) checkbox. Selecting the checkbox adds a currency column in the rate table and you can maintain the currency against the respective scales for the rate values. The combination of scales, currencies, and rates that you select is unique.

i Note

You can see only one currency column even if you select the [Scale Rel. for CrCY Maint.](#) checkbox for all the scales, whether the scales are relative or absolute.

The system creates the scale when you choose the [Create Scale](#) button. You can also select existing scales (see [Scale](#)).

i Note

If you copy a rate table, the system copies the scales that you created from within the rate table or that you referenced in the rate table. If you delete a rate table, the system deletes the scales that you created from within the rate table. However, the system doesn't delete the scales that you referenced in the rate table.

You can edit local scales independently of the rate table they were created within. To enable this editing, you select the [Edit Loc. MD Objects](#) (Edit Local Charge Management Master Data Objects) checkbox in Customizing for [Transportation Management](#) under [Basic Functions](#) > [General Settings](#) > [Define General Settings for TM](#).

You can also allow a scale item with no value to a scale. When calculating the charges, the system uses the scale item with no value in the following scenarios:

- No scale items match the value of the scale item connected to the calculation base in the freight order or forwarding order.
 - The calculation base has no value.
- Conditions

You can specify conditions as scales in a rate table by using the calculation base [CONDITION](#). This enables you to determine the most appropriate rate when you specify more than one rate for the same kind of scale base. You create conditions using BRFplus conditions and use these conditions in rate tables. In the rate table, you specify the value for a condition as true or false. If the system finds this condition in a forwarding order or freight order, it determines a rate from the rate table where the condition is met.

- Scale items

You can maintain scale items directly in a rate table, for example:

- In the [Rates](#) table, you can insert a scale item.
- In the [Rates](#) table, you can generate incremental scale items for a numeric scale if it doesn't already contain any scale items.
- On the [Scale Items](#) tab page, you can add a scale item.

Scale items are validity-dependent, meaning you can have different scale items for each validity period listed in the rate table. When you create a new validity period, the system copies the scale items from the most recently created validity period into the newly created validity period.

- Index rate tables for fuel surcharges

When you have fuel surcharges that change weekly and where you have different fuel surcharges in different zones in your country or region, you can use a standard index rate table that is valid on a particular date to minimize administration. You change the fuel surcharge rates in the index rate table just once each week or according to the time gaps in your validity periods. The system calculates fuel surcharges according to the rate in the zone on the validity date. You can use the following options to calculate fuel surcharges:

- Use the actual rate amounts that you specify in the standard index rate table.

To enable this approach, in the line item of the calculation sheet, specify a standard rate table with a calculation base of **FSC**, and an index rate table.

- Determine an index for the fuel surcharge and apply the index to the amount or percent you specify in a line item of a calculation sheet.

In this approach the index rate table specification stays the same, but you don't use the actual rate amounts in the index rate table. The system uses the fuel surcharge rate for the scale item that is valid on the date you specify in the **Index Base Date** field. It determines an index and uses the index along with the amount or percent you specify in the relevant line item in the calculation sheet to determine the fuel surcharge amount.

To enable this approach using amounts, specify a standard rate table or amount, the index rate table, and the index base date.

To enable this approach, specify two line items in the calculation sheet. In the first line item, enter the standard rate table. In the second line item, enter the amount or percent you want to apply, the index rate table, the index base date, and the reference-from line number for the first line item.

- Calculation rules

You can define a calculation rule and apply it to a rate or scale item. When defining the calculation rule, you must specify the calculation base, unit of measure, and price unit. If you want the system to apply the calculation rule at rate-table level, specify the price unit on the **Calculation Rules** tab page. If you indicate that you want the system to apply the calculation rule at scale-item level, specify the price unit on the **Scale Items** tab page. Note that if the price unit of one of the scale items isn't the same as the others, you must specify the level as **Scale Item**. The system applies the calculation rule only when the calculation type of the scale item for the selected rate value is relative.

You can define calculation bases in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Data Source Binding > Define Calculation Bases**.

i Note

If you specify the validity period for a relative scale that has at least one scale item that is also relative, the system creates a default calculation rule for the scale items. The system applies the default calculation rule at the rate-table level.

- Calculation types

You must specify a calculation type before creating a scale item for a scale, which affects how the system calculates charges with the scale. The available calculation types are absolute and relative. Note that you can specify a relative calculation type for numeric scale bases only. If the calculation type is relative, apply the calculation rule at scale-item level and specify a price unit for each scale item.

You can also specify a default calculation type for a scale type based on the calculation base of the scale. The system then enters the default calculation type when you enter the calculation base for the scale. You can overwrite the default calculation type. You can specify the default calculation type in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Data Source Binding > Define Calculation Bases**.

- Currencies

You can have different currencies for each rate in the rate table. You can also indicate the scales that are relevant for the currency of the rate. The currency of the rate is then the same as the scale currency.

- Through rates

When you trigger charge calculation, the system finds a rate from the source location of the first stage to the destination location of the last stage. The system calculates the charges based on this rate. To use a through rate, maintain a rate for the location pair (source location and destination location). You can enable through rates in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**.

- Rate tables in freight agreement RFQ masters

- Capacity

You can assign capacity driver scales on the **Scales** tab page and specify the capacity dimension at the rate-table level. The capacity driver scales determine the scales that constitute the requested capacity in the rate value table. Freight and forwarding agreement management use capacity scales and capacity dimensions at rate level to specify confirmed capacity. For more information, see [Capacities in Rate Tables for Freight Agreement RFQ Masters](#).

- Target rates

You can define a target rate or a target amount to specify the maximum price you're willing to pay for a carrier's service. You also specify a tolerance as a percentage. If the rate is beyond the target amount or target rate after applying the tolerance percentage, the system doesn't take a carrier's rate into consideration.

- Request rates

You can select the **Request Rate** checkbox to indicate the rate records that the system downloads when downloading an individual freight agreement RFQ as a Microsoft Excel file. When you send this file to a carrier, the carrier responds with rates only for these rate records.

For more information, see [Strategic Freight Procurement](#).

Activities

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

- To define or maintain a rate table for an agreement or calculation sheet, on the launchpad choose one of the following:
 - **Edit Freight Agreement**
 - **Edit Forwarding Agreement**
 - **Edit Internal Agreement**
 - **Edit Calculation Sheet**
- To display or edit a rate table separately from the calculation sheet it was created within, open the **Edit Rate Table Definition** app or **Maintain Rate Table** app from the launchpad.
- To define a rate table, open the **Create Rate Table Definition** app from the launchpad. A rate table can be used in multiple agreements and calculation sheets.
- To maintain a rate table, open the **Maintain Rate Table** app from the launchpad.
- To generate incremental scale items, select the relevant row on the **Dates and Values** tab page. If there's a numeric scale in the rate table that doesn't already contain any scale items, you can choose the **Generate Scale Items** button. Select the scale and the system generates the incremental scale items for that scale. You must enter the values you want the scale items to start and end at, and the step value you want each scale item to increment by.
- To release a validity period so that the rate values can be used for charge calculation, select the validity period on the **Dates and Values** tab page and select **Set Status > Released**.

Creation and Maintenance of Rate Tables and Local Rate Tables Using Microsoft Excel

Use

You can use Microsoft Excel to speed up the process of creating and maintaining large rate tables and local rate tables.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Rate and Scale Maintenance

You can maintain a large rate table, with or without scale items, using Microsoft Excel. You can enter the rates and scales into a Microsoft Excel file and upload the file to your TM system. Alternatively, you can define an empty rate table with validity periods in TM and download it to Microsoft Excel. You can then maintain the rate table in the Microsoft Excel file and upload it to your TM system.

You can choose to download a rate table to Microsoft Excel in one of the following ways:

- Rates for all validity periods in the rate table (scale ranges not included)
- Rates for one validity period and selected scale ranges
- Rates for multiple validity periods (scale ranges not included)

You can also choose to download the rate table in the background and be notified by e-mail when the download to the application server path is complete (see [Background Download of Rate Tables](#)).

You can view non-numeric scale items in the downloaded excel.

You can choose to upload all the rates and scale items of a rate table from Microsoft Excel to TM.

Mass Creation of Rates Tables

You can create multiple rate tables in Microsoft Excel using rate table templates and upload them to TM. You use the [Mass Creation of Rate Tables](#) (/SCMTMS/TCC_RATE_MASS_CREATE) program to download the Microsoft Excel template in which you enter the new rate table details and to upload the new rate table details to TM. In the Microsoft Excel template, you can specify the following for each row in the Microsoft Excel:

- The way you want the name of the rate tables to be generated, that is, by the system, by using the rate table template name, or by using a rate table name that you provide

The system adds a number to the end of the name of each rate table to avoid any duplication. Note that if you specify that you want to provide a rate table name, you must enter the rate table name in the [Rate Table](#) column.

- The rate table template on which to base the rate tables

You must specify the rate table template for the system to be able to create the rate tables.

- The quantity of rate tables

The system creates 1 rate table if you do not enter a quantity.

The system provides you with a list of the created rate tables and also of those that were not created successfully.

Background Download of Rate Tables

When maintaining a large rate table using Microsoft Excel, you can download the rate table in the background. Transportation Management (TM) can notify you by e-mail when the download to the application server path is complete. For more information, see [Background Download of Rate Tables](#).

Background Download of Rate Tables and Local Rate Tables

Use

When maintaining a large rate table or a local rate table using a spreadsheet application, you can download the rate table or the local rate table in the background. Transportation Management (TM) can notify you by e-mail when the download to the application server path is complete.

Process

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

1. You define a logical path for the logical file **Rate Table Background Download** (/SCMTMS/RATE_DOWNLOAD_BKGRND) using transaction FILE and then assign a physical file to the logical file.
2. You download the rate table to the application server path provided in the logical path using the **Rate Excel Integration** (/SCMTMS/TCC_RATES_EXCEL_NEW) program.
3. You enter the rate table that you want to download in the background and specify the download details.
4. You specify that you want to download the rate table in the background.
5. You specify whether you want TM to notify you by e-mail when the download to the application server path is complete.

Alternatively, you can use transaction SM37 to view the status of the download.

6. Once the download to the application server path is complete, you download the rate table from the application server path to your local desktop.

You download the rate table to your local desktop using the program **Rate Table Download from Application Server** (/SCMTMS/TCC_DOWNLOAD_FROM_AS) in transaction SE38.

More Information

[Rate Definition and Maintenance](#)

Update Rates

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can use the **Update Rates** (/SCMTMS/RATE_MASS_UPDATE) program to change the values in a single rate table or in multiple rate tables at the same time. Note that if you want to change the values in a single rate table, you can also access the **Update**

Rates program directly within the rate table by choosing the **Update Rates** button.

You can change the values for one period, multiple periods, or all periods in an existing validity period. You can also increase or decrease a rate by a percentage or an amount.

If the validity period you need to update is not already specified in the existing validity periods, you can create a new validity period in which the rate change applies. You can specify a new validity period within the date ranges already specified in the existing validity periods. You can also specify a new validity period that is outside the existing validity periods in the rate table. By default, the system uses the rate, calculation rule, and scale item settings in the latest validity period when it creates the new validity period. For new validity periods, you can also automatically set the status to **Released** during the rate table update.

Integration

The update of rates is a function in rate maintenance. For more information, see [Rate Definition and Maintenance](#).

Prerequisites

You have specified the settings required for rate maintenance. For more information, see [Rate Definition and Maintenance](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Master Data Cockpit

You can update rate tables for forwarding, freight, and internal agreements from the master data cockpit. To work with the master data cockpit, open the **Charge Management** app from the launchpad, choose the master data cockpit for the relevant agreement type, and enter your search criteria.

In the personal object worklist result, if you select a line item that contains both a rate table and a calculation sheet, the system opens the **Update Rates** function when you choose the **Update Rates/Calculation Sheet Amounts** button. The system automatically includes the rate table from the selected line item in the update criteria when you open the function.

If you select a line item that contains a rate table and a different line item that contains a calculation sheet only, the system opens both the **Update Rates** and **Update Calculation Sheet Amounts** functions on the same screen when you choose the **Update Rates/Calculation Sheet Amounts** button. The system automatically includes that rate table and calculation sheet from the selected line items in the update criteria when you open the function.

For more information, see [Update Calculation Sheet Amounts](#).

Scale Ranges

When you specify a new validity period or when you update a rate in an existing validity period, you can also specify a scale range, for example, source location. This enables the system to change only the rates in a particular scale range. However, you can only specify a scale range that is already specified in an existing validity period of an existing rate.

Scale Items

You can update rates based on the calculation type of the scale items (for example absolute or relative, or both absolute and relative). For example, you can update the rates for relative scale items with one rate change and the rates for absolute scale items with a different rate change. For relative value scale items, you can also update the price unit. For more information, see [Rate Definition and Maintenance](#).

Display Currency

You can display the currency of a rate table so that you are fully informed before you make any updates.

Example

You want to create a new validity period that contains new rates where two validity periods already exist. The following table describes the settings in the first existing validity period of January 1, 2010 to December 31, 2010:

Source Location	Destination Location	Value (USD)
NYC	PHI	100
NYC	CHI	120
NYC	SFO	340

The following table describes the settings in the second existing validity period January 1, 2011 to December 31, 2011:

Source Location	Destination Location	Value (USD)
NYC	PHI	110
NYC	CHI	130
NYC	SFO	350

You need to create a new validity period with new rates to cover the period December 1, 2010 to December 31, 2010. The following table describes your inputs to the **Update Rates** function:

Setting	Attribute
New validity period setting from	12-01-2010
New validity period setting to	12-31-2010
Source location	NYC
Destination location	PHI, CHI
Percentage	+10%

The system creates a new validity period for December 1, 2010 to December 31, 2010 with new rates. It keeps the old validity period for 2010, but changes the validity dates to January 1, 2010 to November 1, 2010. The following tables describe the new validity periods:

Validity 1: January 1, 2010 to November 1, 2010

Source Location	Destination Location	Value (USD)
NYC	PHI	100
NYC	CHI	120
NYC	SFO	340

Source Location	Destination Location	Value (USD)
NYC	PHI	110
NYC	CHI	132
NYC	SFO	340

More Information

[Rate Definition and Maintenance](#)

[Master Data Cockpit](#)

Rate-Level Validity Periods

Use

You can change rates on a short-term basis without needing to update all the rates of the parent validity period in the rate table or the local rate table. This allows you to, for example, offer lower rates for a specific time period.

To specify a rate on a short-term basis, you split the validity period of the rate into a shorter validity period than that of the parent validity period in the rate table or the local rate table. You can split a validity period multiple times. You then enter a rate for each of the shorter validity periods.

i Note

If you enable rate validity split, you can only view rates on the [Rates](#) tab page in a list view. You cannot view rates with a column scale in a grid view.

Prerequisites

You have enabled rate validity split in Customizing for [Transportation Management](#) under [Master Data](#) [Rate Tables](#) [Define Rate Table Types](#).

Activities

To be able to specify rates on a short-term basis, you select the [Rate Validity Split](#) checkbox on the [General Data](#) tab page of the rate table or the local rate table. Note that once you select the [Rate Validity Split](#) checkbox and save your changes, you cannot deselect the checkbox.

On the [Dates and Values](#) [Rates](#) tab page, you can then split the rate validity of a rate by choosing the [Split Rate Validity](#) button and specifying the date on which you want to split the rate validity. The system enters that date as the end date of the rate that you selected to split. The system also creates a new row with a rate whose validity period starts on the rate validity split date and ends on the end date of the parent validity period.

Note that you can also simply copy the rate into another row and split the validity period manually.

i Note

If you change the start date or end date of a parent validity period, by default the system adjusts all its rate-level validity periods that have the same start date or end date.

If you select the **Update Rate Validity Periods** checkbox, the system adjusts any rate-level validity periods that are inconsistent with the parent validity period. This adjustment may include deleting rates that have invalid validity periods or validity periods outside of the parent validity period. If you deselect this checkbox, the system informs you about any inconsistent rate-level validity periods. You can then adjust each inconsistent rate-level validity period manually or by choosing the **Update Rate Validity Periods** button.

Rate Categories

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You can specify the rate category of a rate table to facilitate the calculation of air freight charges. Once you specify the charge type for a rate table, you can specify the rate category. The rate category affects the scale items you should have in the rate table or the local rate table. The rate categories available are as follows:

Unit Load Device (ULD)

If you define a rate table as a ULD rate category, you must define a scale with a weight-based calculation base. Once you define a weight-based scale item, you can choose the **Maintain Scale Items** button to automatically create /A or /B scale items of rate class U (ULD rate) and /C scale items of rate class E (extension rate).

For weight-based scale items, you can choose a ULD charge code type of /A or /B. For ULD charge code types of /A and /B, the system automatically applies rate class U (ULD rate) for pivot weights. For ULD charge code types of /C, the system automatically applies rate class E (extension rate) for over-pivot weights.

If you choose ULD charge code type /A, the system applies charge codes /A and /C for pivot and over-pivot scale items respectively. If you choose charge code type /B, the system applies charge codes /B and /C for pivot and over-pivot scale items respectively. If you choose a rate table view type of **Column**, you use the **Maintain Scale Items** button to create scale items. If you choose the list view, the system uses the ULD charge code types you specify in **General Data > Charge Type Settings** for all scale items.

When creating the scale items for weight-based scales in a ULD rate table, you can create multiple pivot weights for the ULD rate types. You can then use different pivot weights when required, for example, for different origin and destination location pairs, days of the week, or flight codes. The system displays the additional pivot weights in the over-pivot charge column of the rate table and in spreadsheet file downloads.

i Note

Scale items with ULD charge code /A are rates per kg based on the minimum pivot weight per the International Air Transport Association (IATA) ULD rating type. ULD charge code /A represents a relative rate against the scale item combination.

Scale items with ULD charge code /B are minimum flat charges per the IATA ULD rating type with a chargeable weight up to the first defined weight break. ULD charge code /B represents an absolute amount against the scale item combination.

Specific Commodity Rate (SCR)

If you define a rate table as an SCR rate category, you must define a scale with the `COMMODITY_CODE` calculation base and a commodity code type, and a scale with a weight-based calculation base. The rate class S is associated with the scale items of an SCR rate table.

General Cargo Rate (GCR) – M, N, & Q

If you define a rate table as a GCR – M, N, & Q rate category, you must define a scale with a weight-based calculation base and ensure it's mandatory to maintain a minimum value for the calculated rate. Once you define a weight-based scale item, you can choose the **Maintain Scale Items** button to automatically create one scale item of rate class N (normal rate) and one or multiple scale items of rate class Q (quantity rate).

More Information

[Rate Definition and Maintenance](#)

[Charge Management Settings for Air Freight](#)

TACT Rates

Use

You can calculate air freight transportation charges based on TACT rates, as well as using the contract rates available in standard rate tables (both kinds, including local) in Transportation Management (TM). The International Air Transport Association (IATA) publishes TACT rates three times a year, and most of the world's major carriers use TACT rates for air freight. IATA represents more than 100 airlines comprising more than 70% of scheduled international air traffic. TACT rates are published between city pairs and can be specific to carrier and to unit load device (ULD) rate type.

The system automatically determines the relevant TACT rate based on the following mandatory parameters:

- City of departure and destination
- Chargeable weight

You can specify more parameters before the system determines the relevant TACT rate.

Prerequisites

- You have uploaded TACT rates to TM (see [TACT Rate Upload](#)).
- You have defined unit load device (ULD) rate types in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Air Freight Settings** > **Define ULD Rate Types**.
- You have mapped ULD types to ULD rate types in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Air Freight Settings** > **Map ULD Types to ULD Rate Types**.
- You have defined IATA airport codes and mapped them to IATA city codes in Customizing for **Transportation Management** under **Basic Functions** > **Type Codes and Role Codes** > **General Type Codes and Role Codes** > **IATA Codes** > **Define IATA Location Codes**.

Features

- **TACT Rates** personal object worklist

You can view and search for specific TACT rates in the [TACT Rates](#) personal object worklist.

- Contract rates

Generally, contract rates are lower than TACT rates for a specific city pair and weight range. As a logistics service provider, you can agree a contract rate with a carrier but charge a TACT rate to the customer. To enable this, you can specify a TACT rate for printing the settlement document and a contract rate for pricing. You specify which rate type to use for printing and which for settlement.

i Note

If you charge a TACT rate to the customer, the default calculation method is air freight: standard rating. However, you can choose to use a different calculation method.

If you have a TACT rate in the air waybill (AWB) but charge a contract rate to the customer, you should not use a break-weight calculation method. However, if you use a break-weight calculation method, the system takes the chargeable weight in the AWB into account for the settlement document. Note that the system then may use a higher weight break than the one determined by the break-weight calculation method. For more information, see [Calculation Methods](#).

- TACT rate types

The following TACT rate types are available:

- ULD rate
- Specific commodity rate
- General cargo rate

More Information

[Rate Table](#)

[Rate Definition and Maintenance](#)

[Setup of MD for Charge Management and SP Catalogs](#)

[Charge Calculation](#)

TACT Rate Upload

Use

Uploading TACT rates to database tables in Transportation Management (TM) allows TM to use TACT rates when calculating air freight transportation charges. You upload TACT rates each time the International Air Transport Association (IATA) publishes them.

Process

1. You define a logical path for the /SCMTMS/TACT_UPLOAD logical file using the [Logical File Path Definition](#) transaction (transaction FILE) and then assign a physical file to the logical file.
2. You upload the TACT rate file to the application server and trigger the system to process the TACT rates in database tables using the [TACT Rate Upload](#) transaction (transaction /SCMTMS/TACT_UPLOAD).

You can specify selection parameters to restrict the TACT rates you want the system to process. The system processes the TACT rates in the background. You can view the status using the [Simple Job Selection](#) transaction (transaction SM37).

i Note

You can use the [TACT Rate Process](#) transaction (transaction /SCMTMS/TACT_PROCESS) at a later point in time to process TACT rates that you previously uploaded to the application server.

You can use the [TACT Rate Upload](#) (/SCMTMS/TACT_RATE_UPLOAD) report and the [TACT Rate Processing](#) (/SCMTMS/TACT_RATE_PROCESS) report instead of the transactions.

3. The system automatically maps IATA codes to SAP codes using the [Mapping of IATA Codes to SAP Codes](#) (/SCMTMS/TACT_MAP_IATA_TO_SAP) report.

You can also run this report at a later point in time if, for example, more country or region codes are added.

More Information

[TACT Rates](#)

Scale and Local Scale

Definition

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A scale is a dimension of a rate. A scale is used to define a parameter that then defines a rate. For example, if a rate depends upon distance and weight, you must define a separate scale for distance and for weight. The distance and weight scales are then used to define the rate.

Use

You can create a scale (both kinds, including local) manually or automatically by using a scale template with predefined scale bases and calculation type. For more information about master data templates, see [Scale Maintenance](#). For more information about Customizing templates, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Basic Settings](#) > [Templates](#) > [Define Scale Templates](#) .

Structure

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

A scale can contain the following main tab pages and screen areas:

- [General Data](#)

The [General Data](#) tab page contains general information about the scale, for example, the scale unit measure and the scale base, which defines how the system interprets the scale values during calculation. Note that you can assign a default calculation type to the calculation base in Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Data Source Binding](#) > [Define Calculation Bases](#) .

- [Items](#)

The **Items** tab page contains the details of the scale items such as the calculation type, which indicates how the system calculates charges with the scale item.

i Note

If you use a scale in a rate table and use the **Maintain Scale Items** button on the **Scales Items** tab page of the **Dates and Values** tab page, the system automatically enters the default calculation type for the calculation base of the scale item.

If you enter a calculation type for a scale item, the system overwrites the default calculation type with the one you entered. You can also manually change the calculation type for a scale item.

- **Change Documents**

You can view all changes made to a scale on the **Change Documents** tab page or screen area. The changes are listed by date and time. You view the details of the changes at header level by selecting the relevant row in the **Transactions** screen area and at field level by selecting the relevant row in the **Record Level Changes** screen area. You must refresh the scale to view recently-saved changes.

Integration

A scale is integrated with agreements, calculation sheets, local calculation sheets, rate tables, and local rate tables. A scale or a local scale is used by a rate table or a local rate table to define the rates. For more information, see [Agreement](#), [Calculation Sheet](#), and [Rate Table](#).

More Information

[Charge Calculation](#)

[Setup of MD for Charge Management and SP Catalogs](#)

Scale Maintenance

Use

You create and maintain scales (both kinds, including local) as a step in the process to set up master data for charge calculation. For example, you can configure the items of a scale according to one of the following:

- Quantities
- Volumes
- Weights
- Distances
- Points

i Note

You can generate incremental scale items for a scale (both kinds, including local) when you are defining a rate table (both kinds, including local). For more information, see [Rate Definition and Maintenance](#).

Integration

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Scale maintenance is a step in the process to set up master data in the Charge Management and Service Product Catalogs component and is integrated with agreements, calculation sheets, and rate tables. For more information, see [Setup of MD for Charge Management and SP Catalogs](#).

You can use the master data cockpit to access an overview of the charge management master data per organizational unit, business partner, and charge type. The system also shows the relationship between the different types of charge management master data. For more information, see [Master Data Cockpit](#).

Prerequisites

- You have defined a valid scale base with a relevant unit of measure in Customizing for **Transportation Management** under [Basic Functions](#) [Charge Calculation](#) [Data Source Binding](#) [Define Scale Bases](#).
- You have defined number ranges in Customizing for **Transportation Management** under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Number Range Intervals](#) [Assign No. Ranges to Calc. Sheets, Rate Tables, & Scales](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

- Templates

You can use templates to create scales and scale items with predefined scale bases, calculation types, and rate categories.

You can define Customizing templates in Customizing for **Transportation Management** under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Templates](#) [Define Scale Templates](#). You can transfer Customizing templates across different TM systems.

You can define master data templates directly in Transportation Management (TM). If required, you can define a master data template based on a Customizing template.

You can view and update all scale templates in the [Scale Templates](#) personal object worklist.

- Calculation types

You can specify a calculation type for a scale base, which affects how the system calculates charges with the scale. The available calculation types are absolute and relative. Note that you can specify a relative calculation type for numeric scale bases only.

You can also specify a default calculation type for a scale type based on the calculation base of the scale base. You can specify the default calculation type in Customizing for **Transportation Management** under [Basic Functions](#) [Charge Calculation](#) [Data Source Binding](#) [Define Calculation Bases](#).

Activities

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

To define a master data template for a scale, open the [Create Scale Template](#) app from the launchpad.

To create or update a scale for an agreement, calculation sheet, or rate table, on the launchpad choose one of the following:

- [Edit Freight Agreement](#), [Edit Forwarding Agreement](#), or [Edit Internal Agreement](#)
- [Edit Calculation Sheet](#)
- [Maintain Rate Table](#)

To display or edit a scale separately from the agreement, calculation sheet, or rate table it was created within, open the [Edit Scale](#) app from the launchpad. Note that to be able to edit local scales independently of the charge management master data object they were created within, you select the [Edit Loc. MD Objects](#) (Edit Local Charge Management Master Data Objects) checkbox in Customizing for [Transportation Management](#) under [Basic Functions](#) > [General Settings](#) > [Define General Settings for TM](#).

To create or update scales, open the [Create Scale](#) app or the [Edit Scale](#) app from the launchpad. A scale can be used in multiple agreements, calculation sheets, and rate tables.

More Information

[Scale](#)

Master Data Cockpit

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

The master data cockpit provides an overview of the agreements and charge management master data per organizational unit, business partner, and charge type. It also provides links directly to the main master data and shows the relationship between the different types of charge management master data.

The master data cockpit lists the master data by freight agreement, forwarding agreement, or internal agreement and by default, displays the agreements and rate tables that are valid on the current date. You can expand the list to see the rates, calculation sheets, and related information, such as currencies, scale bases, and scale items. You can use the master data cockpit to get specific information, for example:

- The rates for a transport between your sales organization and a specific customer or between your purchasing organization and a specific carrier
- All charges maintained for one item in an agreement that has many rate tables
- The agreements valid for a specific period

Note that the system always includes the agreements and rate tables that are valid on the current date, as well as the agreements and rate tables that are valid for the specific period.

Integration

The master data cockpit is integrated with agreements and with the charge management master data: calculation sheets, local calculation sheets, rate tables, local rate tables, scales, and local scales. For more information, see [Agreement](#), [Calculation Sheet](#), [Rate Table](#), and [Scale](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Selection Criteria

You enter selection criteria, such as status, transportation mode, and source zone, to filter the agreements and master data objects you want to list in the master data cockpit. For freight agreements, you must enter the purchasing organization and carrier as selection criteria. For forwarding agreements, you must enter the sales organization and ordering party as selection criteria. For internal agreements, you must enter the organizational unit name and business partner as selection criteria.

i Note

If an agreement item of an agreement or a calculation sheet item matches the selection criteria but the agreement header does not, the agreement still appears in the master data cockpit. It is sufficient for the selection criteria to match the agreement item or calculation sheet item. The agreement header does not need to match the selection criteria. However, the business partner details specified in the agreement header must be a subset of the business partner details specified by the preconditions at the agreement item level.

You can also enter attributes in the selection criteria for the system to use when determining which calculation sheets to list in the master data cockpit. The attributes for calculation sheet and calculation sheet items are shipment type, stage type, and service level.

You can also enter preconditions in the selection criteria for the system to use when determining which agreements to list in the master data cockpit. The preconditions for agreement and agreement items can be, for example:

- Transportation mode
- Shipment type
- Traffic direction
- Stage type
- Service level
- Air waybill type

i Note

If an agreement item of an agreement or a calculation sheet item matches the preconditions or attributes but the agreement header does not, the agreement still appears in the master data cockpit. It is sufficient for the preconditions or attributes to match the agreement item or calculation sheet item. The agreement header does not need to match the preconditions or attributes.

The system also considers the trade lane preconditions specified for an agreement item. However, the system displays the calculation sheet item irrespective of the trade lane preconditions specified for a calculation sheet item.

Mass Update

You can change the charge item amount in one or more calculation sheets at the same time. For more information, see [Update Calculation Sheet Amounts](#).

You can change the rate values in a single rate table or in multiple rate tables at the same time. For more information, see [Update Rates](#).

Currency Conversion

You can convert the currency of the rate amount using the [Convert Currency](#) button.

Layout

You can add and remove columns to the master data cockpit, and save the changes as a personalized view. In the [Master Data Cockpit for Forwarding Agreements](#), you can choose to use a layout with the [Agreement](#), [Calculation Sheet](#), and [Rate Table](#) columns always visible.

Activities

To access the master data cockpit, open the [Charge Management](#) app from the launchpad. Choose the master data cockpit for the relevant agreement type and enter your selection criteria, including the mandatory criteria.

Charge Calculation

Use

You can calculate the charges for transporting goods between locations based on the rates you maintain in a rate table (both kinds, including local). The system uses master data, such as, agreements, calculation sheets, local calculation sheets, rate tables, and local rate tables, when calculating transportation charges. For more information, see [Charge Management and Service Product Catalogs](#).

You can trigger charge calculation from the following Transportation Management (TM) components:

- [Forwarding Order Management](#)
- [Freight Order Management](#)
- [Planning](#)
- [Forwarding Settlement](#)
- [Freight Settlement](#)

Note that you can manually edit the transportation charges after charge calculation.

Types of Charge Calculation

The system uses a standard charge calculation process to calculate transportation charges for the following:

- Forwarding orders
- Freight orders
- Forwarding settlements
- Freight settlements
- Forwarding quotations
- Service orders
- Tendering
- Carrier selection

For more information, see [Standard Charge Calculation](#).

The charge calculation process can also vary depending on multiple factors, for example:

- Transportation mode category
 - The system uses a slightly different charge calculation process to calculate the charges for air freight. For more information, see [Air Freight Charge Calculation](#).
 - For rail transportation, you can calculate charges based on the number of railcar units. You can also calculate the transportation charges in rule 11 scenario in a rail freight order when the carrier that pays for the stage is the same or different from the carrier that executes the stage. For more information, see [Rail Freight Order](#).
- Services
 - You can charge your customers for the additional charges incurred due to varying fuel costs. For more information, see [Fuel Surcharge Calculation](#).
 - When you create a service order for a freight booking or a freight booking item, you can calculate the charges for the services. For more information, see [Service Order](#).

i Note

We improved how TM determines the rates during charge calculation for agreement items and calculation sheet line items that have preconditions. We improved system performance during charge calculation by ignoring empty instances of precondition-dependent objects. If you upgraded to TM 9.4 or lower, you can use program [Precondition Refactoring for Agreement and Calculation Sheet Items](#) (/SCMTMS/MIGR_140_PRECOND_UPD) to implement the performance improvements.

Features

- Calculating transportation charges enables you to analyze the profitability of a forwarding order. The system takes into account all the costs and revenues associated with the processing of a forwarding order and determines the associated profit.
For more information, see [Profitability Analysis](#).
- You can distribute costs in freight orders or freight settlement documents at an ERP-item level or forwarding-order level.
For more information, see [Cost Distribution Management](#).
- During carrier selection, calculating the transportation charges helps you to determine the lowest total costs.
For more information, see [Strategies](#).
- During tendering, calculating the transportation charges helps you to calculate the absolute price limit.
For more information, see [Tendering](#).
- You can consolidate freight units from multiple forwarding orders into a trailer freight unit and charge the customer based on specific criteria, for example, a standard rate or the weight of the trailer.
For more information, see [Use of Trailers](#).
- The system automatically triggers charge calculation when you create the following:
 - Forwarding settlement document
 - Freight settlement document
 - Internal settlement document
 - Credit memo
- You can calculate charges for multiple forwarding orders and freight orders.

You can calculate the charges for multiple forwarding orders using the [Mass Charge Calculation for Forwarding Orders](#) report. To access the report, on the [SAP Easy Access](#) screen, choose [Logistics](#) [Transportation Management](#) [Administration](#) [Background Processing](#) [Calculate Charges for Forwarding Orders](#).

You can calculate the charges for multiple freight orders using the [Mass Charge Calculation for Freight Orders](#) report. To access the report, on the [SAP Easy Access](#) screen, choose [Logistics](#) [Transportation Management](#) [Administration](#) [Background Processing](#) [Calculate Charges for Freight Orders](#).

- When you create a credit memo, the system recalculates the transportation charges for the relevant forwarding order.

For more information, see [Credit Memos](#).

- You can calculate transportation charges between organizations of the same company code and between organizations of different company codes. If the involved business units and gateway belong to different organizations of the same company code, you perform intracompany settlement. If the involved business units and gateway belong to the organizations of different company codes, you perform intercompany settlement. You can also calculate the internal charges in a freight order when a purchasing organization of the freight order subcontracts resources from another organization.

Note that the system calculates the internal charges in a freight order based on the resources. You use internal agreements to maintain the contract between the different organizations that belong to the same company code or different company code.

For more information, see [Internal Settlement Management](#).

- If you're using an agreement stored in an external system, you can establish a connection from TM to the external system. The system displays the external agreement and the freight agreement item number on the [Details](#) tab page of the [Charges](#) tab page.

For more information, see [External Agreements](#).

- The system calculates the charges based on the settlement basis you specify in the forwarding agreement.

If you specify the settlement basis as trailer-based, the system sets the calculation level as stage level in the forwarding order. During settlement, the system creates a single forwarding settlement document for all the forwarding orders that meet the specified execution criterion and bills the business partners based on certain execution criteria, such as the number of trailers used to fulfill the order. The system also changes the invoicing status of the technical charge item in the [Charges](#) tab of the forwarding order accordingly. If you don't specify the settlement basis, the system sets the settlement basis as forwarding order-based. The system uses the calculation level specified in the calculation profile and settles each of the forwarding orders individually.

For more information, see [Forwarding Settlement](#).

- You can create invoices for partial amounts by manually entering the amounts in forwarding settlement documents.
- You can specify the calculation date type that the system uses as the basis for the rate, exchange rate, and agreement validity, and ultimately the charge calculation.

If the calculation date type for a subcontracting-relevant freight order is order date, the system calculates the charges based on the date that the freight order is subcontracted to the carrier. The system displays the order date on the [Subcontracting](#) tab page of the freight order.

You specify the calculation date type in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Calculation Profiles](#).

- The charge lines created from the calculation sheet item with the SERVICE resolution base inherit the logistical data from the main item in the hierarchy.

When you assign a charge type to a service type, the system defaults the resolution base to SERVICE. You assign charge types to service types in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Assign Charge Types to Service Types](#).

For more information, see [Service Product Catalogs](#).

- You can selectively apply charges for certain charge types (for example, fuel), for active vehicle resources only (for example, truck) and not the passive vehicle resources (for example, trailer). You use the resolution bases `ACTIVE_RESOURCE` and `PASSIVE_RESOURCE` to differentiate between active vehicle resources and passive vehicle resources.
- You can access the details and messages generated during charge calculation on the **Charges** or **Internal Charges** tab page of the relevant business documents, such as a forwarding quotation, forwarding order, and freight order.

For more information, see [Charges Overview](#).

- When you create a follow-on settlement document from a freight order or forwarding order, you can manually change the charge amount for a charge line in the forwarding order or freight order. When you recalculate charges in the settlement document, the system can update the charge amounts in the settlement document based on the manually-changed charge amounts in the forwarding order or freight order.

You enable the system to update charge amounts in the settlement document based on manually-changed charge amounts in the forwarding order or freight order by selecting the **Update Settl. Chrgs from Order** checkbox in Customizing for **Transportation Management** under **►Settlement ► Define Settlement Profiles ►**.

- If you have freight order or a freight booking with items that have different source and destination locations, the system can determine a rate based on the locations in the item. Note that rate determination using locations in the item works only when calculating charges at item level and header level.
- You can use freight terms as the basis of settling forwarding orders and forwarding quotations. You can agree the transportation charges in advance with your sold-to party. In the calculation sheet (both kinds, including local), on the **Freight Terms** tab, you can specify the payer (prepaid or collect) for a charge type for a freight term. That is, the same freight term can have a different payer for different charge types. The system calculates charges based on this information you specify and serves as the basis for settling the overall transportation costs.

You can use a single forwarding quotation or a forwarding order with freight terms to calculate the prepaid and collect charges. The system creates two charge items one each for the prepaid payer and the collect payer. You can create forwarding settlement documents to settle charges between the following parties:

- Prepaid invoicing party and prepaid payer for prepaid charges
- Collect invoicing party and collect payer for the collect charges

For more information, see [Freight Terms in Forwarding Orders](#).

- You can determine the rates you want to provide to shippers or logistics service provider for forwarding agreement quotation items based on the existing service products. In this case, the system doesn't perform charge calculation but provides logistical information of the customer by analyzing the existing and past service products. For more information, see [Rate Determination of Forwarding Agreement Quotation Items](#).

More Information

[Setup of MD for Charge Management and SP Catalogs](#)

Standard Charge Calculation

Use

This process describes how the system calculates transportation charges when you trigger charge calculation from one of the following Transportation Management components:

- [Forwarding Order Management](#)
- [Freight Order Management](#)
- [Forwarding Settlement](#)
- [Freight Settlement](#)

Prerequisites

- In Customizing, you have carried out the activities for **Transportation Management** under **Basic Functions > Charge Calculation**.
- You have set up the master data for the Charge Management and Service Product Catalogs component.

For more information, see [Setup of Service Product Catalogs and Charge Management MD](#).

Process

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

1. You trigger charge calculation.

You can also calculate transportation charges between organizations of the same company code and between organizations of different company codes. For more information, see [Internal Settlement Management](#).

The system can calculate charges at header, item, or stage level, depending on your Customizing settings in the calculation profile.

2. The system determines the agreement within organizational hierarchies (such as company organization or corporate organization) and calculation sheets.

You can specify a leading charge type that must have a value for the system to consider the calculation sheet for charge calculation. You specify leading charge types in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charge Types**. For more information, see [Calculation Sheet Maintenance](#).

You can also enter a zero rate for a mandatory charge line by selecting the **Zero Rate** checkbox in the rate table. The system still considers the calculation sheet for charge calculation.

Note that if you trigger charge calculation from a personal object worklist (POWL) or if you perform batch charge calculation with the agreement determination type as **Display All Agreements**, the system does not calculate charges. Also, the system sets the calculation status as **Automatic Charge Calculation Failed**.

For more information, see [Agreement and Calculation Sheet Determination](#).

3. The system determines rate tables and scales.

The system determines the most appropriate rate when calculating transportation charges, based on certain criteria and attributes. For more information, see [Rate Determination](#).

4. During calculation, the calculation sheet combines charge items while taking into account the sequence in which the system takes these charge items.

5. The system rounds the charges for each charge line before calculating the total charges or rounds the charges once it has calculated the total charge, depending on your settings.

You specify the level you want to round the charges at in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Rounding > Assign Rounding Profiles to Document Currencies**.

i Note

You can access the details and messages generated during charge calculation on the **Charges** or **Internal Charges** tab page of the relevant business documents, such as a forwarding quotation, forwarding order, and freight order. For more information, see [Charges Overview](#).

The system displays the notes, which you have entered in the calculation sheet item, in the corresponding charge line during charge calculation.

6. You can manually change the calculation bases, quantity, and unit of measure for a charge item and recalculate the charges.

When you manually change the quantity, the system determines the appropriate rate from the rate table for the changed quantity. If you have maintained only one calculation base in your rate table, the system displays it in the **Calculation Base** field on the **Charges** tab page by default. If you have maintained more than one calculation base in the rate table, you must choose one from the input help that displays all the calculation bases maintained in the rate table. Note that the system can recalculate the charges only if the charge item has a rate table and the calculation base is numeric.

i Note

If you manually change, insert, or delete a charge line, the system flags the charge line as being manually changed. The system retains manual changes during charge recalculation, provided that the system determines the same agreement and agreement item. However, the system overwrites manual changes during charge recalculation in the following situations:

- You change the business partner, organizational unit, or logistical data and the system determines a different agreement or calculation sheet during recalculation.
- You change the calculation level.
- You split or merge stages in the business document for through rates.
- You change charge items due to through rates.

If you reset the charges in a business document, the system also resets manual changes.

More Information

[Charge Calculation](#)

[Air Freight Charge Calculation](#)

[Fuel Surcharge Calculation](#)

[Calculation Profiles](#)

[Calculation Logic](#)

Calculation Logic

Use

This function enables the system to use the following objects when calculating transportation charges:

- Resolution base

- Calculation base
- Collective rate

Integration

This function is used in the charge calculation process, which is used in the following Transportation Management (TM) components:

- [Forwarding Order Management](#)
- [Freight Order Management](#)
- [Forwarding Settlement](#)
- [Freight Settlement](#)

For more information, see [Charge Calculation](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Resolution Base

Resolution base is a component of a calculation sheet item that assigns a charge type to a specific data source level. This helps to correctly identify the relevant attributes, for example weight, distance, or product quantity that the system needs to resolve the charge element and apply the correct rates. The charge type, for example, FB00, which is a single building block of the calculation sheet, is calculated with regard to the resolution base. During charge calculation, the system resolves the calculation sheet according to the given input data. At runtime, the calculation logic works as follows:

- As a result of the initial resolution step, the system creates a calculation table based on the logistical data source. The system hasn't yet performed any calculations or determined any rates.
- The system processes the calculation table and determines a value for each row given in the calculation sheet. If the system can't determine any value for a certain charge type, the system issues a warning or error and doesn't apply a charge for that row in the calculation sheet. However, the process isn't interrupted.

You can also assign a consignment order as a resolution base to a freight document. During charge calculation, the system retrieves the relevant details from the consignment-based item and further settlement is done. For more information on consignment orders, see [Consignment Order](#).

i Note

The resolution base is defaulted to SERVICE in the calculation sheet, when you maintain a charge type in the calculation sheet that is assigned to a service type in Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Basic Settings](#) > [Assign Charge Types to Service Types](#) .

You can retain the default resolution base of SERVICE, if you maintain the service as a main item in the forwarding order and you want the system to apply the charges for the service only once in the charge item. The system applies the service charges based on the availability of the service type in the forwarding order and the corresponding charge type in the calculation sheet.

You can change the default resolution base of SERVICE, if the charge type entered in the calculation sheet is for charging a service for a container or package or product. For the system to apply these charges, specify the item of the corresponding

service type as the main item. The system applies the service to the appropriate logistical items based on the resolution base of the corresponding charge type in the calculation sheet.

❖ Example

SERVICE resolution base

In the forwarding order, you have three containers for which you have to apply the charges for fumigation service.

You have a charge type CT_FUMIG assigned to a service type SERV_FUMIG. In the calculation sheet, when you insert a charge line with the charge type CT_FUMIG, the system defaults the resolution base as SERVICE. You change the resolution base to CONTAINER in the calculation sheet. In the forwarding order, you insert an item with the service SERV_FUMIG as the main item. During charge calculation, the system creates three charge lines for three containers each and calculates the charges.

You can also charge a service to different logistical items. You must maintain a service type as a sub item under the corresponding container or package or product. You must maintain a charge type with the resolution base as SERVICE. During charge calculation, a service charge line is created for each logistical item that has service as a sub item.

❖ Example

CONSIGNMENT resolution base used for header-level charge calculation in a freight order

In a freight order, **FO1**, you have the following item hierarchy, with two consignment orders :

Item Hierarchy		
Active Vehicle1		
	Consignment Order1	
		Product1
	Consignment Order2	
		Product2

The freight agreement **AGR1** has the following calculation sheet items:

Charge Type	Resolution Base
SF02	ROOT
SG22	CONSIGNMENT
SF24	ROOT

AGR1 is used in **FO1** for charge calculation and the charge details are shown after this:

Charge Type	Logistical Reference	Rate amount	Quantity	Final Amount
SF02	Root	100		100
SG22	Consignment Order1	8	10.5	84
SG22	Consignment Order2	7	10.5	73.5

Charge Type	Logistical Reference	Rate amount	Quantity	Final Amount
SF24	Root	5		5

The charge type **SG22** has the resolution base of **CONSIGNMENT**.

Calculation Base

With the calculation base, you define the instruction to determine the data source. The calculation base represents the configurable link to the data source with which a rate is determined. The system uses the calculation base to automatically calculate the charges.

❖ Example

You have a two-dimensional rate table with the first dimension (scale) based on the destination harbor and the second dimension (scale) based on the weight of the cargo to be shipped. The rate that the system identifies after the two dimensions of the price matrix have been evaluated, for example USD 39.60, is multiplied by the volume to be shipped. In this example, the logistical data source has to provide at least three different pieces of information to calculate the correct charge. There's a calculation base for each of these three entities, describing the path to the certain data source.

❖ Example

You're shipping a container from Europe to various destinations in the United States, such as New York and Los Angeles. You want to charge for the transportation based on the source location (Europe), the destination location (New York and Los Angeles), and the gross weight of the container. To enable this, you define a rate table with three scales (dimension) – source location, destination location, and gross weight. Each of these scales uses the corresponding calculation bases, which are **SOURCELOC**, **DESTLOC**, and **GROSS_WEIGHT**.

The calculation base and the resolution base are coupled in a way that the same calculation base leads to a different data source dependent on the currently processed resolution base.

❖ Example

You have the following master data maintained in a calculation sheet:

Calculation Base	Resolution Base	Calculation Rule
Gross Weight (maintained in rate table A for charge type FB00)	Container (for charge type FB00)	USD 10 per 100 kg
Gross Weight (maintained in rate table B for charge type BASE)	Package (for charge type BASE)	USD 25 per 100 kg

You have an FWO with 2 containers and 1 package with the following gross weights:

- Container 1–1000 kg
- Container 2–2000 kg
- Package 1–200 kg

In this situation, the system applies charge type **FB00** for each container (resolution base = container) and calculates the transportation charges as follows:

- Container 1: $(1000/100) \times 10 = \text{USD } 100$

- Container 2: $(2000/100)*10 = \text{USD } 200$

The system also applies charge type **BASE** for each package (resolution base = package) and calculates the transportation charges as follows:

Package 1: $(200/100)*25 = \text{USD } 50$

Collective Rate

With a collective rate, you apply group discounts to charge types using a grouping rule. You base the grouping rule on a calculation base, such as resource type or destination location. The grouping calculation base is determined by a grouping rule either directly or dynamically using a condition. When you group charge types, the combined quantity of the charge items is used to calculate the rate.

❖ Example

You use a weight scale to charge for transporting containers, and you create a grouping rule to group containers by destination location. The system uses the combined weight of all containers going to the same location to calculate the collective rate, but each individual weight is applied to the grouping rule.

The standard rate for a container is USD 1 per kg. The collective rate is USD 0.7 per kg if the combined weight is over 600 kg. Each container going to one warehouse weighs 150 kg. However, the combined weight for five containers is 750 kg, so the system charges the collective rate of USD 0.7 per kg instead of USD 1 per kg.

Example

Resolution Base

You have an FWO with 3 stages and want to calculate the transportation charges for the entire FWO, not for each stage. Therefore, you specify the header as the calculation level. You want the system to calculate charge type **BAF** once in this FWO. Therefore, you specify the resolution base as **ROOT**. This ensures that there's only one resulting row for the calculated calculation sheet in the **Charges** tab page of the FWO. For more information about the **Charges** tab page, see [Charges Overview](#).

You want the system to calculate charge type **THC** for each container in the FWO. Therefore, you specify the resolution base as **Container**. If you ship three different containers, there are three resulting rows for the calculated calculation sheet – one for each container.

The final sum adds the transportation charges in the four rows:

- Row 1 for the **BAF** with amount **a**.
- Sum of rows 2, 3, and 4 of **THC** with the following amounts:
 - **b**: Charge for container 1
 - **b**: Charge for container 2 (the amount is the same as for container 1, because it has the same attributes)
 - **c**: Charge for container 3 (different than amount **b** because container 3 is bigger)

Charges Overview

Use

This function displays the results of the charge calculation on a **Charges** or an **Internal Charges** tab page for the following business documents:

- [Forwarding Quotation](#)
- [Forwarding Order](#)
- [Forwarding Settlement Document](#)
- [Credit Memos](#)
- [Freight Order](#)
- [Freight Booking](#)
- [Service Order](#)
- [Freight Settlement Document](#)

You can manually edit the charges on the **Charges** or the **Internal Charges** tab page.

Prerequisites

- In Customizing, you've carried out the activities for **Transportation Management** under **Basic Functions > Charge Calculation**.
- You have set up the master data for the Charge Management and Service Product Catalogs component. For more information, see [Setup of MD for Charge Management and SP Catalogs](#).
- You have triggered charge calculation for the respective business document.

Features

You can see the calculation date with a timestamp that specifies the date, time, and zone. You can use the timestamp for agreement validity, agreement item validity, calculation sheet line item validity, and rate table validity. Note that only in newly created business documents, you can see the timestamps.

The system displays the charge calculation results in a hierarchical table consisting of charge items and charge lines. The charge items and charge lines correspond to the level at which the system determined the agreement and calculation sheets (both kinds, including local). For more information, see [Overview of Charges Grid](#).

Charge Calculation Logs

You can see the details of how the system calculated the charges on the **Charge Analysis Log** tab page. The system re-creates these details each time you trigger charge calculation for the business document. You can access all messages generated by the system during charge calculation on the **Charge Calculation Log** tab page.

You can turn off or turn on the charge calculation logs for your user profile. You control this setting using the transaction SU3. On the **Parameters** tab page, if you specify X as the value for the parameter /SCMTMS/CLOG, the system turns on the charge calculation logs. If you don't specify any value for this parameter, the system turns off the charge calculation logs.

Charges Default View

You can choose whether you want the system to display an expanded view or a collapsed view of the charge items as the default view.

In order to make this setting, in the business document, you choose . A dialog box appears in which you can select or deselect the **Expand Charge Items** checkbox. Changes to settings have an immediate effect.

Grouped View

You can view charge lines with the **Container** resolution base, grouped into aggregated rows, if the charge lines have the following attributes in common:

- Charge type
- Equipment type
- Rate amount
- Rate currency

Similarly, you can view charge lines with the **Package** resolution base, grouped into aggregated rows, if the charge lines have the following attributes in common:

- Charge type
- Package type
- Rate amount
- Rate currency

The system totals the final amount, calculated amount, and group count for all of the grouped charge lines. You can expand the aggregated row to display the individual charge lines.

Note that if you change the rate amount or rate currency at the aggregated level, the system changes the rate amount for all the charge lines in the group. For example, if you change the rate amount and rate currency from USD 20 to EUR 25 at the aggregated level, the system passes the change down to all the charge lines in this group.

You can specify that you want the system to group the charge lines by default in the Customizing activity for the respective business document. If you don't specify this in the Customizing activity for the respective business document, the system checks if there's a default view specified in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles**.

If you specify a default view in both the general settings and in the settings for a specific business document type, the business document settings take precedence. However, if you don't specify a default view, the system doesn't group the charge lines by default.

Charge Line Insertion

You can insert blank charge lines one at a time or insert one or more charge lines with the charge type already specified. When inserting charge lines with specific charge types, you can filter by the following attributes:

- Charge type
- Charge subcategory
- Charge category
- Positive/Negative
- Value type
- Description

Inserted charge lines inherit the calculation resolution base and group type of the charge line you inserted them from. If you insert charge lines from the charge item or the aggregated row of a group, the inserted charge lines inherit the ROOT calculation resolution base and the group type is blank. You can change the group type of inserted charge lines.

i Note

If you change the group type of an inserted charge line, the system doesn't group this charge line into the same aggregated row as the charge line you inserted it from. This is because the inserted charge line and the charge line you inserted it from no longer have the same equipment-type attribute.

Show/Hide Access Sequence

If the instruction type is **Line Item Selection**, the system evaluates the line to be considered for the calculation among a set of two or more lines. This evaluation is based on the highest or the lowest value of the affected lines, and on the access sequence.

If you set the operation to **High**, for example, the most expensive charge item remains and the cheaper charge items are deactivated.

In general, the system displays only the charge lines used for charge calculation in case of access sequence. However, you can use this button to view the access sequence of all the evaluated charge items.

Reset Charges

You can revert the manual changes you made to the charge calculation results. When you reset charges, the system follows the usual agreement determination process and performs charge calculation.

Note that when you reset charges, all the manual changes you made to the charge calculation results will be lost.

Local Currency

During internal charge calculation, the system determines the local currency from the currency of the sales organization.

Selective Copy of Charges

You can choose to exclude specific logistics items from the copy process such as, service items or packages when you copy a forwarding order. When you exclude some of the logistics items and copy a forwarding order, the system recalculates the charges and updates the charges accordingly in the new forwarding order. For more information on charge calculation, see [Standard Charge Calculation](#).

More Information

[Charge Calculation](#)

Overview of Charges Grid

Use

In the following business documents, the result of a charge calculation is stored in a hierarchical table on the **Charges** tab page and the **Internal Charges** tab page:

- [Forwarding Quotation](#)
- [Forwarding Order](#)

- [Forwarding Settlement Document](#)
- [Credit Memos](#)
- [Freight Order](#)
- [Freight Booking](#)
- [Service Order](#)
- [Freight Settlement Document](#)

The charge items and charge lines in the hierarchical table correspond to the level at which the system determines the agreement and calculation sheets (both kinds, including local). The system specifies various attributes corresponding to the charge lines and charge items in this table.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

The following table lists the important attributes of charge lines and charge items and the corresponding descriptions:

Attribute	Description
Charge hierarchy	The charges grid has a sum line, charge items, and charge lines. Sum line is an initial mandatory line, which you can't delete. All the charge lines and charge items are part of this sum line. A charge item is a line item of a sum line. There can be more than one charge line depending on the calculation level (header, item, or stage). The charge line is a line item of a charge item. There can be more than one charge line for a charge item depending on various criteria such as, charge type and resolution base.
Execution-based charge item	In trailer-based and truck-based settlement scenarios, the system selects this checkbox in forwarding orders. Selection of this checkbox indicates that the system doesn't calculate charges in the forwarding order. The system creates the charge items and displays them for reference purposes.
Actions	The system displays an icon in the Actions field when you insert a charge line. You can use this icon to manually insert consecutive charge lines after this charge line. Note that in rate split scenario, you can use this button to split the charge lines in the freight booking. For more information, see Rate Split.
Calculation status	The system indicates the status of charge calculation for a charge item or a charge line. The system displays the following statuses: • No calculation error • Calculation warning • Calculation error
Charge type and charge description	Charge type is a classification of a charge line that helps determine how the system calculates the transportation charges for the charge line. A few charge types available are as follows: • Basic handling charges • Delivery charges • Documentation charges

Attribute	Description
	The system displays the description of the charge type in the Charge Description field.
Commodity code and commodity code type	These fields display the commodity code item number and the commodity code type used for charge calculation. Commodity codes enable you to define the categories of products that the parties in the agreements have agreed to transport between them. You can group commodity codes using a commodity code type. For more information on commodity codes, see Classification of Goods.
Logistical reference	Logistical reference displays the item or stage information for a charge line (with the resolution base of container, package, product, or stage) from which the system obtains the logistical data.
Rate amount	Rate amount is the amount that is used as the basis for calculating the calculated amount for a charge line. The system derives the rate amount from the master data. You can manually add or change the rate amount directly on the charges screen area.
Manually-changed rate amount	This field indicates whether a rate amount is changed manually. When you manually change the rate amount, the system selects this checkbox.
Price unit, unit of measure, and quantity	Price unit is the factor with which the system derives the calculated amount. The system derives the price unit from the calculation rule you specify in the corresponding calculation sheet. Unit of measure specifies the unit of measure of the price unit. Quantity specifies the quantity used for calculating the charge amount. i Note When you don't specify any calculation rule, the rate amount and the calculated amounts are the same. ❖ Example You have a forwarding order with the following data: • Rate amount – USD 10 • Quantity – 1000 kg If you specify a calculation rule in the calculation sheet with the price unit as per 100 kg, the calculated amount is USD 100 (rate amount * quantity / price unit).
Calculation base	Calculation base is the actual base or factor with which the system calculates the charges for the scale. There can be various scenarios in which the system must calculate a charge for different forms of weights, for example, gross weight and net weight. In this case, both gross weight and net weight correspond to the scale base Weight (W), whose dimension is mass. However, by definition, gross weight and net weight are two entirely different charge aspects. Therefore, they have two different calculation bases, such as W1 for gross weight and W2 for net weight. To follow the scenario given here, you define a calculation base for every scale base. This calculation base is the final element in the charge determination. This means that the calculation base is used to determine the charges. You can define calculation bases in Customizing for Transportation Management under Basic Functions > Charge Calculation > Data Source Binding > Define Calculation Bases.

Attribute	Description
Calculated amount	Calculated amount is the amount that is used as the basis for calculating charges for a charge line. The calculated amount is derived from the rate amount. Note that when you don't specify any calculation rule, the rate amount and the calculated amounts are the same.
Ignore calculation rule	If you select this checkbox, the system ignores the calculation rule for the charge line you specified in the relevant calculation sheet or rate table, and it doesn't calculate the calculated amount. Instead, it uses the rate amount as the calculated amount. Note that the system deselects this checkbox by default. This checkbox is enabled only for charge lines with a rate table. For more information on calculation rules, see Rate Definition and Maintenance.
Reference-from and reference-to line numbers	These fields specify the first and the last line numbers respectively of a reference range. i Note • If you enter a reference-from line number without entering a reference-to line number, the system interprets this as a single reference percentage value. This means that the percentage value only applies to the single charge item in the reference-from line. • If you enter a reference-to line number without entering a reference-from line number, the system interprets this as a range starting from the first charge item to the charge item in the reference-to line. Items with the Sum and Subtotal instruction types aren't considered in a reference range.
Final amount	The amount that the system calculates based on the following: • Calculation rule • Calculation base • Calculated amount • Document currency
Document currency	You can specify the document currency at agreement level. If you don't specify a document currency at agreement level, the system determines the document currency using the following hierarchy: 1. The currency specified for the business partner 2. The currency specified for the organizational unit function on the Org. Data tab page of the organizational unit 3. The local currency specified in the general settings of the sales or purchasing organization Note that for a single order freight term scenario, you must refer to the corresponding document. You can specify the local currency for the business partner in the Define Business Partner app on the fiori launchpad. You can specify the local currency for the organizational unit in Customizing for Transportation Management under Master Data > Organizational Model > Change Organizational Model. You can specify the local currency in the general settings in Customizing for Transportation Management under Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles.
Exchange rate	Exchange rates are available at charge line level. You can get an overview of exchange rates on the Exchange Rates or Calculated Amounts and Exchange Rates tab page. For example, you can see the currencies, the type of currency conversion, and the exchange rate. If you enter different exchange

Attribute	Description
	rates, the system recalculates the currency conversions. If you change the exchange rates at the sum level, the system propagates the changes to all the charge items and charge lines under the sum level. i Note If you change the exchange rate at the charge item level, the system automatically updates all relevant charge lines. If you change the exchange rate at the charge line level, the change only affects that charge line. If you manually change an exchange rate for a charge item, the system no longer automatically updates the exchange rate at the charge item level. If you manually change an exchange rate for a charge line, the system no longer automatically updates the charge line with changes made at the charge item level.
Statistical charge	This field indicates whether a charge line is for information only and isn't a part of the total amount. If you select this checkbox, on charge calculation, the system selects the Deactivated checkbox automatically for the charge line. As a result, charge line amount isn't included in the total amount.
Deactivated	This field indicates whether a charge line is active or inactive.
Charge item operation	If you select this checkbox, the charge line amount isn't included in the total amount. The system automatically deactivates a charge line when you select the Statistical Charge checkbox for a charge line.
Payer	The system displays whether a charge line is relevant for a prepaid party or collect party.
Charge item operation	This field indicates how the system determines which charge item to include during charge calculation. You use this value in combination with the Line Item Selection instruction type. If you set the charge type to Line Item Selection and the operation to Highest Value. The system selects the most expensive charge line under it. Access sequence selects the first charge line that has been calculated successfully.
Charge calculation base, chargeable quantity, and unit of measure	These fields specify the details of quantitative calculation bases (measurable calculation bases such as volume and weight) used for rate determination. You can manually change the chargeable quantity used for rate determination. Note that the charge line must have a rate table with scales of different dimensions for the system to calculate charges.

More Information

[Charges Overview](#)

Invoicing Status of Freight Documents using Lean Services

The invoicing status in a freight document tells you the status of the posting of accruals and invoice verification for a transportation charge. You can see the invoicing status of a freight document in the charges grid at the charge element level.

When you create a freight document, the invoicing status is **Not Ready for Posting**. If the system calculates the charges successfully, the status is set to **Ready for Confirmation** based on the event reported in the freight order, freight booking, or consignment order. For more information on events, see [Freight Cost Confirmation and Posting](#).

You can use the [Monitor Freight Documents for Accrual](#) app to monitor your freight documents to post accruals. When your freight documents are at the invoicing status [Ready for Confirmation](#) and you've verified that the planned costs are correct, you can confirm the transportation charges for posting to Financial Accounting.

The following invoicing statuses are available:

Invoicing Status	Description
Not Ready for Posting	The charge element is newly created, and the checkout event or an equivalent event isn't reported. This indicates posting of freight cost isn't initiated.
Ready for Confirmation	After the relevant checkout event or an equivalent event is reported charges are successfully calculated, then the charges are ready to be confirmed. After you confirm the charges, the system initiates the posting for accruals.
Ready for Posting	The charges are confirmed and are ready for posting of accruals.
Posted	Charges are posted for accruals.
Invoice Verified	Charges are posted and invoice is received and verified.
Posting Failed	The charge element was posted for accruals but failed due to some reason.

Transition of Invoicing Status

The following table shows the various scenarios for charge elements explaining the status transition of the freight documents :

i Note

You can't change any freight document, charge item, or charge element with the status [Invoice Verified](#).

Scenario	Current Status	Next Status
Initial charge calculation isn't done and Checkout/Departure event not reported	Not Ready for Posting	Not Ready for Posting
Initial charge calculation is done and Checkout/Departure event is reported	Not Ready for Posting	Ready for Confirmation
Checkout / Shipped on Board / Uplift Confirmed event reported in the execution tab	Not Ready for Posting	Ready for Confirmation
The event Checkout is already reported in the freight document and then Not Checkout event is reported	Ready for Confirmation	Not Ready for Posting
Posting is triggered using Confirm Posting	Ready for Confirmation	Ready for Posting
When the freight cost posting to Material Management (MM) is successful	Ready for Posting	Posted
When the freight cost posting to MM failed	Ready for Posting	Posting Failed

Scenario	Current Status	Next Status
When freight document in Ready for Posting status undergoes changes in charges	Ready for Posting	Ready for Confirmation
A freight document with the status Ready for Posting undergoes changes in charges and Not Checkout event is reported	Ready for Posting	Not Ready for Posting
When you resolve the errors that led to Posting Failed and post the freight document again	Posting Failed	Posted
You do Cancel Posting on the Posting Failed document and Not Checkout event is reported	Posting Failed	Not Ready for Posting
When the invoice is processed for the Posted freight document	Posted	Invoice Verified
When freight document is Posted and FSD is created, but Purchase Order (PO) and Service Entry Sheet (SES) creation failed	Posted	Posting Failed
When you do Cancel Posting on the Posted freight document for which Not Checkout event is reported	Posted	Not Ready for Posting
When you do Cancel Posting on the Posted freight document for which accrual is confirmed	Posted	Ready for Posting
When invoice is reversed by MM	Invoice Verified	Posted
When freight document status is Posting Failed , FSD is created, but PO and SES creation failed and FSD is canceled.	Posting Failed	Ready for Posting

The following table explains the transition, if there are changes to any charges in the charge element:

Scenario	Current Status	Next Status/Impact
Changes to any charge element	Not Ready for Posting	No impact
Any change in the amount or charge element is activated/deactivated. Note that the updated amount isn't zero.	Status can be either of the following: • Ready for Confirmation • Posting Failed • Ready for Posting • Posted • Not Ready for Posting	Ready for Confirmation
When you update the charge amount to zero		Not Ready for Posting
Add a new charge element with non-zero amount.		If Checkout/Departure event is reported, then status is Ready for Confirmation .

Scenario	Current Status	Next Status/Impact
		If Checkout/ Departure isn't reported, then status is Not Ready for Posting .
Add a new charge element with zero amount		Not Ready for Posting
Calculation status of the charge element has errors, but you resolve the errors	Not Ready for Posting	If Checkout/Departure event is reported, then status is Ready for Confirmation . If Checkout/ Departure isn't reported, then status is Not Ready for Posting .

i Note

You can't delete a charge element when it has the following status:

- **Posting Failed**
- **Posted**
- **Invoice Verified**

Invoicing Status Roll-up

The invoicing status at the charge element level is rolled up to charge item level and then to the charges header level. This provides an overall invoicing status at the charges header and item level. Roll-up is essential so that freight documents are picked based on the status **Ready for Confirmation** to perform initial posting or to execute the changes.

There can be various scenarios where multiple charge items are created like stage level calculation, additional agreement party, service based charges and so on. In all the cases, the invoicing status is rolled-up from the charge elements to the corresponding charge items and then to the charges header level. For roll-up, the priority of invoicing status is as follows:

1. **Ready for Confirmation**
2. **Posting Failed**
3. **Ready for Posting**
4. **Posted**
5. **Invoice Verified**
6. **Not Ready for Posting**

Calculation Methods

Use

This function enables the system to calculate transportation charges according to a specific logic, for example, by considering the break-weight or the deficit weight. You specify a calculation method for each charge type when defining a calculation sheet or a local calculation sheet. The system applies the calculation method when calculating the charges for the charge type.

Integration

You specify a calculation method for each charge type when defining a calculation sheet or a local calculation sheet.

This function is used in the charge calculation process (see [Charge Calculation](#)).

Prerequisites

You have defined calculation methods in Customizing for **Transportation Management** under **Basic Functions** **Charge Calculation** **Enhancements to Charge Calculation Engine** **Define Calculation Methods**.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Calculation Method Types

You can choose the type of calculation method to use for each charge type. If you don't choose the type of calculation method, the system uses the standard calculation method.

The following calculation method types are available in Transportation Management (TM):

- Standard

The system calculates the transportation charges based on the specified rate, calculation rule, and actual weight. For example, the rate is USD 10 and the calculation rule specifies that the rate is for every 100 kg. The weight is 1800 kg, so the system calculates the transportation charges as USD 180 (1800 kg/100 kg * USD 10).

- Break-Weight

The system calculates the transportation charges by comparing and selecting the lower rate from either the actual rate range or the lower end of the next rate range. Break-weight is the weight from which it's cheaper to calculate the rate as opposed to the actual weight of the goods to be transported. One of the scales in the rate must be weight.

❖ Example

You're shipping 490 kg and are using the following break-weights and rates:

Weight Scale	Rate
>= 100 kg	USD 10 per kg
>= 500 kg	USD 8 per kg
>= 1000 kg	USD 6 per kg

Using the standard calculation method, the system calculates the transportation charges as USD 4900 (490 kg * USD 10).

Using the break-weight calculation method, the system checks if it's cheaper to use the rate for the original rate range (USD 10 per kg for >=100 kg) or the next rate range (USD 8 per kg for >=500 kg). Since the next rate range is cheaper, the system considers the 490 kg as being >= 500 kg and calculates the transportation charges as USD 4000 (500 kg * USD 8).

- Clipping

The system calculates the transportation charges by working through all scales level-by-level, even if the value lies outside the scale. The system then totals up the calculation results from each scale level to produce the overall result. For clipping, one scale item must have a relative calculation type.

❖ Example

The scale levels are defined with the following values:

- Up to 5 tons: USD 50 absolute
- Up to 7 tons: USD 14 per ton
- Up to 10 tons: USD 13 per ton
- Up to 15 tons: USD 12 per ton

Using the clipping calculation method for a load weighing 12 tons, the system calculates the transportation charges as follows:

- 5 tons at USD 50 = USD 50 (first scale level)
- 2 tons at USD 14 per ton = USD 28 (up to the second scale level)
- 3 tons at USD 13 per ton = USD 39 (up to the third scale level)
- 2 tons at USD 12 per ton = USD 24 (remainder in the fourth scale level)

This results in a charge of USD 141. If you don't use the clipping calculation method, the 12 tons fall into the "up to 15 tons" scale level and the transportation charge is USD 12 per ton, resulting in a charge of USD 144.

• Deficit weight rating

The system calculates the transportation charges for a weight greater than the actual weight of the goods to be transported, if it results in a lower transportation charge. The system calculates the difference between the actual weight and the next scale level and adds this difference to the goods to be transported as a deficit weight. The system then calculates the costs for the deficit weight using the cheaper rate. Deficit weight rating (DWR) is primarily used in U.S. land transportation.

❖ Example

You're shipping two freight classes with different rates. For 500 kg of freight class A, the rates are as follows:

Weight Scale	Rate
< 1000 kg	USD 1.50 per kg
>= 1000 kg	USD 1 per kg

For 400 kg of freight class B, the rates are as follows:

Weight Scale	Rate
< 1000 kg	USD 2.50 per kg
>= 1000 kg	USD 2 per kg

Using the standard calculation method, the system calculates the transportation charges as USD 750 for freight class A (500 kg * USD 1.50) and USD 1000 for freight class B (400 kg * USD 2.50). The total charge is USD 1750.

Using the DWR calculation method, the system adds 100 kg to the actual weight to bring it to 1000 kg for both freight classes. The system then calculates the transportation charges as USD 500 for freight class A (500 kg * USD 1) and USD 800 for freight class B (400 kg * USD 2). The system calculates the deficit weight using the lowest rate, resulting in a charge of USD 100 (100 kg * USD 1). The total charge is USD 1400.

- External system

The system connects to an external system to determine the rates.

- Internal charge calculation

The system can calculate transportation charges between organizations of the same company code and between organizations of different company codes. You can use internal agreements for internal charge calculation between two different business units and between business units and gateways. Using internal agreements between business units and gateways, you can maintain the contract between the different organizations that belong to the same company code or different company. You can calculate the external freight charges for each stage or combination of stages in freight orders or freight bookings. During internal charge calculation of a forwarding order, the external freight charges for each stage or combination of stages that belong to the same freight order and freight booking appear in the corresponding forwarding orders. The system can't consolidate charges for two or more stages of the same forwarding order if these stages belong to different freight orders or freight bookings. Note that you can create an internal settlement document only between a business unit and a gateway.

If the involved business units and gateway belong to different organizations of the same company code, you perform intracompany settlement. If the involved business units and gateway belong to the organizations of different company codes, you perform intercompany settlement.

For more information, see [Internal Settlement Management](#).

- Air freight: standard rating

Depending on the rate type, the system calculates the air freight transportation charges based on TACT rates or contract rates. The system uses a combination of the clipping and break-weight calculation methods to calculate the charges using unit load device (ULD) rates. The system uses the standard rate determination process for general cargo rates (GCRs) and specific commodity rates (SCRs).

- Air freight: break-weight rating

Depending on the rate type, the system calculates the air freight transportation charges based on TACT rates or contract rates. The system uses a combination of the clipping and break-weight calculation methods to calculate the charges using ULD rates. The system uses the break-weight determination process for GCRs and SCRs. The break-weight calculation method for air freight compares the higher weight break charges with the sum of lower weight break charges and the over-pivot charges and chooses the minimum rate. The system compares the weight breaks for ULD rates, GCRs, and SCRs using a greater-than scale.

- Fuel surcharge calculation

You can calculate the fuel surcharge using the calculation method FUEL_SURCHARGE. For more information, see [Fuel Surcharge Calculation](#).

- Cost pull

You can transfer distributed external costs from the freight side to the forwarding side, and use the costs as a basis for settling with a customer or an internal organization. To transfer the result of a cost distribution from the freight side to the forwarding side, you use an instruction type of **Cost** at the item level. The system automatically includes an internal charge calculation method type, and the cost pull calculation method COST_PULL. For more information, see [Cost Pull](#).

Calculation Method Modeling

You can model your own calculation method types instead of choosing a calculation method type supplied by TM. Each calculation method type is realized as an implementation of the interface [Calculation Methods Interface \(/SCMTMS/IF_TC_CALC_METHODS\)](#). You can override the implementations of the calculation method types with customer-specific implementations.

Calculation Profiles

Use

This function provides a central location to define the settings that you want the system to use when calculating transportation charges.

Integration

This function is used in the charge calculation process. For more information, see [Charge Calculation](#).

Features

The following are examples of settings you can define in a calculation profile:

- Specify the calculation date type that is used as the basis for the rate and agreement validity during charge calculation.

For example, the order date, the invoice date, or the expected end date of the main carriage.

- Specify the exchange rate date type that is used as the basis for exchange rates during charge calculation.
- Specify the level at which the system calculates the charges, for example, at header level, item level, or stage level.

The system can calculate charges for an agreement item at header level even when calculating the rest of the charges at stage level. To enable header-level charges when calculating charges at stage level, the calculation level must be stage level.

- Specify the determination rule that the system uses to determine the agreement and the calculation sheet (both kinds, including local).
- Specify a single rate, known as a through rate, for multiple stages of a transportation route.
- Specify the default dimensional weight profile for the organizational unit and the condition that the system uses to determine the dimensional weight profile.
- Specify the exchange rate type the system uses for currency conversion during the charge calculation process.
- Specify the source of the data that the system uses for charge calculation and invoicing, for example, actual route, ordered route, or actual and ordered route.

If you use the actual route, the system uses actual route information. If you use the ordered route, the system uses ordered route information from forwarding orders.

Once you've defined the settings, you can assign the calculation profile to an organizational unit or a business partner role.

i Note

If you assign a calculation profile to an organizational unit and to a business partner role, the system applies the calculation profile for the business partner role.

Activities

You can define calculation profiles in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Calculation Profiles](#).

You can assign a calculation profile to an organizational unit in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Charges Profiles](#). You can specify both a sales calculation profile and a purchasing calculation profile if the organizational unit is a forwarding house. You can also assign a calculation profile to a business partner with one of the following roles:

- Sold-to party
- Ship-to party
- Carrier
- Vendor

You can assign a calculation profile to business partner roles in the [Define Business Partner](#) app on the fiori launchpad. You can assign a calculation profile on the [Customer Org. Data](#) tab page of the business partner role. Here you can define individual profiles for each organizational unit of the business partner.

Charges Templates

You use charges templates when a valid freight agreement between the business parties, for example the purchasing organization and carrier, is missing. Charges templates are freight agreements without carriers that contain the possible charge types for the respective business processes of the freight documents (freight orders and freight bookings).

Use

You can use a charges template in the following scenarios:

- Agreement is under negotiation

When the parties are yet to finalize the charges and the rates for a transportation order, they can use a charges template to understand the possible charge types that are required and relevant for the transportation order.

- Spot tendering

In spot tendering, the carrier assignment is done on the spot and there's no possibility to have an agreement with the carrier. In this case, the shipper uses a charges template to propose the valid charge types for the business and the charge amount can be manually entered in the freight document.

- Mass creation of freight agreements

For ocean freight bookings, a shipper can create multiple agreements based on charges templates.

- Validation of charge types based on the business process of the carrier

You can use a charges template where you want to restrict the charge types based on the transportation mode or shipping type of the carrier. For example, if a carrier transports road freights, then they can see the charge types relevant only for the transportation mode Road and not for Rail or Sea.

Prerequisites

- You've selected the [Carrier Optional](#) checkbox in Customizing for [Transportation Management](#) under [Master Data](#) [Agreements and Service Products](#) [Define Freight Agreement Types](#), to use a freight agreement as charges template.

- You've assigned transportation mode and shipping type in Customizing for **Transportation Management** under **Master Data** > >> **Basic Functions** > >> **Charge Calculations** > >> **Basic Settings** > >> **Define Charge Types** .

Features

The following details are mandatory:

- Validity period
- Description
- Purchasing organization
- Transportation mode
- Shipping type
- Manual charges are always allowed, so the **Manual Charge Item** checkbox is always selected.
- When you manually add charges in the charges grid of the freight document, the system validates the transportation mode and shipping type of the charge type with that of the freight document.

For example, when you add unplanned charges to a charges template, the system validates whether the unplanned charges are relevant for the transportation mode and shipping type of the freight document.

- If the charge type has no transportation mode or shipping type assigned to it, then the charge type is valid for all transportation modes and shipping types.
- When you add a calculation sheet or local calculation sheet to a charges template, the system validates the transportation mode and shipping type of the charge type in the calculation sheet or local calculation sheet with that of the freight agreement header or items.
- Rates (rate table or flat rate) are optional.
- During charge calculation, the system first tries to determine a freight agreement between the business parties. If there isn't a freight agreement, the system determines a charges template based on the business organization, for example, the purchasing organization.

Related Information

[Agreement and Calculation Sheet Determination](#)
[Rate Determination](#)

External Agreements

Use

This function allows Transportation Management (TM) to use agreements stored in an external system when calculating transportation charges. When processing the calculation sheet (both kinds, including local) and checking the relevant rates, the TM system connects to the external system to determine the rates. TM can obtain accurate rates or estimated rates from the external system, depending on whether you specify the reference number of the external system's agreement.

Integration

This function is used in the charge calculation process, which is used in the following TM components:

- [Forwarding Order Management](#)
- [Freight Order Management](#)
- [Forwarding Settlement](#)
- [Freight Settlement](#)

For more information about the standard charge calculation process, see [Standard Charge Calculation](#).

Prerequisites

To use this function, you've established a connection from TM to the external system by configuring the following settings:

- You have integrated Web services into your TM system.
- You've defined the bindings or logical ports for the Web service by using transaction SOAMANAGER.
- You've defined your user access by using transaction SE38 and executing the report [Secure Store Logon for SMC3](#) (/SCMTMS/TCC_SEC_STORE_LOGON).
- You've created a material freight group in Customizing for [Transportation Management](#) under [Master Data](#) [Transportation Network](#) [Transportation Lane](#) [Carrier Profile](#) [Define Material Freight Groups](#).
- You've created a freight code set with freight codes and assigned the freight codes to the material freight group in Customizing for [Transportation Management](#) under [Master Data](#) [Transportation Network](#) [Transportation Lane](#) [Carrier Profile](#) [Define Freight Code Sets, Freight Codes, and Determination](#).
- You've defined the connection parameters and charge types and mapped the external system's parameters to the calculation bases in TM by using transaction SM34 and accessing the /SCMTMS/VC_SMC3 view cluster.

Activities

To use an agreement stored in an external system, enter [External System](#) in the [Calculation Method Type](#) field of the agreement's calculation sheet (both kinds, including local).

To obtain accurate rates from the external system, in the agreement specify the reference number of the external system's agreement.

To obtain estimated rates from the external system, don't specify the reference number of the external system's agreement.

More Information

[Agreement](#)

[Calculation Sheet](#)

Agreement and Calculation Sheet Determination

Use

Transportation Management (TM) enables you to create agreements between business partners and sales groups or sales organizations. Agreements are integrated with calculation sheets and rate tables (both kinds, including local) to enable the system

to calculate charges. During charge calculation, the system determines the most appropriate agreement, and thereby the calculation sheet (both kinds, including local), in order to determine a rate.

The system determines the agreement within organizational hierarchies (such as company organization or corporate organization) based on the following:

- Business partner
- Organizational unit (sales organization for forwarding agreements and purchasing organization for freight agreements)
- Organizational unit hierarchy

If the system doesn't find an agreement between the business partner and sales group, it determines an agreement between the business partner and sales organization.

If the system finds multiple agreements between a business partner and a sales organization, the system does one among the following based on the agreement determination type:

- The system finds the final amounts for all the agreement items and chooses the agreement item with the minimum final amount for charge calculation.
- The system finds the final amounts for all the agreement items and chooses the agreement item with the maximum final amount for charge calculation.
- The system displays all the agreements from which you can select an agreement for charge calculation.

i Note

In the following scenarios, the system may not display all agreements even when you specify the agreement determination type as **Display All Agreements**:

- If you select the **Eval. All Agreements** checkbox in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles**, the system displays agreements or agreement items based on the leading charge type. If the charge item has one or more agreement or agreement items with a leading charge type, the system displays only the agreements or agreement items with a leading charge type. If the charge item has agreement or agreement items without a leading charge type, the system displays all the relevant agreements and agreement items.
- If you enable automatic charge calculation for forwarding orders and freight orders, the system doesn't determine multiple agreements. Instead, the system sets the calculation status as **Automatic Charge Calculation Failed**.
- If you trigger charge calculation from a personal object worklist (POWL) or if you perform batch charge calculation with the agreement determination type as **Display All Agreements**, the system doesn't calculate charges. Also, the system sets the calculation status as **Automatic Charge Calculation Failed**.

You specify an agreement determination type in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**. If you don't specify an agreement determination type, the system doesn't consider the agreement determination type for charge calculation and follows the standard agreement determination process.

Once the system determines an agreement, it determines a calculation sheet (both kinds, including local) based on various attributes specified in the forwarding order or freight order such as, shipping type and stage category. If you haven't specified any attribute in the forwarding order, the system arbitrarily determines a calculation sheet (both kinds, including local) for charge calculation.

Integration

The system determines agreements and calculation sheets (both kinds, including local) during charge calculation. For more information, see [Charge Calculation](#).

Prerequisites

You have set up the master data for the Charge Management and Service Product Catalogs component. For more information, see [Setup of MD for Charge Management and SP Catalogs](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

The agreement and calculation sheet determination process can slightly differ from the usual way when you specify additional attributes in the agreement, forwarding order, freight order, or calculation profile. The system then determines the most appropriate agreement and calculation sheet based on the following:

- Leading charge type

The system takes the leading charge type into account by skipping agreements that use calculation sheets without a leading charge type. The system then skips agreements if the leading charge type doesn't have a rate. If there are multiple leading charge types in a calculation sheet, all the leading charge types must have a rate defined for the system to consider the agreement and calculation sheet for charge calculation. We recommend that you maintain a single leading charge type in a calculation sheet.

To allow the system to take leading charge type into account, select the **Eval. All Agreements** checkbox in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles**.

- Incoterms

In forwarding orders with Incoterms, the system determines the charges from the calculation sheet item based on the Incoterm and the payer role. For the system to be able to determine a calculation sheet item for charge calculation, the Incoterm and the payer role specified in the forwarding order must match the Incoterm and payer role specified in the calculation sheet item. The system then adds a charge line to the forwarding order. For more information, see [Incoterms in Forwarding Orders](#).

- Manually-specified agreement

You can specify a freight agreement or freight agreement item for a freight order stage. The system calculates the charges based on the specified freight agreement or freight agreement item and doesn't determine any other agreement.

- Default route

You can specify a default route in a forwarding order and a list of default routes at the agreement-item level. If the default route specified in a forwarding order or freight order matches one of the default routes specified in an agreement item, the system uses the calculation sheet associated with the agreement item during charge calculation.

i Note

You can also specify a blank default route in the agreement item. For a forwarding order with or without default route, the system may select this agreement item for charge calculation.

- Stage type

You can specify a stage type for a stage in a forwarding order or freight order and also in an agreement item. If the stage type specified in a forwarding order or freight order stage matches the stage type specified in an agreement item, the system uses the calculation sheet associated with the agreement item during charge calculation.

- Resource ownership type

You can specify a resource ownership type for a resource item in a forwarding order or freight order and also in an agreement. If the ownership type specified in a forwarding order or freight order matches the ownership type specified in an agreement item, the system uses the calculation sheet associated with the agreement item during charge calculation.

- Forwarding agreement quotation

In scenarios such as forwarding agreement quotation, you may want the system to determine agreements of all validity periods including expired agreements and service products.

To enable this, you must set the parameter SEL_SP as X in the back-end and deselect the **Eval. All Agreements** checkbox in the **Define Charges Profiles** Customizing activity.

For more information this scenario, see [Rate Determination of Forwarding Agreement Quotation Items](#).

i Note

If you're using an agreement stored in an external system, you can establish a connection from TM to the external system. The system displays the external agreement and the freight agreement item number on the **Charges > Details** tab page. For more information, see [External Agreements](#).

You use internal agreements to maintain the contract between the different organizations that belong to the same company code or different company code. For more information, see [Internal Settlement Management](#).

More Information

[Agreement Maintenance](#)

[Calculation Sheet Maintenance](#)

[Rate Definition and Maintenance](#)

Freight Agreement Determination of Freight Bookings for Stage Invoicing Carriers

For freight bookings, when you've enabled invoicing carrier and stage invoicing carrier at stage level, the system determines the freight agreement for each stage level invoicing carrier and the subsequent charge calculation and payment are settled for the individual invoicing carriers. Along with the stage-level invoicing carrier, freight agreement determination is based on the transportation mode, shipping type, and other mandatory attributes, such as validity period and involved parties.

In a freight booking, if you specify **Stage Invoicing Carrier Level** as **Header Carrier**, the system determines the valid agreement based on the carrier, transportation mode, and shipping type at the header level and settles the final charges with the respective header carrier.

If you've specified **Stage Invoicing Carrier Level** as **Stage Carrier**, then the system determines the valid agreement based on the invoicing carrier, and the transportation mode and shipping type for that stage (pre-carriage, main carriage, and on-carriage). The system settles the final charges with the respective stage invoicing carrier.

For more information, see [Freight Settlement](#).

You can see the freight charges for the stage invoicing carrier and the header carrier on the **Charges** tab of the freight booking. For more information on freight agreements, see [Agreement](#).

i Note

Transportation mode and shipping type are mandatory attributes of a freight agreement. The system determines a freight agreement for a freight booking based on the appropriate transportation mode and shipping along with other mandatory attributes, for example lifecycle status as **Released**.

Prerequisites

- You've assigned the appropriate **Application Configuration Settings** in the Customizing for **Transportation Management** under **Freight Order Management > Freight Booking > Define Freight Booking Types**. Assign **/SCMTMS/FRE_BOOK_OCEAN_ES** for ocean freight bookings and **/SCMTMS/FRE_BOOK_AIR_ES** for air freight bookings.
- You've defined a freight agreement with transportation mode and shipping type that matches with the transportation mode and shipping type of the stage that you assign to the invoicing carrier in the freight booking.
- For a freight booking, you've made the following settings:
 - Assigned a transportation mode and shipping type for the header and each stage of the freight booking.
 - Entered the **Stage Invoicing Carrier Level** as **Stage** or **Header** on the **Stages** tab of the freight booking.
 - Assigned an invoicing carrier to each stage of the freight booking.

Features

- Freight agreement determination without shipping type in the stage of a freight booking

When you don't assign shipping type for a particular stage, the system determines the freight agreement with the transportation mode for the stage invoicing carrier. If there's no such freight agreement, the system determines the freight agreement based on the header-level transportation mode and shipping type.

- Agreement determination without transportation mode and shipping type in the stage of a freight booking

If you don't assign transportation mode and shipping type to a stage, the system determines the freight agreement based on the header-level transportation mode and shipping type for the invoicing carrier. If there's no such freight agreement, the system can't determine a freight agreement for the stage.

- Same freight agreement determination for freight order or freight booking based on carrier, transportation mode, and shipping type

While determining a freight agreement whether for a freight order or freight booking (stage-level), the system checks the combination of the carrier, transportation mode, and shipping type and picks the same agreement for a freight order or freight booking (stage-level) that matches that combination. This ensures that you don't create multiple freight agreements with the same carrier, transportation mode, and shipping type.

Example

You've created an ocean freight booking with multiple stages and the following values for transportation mode, shipping type, and carrier:

Freight Booking	Transportation Mode	Shipping Type	Header Carrier
Freight_Booking	Sea	FCL	Main_Carrier

You've also created a freight order with the following transportation mode and shipping type:

Freight Order	Transportation Mode	Shipping Type
Freight_Order	Road	FTL

The freight booking **Freight_Booking** has multiple stages and the following values for transportation mode, shipping type, and carrier:

Stages	Transportation Mode	Shipping Type	Invoicing Carrier	Stage Invoicing Carrier Level
Stage1	Road	FTL	Carrier1	Stage
Stage2	Sea	FCL	Carrier2	Stage
Stage3	Road	N/A	Carrier3	Stage
Stage4	N/A	N/A	Carrier3	Stage

You've created and released the freight agreements with the following values for transportation mode, shipping type, and carrier:

Freight Agreement	Carrier	Transportation Mode	Shipping Type
Agr1	Carrier1	Road	FTL
Agr2	Carrier2	Sea	FCL
Agr3	Main_Carrier	Sea	FCL
Agr4	Carrier3	Road	FTL
Agr5	Carrier3	Sea	FCL

During charge calculation for the freight booking, the system determines the following freight agreements for the header and stage level:

- Header level and header carrier **Main_Carrier**

The system determines agreement **AGR3** because the transportation mode and shipping type of **AGR3** match with the transportation mode and shipping type at the header-level of the freight booking.

- Stage1** and invoicing carrier **Carrier1**

The system determines freight agreement **AGR1** because the transportation mode and shipping type of **AGR1** match with the transportation mode and shipping type for **Stage1** of the freight booking.

- Stage2** and invoicing carrier **Carrier2**

The system determines freight agreement **AGR2** because the transportation mode and shipping type of **AGR2** match with the transportation mode and shipping type for **Stage2** of the freight booking.

- Stage3** and invoicing carrier **Carrier3**

The system determines freight agreement **AGR4** because the transportation mode of **AGR4** matches with the transportation mode of **Stage3** of the freight booking.

- **Stage4** and invoicing carrier **Carrier3**

The system determines the freight agreement **AGR5** because, in the absence of a stage-level transportation mode and shipping type, the system considers the transportation mode and shipping type of the header level.

During charge calculation for the freight order, **Freight_Order**, the system determines the freight agreement **AGR1** as it has the same carrier, transportation mode, and shipping type as that of the freight order.

Related Information

[Freight Booking Agreement](#)

Rate Determination

Transportation Management determines the most appropriate rate when calculating transportation charges, based on certain criteria and attributes. The system determines a rate from the rate table (both kinds, including local) automatically during charge calculation based on the attributes specified in the business document and the agreement.

With rate determination, you can also determine appropriate rates for various scenarios, for example:

- You need to charge customers additional charges for providing a value added service, such as overnight service, first priority service, and service provided on a holiday.
- You need to charge customers for a single transportation rate instead of multiple rates.

Rate Determination Techniques

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

- [Through Rates](#)

The system can determine a through rate from the source location of the first stage to the destination location of the last stage in a forwarding order or freight order. You can also determine a through rate for two or more stages.

- [Rate Determination with Partial Information](#)

The system can determine the correct rate even when relevant information is missing from the forwarding order, freight order, or rate table.

- [Rate Determination with Scales of Different Dimensions](#)

The system can determine the correct rate even when you maintain a rate table with scales that contain different dimensions.

- [Rate Determination Based on Holiday Calendar](#)

You can charge customers additional charges incurred for providing services on non-working days in a particular country or a region, such as weekend days and public holidays.

- [Rate Determination for Direct Shipment of Parcels](#)

During the direct shipment of parcels, you can charge customers additional charges incurred for providing an overnight service or a first priority service.

- [Rate Determination Based on Zone Hierarchy and Postal Codes](#)

Based on zone hierarchy, the system can determine the correct rate even when you don't maintain rates for a specific zone. Also, you can maintain a single rate in the rate table for multiple postal codes within a specific range by using the wildcard character (*) in place of numerals for the postal code.

- [Collective Rate Determination](#)

When you create a collective settlement document from multiple forwarding orders or freight orders, you can choose to sum up the values of the calculation bases with scales of different dimensions (as specified in the grouping rule) in multiple forwarding orders and determine a rate for the collective value.

- [Rate Determination Based on Transportation Duration](#)

As a logistics service provider or shipper, you can charge your customers or pay your carriers based on how long it takes to transport the order.

- [Rate Determination Based on Planned Distance](#)

As a logistics service provider or shipper, you can charge your customers or pay your carriers based on the planned distance of each stage or of the entire freight order or freight booking.

- [Rate Determination Based on Transportation Stops](#)

As a logistics service provider or shipper, you can charge your customers or pay your carriers based on all stops and intermediate stops also including substops of a transportation order.

- [Rate Determination Based on Normalized and Additional Normalized Quantities](#)

Normalized and additional normalized quantities are units that are used to determine the number of pallets and cargo space occupied by transported cargo items.

- [Rate Determination based on Commodity Codes Assigned to Material Master](#)

You use commodity codes based on SAP Governance, Risk and Compliance (GRC), that is, a source in the material master, to calculate the transportation charges.

- [Rate Determination based on Units of Measures \(UoMs\)](#)

You use non-dimensionless units of measure (UoM) in your scale base to calculate the transportation charges.

Through Rates

Use

This function allows the system to use stage categories to calculate through rates in a forwarding order or freight order. You can have a through rate for each stage category (for example, pre-carriage, main carriage, or on-carriage), a combination of successive stage categories (for example, pre-carriage and main carriage), or an overall through rate. An overall through rate is a single rate that can be used from the source location of the first stage category to the destination location of the last stage category.

You can also enable through rates for internal transfer pricing (ITP) while calculating internal charges in a forwarding order. Depending on the agreement settings, you can calculate the internal charges for each stage or combination of stages of a forwarding order in a freight order or a freight booking.

Integration

This function is used in the charge calculation process. For more information, see [Charge Calculation](#).

Prerequisites

- You've enabled the calculation of through rates in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Calculation Profiles**.
- You've entered a rate in the rate table (both kinds, including local) for the location pair (source location and destination location).

Alternatively, if the leading charge type has a rate specified and you assign a rate directly to a calculation sheet line item without using a rate table (both kinds, including local), you must enter a trade lane precondition for the calculation sheet line item.
- To enable internal charge calculation between the business unit and the gateway, you've performed the following:
 - You've maintained an internal agreement between the business unit and the gateway.
 - You've maintained a calculation sheet line item with the calculation method type **Internal Charge Calculation**.
- To enable through rates for ITP, you've selected the **Through Rate** checkbox and you've maintained the calculation level as **Calculation at Stage Level** in the calculation profile for the gateway. You define calculation profiles in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Calculation Profiles**.

Features

While calculating through rates, the system evaluates through rates for a single stage category or a combination of stage categories. The system also evaluates partial through rates when a through rate is available for the main carriage but not for the pre-carriage and on-carriage. The system doesn't consider through rates that split a stage category.

The use of stage categories simplifies and speeds up the calculation of through rates as the system checks for rates for every location pair (source location and destination location).

You can see in the application log how the system evaluated and determined the through rates. You can use transaction **SLG1** to access the log object **/SCMTMS/TMS** with sub-object **TCM**. For the external ID, search with a wildcard and the user name.

Example

You want to calculate the charge of a carriage on a route with the following stage categories:

- Pre-carriage stage category with two stages: pick-up and pre-carriage 1
- Main carriage stage category with two stages: main carriage 1 and main carriage 2
- On-carriage stage category with two stages: on-carriage 1 and delivery

While calculating the through rates, the system evaluates the through rates for a single stage category or a combination of stage categories, as follows:

- Pre-carriage, main carriage, and on-carriage together
- Pre-carriage and main carriage together
- Main carriage and on-carriage together
- Main carriage only

- Pre-carriage, main carriage, and on-carriage separately

The system doesn't consider through rates that split a stage category. For example, the system doesn't consider a through rate for pre-carriage and main carriage 1. The system doesn't split the main carriage stage category even if you maintain the rates in a split manner. But you can have a partial through rate where the through rate is maintained for the main carriage only but no through rate is available for the pre-carriage and on-carriage.

In the application log, you can see the through rates that the system used to calculate the charges and how the system determined those through rates.

More Information

[Rate Determination](#)

[Charge Calculation](#)

Rate Determination with Partial Information

Use

The system can determine the correct rate even when relevant information is missing from the forwarding order, freight order, rate table, or local rate table. When one or two attributes are missing from the forwarding order or freight order but the rates are present, the system determines multiple rates. Then you can select the applicable rate manually or the system can select it automatically based on the multiple rate default specified for the calculation sheet line item. Note that the system can determine multiple rates only if the calculation base that is missing in the forwarding order or freight order isn't numeric. When a carrier is assigned to a stage in the forwarding order or freight order, but there's no rate for the carrier in the rate table or the local rate table, the system determines a generic rate.

Note that the system can also determine multiple rates for mandatory charge lines.

Integration

The system determines the rate during the charge calculation process. For more information, see [Charge Calculation](#).

Prerequisites

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

For the system to determine multiple rates, you've assigned a rate table with two or more attributes to the calculation sheet. Depending on the attributes, you've also specified the resolution base for the charge type.

For the system to determine a generic rate, you have a rate table containing a scale item with no value. To allow a rate table to contain a scale item with no value, choose the calculation base of the carrier on the **Scales** tab of the rate table and select the **Item with No Value Allowed** checkbox. The system adds a scale item with no value to the scale.

Features

- When the IATA note or general note information isn't available in the business document and when you maintain the calculation base as IATA note or general note in the calculation sheet (both kinds, including local), the system determines

multiple rates. If the multiple rate determination results in a single rate, the system performs an automatic rate lookup and calculates charges.

- For each calculation sheet line item, you can specify on the **Rates** tab what you want the system to do when the system finds multiple rates.

For example, the system can automatically select one of the following:

- The lowest rate
- The highest rate
- The only rate (if only one rate is found)

If you enter **All** or don't specify what you want the system to do when the system finds multiple rates, the system doesn't automatically select the applicable rate. Instead, you must select the applicable rate manually. Specifying the multiple rate default allows the system to select the rate automatically without you needing to intervene manually.

- If you specify commodity codes in a forwarding order or the associated master data, the system determines multiple rates in the following cases:
 - When you have a forwarding order with no commodity codes and the associated rate table (both kinds, including local) has rates for multiple commodity codes. On triggering charge calculation, the system displays multiple rates for the charge line.
 - When you have a forwarding order with multiple commodity codes and the associated rate table (both kinds, including local) has the calculation base as **COMM_CODE_MUL**. On triggering charge calculation, the system displays multiple rates for the charge line.

For more information on commodity codes, see [Classification of Goods](#).

Activities

The system determines multiple rates when you choose **Multiple Rates** and you can choose the applicable rate. Note that you can choose to show or hide scale value descriptions and rate table notes in the **Multiple Rates** dialog box.

i Note

The system determines a generic rate instead of displaying multiple rates when the charge type under the instruction type **Line Item Selection** and the corresponding rate table (both kinds, including local) have more than one scale base with a generic value.

For the system to determine a generic rate when calculating the charges, the system uses the scale item with no value if no scale items match the value of the calculation base in the forwarding order or freight order or if the calculation base has no value.

Example

The following example is applicable to both kinds of rate table and calculation sheets, including local rate tables and calculation sheets.

You assign the following rate table to the calculation sheet:

Source Location	Destination Location	Transshipment Location	Rate
HAM	HOU	CY_NUE	USD 100 per 10 kg
HAM	HOU	CY_NUE	USD 200 per 10 kg

The transshipment location is missing in the forwarding order or freight order. The system determines multiple rates. As you specified **All** as the multiple rate default for the calculation sheet line item, you must select the applicable rate manually in the charges grid on the **Charges** tab of the business document.

When entering the data for another rate table, you enter the following data:

Source Location	Destination Location	Carrier	Rate
HAM	HOU	A	USD 100 per 10 kg
HAM	HOU	*	USD 200 per 10 kg

The system determines a generic rate as follows:

- If the carrier is A in the forwarding order or freight order, the system applies USD 100.
- If the carrier is anything other than A in the forwarding order or freight order, the system picks up the scale item with no value and applies USD 200.

If no carrier is assigned to a stage in the forwarding order or freight order, the system determines both rates and you select the applicable one.

More Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

[Scale Maintenance](#)

Rate Determination with Scales of Different Dimensions

Use

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

When you maintain a rate table with scales of different dimensions, the system can determine the correct rate based on the information in the forwarding order (FWO) or freight order (FO). The system performs the following tasks to determine the correct rate:

1. Selects the rate table item in which different attributes of the same scale match the attributes in the FWO or FO.
2. Uses this rate table item to determine the correct rate that matches the other attributes in the FWO or FO.

Integration

The system determines the rate during the charge calculation process. For more information, see [Charge Calculation](#).

Activities

To define or change a rate table or a local rate table for an agreement, a calculation sheet, or a local calculation sheet, choose one of the following apps on the fiori launchpad:

- [Freight Agreements](#) or [Forwarding Agreements](#)
- [Create Calculation Sheet](#) or [Edit Calculation Sheet](#)

Example

You specify the following attributes in a rate table or a local rate table:

Source Location	Destination Location	<=200 kg	<=400 kg	<=600 kg
A	B	USD 50	USD 90	USD 130
A	C		USD 75	USD 100

In the FWO, you specify the source location as A and the destination location as C for a freight unit of 150 kg.

During charge calculation, the system determines the rate table item for source location A and destination location C. The system then determines the less than 400 kg rate of USD 75 as the correct rate from the rate table or the local rate table because you haven't entered a rate for the less than 200 kg.

More Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

Rate Determination Based on Holiday Calendar

Use

As a logistics service provider or carrier, you can charge your customers additional charges incurred for providing services on nonworking days, such as weekends and public holidays, in a particular country or region.

Prerequisites

You've defined factory calendars in Customizing for [Transportation Management](#) under [Master Data](#) [Calendar](#) [Maintain Factory Calendar](#).

You've defined a location calendar for a specific country and a region using a factory calendar in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Location Calendar](#).

Integration

The system determines the rate during the charge calculation process. For more information, see [Charge Calculation](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

In a rate table, you can maintain the calculation base as either **PICKUP_HOLIDAY** or **DELIVER_HOLIDAY**. If you specify the calculation base as **PICKUP_HOLIDAY**, the system uses the source location specified in the forwarding order or freight order for rate determination. If you specify the calculation as **DELIVER_HOLIDAY**, the system uses the destination location specified in the forwarding order or freight order for rate determination.

Based on the calculation date type specified at the charge type level in the calculation sheet, the system chooses the calculation date as order date, system date, or invoice date. If you haven't maintained any calculation date type for a calculation sheet item, the system uses the calculation date specified in the calculation profile to determine rates for charge calculation. Using the location and date information, the system determines from the factory calendar whether the date used for charge calculation is a nonworking day or a working day in a specific region.

More Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

Rate Determination for Direct Shipment of Parcels

Use

As a logistic service provider or carrier, you may need to transport parcels to the customer as soon as possible and hence you forward them directly without consolidation. In such cases, you can charge the customer for the additional charge incurred for providing an overnight service or a first priority service.

Transportation Management enables you to determine rates for direct shipments from either the Web services provided by an external parcel service provider or from a standard rate table (both kinds, including local).

Integration

This function is a part of the charge calculation process. For more information, see [Charge Calculation](#).

Prerequisites

- You have defined a calculation profile with the calculation level as **Calculation at Item Level** in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**.
- You have assigned the calculation profile to a charges profile and maintained a local currency in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles**.
- You have assigned the charges profile to an organizational unit in the transaction PPOME.
- You have defined a manifest or express freight order type in Customizing for **Transportation Management** under **Freight Order Management > Freight Order > Define Freight Order Types**.
- You have maintained the direct shipment option settings in Customizing for **Transportation Management** under **Planning > Freight Units > Define Freight Unit Types**.

- You have maintained a master waybill stock between the purchase organization and the carrier in the [Waybill Number Stocks](#) app on the fiori launchpad.

Features

You can determine rates from a standard rate table (both kinds, including local). You can create an agreement with a carrier and assign each agreement item to one carrier product. You can specify the attributes, such as length, width, height, and girth, on the [Parcel](#) tab page of the agreement. When you perform direct shipment option determination, the system compares the attributes of the parcel specified in the freight unit with that specified in the agreement. For the system to consider an agreement for charge calculation, all the attribute values specified in the freight unit must be less than or equal to the attribute values specified in the agreement. If there are multiple agreements that meet this condition, the system displays a list from which you can choose one.

Activities

You can create a parcel freight order in the [Create Road Freight Order](#) app on the fiori launchpad. Note that a parcel freight order is created as a road freight order that has a corresponding freight order type and the shipping type [Parcel](#). For more information see [Parcel Shipment](#).

More Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

[Agreement Maintenance](#)

Rate Determination Based on Zone Hierarchy and Postal Codes

Use

The system determines the correct rate in the following scenarios:

- If you don't maintain the rates for a specific zone, the system determines the appropriate rate based on zone hierarchy.
- If you don't maintain the rates for a specific postal code, the system determines the rate based on wildcard search.

Integration

The system determines the rate during the charge calculation process. For more information, see [Charge Calculation](#).

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Zone Hierarchy

The system calculates the charges based on the transportation zone hierarchy you maintain. For example, you've maintained the following zones in the following hierarchy:

1. Zone A

2. Zone B

3. Zone C

4. Zone D

During charge calculation, for a forwarding order with zone D, the system determines a rate for zone D from the rate table. If you haven't maintained any rates for zone D, the system determines the next appropriate rate from the rates maintained for the superordinate zones in the hierarchy as follows:

1. Zone C

2. Zone B

3. Zone A

Postal Codes

You can maintain a single rate for multiple postal codes within a specific range by using the wildcard character (*) in place of numerals in the postal code and maintain rates in the rate table. For example, you maintain a rate table as follows:

Postal Code/ Weight	>100 kg	>200 kg
90210	10 USD	-
9021*	12 USD	-
902*	15 USD	25 USD

You have a forwarding order with a postal code of 90210 and gross weight of 250 kg. During charge calculation, the system determines 25 USD as the rate from the rate table and calculates charges. The system looks for the most specific rate (for postal code 90210) and when not found, it determines the next appropriate rate with a wildcard-based search. Here, the rate determination is based on the following sequence:

1. 90210

2. 9021*

3. 902*

More Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

Collective Rate Determination

When you create a collective settlement document from multiple forwarding orders or freight orders, you can choose to sum up the values of the calculation bases with scales of different dimensions in multiple orders and determine a rate for the collective value.

The system groups the rates based on a grouping rule and also on the resolution base you specify in the calculation sheet. The resolution base determines the level (for example, item level or container level) at which the system must determine the rates.

The collective rate can often be less than the rate for the individual values.

Note that you can specify that the system performs standard charge calculation when you don't maintain a rate for a collective quantity.

Prerequisites

- You've defined grouping rules in the view [Define Grouping Rules for Charge Calculation Resolution](#) (/SCMTMS/V_RUL113) using transaction SM30.

The system groups the calculation base values if you specify **Yes** in the [Group Across Orders](#) field and specify the calculation base that the system must use for grouping in the [Group By](#) field. You must deselect the [By Calculation Level](#) checkbox for the system to determine a rate for the collective value.

- You've created a collective forwarding settlement document or a collective freight settlement document using the [Forwarding Settlements](#) app or the [Freight Settlements](#) app on the launchpad.

Example

A customer asks you to transport products from the same pick-up location in two separate forwarding orders. According to the forwarding agreement you have with the customer, you must charge a collective rate based on the source location across multiple forwarding orders. A collective rate helps ensure you charge the better rate for the customer.

You make the following settings:

voucher

- Using transaction SM30, you define the rule for grouping the rates in the view [Define Grouping Rules for Charge Calculation Resolution](#) (/SCMTMS/V_RUL113).

You specify the following settings:

- Group by the [Source Location](#) calculation base
- Group across orders

- In the agreement you have with the customer, you create a consolidation charge line and specify this grouping rule under [General Details](#) > [Items](#) > [Calculation Sheet Overview](#).

- In the agreement, you use a rate table that specifies the following rates per KG:

- \$10 for < 10 KG
- \$8 for between 10 KG and < 25 KG
- \$6 for between 15 KG and < 45 KG

In the two forwarding orders, the gross weight of the transported products is 8 KG and 13 KG. The system determines the collective rate based on the consolidated gross weight as follows:

$$8 \text{ KG} + 13 \text{ KG} = 21 \text{ KG} \times \$8 = \$168$$

If the system determined the rate separately for each forwarding order, the rate would be as follows:

$$8 \text{ KG} \times \$10 = \$80$$

$$13 \text{ KG} \times \$8 = \$104$$

$$\$80 + \$104 = \$184$$

The collective rate is the better rate for the customer.

More Information

[Freight Settlement Integration using MM External Services Management](#)

[Forwarding Settlement](#)

[Charge Calculation](#)

Rate Determination Based on Transportation Duration

As a logistics service provider or shipper, you can charge your customers or pay your carriers based on how long it takes to transport the order. For the system to determine the rate based on the transportation duration, you specify the calculation base as Gross Duration (**DURATION_GROSS**) or Net Duration (**DURATION_NET**) in a rate table or a local rate table. When you specify the calculation base as **DURATION_GROSS**, the system uses the total time taken to transport the order. The total time includes the actual transportation duration and the time taken for other tasks needed to transport the order, such as the time taken to load and unload. When you specify the calculation base as **DURATION_NET**, the system uses only the actual transportation duration. Note that when you maintain the scales or define the scale values, the default unit of measure for the calculation bases is HH: MMMM. You can change the unit of measure to any time-based unit of measure when you define the calculation rules in rate tables. You can change the unit of measure in rate tables under **►Rate Table ► Dates and Values ► Validity ► Calculation Rules ►**.

Integration

The system determines the rate during the charge calculation process. For more information, see [Charge Calculation](#).

Example

You have the following rates for the base freight charge type:

Source Location	Destination Location	Duration >=	Rate	Currency
Nürnberg	Hamburg	16:00	100	EUR
Nürnberg	Hamburg	20:00	120	EUR

The calculation rule is as follows:

Calculation Base	Price Unit	UoM
DURATION_NET	50	Min

The net duration for the freight order is 18 hours. The system calculates the price per minute: 18 hours = 1,080 minutes.

The system calculates the charge for the freight order as follows:

Charge Type	Rate	Currency	Price Unit	UoM	Qty	Calc. Base	Chargeable Qty	Final Amount
BASE_FREIGHT	100	EUR	50	Min	1080	DURATION_NET	18:00	2160

Related Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

Rate Determination Based on Planned Distance

As a logistic service provider or shipper, you can charge your customers or pay your carriers based on the planned distance of each stage or of the entire freight order or freight booking.

To calculate the planned distance-based charges, add the relevant charge type and the calculation base PLAND_DIST in the calculation sheets or scales of the rate table.

Integration

The system determines the rate during the charge calculation process. For more information, see [Charge Calculation](#).

Example

You've a freight order that you're shipping from Munich to Berlin. The planned distance between Munich (source location) and Nürnberg (destination location) is based on the number of stages, if any. Your freight order has two stages, which are the following:

Source Location	Destination Location	Distance
Munich	Nürnberg	169.2 KM

The planned distance from Munich to Nürnberg is 169.2 KM.

You've defined the following calculation bases in the rate table:

Calculation Base	Calculation Base Description	Calculation Type	Scale Type	Scale UoM
SOURCELOC	Source Location	Absolute	Same Scale (=)	
DESTLOC	Destination Location	Absolute	Same Scale (=)	
PLAND_DIST	Distance	Relative	Base Scale (>=)	KM

You've defined the following rate lines:

Source Location (=)	Destination Location (=)	Rate (Distance >=100KM)	Currency
Munich	Nürnberg	3	EUR
Nürnberg	Berlin	5	EUR
Munich	Berlin	7	EUR

The system calculates the charge (**Base FTL - Minimum**) for the entire freight order based on the formula:

$((\text{Rate amount} * \text{Quantity}) / \text{Price per unit})$

In this case, the rate amount is 3 EUR per KM and the quantity (total distance) for the freight order is 169.2 KM. Therefore, the final amount is $((3 \times 169.2) / 1) = 507.60$ EUR.

Charge Item	Charge Type	Rate Amount	Currency	Price per Unit	UoM	Quantity	Final Amount
Base FTL - Minimum	BASE_FTL_MIN	3	EUR	1	KM	169.2 KM	507.60 EUR

Related Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

Rate Determination Based on Transportation Stops

As a logistic service provider or shipper, you can charge your customers or pay your carriers based on all stops and intermediate stops also including substops of a transportation order.

For the system to determine the rate based on the number of stops, you can specify the following calculation bases in the calculation sheets or scales of the rate table:

- **NUMBER_OF_STOPS** (Number of Stops)
- **NO_OF_INT_STOPS** (Number of Intermediate Stops)
- **STOPS_SUBSTOPS** (Number of Stops Including Sub-Stops)
- **INTSTOP_SUBSTOP** (Number of Intermediate Stops Including Sub-Stops)

You can see the stops in the **Stages** tab of freight documents or the **Route** tab of forwarding documents.

You can see the sub-stops (for example the loading or unloading stops, for both inbound and outbound deliveries) for Advanced Shipping and Receiving (ASR) enabled shipping or receiving points in the **Overview** tab of freight orders. For more information and prerequisites, see [Advanced Shipping and Receiving](#)

i Note

- You can use the calculation bases **NUMBER_OF_STOPS** (Number of Stops) and **NO_OF_INT_STOPS** (Number of Intermediate Stops) for both freight and forwarding documents.
- You can use the calculation bases **STOPS_SUBSTOPS** (Number of Stops Including Sub-Stops) and **INTSTOP_SUBSTOP** (Number of Intermediate Stops Including Sub-Stops) only for freight orders.

Integration

The system determines the rate during the charge calculation process. For more information, see [Charge Calculation](#).

Example

You've a transportation order whose source location is Nürnberg and destination location is Hamburg. There's a shipping/receiving point (industry park) in Hamburg with two warehouses. There're two substops Warehouse1 – Unloading Point1 (UL001) and Warehouse2 – Unloading Point2 (UL002).

You've the following stages and locations:

Stages	Source Location	Destination Location	Sub Stops
Stage 1	Nürnberg	Frankfurt	None
Stage 2	Frankfurt	Hamburg	Warehouse1 – Unloading Point1 (UL001)
			Warehouse2 – Unloading Point2 (UL002)

Following are the values for header-level charge calculation:

- **NUMBER_OF_STOPS** is 3 (Nürnberg, Frankfurt, and Hamburg)
- **NO_OF_INT_STOPS** is:
 $(\text{NUMBER_OF_STOPS} - 2) = (3 - 2) = 1$
- **STOPS_SUBSTOPS** is 4 (Nürnberg, Frankfurt, UL001, and UL002)
- **INTSTOP_SUBSTOP** is:
 $(\text{STOPS_SUBSTOPS} - 2) = (4 - 2) = 2$

For item-level charge calculation, **STOPS_SUBSTOPS** is always 2 and **INTSTOP_SUBSTOP** is 0.

i Note

The system doesn't support stage-level charge calculation based on the number of sub-stops.

Related Information

[Rate Determination](#)

[Rate Definition and Maintenance](#)

Rate Determination Based on Normalized and Additional Normalized Quantities

Normalized and additional normalized quantities are units that are used to determine the number of pallets and cargo space occupied by transported cargo items.

Use

Additional normalized quantities allow shippers and trucking companies to determine the size of the trailer for transporting cargo items. You can calculate transportation charges based on the normalized quantity and additional normalized quantity.

The normalized quantities and additional normalized quantities are available in the **General Data** tab of the freight order or freight booking.

Based on the agreed rates and freight agreement, the system determines the freight charges based on the normalized quantities and additional normalized quantities. You use calculation bases NLQ for normalized quantities and ALQ for additional normalized

quantities in a rate table to add the rates. These rates are then used to determine the charges for the freight order or freight booking.

For more information, see [Use of Normalized Quantities and Capacities](#).

Rate Determination based on Commodity Codes Assigned to Material Master

You use commodity codes based on SAP Governance, Risk and Compliance (GRC), that is, a source in the material master, to calculate the transportation charges.

You can base transportation charges on commodity codes for a particular numbering scheme.

For example, for rail freight orders in North America, the rates and handling of freight depend on the type of commodity that is shipped by the rail carrier. As carriers aren't aware of the type of freight that is subcontracted by the shipper, standard commodity codes are used to classify the freight for transportation.

There are sets of commodity codes for a particular numbering scheme that you assign to the products of a freight order. You can see the numbering scheme and commodity code for a product on the **Items** > **Commodity Codes** tab of the freight order.

Based on agreed rate and freight agreement between the business parties, the system determines the transportation charges based on the numbering scheme and commodity code. You use calculation base **COM_CODE_SCHM** in the rate table for charges based on commodity codes. For more information see, [Commodity Codes Assigned to Material Master](#).

Rate Determination based on Units of Measures (UoMs)

As a logistics service provider or shipper, you can calculate transportation charges based on units of measures (UoMs) for the product quantities. For the system to determine the rate based on the UoMs, you specify the scale base as **QTYALL** and calculation base as **QTY_ALL_VAL** in a rate table or local rate table.

UoMs can be of the following types:

- Dimensionless where the quantity of the item/product in freight documents have no specific dimensions, for example, **Each Piece**, **Carton**, and so on.
- Non-Dimensionless where the quantity of the item/product in freight documents have specific dimensions, for example, **Meter**, **Millimeter**, and so on.

UoMs are defined in the sales order or purchase order (in the material master of the product items). During charge calculation, you can use the same UoM to calculate the charges. For example:

- If the sales order has the non-dimensionless UoM Meter (M), then the same UoM is used in the subsequent freight units and freight orders. Charge calculation considers Meter (M) as the UoM at the product level.
- If the sales order has the dimensionless UoM Each (EA), then the same UoM is used in the subsequent freight units and freight orders. Charge calculation considers Each (EA) as the UoM at the product level.

Scale and Calculation Bases

For UoM relevant charge calculation, the scale base is **QTYALL**. In the Customizing for the scale base, the **UoM Relevant** checkbox is checked. You can see that in the Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Data Source Binding** > **Define Scale Bases**.

The system considers all types of UoMs, including dimensionless and non-dimensionless UoMs, for calculating the transportation charges when there's no value in the [Dimension](#) field.

If you enter any valid dimension in the [Dimension](#) field for the scale base [QTYALL](#), then the system considers that dimension for charge calculation. You can do that in the Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Data Source Binding](#) > [Define Scale Bases](#).

The calculation base for using all kind of UoMs, including dimensionless and non-dimensionless, is [QTY_ALL_VAL](#).

i Note

There're no changes to the settlement processes when you use dimensionless or non-dimensionless UoMs for charge calculation.

Charge Estimation

Use

This function allows you to estimate freight charges, forwarding charges, and internal charges for a single route based on minimum input. You can estimate charges for the entire route (header-level calculation) or for individual stages of the route (stage-level calculation). Note that the more data you enter, the more accurate the charge estimation.

Using this function, the system allows you to provide an immediate response to a query. This is useful, for example, to allow a sales person to respond promptly to a query from a business partner about the charge for a transport over a particular route.

i Note

The system does not save the data that you enter, or the charge estimation.

Integration

This function is available as part of the charge calculation process. For more information, see [Charge Calculation](#).

Prerequisites

- In Customizing, you have carried out the activities for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#).
- You have set up the master data for the Charge Management and Service Product Catalogs component (see [Setup of MD for Charge Management and SP Catalogs](#)).
- If you want to estimate freight and forwarding charges for air transportation, you must make the charge management settings that are relevant for air freight (see [Charge Management Settings for Air Freight](#)).

Features

You can estimate charges for existing or new business partners. You must enter the following information:

- Sales organization
- Source location

- Destination location

For existing business partners, you must also enter the sold-to party.

i Note

During the charge estimation of freight and forwarding charges for air transportation, the system determines the air waybill type, printing, and settlement based on TACT rates or contract rates as specified in the calculation profile settings. If the system does not determine the air waybill type, printing, and settlement, you can also manually enter these values.

During charge estimation of ocean freight charges, you can calculate the transportation charges either from port-to-port or door-to-door. You specify the consolidation settings to estimate charges for a freight booking in Customizing for **Transportation Management** under **Freight Order Management > Freight Booking > Define Freight Booking Types**.

During strategic freight procurement, the system can estimate the cost of using a carrier to provide transportation services in a trade lane. For more information, see [Strategic Freight Procurement](#) and [Post-Bid Analysis](#).

When estimating the charges for existing business partners, the system uses the specific agreement between the business partner and the sales organization. When estimating the charges for new business partners, the system uses a generic agreement that is not specific to any business partner.

Note that if you do not enter the pick-up date, the system uses the system date as the pick-up date.

Activities

You can estimate freight charges in the **Estimate Freight Charges** app on the fiori launchpad.

You can estimate forwarding charges in the **Estimate Forwarding Charges** app on the fiori launchpad.

Service Charges

Use

You can charge customers for additional services, such as fumigation, packing and labelling, and insurance, even when you do not specify these additional service items in the freight agreement attached to a freight order. You can include the charges for the charge types associated with the service items in the freight order and perform charge calculation.

Integration

You use additional service charges during charge calculation process (see [Charge Calculation](#)).

Prerequisites

- In Customizing, you have carried out the activities for **Transportation Management** under **Basic Functions > Charge Calculation**.
- You have set up the master data for the Charge Management and Service Product Catalogs component (see [Setup of MD for Charge Management and SP Catalogs](#)).
- You have assigned charge types to a service type in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Assign Charge Types to Service Types**.

Features

When you select the **Additional Service Charges** checkbox in a freight agreement item, the system automatically inserts a blank charge line on the **Charges** tab page in the freight order corresponding to each additional service item specified in the freight order. The system inserts these charge lines using the charge types assigned to the service items. The system specifies the instruction type as **Standard** in these newly-added charge lines. You can manually enter information, such as rates and calculation rules, for charge calculation. The system defaults the resolution base to **SERVICE** for these newly-created charge lines.

i Note

If you have assigned multiple charge types to a service type in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Assign Charge Types to Service Types**, the system selects only one of these charge types corresponding to the service to create a blank charge line on the **Charges** tab page.

If you deselect the **Additional Service Charges** checkbox, the system does not automatically insert charge lines for additional services and the system does not consider the additional service items in the freight order for charge calculation.

More Information

[Calculation Logic](#)

Cost Pull

You can use execution costs as a basis for settling with your customers or internal organization by transferring distributed costs from the freight side to the forwarding side. You can charge the customer for the total or partial execution charges and the additional charges using the cost pull calculation method.

Note that cost pull only works with header-level and stage-level charge calculation; it doesn't work with item-level charge calculation.

Integration

Cost pull calculation method is used in the cost distribution process and charge calculation process. For more information, see [Cost Distribution Management](#) and [Charge Calculation](#).

Prerequisites

- You've specified the cost distribution settings in Customizing for **Transportation Management** under **Basic Functions > Cost Distribution**.
- You've assigned the distribution profiles to the relevant organization in the Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charges Profiles**.
- For logistics service providers that want to use the freight cost in a freight order or freight booking for internal settlement, you must enable freight orders or freight bookings for cost distribution. For more information, see the following Customizing activities:
 - For freight orders, see Customizing for **Transportation Management** under **Freight Order Management > Freight Order > Define Freight Order Types**.
 - For freight bookings, see Customizing for **Transportation Management** under **Freight Order Management > Freight Bookings > Define Freight Booking Types**.

- To transfer the result of cost distribution from the freight side to the forwarding side, in the calculation sheet or the local calculation sheet, you must specify **Customer and Service Provider** in the **Charge Usage** field. For internal charge calculation, you must specify **Internal** as the charge usage. You must specify the instruction type as **Cost** when defining a calculation sheet or a local calculation sheet. For more information, see [Calculation Sheet](#).
- You've specified the data source as **Actual Route**. You specify the data source in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Cost Instruction Type

You use an instruction type of **Cost** at item level. When you use a cost instruction type, the system automatically includes an internal charge calculation method type, and the cost pull calculation method **COST_PULL**. You can insert a markup charge line for a percentage value with reference to the cost charge line. For a cost charge line, you can specify a cost pull strategy based on which the system transfers the results of cost distribution to the forwarding side.

Cost Pull Strategy

You specify the cost pull strategy in the calculation sheet. Based on the cost pull strategy you specify, the cost charge line appears in the forwarding order with or without standard charge lines. Based on the cost pull strategy, the system can either add the costs and aggregate it in a standard charge line or just copy the costs under the cost charge line or the standard charge line.

You can specify one of the following settings:

- **Aggregation**

The system adds the charges in all the distributed charge lines in the freight order and displays the totals in a standard charge line in the forwarding order. Note that a standard charge line is mandatory when you specify the cost pull strategy as **Aggregate**. For more information, see [Internal Settlement Management](#).

- **Active Copy**

The system copies the individual distributed charge lines in the freight order to the forwarding order under a cost charge line as active charge lines.

- **Inactive Copy**

The system copies the individual charge lines from the air waybill charge item to the forwarding order under a cost charge line as inactive charge lines. For more information, see [Air Waybill Direct Cost Pull](#).

On the **Cost Pull** tab page in the calculation sheet, you can specify the charge types the system transfers to the forwarding side.

Note that you can't edit the **Amount** or **Currency** fields for the cost charge line and the standard charge lines under the cost charge line.

Source Charge

You can specify the source of the charges in the calculation sheet. Based on the source charge you specify, the system pulls the charge lines from one of the following areas to the forwarding order:

- From the settlement charge item of the freight order or freight booking
- From the air waybill charge item

- From the **Internal Charges** tab page of the freight order
- From the **Charges** tab page and the **Internal Charges** tab page of the freight order
- From the **Internal Charges** tab page of the forwarding order to the **Charges** tab page of the forwarding order

i Note

- The system defaults the source charge as **External**. The system enables the **Source Charge** field in the calculation sheet only when you specify the cost pull strategy as **Active Copy**. If you don't enter a value in the **Source Charge** field, the system pulls the distributed costs in the freight order to the forwarding order. The system copies all the charge lines as active charge lines in the forwarding order.
- When you specify the cost pull strategy as **Active Copy** and the source charge as **Air Waybill**, the system copies the calculation base, the calculation rule, and the rate details, for each charge line from the freight booking to the forwarding order.

Automatic Recalculation of Cost Pull Charges

If charges are changed in a freight order or freight booking in a cost pull scenario, the system recalculates the cost pull charges automatically in the relevant forwarding orders. For more information, see [Automatic Recalculation of Cost Pull Charges in Forwarding Orders](#).

Related Information

[Examples: Cost Pull](#)

[Cost Distribution for LSPs](#)

[Calculation Sheet and Local Calculation Sheet Maintenance](#)

[Calculation Methods](#)

Automatic Recalculation of Cost Pull Charges in Forwarding Orders

If charges are changed in a freight order or freight booking in a cost pull scenario, the system recalculates the cost pull charges automatically in the relevant forwarding orders.

If the forwarding order is in process, the system recalculates the cost pull charges automatically and updates the invoicing status accordingly.

If the forwarding order is complete, the system reopens the forwarding order before recalculating the cost pull charges and updating the invoicing status. The system then updates the status of the forwarding order in alignment with the invoicing status. That is, if the forwarding order is completely invoiced, the status is set to complete. However, if the forwarding order isn't completely invoiced, the status is set to in process.

Note that you can check existing forwarding orders for cost pull charges using the program **Check for Cost Pull Charges in Existing Business Documents** (/SCMTMS/SET_COST_CHARGE_EXIST). If the system finds existing forwarding orders with cost pull charges and charges were changed in the relevant freight order or freight booking, the system recalculates the charges.

i Note

- The system doesn't redetermine the agreement, calculation sheet, or local calculation sheet.

- When automatically recalculating the cost pull charges in forwarding orders, the system also recalculates all other charge lines.

Therefore, any changes to rate tables or local rate tables are reflected in all relevant charge lines.

- If the settlement document already exists, the system can create a new settlement document to settle the updated charges in the business document.

The system creates this new settlement document automatically using the program [Creation of Forwarding Settlement Documents](#) (/SCMTMS/CFIR_CREATE_BATCH). For more information, see [Creation and Transfer of Forwarding Settlement Documents](#).

Prerequisites

You have made the prerequisite settings for cost pull (see [Cost Pull](#)).

Examples: Cost Pull

Use

The following examples describe the various possibilities of the cost distribution function in Transportation Management (TM). For more information, see [Cost Pull](#).

Prerequisites

You plan two forwarding orders in a freight order. The gross weights of forwarding order 1 and 2 are 560 kg and 450 kg respectively.

You have the following values in the freight order and you perform charge calculation:

Charge Type	Rate Amount	Calculation Rules	Total Gross Weight in kg	Calculated Amount in USD
Loading charges	10	On gross weight per 10 kg	1010	1010
Basic charges	200	NA	NA	200

External cost distribution scenario:

After calculating charges, you distribute the costs in the associated forwarding orders. The externally-distributed costs are as follows:

Forwarding Order	Loading Charges	Basic Charges	Total Charges in USD	Gross Weight	Distribution Percentage
Forwarding order 1	560	110.89	670.89	560	55.40
Forwarding order 2	450	89.11	539.11	450	44.60

Internal cost distribution scenario:

You have a vehicle resource DI_TRUCK for shipping the cargo. You maintain an agreement between the organization of the vehicle resource and the business partner of the purchasing organization.

You have the following values in the freight order and you perform internal charge calculation:

Charge Type	Rate Amount	Calculation Rules	Total Gross Weight in kg	Calculated Amount in USD
Loading charges	10	On gross weight per 10 kg	1010	1010
Haulage charges	3	On gross weight per 1 kg	1010	3030

After calculating internal charges, you distribute the costs in the associated forwarding orders. The internally-distributed costs are as follows:

Forwarding Order	Haulage Charges	Loading Charges	Total Charges in USD	Gross Weight	Distribution Percentage
Forwarding order 1	1680	560	2240	560	55.40
Forwarding order 2	1350	450	1800	450	44.60

Example

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Example 1:

You specify the cost pull strategy as **Active Copy** and the source charge as **External** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order under the cost charge line.
- The system creates two charge lines; one for the loading charges and one for the basic charges. From the freight order, the system pulls and displays the calculated amount and final amount as USD 560 and USD 110.89 for the loading charges and basic charges respectively.
- As you've specified the source charge as **External**, the system pulls the costs from the external cost distribution of the freight order. The pulled execution cost has all the charge lines from freight order 1.
- As you've specified the cost pull strategy as **Active Copy**, you can edit the rate amount and currency.

Example 2:

You specify the cost pull strategy as **Inactive Copy** and the source charge as **Internal** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order under the cost charge line.

- The system creates two charge lines; one for the loading charges and one for the haulage charges. From the freight order, the system pulls and displays the calculated amount and final amount as USD 560 and USD 1680 for the loading charges and haulage charges respectively.
- As you've specified the source charge as **Internal**, the system pulls the costs from the internal cost distribution of the freight order. The pulled execution cost has all the charge lines from freight order 1.
- As you've specified the cost pull strategy as **Inactive Copy**, you can't edit the charge lines.

Example 3:

You specify the cost pull strategy as **Inactive Copy** and the source charge as **External & Internal** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order under the cost charge line.
- The system creates four charge lines for the following charges:
 - Haulage charges – USD 1680
 - Loading charges – USD 560
 - Loading charges – USD 560
 - Basic charges – USD 110.89

From the freight order, the system also pulls and displays the above values for the calculated amount and final amount.

- As you've specified the source charge as **External & Internal**, the system pulls the costs from both the internal and external cost distribution of the freight order. The pulled execution cost has all the charge lines from freight order 1.
- As you've specified the cost pull strategy as **Inactive Copy**, you can't edit the charge lines.

Example 4:

You specify the cost pull strategy as **Active Copy** and the source charge as **Own Internal** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order under the cost charge line.
- The system creates two charge lines; one for the loading charges and one for the haulage charges. From the freight order, the system pulls and displays the calculated amount and final amount as USD 560 and USD 1680 for the loading charges and haulage charges respectively.
- As you've specified the source charge as **Own Internal**, the system pulls the costs from the **Internal Charges** tab page to the Charges tab page within the same forwarding order. The pulled execution cost has all the charge lines from internal charges of the forwarding order.
- As you've specified the cost pull strategy as **Active Copy**, you can edit the rate amount and currency.

i Note

- You can use the own internal source charge only for external charge calculation in a forwarding order.
- The own internal source charge works irrespective of the source of internal charges in a forwarding order.

That is, you can use own internal source charge even when the internal charges aren't based on the cost pull scenario.

Example 5:

You specify the cost pull strategy as **Aggregate** and the source charge as **External** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order in a standard line under the cost charge line.
- The standard line has the calculated amount and final amount as USD 670.89, which is the sum of loading charges and basic charges (USD 560 + USD 110.89).
- As you've specified the source charge as **External**, the system pulls the costs from the external cost distribution of the freight order.
- As you've specified the cost pull strategy as **Aggregate**, the pulled execution cost has the total distributed cost from the freight order.

Example 6:

You specify the cost pull strategy as **Aggregate** and the source charge as **Internal** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order in a standard line under the cost charge line.
- The standard line has the calculated amount and final amount as USD 2240, which is the sum of haulage charges and loading charges (USD 1680 + USD 560).
- As you've specified the source charge as **Internal**, the system pulls the costs from the internal cost distribution of the freight order.
- As you've specified the cost pull strategy as **Aggregate**, the pulled execution cost has the total distributed cost from the freight order.

Example 7:

You specify the cost pull strategy as **Aggregate** and the source charge as **Internal & External** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order in a standard line under the cost charge line.
- The standard line has the calculated amount and final amount as USD 2910.89, which is the sum of loading charge, basic charge, haulage charge, and loading charge (USD 560 + USD 110.89 + USD 1680 + USD 560).
- As you've specified the source charge as **Internal & External**, the system pulls the costs from the internal and external cost distribution of the freight order.
- As you've specified the cost pull strategy as **Aggregate**, the pulled execution cost has the total distributed cost from the freight order.

Example 8:

You specify the cost pull strategy as **Aggregate** and the source charge as **Own Internal** in the calculation sheet.

When you calculate charges in the forwarding order 1, the following happens:

- The system calculates the charges and displays them on the **Charges** tab. The system displays the costs pulled from the freight order in a standard line under the cost charge line.
- The standard line has the calculated amount and final amount as USD 2910.89, which is the sum of loading charge, basic charge, haulage charge, and loading charge (USD 560 + USD 110.89 + USD 1680 + USD 560).
- As you've specified the source charge as **Own Internal**, the system pulls the costs from the **Internal Charges** tab page to the **Charges** tab page within the same forwarding order.
- As you've specified the cost pull strategy as **Aggregate**, the pulled execution cost has the total distributed cost from the freight order.

i Note

- You can use the own internal source charge only for external charge calculation in a forwarding order.
- The own internal source charge works irrespective of the source of internal charges in a forwarding order.

That is, you can use own internal source charge even when the internal charges aren't based on the cost pull scenario.

Rate Split

Use

After calculating transportation charges, you can split the charge lines in the freight booking for air waybill printing purposes.

Integration

This function is used in the air freight charge calculation process. For more information, see [Air Freight Charge Calculation](#).

Features

The **Air Waybill View** displays only the weight-based charge lines under the air waybill charge item. You can split weight-based charge lines under the air waybill charge item.

When you insert a split line, the system copies certain parameters from the main charge line. For the inserted charge line, you can enter values for the following attributes:

- Rate category
- Chargeable weight
- Gross weight
- Number of pieces
- Commodity code
- Reference percentage of class rate

i Note

You can only edit the chargeable weight and not the number of pieces and gross weight information in the main charge line. Also, you can only insert a maximum of eleven split charge lines.

The system subtracts the logistical data values of the inserted charge lines from the corresponding logistical values of the main charge line.

A summary line displays the total values of quantity, gross weight, and chargeable weight of all the charge lines under the air waybill charge item. A header line displays the initial values of quantity, gross weight, and chargeable weight from the logistical data and charges data.

In the **Air Waybill View**, you can choose to switch from ULD rate category to general cargo rate category or specific commodity rate category. When you do so, the system refers to the logistical data of loose cargo. Note that you cannot revert back to the ULD rate category after you switch to the general cargo rate category or specific commodity rate category. However, if you use the **Reset** button in the **Air Waybill View**, the system reverts back to the original copy and resets all the changes you have made in the **Air Waybill View**.

Activities

You can navigate to the **Air Waybill View** from the **View** button on the **Charges** tab page to split charge lines.

More Information

[Charge Calculation](#)

Air Freight Charge Calculation

Use

You can use Transportation Management (TM) to calculate air freight charges based on the following:

- TACT rates published by the International Air Transport Association (IATA)

Air freight transportation charges are calculated based on TACT rules and regulations. For more information, see [TACT Rates](#).

- Contract rates available in standard rate tables in TM

For more information, see [Rate Table](#).

Prerequisites

- You've defined a calculation profile with the calculation level at header level in Customizing for **Transportation Management** under ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Define Calculation Profiles** ►.
- You've defined a settlement profile with the calculation option as **Copy All Charges** **Transportation Management** under ► **Settlement** ► **Define Settlement Profiles** ►.
- You've assigned the calculation profile and settlement profile defined above to an organizational unit in Customizing for **Transportation Management in Customizing for** under ► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Define Charges Profiles** ►.

- You've defined the transportation mode category as **Air** for the charge types you want to use for air freight.

You define charge types in Customizing for **Transportation Management** under ► **in Customizing for Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Define Charge Types** ►.

- You have made the master data settings for the Charge Management and Service Product Catalogs component (see [Charge Management Settings for Air Freight](#)).

Process

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

1. You trigger charge calculation for the air freight booking from the Freight Order Management component of Transportation Management (TM).

For more information, see [Management of Air Freight Bookings for Airlines](#).

The system determines the air waybill (AWB) printing and settlement based on TACT rates or contract rates, depending on the calculation profile settings. If the system doesn't determine the AWB printing and settlement, you can also manually enter the values on the **Charges** tab page of the air freight booking. The system creates two charge items, one for AWB printing and the other for the settlement with the carrier. The system prints the charge calculation details, based on the settings that you maintain on the **Charges** tab page of the air freight booking for printing freight charges and other charges.

i Note

Based on the calculation sheet settings, the system displays the IATA charge code and the charge due for other charges.

The system displays the additional charges and commission only under the settlement charge item. Since these charges aren't printed on the AWB, these charges don't appear in the AWB charge item.

You can enter a description for the nature of goods, which is printed in the AWB.

The system determines whether the payment is on the import side or export side based on the charge type classification and the freight term.

2. The system either activates or deactivates the charges based on the traffic direction, charge due, and payer, for example:
 - Due agent charges are activated on the export freight booking irrespective of the payer.
 - Due carrier prepaid charges are active on the export freight booking and inactive on the import freight booking.
 - Due carrier collect charges are deactivated on the export freight booking and activated on the import freight booking.
3. Depending on whether you're using TACT rates or contract rates, the process continues as follows:

Rate Type	Next Steps
TACT	a. The system chooses the unit load device (ULD) rate table, specific commodity rate (SCR) table, or general cargo rate (GCR) table based on the rate category you specify on the Charges tab page of the freight order. The system determines the rate class based on the rate category and chargeable weight. Note that the system determines the TACT rates based on chargeable weights only. If you specify the rate category as a class rate, you can specify either a percentage reduction or a percentage surcharge to a reference rate class depending on the type of goods.

Rate Type	Next Steps
	If you use no rate lookup, rates are not determined or calculated for the charge line and you can maintain blank or zero value for the chargeable weight. b. If you don't specify a rate category, the system follows the access sequence to determine the charges. Depending on the logistical item type, the system determines the economical rate. For ULDs, the access sequence is as follows: i. ULD ii. SCR iii. GCR For packages and loose cargo, the access sequence is as follows: i. SCR ii. GCR Note that the system follows the access sequence only if you maintain the resolution base as Main_Item. If the system chooses a GCR, the system compares the calculated charge with the minimum charge. If the calculated charge is less than the minimum charge, the system chooses the minimum charge. Note that for the system to choose an SCR rate table, you must have maintained the commodity item code on the Capacity and Cargo tab page of the air freight booking.
Contract	a. You manually define the access sequence in the calculation sheet using the Line Item Selection instruction type. For more information, see Calculation Sheet. b. You can't specify the rate category since the system determines the rate category based on the charge calculation. As a result, rate class is also an output of charge calculation. c. If there are special handling requirements for the shipment, you can specify a handling code in the air freight booking. For example, handling code DGR specifies dangerous goods. You can calculate the charges for charge types based on specific handling codes. If there are multiple handling codes and corresponding rates in the same air freight booking, the system displays all the available rates. You can manually choose a handling code with which you want to charge the customer.

i Note

You can manually insert a charge line. For a manually-inserted charge line, you can enter the rate category, rate class, rates, calculation rule, and chargeable weight. If you've specified the rate class as **B** rate or **M** rate, the calculated amount is the same as the rate you've specified. The system doesn't calculate the charges based on the chargeable weight.

4. You specify the calculation method as either [Air Freight: Standard Rating](#) or [Air Freight: Break-Weight Rating](#).

Based on the calculation method you specify, the system uses the break-weight or standard rate determination process as appropriate. Note that for ULD rates, the standard rate determination process supports break-weight calculation methods. For more information, see [Calculation Methods](#).

i Note

If you selected to group the charge types, the system consolidates multiple packages and ULDs, based on the following criteria.

The system consolidates multiple packages if the packages have the following attributes in common:

- Commodity code (valid for SCR)
- Rate category (valid for TACT rates)
- Rate table (valid for contract rates)

The system consolidates multiple ULDs if the ULDs have the following attributes in common:

- Commodity code (valid for SCR and ULD rates)
- Rate category (valid for TACT rates)
- Rate table (valid for contract rates)
- ULD rate type

If the ULDs and packages have the rate category as GCR, the system consolidates multiple packages and ULDs. Note that the system uses the GCR rates for ULDs only when you don't maintain ULD rates. Also, for the system to consolidate packages and ULDs, they must have the following attributes in common:

- Rate category
- ULD rate type
- Commodity code (valid for SCR and ULD rates)

You can group charge types in Customizing for [Transportation Management](#) under [Basic Functions](#) [Charge Calculation](#) [Basic Settings](#) [Define Charge Types](#). If you didn't select this option, the system determines the rates for individual packages and ULDs.

The system determines multiple rates when one or more attributes are missing in the air freight booking. For example, if the note information isn't maintained in the air freight booking and you're maintaining rates for carrier-specific notes, the system determines multiple carrier-specific rates from which you can select the appropriate rate. Note that you can't determine multiple rates for a B rate, a K rate, or a ULD rate. For more information, see [Rate Determination with Partial Information](#).

During contract-based charge calculation, the system can determine the correct rate even when you maintain a rate for a specific weight break that belongs to a combination of same scale. Note that for a combination of same scale values, if you maintain rates for more than one weight break, the system doesn't calculate the correct charges.

❖ Example

You specify the following attributes in a rate table:

Source Location	Destination Location	ULD Rate Type	<=200 kg	>201 kg	<=300 kg	>301 Kg	<=400 kg	>401 Kg
A	D	08	USD 250	USD 1 per kg				
B	D	08			USD 650	USD 2 per kg		
C	D	08					USD 1250	USD 3 per kg

In the FWO, you specify the ULD rate type as 08, source location as B, and the destination location as D for a freight unit of 350 kg.

During charge calculation, the system determines the rate table item for ULD rate type 08, source location B, and destination location D and calculates the transportation charges as USD 700 (350 kg * USD 2). Note that the system displays the charges for the pivot weight and the over-pivot weight in two separate lines. You can view the air freight messages on the [Charge Analysis](#) tab page.

5. If the main item in the cargo is a ULD, the system calculates the chargeable weight by comparing the net weight with the dimensional weight. If the main item in the cargo isn't a ULD, the system calculates the chargeable weight by comparing the gross weight with the dimensional weight.

The system chooses the highest weight as the chargeable weight.

You maintain the gross weight and gross volume on the [Capacity and Cargo](#) tab page of the air freight booking. The system calculates the dimensional weight using the gross volume and the dimensional weight factor maintained in the dimensional weight profile. You maintain a dimensional weight profile either in the calculation sheet, agreement, or calculation profile. You can also modify the chargeable weight manually. If the chargeable quantity is less than the first weight break of 1 kg, the system applies the rate for 1 kg and calculates the charges. Note that if you modify the chargeable weight, the system recalculates the charges.

You define dimensional weight profiles in Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Data Source Binding](#) > [Define Dimensional Weight Profiles](#).

6. The system rounds the chargeable weight based on the rounding profile you maintained in the rate table.

The system rounds the dimensional weight in each charge line before calculating the total charges. Dimensional weight rounding is based on the rounding rule you specify in the dimensional weight profile.

i Note

The air freight charge calculation works in a different manner in the following scenarios:

- Direct Shipments

For direct shipments that aren't consolidated and have to be forwarded as soon as possible, you can charge the customer based on the AWB printing charges in the air freight booking.

Using the `air_cost` calculation method, the system can copy the AWB printing charges in the air freight booking as the settlement amount for the main carriage in the forwarding order. The system deactivates the sub-charge lines in the forwarding order and uses only the total charge for settlement. Note that you can determine the AWB printing charges irrespective of whether the printing calculation is TACT-based or contract-based. For more information, see [Air Waybill Direct Cost Pull](#).

- Air freight bookings with general sales agents (GSAs)

If the airline (issuing carrier) has a GSA operating at the source location, you can calculate the charges based on the agreement with the GSA. You can maintain a freight agreement with the general sales agent (GSA) specifying the GSA as the business partner in the freight agreement. In the [Items](#) screen area, you can specify the airline code that the GSA represents. In the freight booking, on the [Booking](#) tab page, you can specify the airline code that the GSA represents, and you can specify the local carrier as GSA in the [Issuing Carrier Airline Code/Local Carrier](#) field.

During charge calculation, the system determines a rate based on the agreement with the GSA. Note that if you haven't maintained an agreement with GSA, the system determines the agreement with the airline.

- Charge calculation based on volumetric weight factor

Sometimes, lightweight packages and unit load devices (ULDs) occupy large volumes of space in an aircraft. Therefore, the space in the aircraft becomes fully occupied even before reaching its maximum weight capacity. In such cases, the system calculates the charges based on volumetric weight factor. The volumetric weight factor is calculated using the following formula:

Volumetric weight factor = Volume/Weight

You can maintain rates for the volumetric weight factor in cubic centimeter per kilogram (ccm/kg) in the rate table.

i Note

- Before calculating the volumetric weight factor, the system converts the unit of measure (UoM) for volume into ccm and the UoM for weight into kg if the volume and weight are specified in any other UoM.
- To enable the system to calculate the volumetric weight factor, you must specify the volume of the package or ULD in the order.
- For ULDs, the system uses the net weight to calculate the volumetric weight factor. For packages, the system uses the gross weight to calculate the volumetric weight factor.

More Information

[Standard Charge Calculation](#)

[Forwarding Order Management](#)

Air Waybill Direct Cost Pull

Use

Transportation Management enables you to use execution costs as a basis for settling with your customers or internal organization. In this scenario, you can transfer the charge lines from the air waybill charge item to the forwarding side. You can charge customers for total or partial execution charges and additional charges using the cost pull calculation method.

Integration

The system pulls the costs from the air waybill to the forwarding order during the air freight charge calculation process. For more information, see [Air Freight Charge Calculation](#).

For more information on standard cost pull feature, see [Cost Pull](#).

Prerequisites

You have made the master data settings for the Charge Management and Service Product Catalogs component (see [Charge Management Settings for Air Freight](#)).

Features

You can copy charge lines from the air waybill charge item to the forwarding order as either active or inactive charge lines. Copying charge lines as active enables you to edit charge lines in the forwarding order. You specify the cost pull strategy as **Active Copy** or **Inactive Copy** in the calculation sheet or the local calculation sheet.

i Note

When you specify the cost pull strategy as **Active Copy**, the system enables the **Source Charge** field. For the system to copy individual charge lines from the air waybill charge item to the forwarding order, you must use air waybill as the source charge.

Also, when you use air waybill as the source charge, the system considers the following freight order attributes to calculate the final amount in the forwarding order:

- Calculation rule
- Calculation base
- Calculated amount

If you don't specify a value in the source charge field, the system copies the distributed charge lines in the freight order to the forwarding order.

Based on the freight term and the Incoterm you specify in the forwarding order, the system settles the charges to the shipper, ship-to party, import business unit, or export business unit as appropriate.

Based on the freight term, you determine whether the payment is on the import side or export side. Based on the Incoterm, you identify who pays for the individual stages of a transport.

Example

You have a forwarding order with the Incoterm EXW (Ex Works). According to EXW, the ship-to party pays for all the stages in the forwarding order. However, the precarriage and the main carriage stages are executed by the export business unit on behalf of the import business unit. So, there's an internal settlement between the export business unit and the import business unit.

Based on the freight term you specify in the forwarding order, one of the following happens:

- Freight term PP

If you specify the freight term as **PP**, all the charges are prepaid in both the export freight booking and the import freight booking. Prepaid charges are active in the export freight booking and collect charges are active in the import freight booking.

Export Freight Booking

Charge Type	Amount	Freight Term	Status
FB00	100	Prepaid	Active
SCC1	20	Prepaid	Active
DOCF	30	Prepaid	Active

Import Freight Booking

Charge Type	Amount	Freight Term	Status
FB00	100	Prepaid	Inactive
SCC1	20	Prepaid	Inactive
DOCF	30	Prepaid	Inactive

During charge calculation, the following happens:

1. The charge types in the export freight booking are pulled into the export forwarding order and the export business unit internally charges the import business unit 150 USD for the prepaid charges.
2. The charges in the import freight booking are pulled into the import forwarding order and the import business unit charges the ship-to party 150 USD.

- Freight term CC

If you specify the freight term as **CC**, all the charges are collect in both the export freight booking and the import freight booking. Collect charges are active in the import freight booking.

Export Freight Booking

Charge Type	Amount	Freight Term	Status
FB00	100	Collect	Inactive
SCC1	20	Collect	Inactive
DOCF	30	Collect	Inactive

Import Freight Booking

Charge Type	Amount	Freight Term	Status
FB00	100	Collect	Active
SCC1	20	Collect	Active
DOCF	30	Collect	Active

The system doesn't calculate the charges in the export forwarding order since all the charges are collect in the export freight booking.

During charge calculation in the import forwarding order, the charges in the import freight booking are pulled into the import forwarding order and the import business unit charges the ship-to party 150 USD.

- Freight term PC

If you specify the freight term as **PC**, the weight charges are prepaid and the other charges are collect in both the export freight booking and the import freight booking. Prepaid charges are active in the export freight booking and collect charges are active in the import freight booking.

Export Freight Booking

Charge Type	Amount	Freight Term	Status
FB00	100	Prepaid	Active
SCC1	20	Collect	Inactive
DOCF	30	Collect	Inactive

Import Freight Booking

Charge Type	Amount	Freight Term	Status
FB00	100	Prepaid	Inactive
SCC1	20	Collect	Active
DOCF	30	Collect	Active

During charge calculation, the following happens:

1. The charge types in the export freight booking are pulled into the export forwarding order and the export business unit internally charges the import business unit 100 USD for the prepaid charges.
2. The charges in the import freight booking are pulled into the import forwarding order and the import business unit charges the ship-to party 150 USD.

Fuel Surcharge Calculation

Due to uncertainties in fuel prices, it is significant to charge the customer appropriately based on the fuel surcharge index published by the government. As a logistics service provider or carrier, you need to recover, the additional charges incurred due to varying fuel costs. Transportation Management simplifies the fuel surcharge calculation by providing an easy-to-maintain fuel surcharge calculation method.

You can determine fuel surcharge based on two different models. For more information, see [Fuel Surcharge Calculation: Model 1](#) and [Fuel Surcharge Calculation: Model 2](#).

Fuel Surcharge Calculation: Model 1

Use

You can calculate the fuel surcharge based on a fuel surcharge model widely used in North America. In this model, you calculate the fuel surcharge based on the fuel surcharge index published by the government. The fuel surcharge index is based on the origin location or zones.

Prerequisites

- You've defined a calculation profile in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Calculation Profiles**.
- You've assigned the calculation profile to a charges profile in Customizing for **Transportation Management** under **Basic Functions** > **Charge Calculation** > **Basic Settings** > **Define Charges Profiles**.
- You've assigned the charges profile to an organizational unit in the transaction PPOME.

Process

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

To calculate the fuel surcharge, you must maintain two rate tables; a standard rate table with the fuel surcharge rates for corresponding fuel surcharge values, and an index rate table with the location, dates or validity, and the respective fuel surcharge

values. You proceed as follows:

1. Define an index base rate table.

Define an index rate table with the fuel surcharge values based on scale items such as location and calculation date. You can define a calculation rule on the [Calculation Rules](#) tab page.

2. Define a standard rate table.

Define a standard rate table with the calculation base as **FSC** and scale type as either **Base Scale** or **To Scale**. Define the rates for fuel surcharge values.

i Note

You define rate tables in the [Create Rate Table Definition](#) app on the launchpad. For more information, see [Rate Table](#).

3. Define a calculation sheet with a charge item for fuel surcharge and define the following attributes:

- Enter the charge type as **Fuel**.
- Enter the standard rate table in the **Rate Table** field and the index rate table in the **Index Rate Table** field.
- Enter the calculation method type as C and calculation method as FUEL_SURCHARGE in the **Basic Data** tab page.

i Note

You define calculation sheets in the [Create Calculation Sheet](#) app on the launchpad. For more information, see [Calculation Sheet](#).

4. Trigger charge calculation.

You can trigger charge calculation from a forwarding order or freight order.

5. The system determines the fuel surcharge and calculates the transportation charges along with the fuel surcharge.

Based on the location and date specified in the forwarding order or freight order, the system determines a fuel surcharge value from the index rate table. The system then uses this fuel surcharge value to determine a rate from the standard rate table.

Example

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You maintain an index rate table with the following fuel surcharge values:

Calculation Date	Source Location: A	Source Location: B	Source Location: C
09-05-2013	3.12	3.99	3.45
09-12-2013	3.24	3.78	3.65
09-19-2013	3.15	4.12	3.55

You maintain the scale type as **To Scale**.

You maintain a standard rate table with the following rates for fuel surcharge values:

Fuel Surcharge Value	Rate (USD)
3.12	0.33
3.15	0.35
3.24	0.41
3.42	0.43
3.55	0.47
3.62	0.49
3.75	0.53
3.97	0.62
4.10	0.68

You have a calculation profile with the calculation date type as **Order Date**. You have a forwarding order with the **Order Date** as **09.10.2013** and the **Source Location** as **B**.

As you've specified the calculation date as **To Scale** (less than or equal to), the system determines a fuel surcharge value of 3.78, which you've maintained for the date 09.12.2013 for location B.

Now, the system determines a rate for the fuel surcharge value 3.78 from the standard rate table. If you maintain the scale type as **Base Value** (greater than or equal to) in the standard rate table, the system picks the rate of 0.53, which is maintained for 3.75 (as 3.78 is ≥ 3.75) to calculate fuel surcharge. If you've maintained the scale type as **To Value** (less than or equal to) in the standard rate table, the system picks the rate of 0.62, which is maintained for 3.97 (as 3.78 is ≤ 3.97) to calculate fuel surcharge.

More Information

[Fuel Surcharge Calculation: Model 2](#)

[Rate Definition and Maintenance](#)

[Agreement Maintenance](#)

Fuel Surcharge Calculation: Model 2

Use

You can calculate the fuel surcharge based on a base fuel surcharge value. The logistic service provider or carrier agrees with the customer on an index base date based on which the system determines the base fuel surcharge value. The actual fuel surcharge is calculated based on the current fuel surcharge value and base fuel surcharge value.

Prerequisites

- You've defined a calculation profile in Customizing for **Transportation Management** under **Basic Functions** **Charge Calculation** **Basic Settings** **Define Calculation Profiles**.
- You've assigned the calculation profile to a charges profile in Customizing for **Transportation Management** under **Basic Functions** **Charge Calculation** **Basic Settings** **Define Charges Profiles**.

- You've assigned the charges profile to an organizational unit in the transaction PPOME.

Process

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

To calculate the fuel surcharge, you define an index base rate table with the fuel surcharge values for specific calculation dates or validities. Optionally, you can define a standard rate table with the fuel surcharge values for scale items such as location and calculation dates or validity periods. You proceed as follows:

1. Define an index base rate table.

Define an index rate table with the calculation base as **Calculation Date**, scale type as **To Scale** or **Base Scale**, and the value type as **Absolute** or **Percentage**. Note that you must not maintain a calculation rule in this rate table.

i Note

You define rate tables in the **Create Rate Table Definition** app on the launchpad. For more information, see [Rate Table](#).

2. Optionally, you can define a standard rate table.

Define a standard rate table with a calculation base and scale type as either **Base Scale** or **To Scale**. Define the fuel surcharge values for scale items such as source location and destination location.

i Note

If you don't define a standard rate table, the system determines the fuel surcharge using the base rate specified in the calculation sheet.

3. Define a calculation sheet with a charge item for fuel surcharge and define the following attributes:

- Enter the charge type as **Fuel**.
- Enter the index rate table in the **Index Rate Table** field.
- Enter an index base date in the **Index Base Date** field.
- If you've defined a standard rate table, enter it in the **Rate Table** field. Otherwise, you must enter a rate in the **Base Rate** field as a percentage or absolute value.
- Enter the calculation method type as C and calculation method as FUEL_SURCHARGE on the **Basic Data** tab page.

i Note

You define calculation sheets in the **Create Calculation Sheet** app on the launchpad. For more information, see [Calculation Sheet](#).

4. You trigger charge calculation.

You can trigger charge calculation from a forwarding order or freight order.

5. The system determines the current fuel surcharge value.

Based on the calculation date specified in the forwarding order or freight order, the system finds the current fuel surcharge value from the index rate table.

6. The system determines the base fuel surcharge value.

The system finds the fuel surcharge value for the index base date specified in the calculation sheet from the index rate table.

Note that for the system to determine the base fuel surcharge value, you must specify an index base date in the **Index Base Date** field and specify a fuel surcharge value for the index base date in the index base rate table.

7. The system determines the fuel surcharge and calculates the transportation charges along with the fuel surcharge.

If you don't define a standard rate table, the system calculates the fuel surcharge using the following formula:

Fuel surcharge in percentage = (Current fuel surcharge value/Base fuel surcharge value)* Base rate in calculation sheet

If you define a standard rate table, the system determines a fuel surcharge value for the scale items in the forwarding order or freight order and calculates the fuel surcharge using the following formula:

Fuel surcharge in percentage = (Current fuel surcharge value/Base fuel surcharge value)* Fuel surcharge value from the standard rate table

Example

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You maintain an index rate table with the following fuel surcharge values:

Calculation Date	Fuel Surcharge Value
09-05-2013	3.45
09-12-2013	3.65
09-19-2013	3.55
09-25-2013	3.82
09-30-2013	3.91

You maintain the scale type as **To Scale**.

You maintain a base rate of 1.19% in the calculation sheet.

You have a calculation profile with the calculation date type as **Order Date** and in the forwarding order, the order date is 09.18.2013 and the **Index Base Date** maintained in the calculation sheet is 09.28.2013.

The system determines the current fuel surcharge using the order date. The order date 09.18.2013 occurs earlier than 09.19.2013 in the index rate table. Hence, the value 3.55 is determined as the current fuel surcharge value.

The system determines the base fuel surcharge using the index base date of 09.28.2013. The index base date 09.28.2013 occurs earlier than 09.30.2013 in the index rate table. Hence, the value 3.91 is determined as the base fuel surcharge value.

The fuel surcharge is calculated as follows:

Fuel surcharge in percentage = (Current fuel surcharge value/Base fuel surcharge value)* Base rate in calculation sheet

Fuel surcharge in percentage = (3.55/3.91)*1.19 = 1.08

Alternatively, you can also use standard rate tables to calculate the fuel surcharge. You maintain a standard rate table as follows:

Source Location	Destination Location	Fuel Surcharge Value
A	B	3.12
B	C	3.15
C	D	3.24
D	E	3.42

Once the system determines the current fuel surcharge value and the index fuel surcharge value as mentioned above, the system determines the fuel surcharge value for the source location and destination location specified in the forwarding order. In the forwarding order, if you've maintained the source location as **B** and the destination location as **C**, the system determines the fuel surcharge value of 3.15 from the standard rate table.

The fuel surcharge is calculated as follows:

Fuel surcharge in percentage = (Current fuel surcharge value/Base fuel surcharge value)* Fuel surcharge value from standard rate table

Fuel surcharge in percentage = (3.55/3.91)*3.15 = 2.86

More Information

[Fuel Surcharge Calculation: Model 1](#)

[Rate Definition and Maintenance](#)

[Agreement Maintenance](#)

Event-Based Charge Calculation

Use

Transportation Management (TM) enables you to coordinate between your transportation planning and the actual execution by exchanging information with SAP Event Management. You can track various events that occur during transportation execution, such as delays, arrivals, and departures, and enables you to charge the customer appropriately.

With the help of SAP Event Management, you can also charge the customer for additional services, such as fumigation and container cleaning by maintaining events for these services. The system also notifies the progress of the additional services.

Integration

TM is integrated with SAP Event Management to coordinate between the transportation planning and the actual execution. For more information, see Help Portal at <http://help.sap.com/em>.

Prerequisites

You've specified the data source as **Actual Route**. You specify the data source in Customizing for Transportation Management under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

Event Profile for Charge Calculation

You can control the event handler through the event profile. An event handler represents business processes or business objects (for example, purchase order, shipment, production order, or handling unit) that are to be tracked within the supply chain by SAP Event Management. An event handler contains expected, reported, or unexpected events and attributes to identify and track the related business processes. With the event profile, you can configure events for charge calculation. You can also define the following rules based on which the system performs the charge calculation in the forwarding order or freight order:

- Event code
- Event reason
- Event status
- Reference event code
- Reference event reason
- Reference event status
- Business documents in which the event needs to be charged

For example, forwarding orders, freight orders, internal forwarding orders, and internal freight orders in which the event needs to be charged.

You can also link service types to the events, which enable you to bill the customer based on services. If you assign a specific service type to an event code, the system charges the customer for the service when the assigned event code occurs in the forwarding order or freight order.

You can also maintain different event profiles for the customer and carrier by assigning different event profiles to the forwarding order type, freight order type, or freight booking type.

i Note

You can also report events at the item level in a freight order. For example, during buyer's consolidation (consolidation of multiple forwarding orders), you can report an event for one of the containers and the system calculates the charges for the event only on related forwarding order that has the container.

Related Calculation Bases

In a calculation sheet, if you enter a calculation rule based on a calculation base that has related calculation bases, you can directly enter values for the related calculation bases on the [Related Calculation Bases](#) tab page.

For example, to calculate charges in the event of demurrage, for the calculation base `delay_days`, you maintain a related calculation base `grace_days` in Customizing for Transportation Management under [Basic Functions > Charge Calculation > Data Source Binding > Define Calculation Bases](#). In the calculation sheet, you can specify the number of grace days allowed to the customer.

You can also specify your own calculation rule and charge type using the BRFplus (Business Rule Framework plus) condition. You can assign the BRF plus rules to the calculation base. The system processes the condition for all the data access definitions based on the condition you specify. For more information, see [BRFplus for Charge Management and SP Catalogs](#).

Activities

You define event profiles for charge calculation in Customizing for **Transportation Management** under ►► **Basic Functions** ► **Charge Calculation** ► **Basic Settings** ► **Define Charges Profiles** ►. You maintain the event type, event reason, and charge type on the **Define Event Assignment to Charge Type** tab page of the **Define Event Profiles for Charge Calculation** Customizing.

You assign event profiles for charge calculation to forwarding order type in Customizing for **Transportation Management** under ►► **Forwarding Order Management** ► **Forwarding Order** ► **Define Forwarding Order Types** ►.

You assign event profiles for charge calculation to freight order type in Customizing for **Transportation Management** under ►► **Freight Order Management** ► **Freight Order** ► **Define Freight Order Types** ►.

You assign event profiles for charge calculation to freight booking type in Customizing for **Transportation Management** under ►► **Freight Order Management** ► **Freight Booking** ► **Define Freight Booking Types** ►.

You define calculation bases and the related calculation bases in Customizing for **Transportation Management** under ►► **Basic Functions** ► **Charge Calculation** ► **Data Source Binding** ► **Define Calculation Bases** ►.

Example

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

To calculate demurrage charges, you perform the following:

1. You define the calculation base (for example, `delay_days`) and the related calculation base (for example, `grace_days`) in the Customizing activity **Define Calculation Bases**.
2. You maintain a calculation sheet with a calculation rule (for example, `delay_days`) and a charge type (for example, `delay`) on the fiori launchpad. `Delay_days` is calculated by the following formula:

$$\text{Delay days} = \text{Actual data} - \text{Planned data} - \text{Grace days}$$

In the calculation sheet, you can define the number of grace days provided to the customer on the **Related Calculation Bases** tab page.
3. You define event profiles with a charge type (for example, `delay/demurrage`) in the Customizing activity **Define Event Profiles for Charge Calculation**.
4. The system calculates the demurrage charges based on the calculation rule and the rates you specify in the calculation sheet.

Calculation of Charges for Empty Provisioning and Empty Return

Use

In **Transportation Management**, you can request provisioning of empty containers or railcars for providing transportation service in the future. You can also request for return of the empty containers or railcars. You can then calculate the charges for empty provisioning, empty return, or empty provisioning with a transportation service.

Prerequisites

- In Customizing, you have carried out the activities for Transportation Management under ► **Basic Functions** ► **Charge Calculation** ►.

- You have set up the master data for the Charge Management and Service Product Catalogs component.

For more information, see [Setup of MD for Charge Management and SP Catalogs](#).

- You have specified the resolution base as **CONTAINER** for container items so the system can calculate charges at header level.

Features

Note that the following information is relevant whether the calculation sheets, rate tables, or scales are created locally within an agreement, calculation sheet, or rate table or are created independently of an agreement, calculation sheet, or rate table.

You request empty provisioning or empty return for a container in the forwarding order and specify the charge types for empty provisioning and empty return in the calculation sheet.

The system compares the empty provisioning and empty return values specified in the forwarding order and calculation sheet, based on which the system calculates the charges.

Empty provisioning can have one of the following values in the forwarding order and calculation sheet:

- **Requested (X)**
- **Not Requested**
- **Provisioning Only (P)**

Similarly, empty return can have one of the following values in the forwarding order and calculation sheet:

- **Requested (X)**
- **Not Requested**
- **Return Only (R)**

When you create a request for empty provisioning or empty return with the value as **Requested (X)**, the system creates a technical item for either empty provisioning (P) or empty return (R) as appropriate in the back end. The system creates a technical item to include empty provisioning charges or empty return charges along with the freight charges.

You can agree with the shipper on whether you will arrange for the return of containers or you require the shipper to return the empty containers. Charge calculation varies based on the agreement.

Example

You have an empty provisioning request. In the forwarding order, you specify the following values for provisioning:

Charge Item	Empty Provisioning
10	Requested (X)
80	Provisioning Only (P, technical)
30	Not Requested
40	Provisioning Only (P)

Note that the system creates a technical item (P) in the back end for provisioning requested (X).

In the calculation sheet or the local calculation sheet, you specify the following values for the following charge types:

Charge Type	Description	Empty Provisioning
C1	Basic freight charges	Not Requested
C2	Freight charges and provisioning charges	Requested (X)
C3	Provisioning only charges	Provisioning Only (P)

When you calculate the charges at header level with the resolution base as **CONTAINER**, the system applies the following charges:

Charge Element	Charges	Charge Item
CH1		
	C1	10
	C1	30
	C2	10
	C3	80
	C3	40

The system applies basic freight charges (C1) to the containers when the empty provisioning is **Requested (X)** and **Not Requested**.

The system applies basic freight charges and the provisioning charges (C2) to the containers when the empty provisioning is **Requested (X)**.

The system applies provisioning charges (C3) to the containers when the empty provisioning is **Provisioning Only (R)** and the technical item (P) that is created in the back end for provisioning requested.

More Information

[Empty Provisioning and Empty Return](#)

Additional Agreement Parties

You can calculate transportation charges for each of the additional agreement parties in a freight order or freight booking.

Use

In a freight order or freight booking, along with the main carrier, you use additional agreement parties for other logistics services such as export clearance and fumigation of containers.

When you calculate charges for a freight order or freight booking with additional agreement parties, the system calculates the charges based on the valid agreement between the organization and the main carrier and between the organization and the additional agreement parties.

You can calculate charges at header, item, or stage level, depending on your Customizing settings in the charges or calculation profile assigned to the organization, main carrier, or additional agreement parties.

However, if you want to calculate the charges for the main carrier as well as any additional agreement party only at the header level you've to do the following settings in the Customizing for **Transportation Management**. Under **Basic Functions** > **Define General Settings for TM** , select the checkbox **Header Level Charge Calc. in Freight document with Add. Agreement Party**.

i Note

- In the **Business Partner** tab of the freight order or freight booking, under **Partner Function** you add your additional agreement parties as **Addtnl. Agrmt. Party**.
- You add the main carrier in the **General Data** tab of the freight order or freight booking.
- The system doesn't support quotations, carrier selection, and tendering for additional agreement parties.

Example

In the following example, you want to transport food items in a container from Rotterdam to New York. The main carrier is CARR01 and the additional agreement party for import/export clearance is AAP01. You pay CARR01 for the transportation service and AAP01 for the import/export clearance.

The setup for the cargo and service items is:

Item Hierarchy			Item Type	Quantity	Quantity UoM	Service Type
Active Vehicle 10			TRUC	1	EA	
	Container 1000		CONT	1	EA	
		Packages	PKG	10	EA	
Service Item	Custom Clearance					Custom Service

Following are the charges in the calculation sheet of the valid freight agreement between the main carrier CAR001 and the business organization.

Item Hierarchy	Line No.	Charge Type	Amount	Currency	Rate Table
Base Freight	10	BASE_FRT	100	EUR	RT_01
Fuel Surcharge	20	FSC	10	% of Base Freight	

Following are the charges in the calculation sheet of the valid freight agreement between the additional agreement party AAP01 and the business organization.

Item Hierarchy	Line No.	Charge Type	Amount	Currency	Rate Table	Calculation Resolution Base
Custom Fee	30	CUSTOM_FEE		EUR	RT_01	Service

Header level charge calculation

When you calculate charges at the header level, then the charges for the entire shipment for both the main carrier and additional agreement party are calculated at header.

Charge Hierarchy	Charge Type	Rate Amount	Rate Currency	Final Amount	Doc Currency
CARR01				110	EUR
	BASE_FRT	100	EUR	100	EUR
	FSC	10	EUR	10	EUR
AAP01				50	EUR
	CUSTOM_FEE	50	EUR	50	EUR

Related Information

[Charge Calculation](#)

[Freight Settlement](#)

[Features of Dispute Management](#)

[Freight Settlement for Additional Agreement Parties](#)

[Dispute Management with Additional Agreement Parties](#)

Freight Settlement for Additional Agreement Parties

You can create a freight settlement document (FSD) for each of the additional agreement parties that are contained in a freight order or freight booking.

Use

When you create a freight settlement document from the freight order or freight booking, the system displays a dialog box if additional agreement parties exist in the freight order or freight booking. You can select one or more parties and continue with the settlement process.

i Note

You can't enter the details of the services provided by an additional agreement party in a freight order or freight booking. Instead, you can use the charge type in the freight agreement to record the service.

If you've overlapping services between the carrier and an additional agreement party, the system issues an FSD for the overlapping services to both parties. For example, you've a freight agreement with a carrier in which the carrier provides both transportation and fumigation services. You also have an additional agreement party that provides a fumigation service. You've a freight order in which the carrier provides transportation services and the additional agreement party provides a fumigation service. The system creates an FSD that includes a charge for the fumigation service for both parties. You must manage this situation outside of the system.

If you've more than one calculation sheet to cover different services from the same additional agreement party or carrier, the system can't differentiate between the different calculation sheets. It uses just one calculation sheet when it creates an FSD. For example, you've a carrier that provides both carrier and fumigation services. In your system, you've one calculation sheet for transportation charges and a second calculation sheet for fumigation charges. You've a freight order in which the same carrier provides transportation and fumigation services. The system uses just one calculation sheet to create the FSD, and you can't create an FSD that includes both charges. Again, you must manage this situation outside of the system.

The system doesn't support quotations, carrier selection, and tendering for additional agreement parties.

Related Information

[Charge Calculation](#)

[Freight Settlement](#)

[Features of Dispute Management](#)

[Dispute Management with Additional Agreement Parties](#)

Dispute Management with Additional Agreement Parties

You can maintain multiple disputes in a dispute management scenario, if you specify one or more additional agreement parties in your freight document. In some of the transportation scenarios, you as a requester of transportation services, can have multiple service providers execute a single freight document.

For example, you've a transportation requirement to transport goods from one location to another. You also require additional services such as export clearance and fumigation from a service provider. In Transportation Management (TM), you can capture these requests in a single freight document. You can specify the details of the executing carrier and the additional agreement party who performs the additional services in the freight document.

In such scenarios, you've to maintain multiple disputes with multiple service providers. TM enables you to create and maintain multiple disputes in a single freight document. You can maintain multiple disputes in a freight document that has one or both of the following:

- One or more additional agreement parties
- Multiple invoicing carriers

For more information on freight documents and non-linear stages, see [Freight Orders with Non-Linear Stages](#).

Related Information

[Charge Calculation](#)

[Freight Settlement](#)

[Features of Dispute Management](#)

[Freight Settlement for Additional Agreement Parties](#)

Freight Charges for Service Providers

In freight transportation, the main carrier is responsible to provide the transportation service. Apart from the main carrier, there can be other service providers who are responsible for additional services, such as fumigation, custom clearance, insurance, and so on. These additional services are service items in a freight order or freight booking. For a freight order or freight booking, you can calculate transportation charges for the transportation service by the main carrier and the additional services by the service providers.

Use

For each service item, you can assign a service provider and also can add details about the service location and duration, which are used with other logistical information to determine the rates for specific service charges. The system calculates the charges based

on the valid freight agreements between the main carrier and the business organization and the service provider and business organization.

For service providers, the system can calculate charges at service item or header level. But for main carrier, the system calculates the charges depending on the Customizing settings in the charges profile or calculation profile assigned to the organization or business partner.

For calculating service charges based on the service location and duration, you use calculation bases **SRV_LOCATION** and **SRV_DURATION** in the rate table along with resolution base **SERVICE** in the calculation sheet.

For a freight order or freight booking, you add service provider under the **Items** tab in a freight order or freight booking. Select **Charge Hierarchy** as **Service Items Only**. If you require a service that is provided by a service provider that isn't the carrier, select the **Item Specific Service Provider** checkbox, enter the name of the business partner under **Business Partner**, and select the service type.

For more information, see [Service Items](#).

i Note

- The system doesn't support quotations, carrier selection, and tendering for service providers.
- For service items with no service providers, the service charges are included with main carrier.

Example

You've a freight order with main carrier **CAR001** with the following cargo and service items and service provider **SRV_PR01** assigned to the service items:

Item Hierarchy			Item Type	Quantity	Quantity UoM	Service Type	Business Partner	Item Specific Service Provider
Active Vehicle 10			TRUC	1	EA			
	Container 1000		CONT	1	EA			
		Product 10	PRD	10	EA			
		Service 10	SRV	3	EA	Inspection	SRV_PR01	X
		Service 20	SRV	1	EA	Fumigation		

Following are the charges in the calculation sheet of the freight agreement between the main carrier **CAR001** and the business organization.

Item Hierarchy	Line No.	Charge Type	Amount	Currency	Rate Table	Calculation Resolution Base
Base Freight	10	BASE_FRT	100	EUR		
Fumigation Charges	20	FUMI_CHRG	75	EUR		Service

Following are the charges in the calculation sheet of the freight agreement between the service provider SRV_PR01 and the business organization.

Item Hierarchy	Line No.	Charge Type	Amount	Currency	Rate Table	Calculation Resolution Base
Inspection Charges	30	INSP_CHRG	50	EUR	RT_01	Service

Following are the charge calculation scenarios based on the level of charge calculation:

Header level charge calculation

When you calculate charges at the header level, then the charges for the entire shipment for both the main carrier and service provider are calculated at header level, which is $(175 + (50 \times 3)) = 325$ EUR.

Fumigation charge is calculated for main carrier CAR001 as there's no specific service provider for this additional service. The inspection charges are calculated for the service provider SRV_PR01.

Charge Hierarchy	Charge Type	Rate Amount	Rate Currency	Price Unit	UoM	Quantity	Calculation Base	Final Amount	Doc Currency
CAR001								175	EUR
	Base Freight	100	EUR					100	EUR
	Fumigation Charges	75	EUR					75	EUR
SRV_PR01								150	
	Inspection Charges	50	EUR	1	EA	3	QUANTITY_VAL	150	EUR

Item level charge calculation

When you calculate charges at the item level, then the charges for the entire shipment are broken down for individual charge items. There are three items for the main carrier and one service item for the service provider. For the main carrier, the total amount is $(100+75+100) = 275$ EUR and for the service provider the amount is $(50 \times 3) = 150$ EUR. Thus, the final amount is $(275+150) = 425$ EUR.

Charge Hierarchy		Charge Type	Rate Amount	Rate Currency	Price Unit	UoM	Quantity	Calculation Base	Final Amount	Doc Currency
CAR001									275	EUR
	Container 1000								175	
		Base Freight	100	EUR					100	EUR
		Fumigation Charges	75	EUR					75	EUR

Charge Hierarchy		Charge Type	Rate Amount	Rate Currency	Price Unit	UoM	Quantity	Calculation Base	Final Amount	Doc Currency
	Service 20								100	
		Base Freight	100	EUR					100	EUR
SRV_PR01									150	
		Inspection Charges	50	EUR	1	EA	3	QUANTITY_VAL	150	EUR

Stage level charge calculation

When you calculate charges at the stage level, then the charges for the entire shipment are broken down based on the stages of the main carrier and the service provider. There's one stage for the main carrier, so the total amount is $(100+75) = 175$ EUR and for the service provider charge is always calculated at the header level, so the amount is $(50*3) = 150$ EUR. The final amount is $(175+150) = 325$ EUR.

Charge Hierarchy		Charge Type	Rate Amount	Rate Currency	Price Unit	UoM	Quantity	Calculation Base	Final Amount	Doc Currency
CAR001										
	Loc A to Loc B								175	
		Base Freight	100	EUR					100	EUR
		Fumigation Charges	75	EUR					75	EUR
SRV_PR01									150	
		Inspection Charges	50	EUR	1	EA	3	QUANTITY_VAL	150	EUR

Additional Information

You can also settle freight charges for service providers.

Related Information

[Service Items](#)

[Freight Settlement](#)

Waybill Stock Definition

Use

You can define number stocks for business documents to uniquely identify and track transports. The system enables you to use these stocks to allocate waybill numbers to business documents, such as forwarding orders, freight orders, freight bookings, and freight units.

You can create, edit, and delete waybill stock number ranges for business documents of all modes of transport in using the **Waybill Number Stocks** app, which you can open from the launchpad. Furthermore, you can create waybill stock number ranges without assigning a mode of transport yet.

Prerequisites

You have created number stock types in Customizing for **Transportation Management** under ►►**Master Data** ► **Waybill Stock** ► **Define Waybill Number Stock Types** ►.

If you want to define master air waybill stocks, you have defined airline codes for IATA carriers or non-IATA carriers in Customizing for **Transportation Management** under ►►**Master Data** ► **Business Partners** ► **Define IATA Airline Codes** ►. You can also enter a prefix for the carrier.

To specify a local carrier for a master waybill stock, you have performed the following steps:

1. Go to the **Maintain Business Partner** transaction (transaction BP).
2. Enter the business partner for the local carrier.
3. Enter Carrier in the **Display in BP role** field.
4. On the **Vendor Data** tab page, enter the airline code in the **Airline Codes** screen area.
5. Save your entries.

Features

For business documents with a transportation mode of **Parcel**, you can specify tracking number stocks for multiple purchasing organization and carrier combinations. You can draw a number from this stock if the combination of purchasing organization and carrier in the business document matches a purchasing organization and carrier combination in the tracking number stocks.

For house waybills, you can specify a waybill number range stock for multiple sales organizations and ordering parties. You can draw a number from this stock in a house waybill. When you draw a number in the waybill, the system matches the attributes in the business document with the attributes of the waybill stock until it recognizes a matching attribute. The system checks the attributes of the stock in the following order:

1. Sales organization and ordering party combination
2. Ordering party
3. Sales organization

Note that a forwarding order with a transportation mode of air must contain an air waybill type before you can draw a waybill number.

❖ Example

You have created a waybill stock for sales organization 245006 and another waybill stock for ordering party FL_1. When you draw a waybill number in a business document with sales organization 235006 and ordering party FL_1, the system first searches to see whether there is any existing stock that possesses both these attributes. As it does not find a match, it searches for a stock with the ordering party and draws the number from the stock with ordering party FL_1.

For master waybills, you can specify a stock for multiple purchasing organizations, and for local carriers or global carriers. You can draw a number from this stock in a master waybill if the combination of purchasing organization and carrier in the business document matches the combination of purchasing organization and carrier of the waybill stock.

You must specify the airline code for a master air waybill (MAWB). For master bills of lading with transportation mode of ocean or land, you have the option to specify a SCAC.

❖ Example

You have created a waybill stock for an airline code LUF, a purchasing organization 560011, and a carrier SD_01. When you draw a waybill number in a MAWB, the system compares the airline code, purchasing organization, and ordering party of the business document with the airline code, purchasing organization and ordering party of the stock. If the entities match, it draws a number.

In master waybill stocks, you can delegate a waybill number to a purchasing organization. When you draw a waybill number in a business document with this purchasing organization, the system makes the waybill number that you have delegated available in the document.

❖ Example

A business unit in Frankfurt with purchasing organization FRA001 needs a MAWB number for a MAWB document. A business unit in Berlin can delegate a number to the purchasing organization FRA001. When the Frankfurt business unit draws a waybill number in the MAWB, the system makes the delegated waybill number available in the document.

i Note

If the system does not find a waybill stock or tracking number stock from which you can draw a waybill number or tracking number in a business document, it performs the following steps:

1. Checks the transaction PPOME for parent organizations of the organization you have entered in the business document
2. Traverses through the parent organizations until it finds a parent organization that has a waybill stock.
3. Makes a waybill number from this parent stock available in the business document.

More Information

[Management of Air Freight Bookings for Airlines](#)

[Building and Printing of House Bills of Lading and House Air Way](#)

Create and Update Service Products for Independent Charge Types

You use this report to schedule a background processing function to replicate charge types as service products. You use this report when you first set up your system, when you use the settlement of transportation cost features. You can also use this report to schedule a background processing function when you extend the posting of transportation cost settlements to other company codes and plants. You can also use this report when you choose to use a new charge type in your business for which you may not have created a corresponding service product.

Key Features

This report provides the following key features:

- Scheduling settings to schedule the background processing function
- Parameters that the system uses as a default when it creates service products

Additional Details

This is a prerequisite step for the posting of transportation cost settlement features. You must replicate all the charge types you use in your settlement process as service products. During the posting process, the system creates a purchase order and a service entry sheet in the background. For each of the independent charges in the freight order, the system creates a purchase order item with the service product that is linked to the charge type. The system then posts a service entry sheet for each purchase order item that has a linked service product.

When you schedule the background processing function, you can specify an immediate start or a later start date and time.

You must specify the following parameters:

- Material group
- Purchasing group
- Material type of **Service Product**

The system uses the parameters as the default when it creates service products. You must specify the plant that the system uses when it posts transportation costs. You also specify the plant to which you extend a service product.

You can specify independent charge types. If you don't specify a charge type, by default the system creates the service product for all the independent charge types in the system.

i Note

- If new calculation bases are made available in Transportation Management, there can be new units of measure (UoMs) used for charge calculation quantities in freight documents. To be able to use the new UoMs, you must use this function before starting a settlement process involving the new calculation bases. This ensures the system updates existing service products to include the new UoMs.
- Alternately you can also run the program /SCMTMS/MANAGE_SERV_MAT from the background to perform the same activities that you can do using this report from the frontend.

Supported Device Types

- Desktop
- Tablet